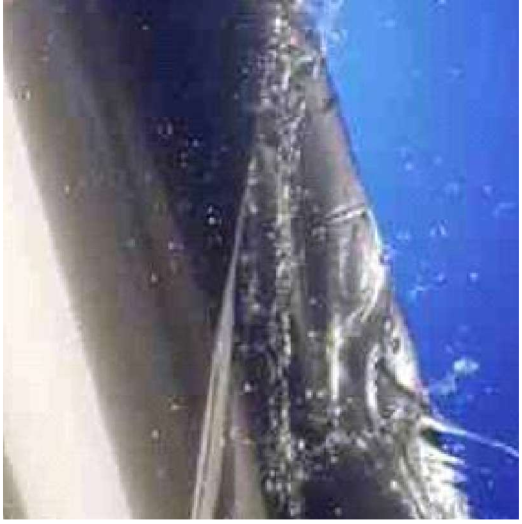
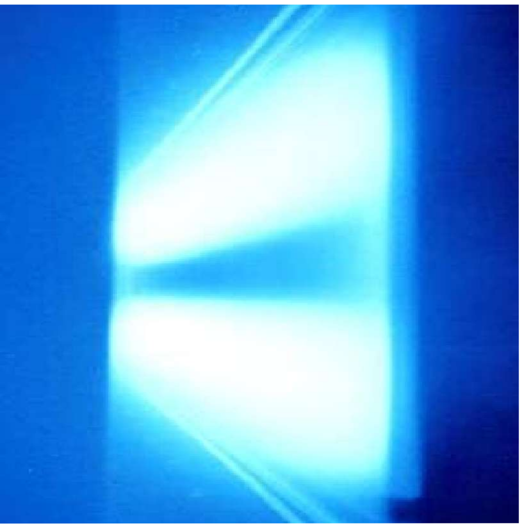
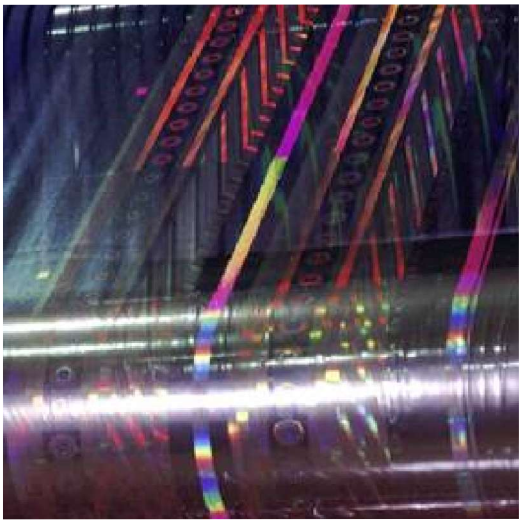


K4000



UFLEX
INDIA



MACHINE SERIAL NUMBER (V06G004)

TO BE QUOTED IN ALL CORRESPONDANCE

BOBST GROUP

Operation And Instruction Manual

GENERAL

General Vacuum Equipment Ltd
Pennine Business Park, Heywood
Lancashire, OL10 2TL, UK

IMPORTANT !

Note!

Do not attempt to operate or maintain this machine until you have read this manual thoroughly and any relevant manuals supplied by the equipment vendors.

If further information is required, please contact General Vacuum Equipment.

At the time of writing, this manual was completely up-to-date. However, due to continual improvements, made possible by the versatility of modern control methods, the final operating procedures may vary slightly from those described herein. This merely implies that the machine has been improved to better fulfil your requirements. If there are any questions, you are encouraged to contact General Vacuum Equipment for clarification.

Whilst every care is taken to ensure that the information contained in this publication is correct, General Vacuum Equipment will assume no responsibility for loss, damage or injury for any inaccuracy.

The company reserves the right to alter or modify the information contained herein at any time in the light of technical or other developments.

Areas of responsibility of the system owner/operator

The system owner (operator) is responsible for ensuring:-

- that each person who deals with starting up, operating, maintaining and servicing the machine system has read and understood the relevant sections of these operating instructions of relevance to him or her.
- that all persons who deal with the machine are technically qualified, appropriately trained and instructed, and have been informed of the potential hazards and risks.
- that unauthorized persons are not able to access the machine.
- that the user always complies with the general safety regulations and all notes related to safety and warnings contained in the following sections.
- that person's who load and unload the machine with substrates are adequately trained and sufficiently informed about potential hazards.

Copyright Information

K4000 M4 Edition

TABLE OF CONTENTS

Preface

Machine Safety

Forward Notes

Section 1 - General Description

Section 2 - Operation of the Machine Winding System

Section 3 - Machine Evaporation Source(s) and Processes

Section 4 - Vacuum Pumping System

Section 5 - Vacuum Pressure Level Monitoring Systems

Section 6 - Operation of Machine Auxiliary Systems

Section 7 - Machine Operational Sequences

Section 8 - Process Troubleshooting and Cleaning Processes

Section 9 - Process Consumables

Section 10 - Viewing of Process Operation

Section 11 - Control/Interlock Descriptions

Section 12 - Preventative Maintenance of Machine

Section 13 - Machine Lubrication

Section 14 - Component Setting Up and Troubleshooting

Section 15 - Machine Documentation

Section 16 - Machine Spare Parts

Appendices

Appendix A -Bill of Materials

Appendix B - Glossary of Terms

Appendix C - Measurement of Aluminium Deposit Thickness

Appendix D - Aimcal - Metallizing Technical Reference

PREFACE

This manual has been written to provide the user with the basic knowledge needed to operate the machine

Safety Warnings:

All large semi-automatic machines can be extremely dangerous if safety precautions are not taken. It is essential that this section of the manual is read and complied with.

Section 1: General Description

The major sections/components of the machine are described in brief in this section

Section 2: Operation of the Machine Winding Systems

The user is instructed on the purpose of various systems related to the different winding systems in this section.

Section 3: - Machine Evaporation Source(s) and Processes

The user is instructed in the operation of the coating processes and how its components are used.

Section 4: Vacuum Pumping System

This section describes the operation of the various types of pumps that are supplied to evacuate the chamber down to the required operating pressure level. It also describes the various components used to evacuate the machine, and leak-detecting methods are described.

Section 5: Vacuum Pressure Level Monitoring Systems

This section describes the type and purposes of the different gauges used to monitor the pressure levels at various sections of the machine.

Section 6: Operation of the Machines Auxiliary Systems

This section describes the various auxiliary equipment on the machine.

Section 7: Machines Operational Sequences

This section takes the user through the normal operating procedures in a block diagram format.

Section 8: Process Troubleshooting and Cleaning Processes

This section relates various process problems that may occur and how they are overcome on the machine. It also contains the normal cleaning procedures that should be used in a normal production environment.

Section 9: Process Consumables

This section describes the various consumable items that will be used within the machine.

Section 10: Viewing of Process Operation

This section describes the various viewing positions around the machine

Section 11: Control Systems and Panel Layouts

The various control interfaces on the machine are listed and the functions of which, are described in this section.

Section 12: Preventive Maintenance

This section of the manual contains the suggested preventive maintenance schedule that should be followed to ensure normal machine reliability and typical designed life cycles of components used within it.

Section 13: Machine Lubrication

The recommended oil and grease types required to maintain the machine are specified in this section. Equivalent types are also given.

Section 14: Machine Component Setting Up Instructions

Contained within this section of the manual, are the set up instructions, maintenance procedures and replacement instructions of components.

Section 15: Machine Documentation

The machine is built to a technical specification as listed within this section. It can be used for reference purposes.

This section contains a listing of all the relevant documents supplied with this machine.

This will include the proprietary manuals supplied with the machine from the component vendors and located within the proprietary manual binders supplied.

Section 16: Spare Parts

To order spare parts for the machine can be sometimes be complicated. Therefore this section of the manual explains where to find the spare part listings and how to recognize these parts.

It also contains a list of serial numbers of major sub components to help in ordering parts

Section 17: Appendices

This section of the manual contains various supplemental information, which may relate to this manual. Including data sheets, training manuals and product updates:

Appendix A – Bill Of Materials - All proprietary information given by OEM's will be in the other binders provided.

Appendix B - Glossary Of Terms - Explanation of terms used within the manual

Appendix C - Measurement of Aluminium Deposit Thickness- Explains different methods of measuring the aluminium coating

Appendix D - Aimcal - Metallizing Technical Reference – Trade association Reference Document

Manual Conventions

Structure of safety precautions

The following words have specific meanings in this and other General Vacuum Equipment manuals:

	Note!
	**Indicates machinery damage can result if correct precautions are not followed**

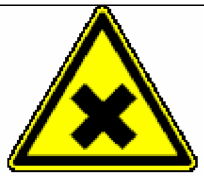
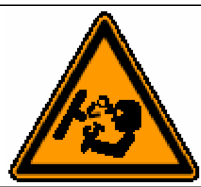
	Caution !
	**Indicates machinery damage can result if correct precautions are not followed**

	Warning !
	**Indicates loss of life, severe personal injury, or substantial machinery damage will result if proper precautions are not taken.**

	DANGER
	**Stands for work or operational procedures which directly cause serious injuries or death if not adhered to.**

	DANGER
	**Beware Live Electrical Circuits, Isolate before working in this area**

Common Safety Symbols

				
Danger electricity, electric shock (Arc Flash Labels)	Attention Notice	General Warning	Hazardous materials, toxic or very toxic Substances	Magnetic fields influence the functioning of pacemakers.
				
Harmful or irritant materials	Flammable Materials	Hot Surface Hazard	Hazard Warning Sign: of harming your hands	Implosion Hazard
				
Crushing Hazard Danger of entrapment	Entrapment Hazard	Danger, overhead crane or overhead load	Compressed Gas Bottle Hazard	Potential Hazard to Pressurised system
				
Fork Lift Trucks Operating	Hazardous to the Environment	Beware of moving machinery	Explosion risk, Explosive Material or Compressed Gases	Corrosive materials
				
Strong Magnetic Field Hazard	Fork Trucks Working in this area	Trip Hazard	Low temperature Hazard	Cutting hazard, Sharp edges watch your fingers
				
Magnetic fields influence	Slippery floor, slippery surface			

Equipment Identification

When contacting General Vacuum Equipment, please have the information on the equipment identification label.

The identification label is located on the rear side of the chamber. It states the specifications of the system

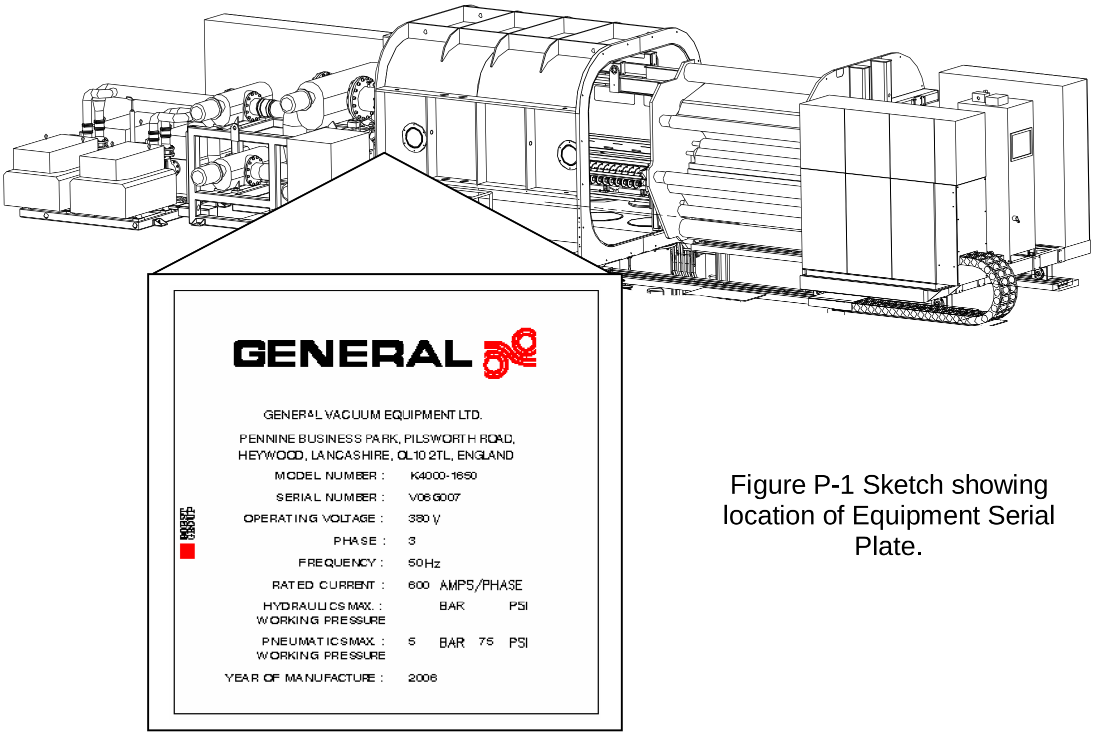


Figure P-1 Sketch showing location of Equipment Serial Plate.

Model Number:	Is the model type reference
Serial Number:	Is the reference of the overall system
Operating Voltage:	Is the designed operating supply voltage
Phase:	Is the designed power supply type.
Frequency:	Is the designed power supply frequency.
Rated Current:	Is the full load installed current per phase
Hydraulic Max. Working Pressure:	Is the rated maximum working pressure in Bar and PSI
Pneumatic Max. Working Pressure:	Is the rated maximum working pressure in Bar and PSI
Year of Manufacture	Is the year of manufacture

Customer Support

The machine is supported by a customer support system designed to offer the user, a fast, courteous service. If you need assistance beyond the scope of the manuals, spare parts or a site visit by a service engineer, please contact General vacuum equipment at the following headquarters address.

GENERAL VACUUM EQUIPMENT LTD,
Pennine Business Park,
Heywood,
Lancashire,
United Kingdom

Telephone: (44) 1706 622442
Facsimile(44) 1706 622772

E-mail Enquires:-service.general@bobstgroup.com

Spares & Service Contacts:

For all your requirements concerning Warranty, Technical Support & Spares & Service please direct your correspondence to:-

E-mail Enquires:- service.general@bobstgroup.com

MACHINE SAFETY

This section of the manual provides a list of safety precautions that should be strictly adhered to by all personnel, operating or maintaining the machine.

Safety is everyone's responsibility! Everyone must do his or her part to ensure a safe working environment.

General Safety

These general precautions will help to create an environment more conducive to safe, defect-free work and optimum productivity. These precautions should be adhered to at all times by personnel in the vicinity of the machine.

Warning !

The following safety precautions should be adhered to at all times.

- Always wear safety shoes.
- Do not wear neckties, long sleeves, wristwatches, rings, gloves, etc., when operating or maintaining this machine.
- The area around the machine should be kept dry and free from obstructions. Keep the area in a good housekeeping order.
- When handling machine parts, rolls, etc., always follow your company's procedures on lifting. Keep your back straight and lift with your legs. Do not over exert yourself, get assistance if necessary.
- When using access platforms, etc., around the machine, ensure that they are extremely sturdy, safe and have anti-slip surfaces.
- Tools and other miscellaneous parts should always be kept off the machine.
- Do not use the machine as a workbench.
- Use approved eye protection (safety glasses, face shield, goggles) when cleaning the machine.
- Use approved air filtration equipment (filter masks) when cleaning the machine.

Proper use

The machine is designed to coat film-shaped, flexible materials (in the following referred to as substrates) on one side with aluminium.

The safety precautions stated in of these instructions must be complied with for all processes performed using the machine.

If the named processes are carried out, then additionally the safety precautions in the corresponding process instructions must be adhered to.

Any and every use beyond these specifications is considered improper and can result in serious injuries or equipment damage. General Vacuum Equipment will assume absolutely no liability for any such injuries or damage.

Note!

The process materials listed below are specified for operation of this machine.

The vacuum pumping system is designed only for the listed specified gases.

The vacuum pumping system is not designed for processing for toxic or explosive gases.

No other process materials should be evaporated nor may other process gases should be used.

Use of other materials or gases is considered to be improper.

No liability will be assumed for any resulting damage.

Specified process materials:

Process materials	Designation	Chemical Symbol
Gases:	Argon Oxygen	Ar O_2
Evaporating materials:	Aluminium	Al
Plasma Target material	Aluminium	Al

Operational Safety

The process cycle is mainly automatic. However, there are certain **Mandatory Safety Features** on the machine that must be observed under all circumstances.

Warning !

The following safety precautions should be adhered to at all times.

- The machine must not be operated with any guards, panels or doors removed or unsecured.
- When the machine is being opened or closed, the operator must ensure that it is safe to do so. He must ensure that there is no person in the line of the movement and that any other obstacles such as access steps, etc. are clear of the moving ends.
- Before the chamber can be opened or closed, audible and visual alarms are given automatically for a predetermined time.
- The winding carts linear open or close and enable push buttons must be depressed for the whole period of opening or closing.
- Before unloading a roll from the machine the operator must ensure that the winding cart is located in its fully opened position.
- The winding cart, when withdrawn from the vessel must only be operated in an "inch" mode (i.e., Jog speed).
- Audible or visual alarms are given automatically before and during the period while the winding mechanism turns in the inch mode. The inch section of the DRIVE screen must be pressed throughout this inch running condition. Removing the finger automatically stops the winding mechanism turning. An optional remote inching push-button may be supplied within the loading area of the machine.
- Under no circumstances must an operator attempt to clean the machine rollers when the winding mechanism is rotating or when tension is applied to the substrate.
- A **LOCKOUT SWITCH** is located at the front of the machine (Clearly Labelled). The action of locking out the isolator will isolate the following functions. Isolates the electric motor that moves the winding cart in linear directions and isolates the winding system drive power supply and places the 'Guard Stop' circuit into a safe condition.
- Know the location of the 'E STOP' button. When activated this control circuit will stop the drives quickly in a controlled manner. Normally within 10 seconds. After this period of time the drive system will automatically de-energize all the power to the winding system drive circuits. The rollers and rolls of substrate will slow down under natural inertia restraints. All other rotating equipment will be powered down instantly.

  In any situation where the alarm siren is sounding, be aware that all personnel should stand away from the machine, as potential movement will take place.

Warning !

Further precautions for machines fitted with a plasma treater.

- This equipment is designed to run on a process gas containing less than 20% oxygen. Use of process gases containing more than 20% require specialised pumping systems for safe operation. Therefore always use the correct gas mixture.

- Any Persons who have active or passive health aids upon their person must stay away from the equipment, when it is in operation. This is because the plasma power supply can create electromagnetic fields when operated. The risk is minimised because the electrodes are located inside the vacuum chamber and is only energised when it is closed and under a vacuum condition.

Warning !

The pumping system has been designed to evacuate gas mixtures that should not contain more than 20% oxygen by volume.

Locking Out the Movement Isolator

Warning !

Under no circumstances must any person enter the chamber unless they have locked out the isolator.

To lockout the movement lockout isolator, turn it through 90 degrees and place a padlock on the switch.

Figure S - 1 Locking Out Of Movement Isolator

Warning Siren and Beacons

The audible siren and visual warning beacons will operate in a number of conditions. Ensure that you are familiar with these various conditions as explained in the table below.

Visual Beacon Condition	Audible Alarm Condition	Meaning
Off	Intermittent 2 sec on, 2 sec off	Lock out isolator has not been rotated to safe condition
Flashing	On	Stand Away from the machine, movement is about to take place. one of the following conditions are to take place; • Tension to be applied to the web • Traverse cart movement • Inching of the winding mechanism • Movement of lay arm assemblies

Table S - 1 Warning Siren And Beacons

Now that you have read these general and operational safety precautions, be sure to read and understand all other warnings and cautions in the rest of this manual, before attempting to operate or maintain this machine.

General Vacuum Equipment manufactured this equipment to meet and exceed all codes and industry standards for machine safety.

The Health and Safety At Work Act 1974 UK

Under Section 6(1) and 6(4) of the above Act, we make available certain information on the product we supply. The following information must be brought to the attention of all personnel involved with this equipment as required by the Act.

All users should be aware of the hazards present and act accordingly.

Important!

Only fully accredited personnel should carry out inter-connection of wiring on any electrical equipment. All electrical equipment should be switched off at the main's isolator switch before any adjustment or connections are made to the control panels or electrical components.

- The equipment is electrically powered at mains supply voltage. It must be earthed and all electrical supply regulations complied with.
- All guards, electrical or mechanical interlocks and other safety devices that were fitted when the machine was delivered must be kept in place. These safety devices must be inspected for effective operation at regular intervals by competent maintenance personnel.
- The equipment must be completely isolated from the electrical supply before any terminal covers are removed.
- The equipment should not be tampered with or changes incorporated, or used in a non-standard manner without first seeking advice from General Vacuum Equipment, the manufacturers of this equipment.
- All hoses and ancillary devices that are connected to the equipment should be of suitable quality to withstand the operating temperature and internal pressure specified in any data sheets provided.
- Flexible hoses connected by the user should be installed to conform to the recommendations of the hose manufacturers especially with regard to minimum bending radii, and avoidance of wear by friction of moving hose or adjacent parts.
- All pipe work, valves, flexible hoses, etc. should have regular inspection by qualified personnel and the equipment taken out of service until rectified, should any wear or leakage occur.
- Do not deface, remove or paint over warning plates that are mounted upon the machine.

Safety Lockout Procedure

The majority of companies operate a safety LOCKOUT scheme.

Read and understand your company's lockout procedure and all applicable local safety regulations.

The following is a guide of lockout procedure; General Vacuum Equipment does not accept any responsibility for loss, damage or injury for any inaccuracy.

Warning !

Never remove another person's lockout (padlock) or tag. Never assume that a machine is locked out. Always check !!

Lockout Procedure - Guidelines

- Remove all product rolls
- Lockout the main power supply isolator with your padlock
- Attach a "DANGER - PERSONNEL WORKING" tag on the lock with your name on it.
- Shut off all sources of power and flow of material and services. This includes;
    - Electrical supply
    - Pneumatics
    - Hydraulics (Option)
    - Liquid Nitrogen (Option)
    - Process gases (Option)
- Block any rotating elements to prevent accidental rotation. In some cases, it may also be necessary to block components that may lower under their own weight.

Pneumatic Isolation Valve

A single main air supply isolation valve is located on the booster group framework. It is located before the air regulator (lubricator).

Before isolating the air supply, ensure that any equipment on the machine that is supported by a pneumatic device is in a safe condition.

Warning !

All diffusion pumps used on the machine uses silicone based oils. On no account must mineral based oils be used.

Potential Health Hazards

The condensates from the process deposited on the chamber walls of a vacuum system are generally as extremely fine particles. The nature, as well as the form, of the material may pose the following health hazards:

a) Inhaling fine particles (powder) may cause damage to the lungs. To help prevent this, wear a protective respirator mask with fine filter that has been approved by a relevant safety organization

b) Some substances are toxic and inhaling them should be avoided. Take steps to ascertain whether or not the material being deposited is a known toxic substance. Refer to the Material Data Sheet(s) covering the evaporant(s) in question.

c) Certain powers, titanium for instance, can cause flash fires when exposed to oxygen or other oxidizers. Therefore, when opening the chamber after a deposition cycle, exercise extreme caution and allow time for the coating surface to oxidize. Breakage of some of the more reactive condensates may be hazardous, even when the above precautions are observed. In this situation, fire protective clothing should be worn. Reference must be made to relevant coating material safety sheets.

This machine has been designed to process the following evaporants and gases:-

Process materials	Designation	Chemical Symbol
Gases:	Argon Oxygen	Ar O_2
Evaporating materials:	Aluminium	Al
Plasma Target material	Aluminium	Al

Table S - 2 Machine Evaporants and Gases

Warning !

NO OTHER MATERIALS THAN THOSE MENTIONED ABOVE SHOULD BE EVAPORATED IN THIS MACHINE.

Noise Reduction

The various national health and safety organizations have established guidelines mandating that hearing protection must be worn in environments with noise levels exceeding 82db, weighted average, over an 8 hour period.

General Vacuum Equipment strives toward keeping the sound level of the equipment within this range so that hearing protection is not required. Noise level, however is also a function of the reflective surfaces of the room in which the machine is housed.

Generators of noise on the machine are;

- Drive motor blowers:- Due to the nature of the air passing through the blowers at high speed, noise is generated. Do not place restrictors in the inlet or outlets grills that may increase this noise. Check regularly for obstructions and debris.

- Chiller Unit compressors:- The covers on the chiller and cryogenerator units must be located in place at all times except when maintenance is being carried out.

- Belt Drives: - The largest cause of drive belt noise is improperly tensioned belts. Follow the instructions within this maintenance manual to overcome this problem. High-pitched squealing will be heard if the guards over the belts are touching a rotating member. Check for reason of noise source, when heard.

- Pumps: - The pumps supplied on the machine have been designed to operate under the recommended level. Check for reasons of abnormal noise such as slipping of drive belts.

- Venting of the machine from the surrounding atmosphere will cause noise, inlet filters are fitted to reduce this noise.

- Auxiliary equipment such as compressors and water supply pumps can also produce noise. These sources should be housed within a closed acoustic enclosure.

Risk Analysis

A risk analysis of potential hazards is made on all equipment manufactured by General Vacuum Equipment.

A copy maybe requested by contacting General Vacuum Equipment.

	Note!
	**The design of the equipment is under constant review and changes may be made to improve its quality. If any other information is required, or if there is any doubt regarding the safe application or installation of the equipment, please contact General Vacuum Equipment.**

FORWARD NOTES

The equipment covered by this manual has been designed and manufactured to give long and satisfactory service provided that the operating and maintenance guidelines contained herein, are closely adhered to.

It is important when carrying out maintenance work to ensure that all previously mentioned safety precautions are observed, and particularly that electrical power, hydraulic, air and water are turned off.

All personnel prior to operating the machine, or carrying out any repairs or maintenance work should read this manual and have been trained in its use.

Please note that, whilst every effort is made to ensure the accuracy of the information contained in the manual, we cannot accept responsibility for instructions supplied by manufacturers of equipment, which has been incorporated into our machine.

Definition of Personnel

Only authorised, trained personnel are permitted to operate and maintain the system. The system personnel are grouped into three categories according to their training and ‘logon’ rights. Those assigned to a given group are required as a minimum, to have the knowledge and skills of the lower groups:

- Operating personnel
- Maintenance and service personnel
- Production Supervisors

The rights of personnel to access the various levels of the machine control are defined by assigning ‘login’ passwords to them.

Access Rights	Personnel		
	Operator	Production Supervisor	Maintenance Personnel
Vacuum System – Automatic Control	Yes	Yes	Yes
Vacuum System – Manual Control	No	Yes	Yes
Recipe Setup – Normal Access	Yes	Yes	Yes
Recipe Setup - File Deletion	No	Yes	No
Maintenance Timers/configuration	No	No	Yes
Deposition System Calibration	No	Yes	Yes
Pump down Reference Curve Save	No	Yes	Yes
Shutdown WinCC - Interface	No	Yes	Yes
Calibrate Monitor Systems	No	Yes	Yes
Change maintenance Timers	No	No	Yes
Sign off Maintenance Items	No	No	Yes
Change Configuration	No	No	Yes

Table F-1 Levels of Access to Control Systems

Definition of Operating personnel

Operating personnel is permitted to operate the machine in normal operation (automatic coating).

This comprises the following activities:

- Carrying out coating process via the Machine Man Interface (MMI) and other parts of the machine.
- Loading and unloading the winding system with substrate.
- Cleaning the machine as described in this manual.
- Only those persons are permitted to operate the machine that have successfully been trained in the operation of the machine by the customers experienced supervisor, that have been previously been trained by an employee of General Vacuum Equipment.
- Replacing the pressurized process gas bottles (Plasma Option)

Definition of Operating Supervisor Personnel

Operation supervisors must have the operation knowledge of operating personnel.

The supervisor may operate the machine and has access to all the normal functions of the operator station. The supervisor is entitled to create, store and delete process recipes.

They also have access to check the calibration of the machine systems.

Supervisors are employees the machine owner.

Definition of Maintenance Personnel

Maintenance or service personnel will generally have a limited knowledge of the machine operation.

Warning !

Reduced level of safety.

Improper operation can cause serious injuries of the personnel and considerable equipment damage.

The maintenance and service personnel may operate the machine in normal operation, regularly perform necessary maintenance work and additionally perform all further service work which is necessary for proper operation. For this purpose, the machine may be operated in certain manual modes.

Maintenance of the machine comprises the following activities:

- Switching the system on and off.
- All work mentioned in the maintenance schedule (see chapter 12).

Service of the machine comprises the following activities:

- Shutting down and dismantling the machine for access to the system components.
- Repair work on mechanical and electrical components of the machine.

Maintenance and service of the machine may only be carried out by trained competent employees of the system owner.

For work on electrical components, training as an electrical technician or comparable technical training is recommended

Faulty Parts Return

If a part is to be returned the GENERAL VACUUM EQUIPMENT factory, contact must be made to obtain a returns authorisation number (RA) for the part to be returned. This identification number is to be clearly written on the paperwork accompanying the part. All communications by telephone or facsimile should be made to the spare parts manager.

The package should be clearly labelled with the address of the sender, enclosing your return note, and forwarded to:

GENERAL VACUUM EQUIPMENT Limited,
Pennine Business Park,
Heywood, OL10 2TL, UK.

Illustrations, Charts and Figures

Machine Safety

Section 1 - General Description

Section 2 - Operation Of The Metallizer Winding System

Section 3 - Operation Of The Metallizer Evaporation Source

Section 4 - Vacuum Pumping System

Section 5 - Vacuum Pressure Level Monitoring Systems

Section 6 - Operation Of Metallizer Auxiliary Systems

Section 7 - Metallizer Operational Sequences

Section 8 - Metallizer Troubleshooting And Cleaning Processes

Section 9 - Process Consumables

Section 10 - Viewing Of Operation

Section 11 - Control/Interlock Descriptions

Section 12 - Preventative Maintenance

Section 13 - Machine Lubrication

Section 14 - Machine Component Setting Up Instructions

Section 15 - Machine Documentation

Section 16 - Machine Spare Parts

Appendix B - Glossary Of Terms

Appendix C - Measurement Of Aluminium Deposit Thickness

SECTION 1 GENERAL DESCRIPTION

Section 1.1 Evaporation Process
Section 1.2 Vacuum Chamber
Section 1.3 Winding Mechanism and Drive System
Section 1.4 Vacuum Pumping System
Section 1.5 Evaporation Source
Section 1.6 Cryogenic System
Section 1.7 Machine Control System
Section 1.8 Pneumatic Services
Section 1.9 Water Systems
Section 1.10 Process Chiller Unit
Section 1.11 Plasma Treater Unit

This equipment produces products that can be used for many end products including stamping foils, decorative and barrier packaging. This section of the manual gives a brief description of the major sub-sections of the equipment.

1.1 Evaporation Process

Briefly, the evaporation process consists of thermally evaporating a metal, usually aluminium, in a high vacuum environment (5 x 10-4 mbar). This is carried out within a vacuum chamber and allowing it to condense on a moving web of plastic or paper.

The web, although moving past the evaporators at high speed, receives both radiated heat from the evaporators and latent heat from the condensing aluminium. Therefore, it is important that the substrate is tensioned between rollers as it passes through the coating area.

By the use of carefully controlled winding system, the substrate is taken from the unwind roll, over tension measuring rollers, over spreader rollers, through the coating area to the rewind roll without stretching or otherwise damaging it.

This is a batch-type process and consists of the following equipment listed in the following sub-sections.

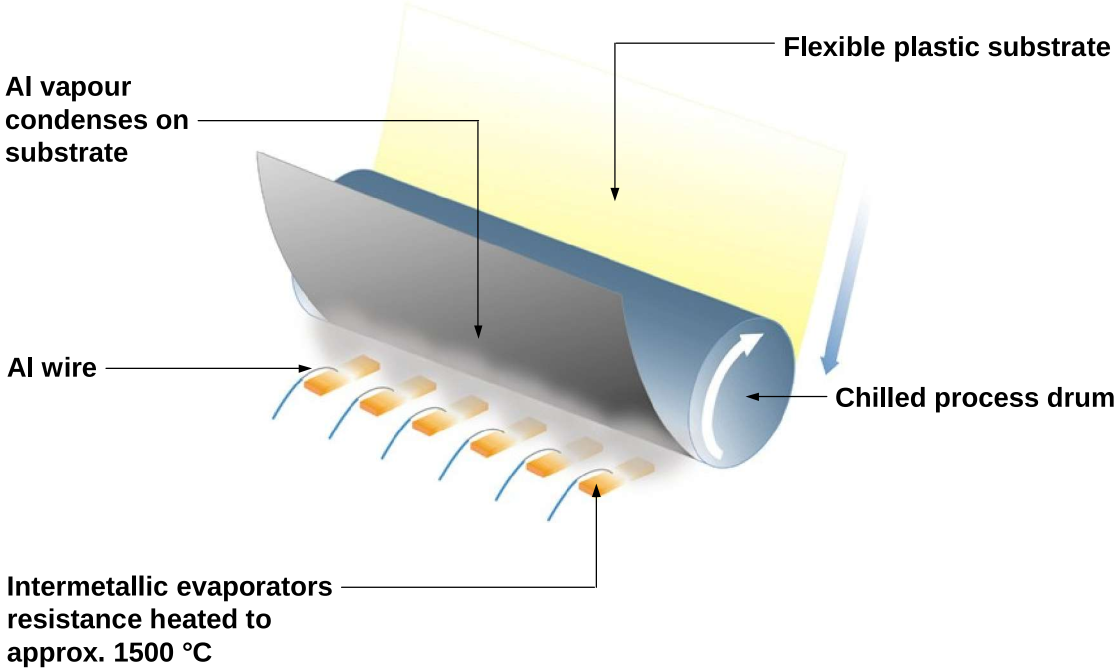


Figure 1.1 - 1 Evaporation Process

The individual evaporators can only evaporate a limited amount of aluminium, therefore a number of them are supplied. The evaporators are positioned at 90 degrees to the substrate movement. They are evenly spaced 100mm apart and extend across the width of the machine. This is to allow even coverage across the width of the substrate.

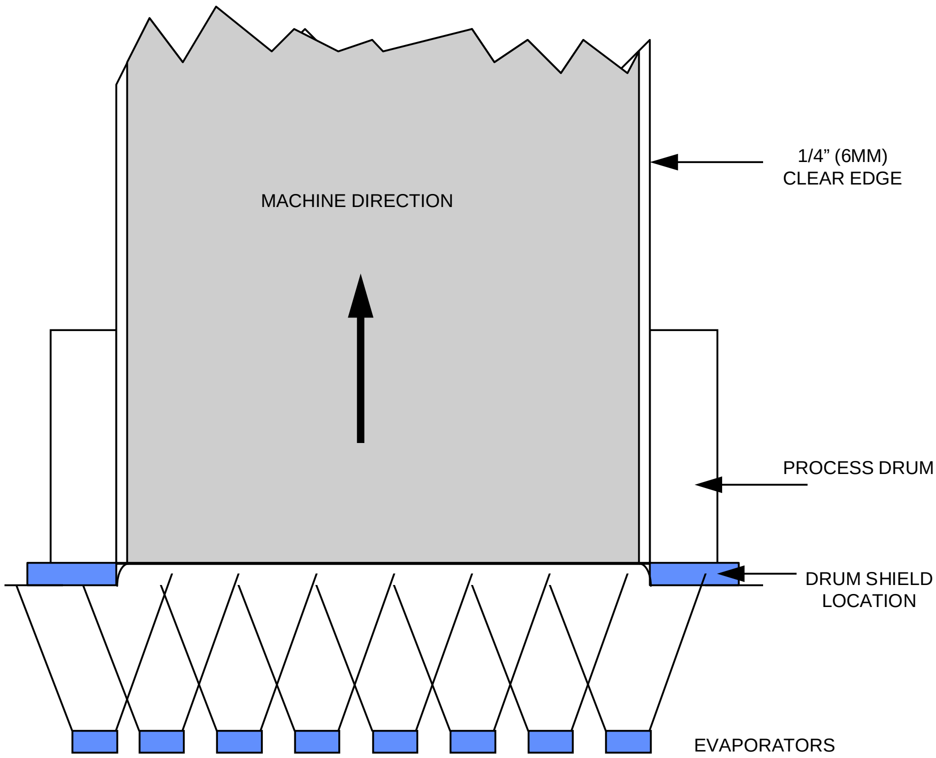


Figure 1.1 - 2 Diagram Of Evaporator Position Relationship to the Substrate .

1.2 Vacuum Chamber

The complete process is carried out within the vacuum chamber. This chamber is generally constructed out of mild steel in a re-enforced rectangular shape. This shape design ensures that there is an even distribution of atmospheric pressure on the outside wall during the process and minimizes any distortion.

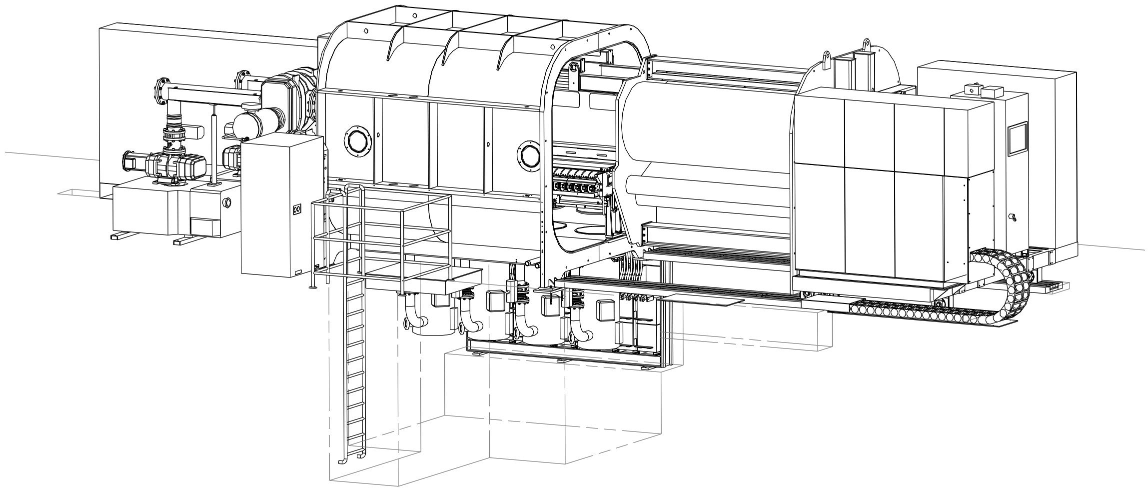


Figure 1.2 - 1 Dimensional View of Vacuum Chamber

1.3 Winding Mechanism and Drive System

The winding mechanism cart is moved out of the vacuum chamber for loading and unloading the process rolls. This cart carries the following;

- Web transport mechanism
- Process chiller unit
- Web transport drive system
- Plasma treater system

The cantilevered web transport/winding mechanism is supported on rails within the chamber.

A machine's winding mechanism comprises of an unwind roll, spreader rollers, load cell rollers, nip rollers, water cooled rollers, tensioning rollers and a rewind roll. See the winding mechanism drawing for the exact arrangement.

A number of the aforementioned rollers may be driven directly by an AC drive system, i.e.: unwind roll, cooling drum, rubber covered draw roller and the rewind roll. The drive system used is composed of standard hardware components and uses industry standard software for accurate winding conditions.

Other rollers within the mechanism are tendency driven by a timing belt system (Located on the outside of the outboard winding mechanism support plate).

A roll of material is loaded into the unwind chucks using a bridge crane. The chucks are designed to support the roll shaft. The material is threaded over the rollers and around to an empty rewind core (see Figure 1.3 - 1)

A length counter is provided and length is displayed upon the MMI interface screen.

Inter-zone seals of two types are located on the winding cart. Rigid low conductance concaved mild steel seals are located close to the drum where the substrate enters and exits the evaporation zone. Flexible sealing strips inside the chamber to provide flexible zone seals with the winding cart

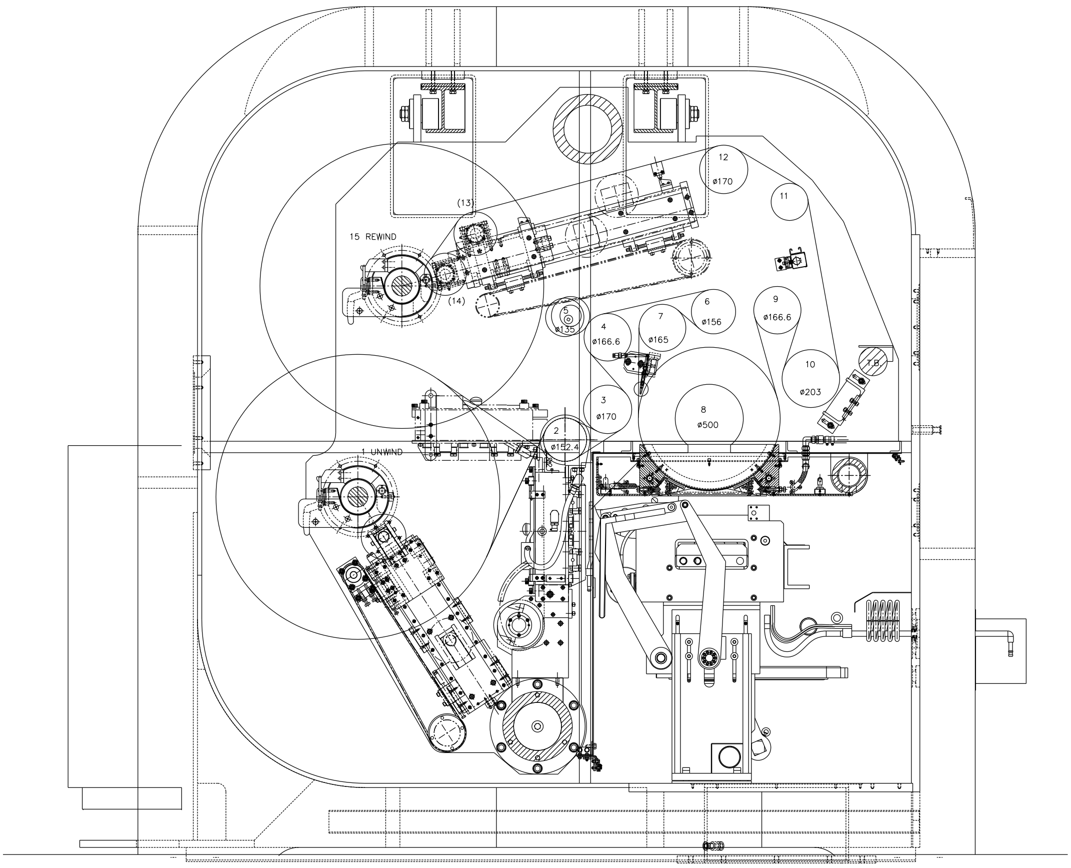


Figure 1.3 - 1 Winding Mechanism Layout

1.4 Vacuum Pumping System

The vacuum chamber is divided into two vacuum zones by various sealing methods within the chamber.

Benefits include lower power consumption due to smaller pumping group requirements, lower pressure levels in the evaporation zone and improved cleanliness of the winding zone.

A number of different types of pumps are used to evacuate the chamber and provide the vacuum level required.

The pumping system varies according to the machine type and the required pumping requirements. A description of how the various pumps operate, is explained in a later section of this manual.

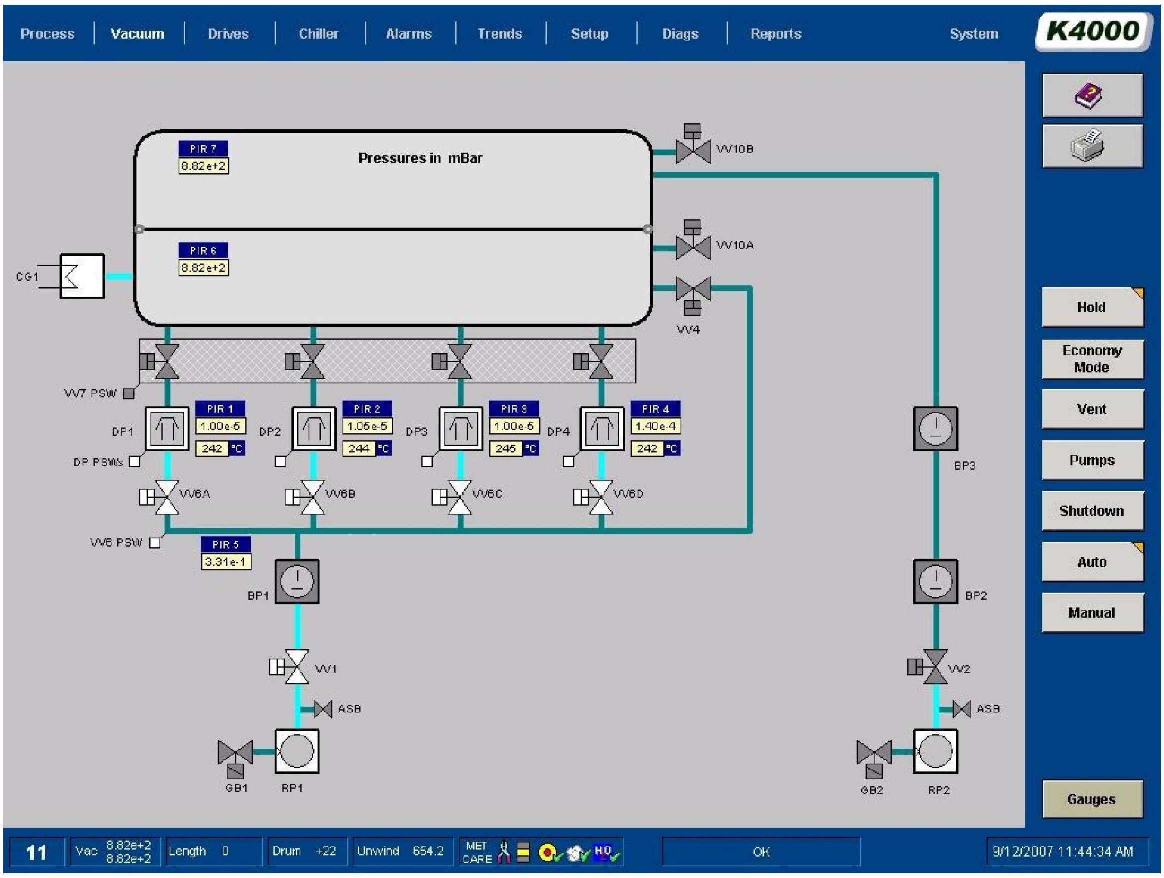


Figure 1.4 - 1 Vacuum Pumping System Mimic Configuration

1.5 Evaporation Source

- Aluminium

Generally aluminium is evaporated from electrically heated evaporators (i.e. boats) and condenses on the substrate as it passes the source.

The evaporators are located within an enclosure called the source box, which is manufactured from mild steel and is water-cooled.

The electrical resistance of the evaporators is such that, at approximately, 750 amps and 10 volts is required to raise them to their operating temperature of around 1400 C.

The aluminium is delivered to each of the evaporators in a wire form from stepper motor driven wire feeders.

This machine has a single source fixed within the chamber. It designed to allow easy access for cleaning and maintenance.

The deposition control system monitors the deposition uniformity across the web and sends signals to the stepper motors automatically to achieve uniform deposition across the web.

Viewing windows in the chamber wall allow the operator to monitor the evaporators whilst the metallizer is running, so that adjustments, if required, can be made.

- Zinc Sulphide

This type of machine can also evaporate zinc sulphide also using a special adaptor box which is filled with pellets.

- Silicon Dioxide

This type of machine can also evaporate silicon dioxide also using a special adaptor box which is filled with pellets.

1.6 Cryogenic System – Cryo-generator

One or more cryogenic coil systems are built within the chamber to assist the other vacuum pumps within the system to "pump" the large amounts of water vapour that are given off by the substrate.

Water vapour, with a high volume and high vapour pressure, is frozen to ice, with a low volume and low vapour pressure, by the cryogenic coil. The cold coils have a very high pumping rate specifically for water vapour.

The refrigerated coils are an integral part of the pumping system and both manual and remote automatic control are provided.

To supply the refrigerated coils, cryo-generator units are used to supply a refrigerant gas capable of cooling the coils to temperatures less that -110 degrees C.

Figure 1.6 - 1 Photograph Of Cryogenic Coils In A Machine

Figure 1.6 - 2 Photograph of Cryogenic Unit

1.7 Machine Control Systems

Machine Components

This technology based control system provides the ability to the machine to produce high quality products on a constant basis. It is based on using a combination of computer and programmable logic controllers (PLC) and various communication links and monitoring devices to gather information and then act upon it.

Information is passed down its communication routes about the following parameters;

- Measurement of coating being applied
- Power levels being supplied to the evaporators
- Vacuum levels
- Wire feed rates
- Drive system parameters
- Chiller system

From this information the computer will modify parameters such as evaporator power levels and wire feed rates to achieve high quality product.

Figure 1.7 - 1 MMI Interface Screen

At the end of a coating cycle, the computer will provide a data report on the previous cycle. This data can be outputted in a print format.

The computer makes use of the Profibus protocol communication systems for transmitting information between the various system components.

Interface between the user and the computer is provided by the provision of touch screen facilities.

More details of the operation of this equipment are described within the operating section of this manual.

Man- Machine Interface (MMI)

This is the term given to the screen that the operator uses to control the machine.

Boot Screen

After switching on the main isolator, the computer will power up and display the BOOT screen.

Figure 1.7 - 2 Machine Boot Screen

This screen has to be activated by touching the surface of the touch screen or alternatively using keyboard or mouse.

MMI Operational Levels

The MMI software has been compiled from industry standard software packages. To ensure that the possibility of corruption to the operation program is minimised, access to certain software programs is limited by the use of passwords.

Login In Screen

The computer has been set up to allow monitoring of usage by different operators and for production purposes, shift patterns.

On start up a SYSTEM screen will be shown and the user must touch the flashing LOGIN button to allow them to login.

At this time of this login sequence, the user will be asked to enter their login name and password.

Login names, passwords and restrictions are documented in the maintenance manual.

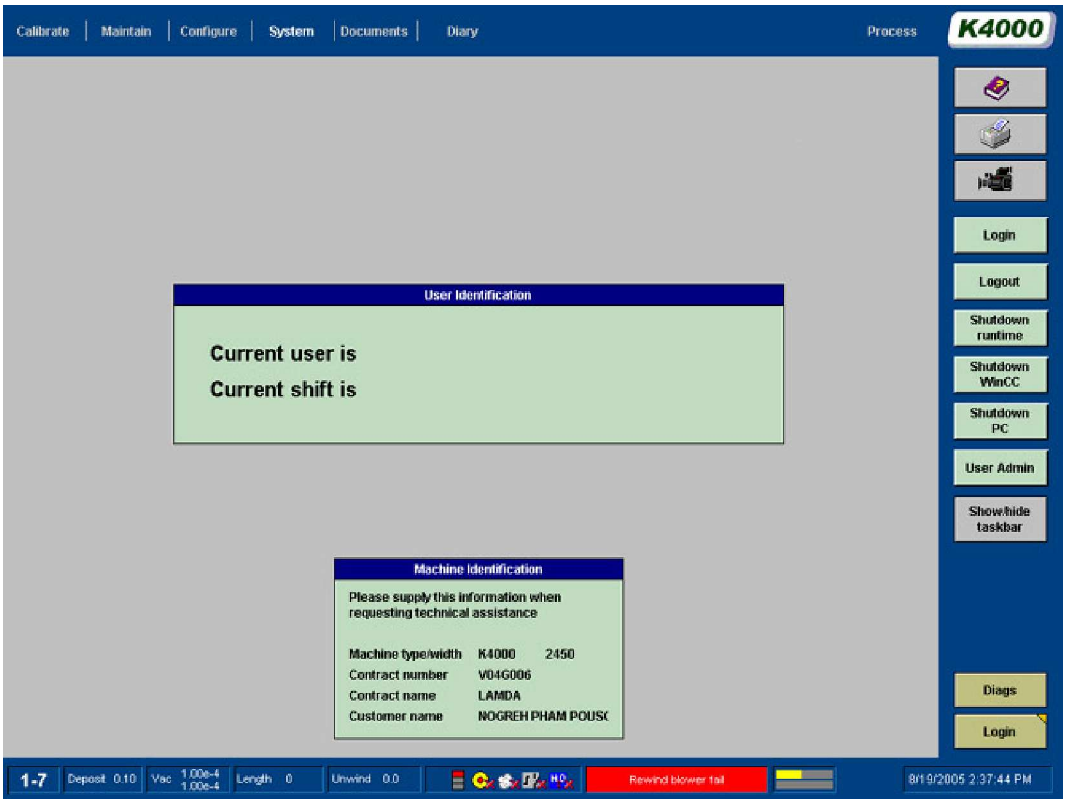


Figure 1.7 - 3 System Screen

All other normally used screens are accessible from this screen. There are some screens that are infrequently used and these are only available via the specific screens that they are attached.

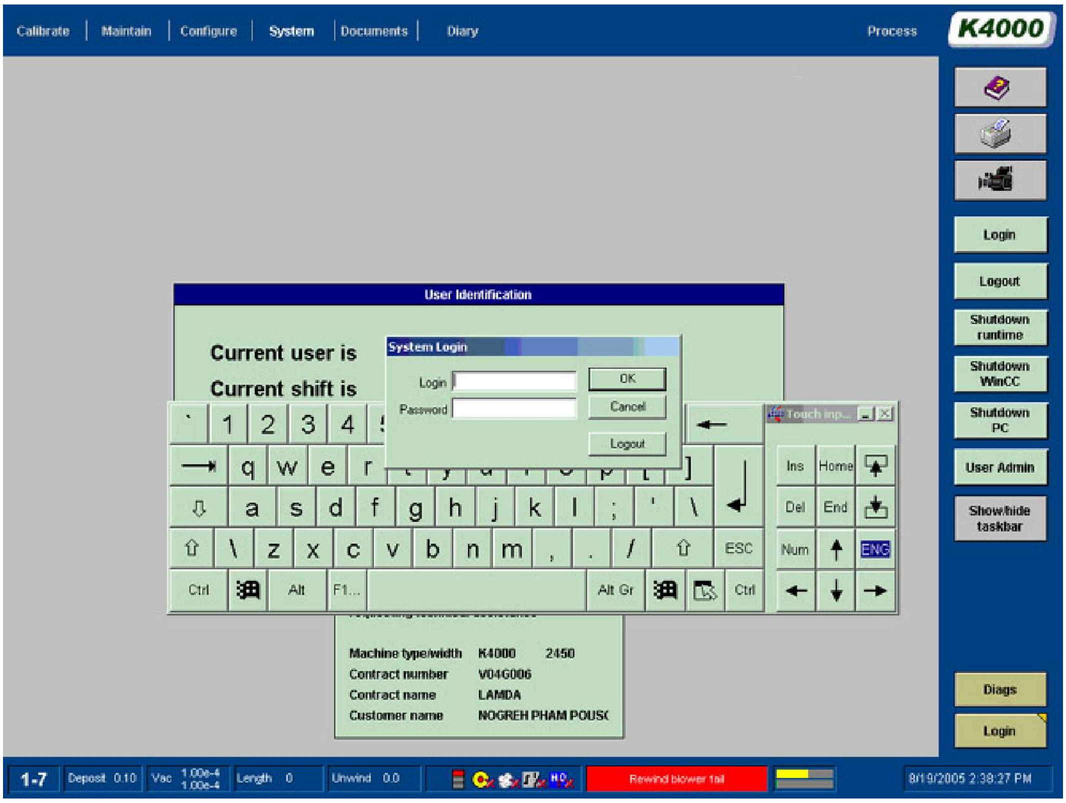


Figure 1.7 - 4 Log In Screen

Use the keyboard to enter the user's login name and password.

The factory set passwords for operators are.

Login name:	Login Password:
Operator 1	Operator 1
Operator 2	Operator 2
Operator 3	Operator 3
Operator 4	Operator 4

From this screen the operator has a choice of the following can choose the following directions to access other screens:-

SYSTEM	*Screens*	PROCESS	*Screens*
	• CALIBRATE Screen		• PROCESS Screen
	• MAINTAIN Screen		• VACUUM Screen
	• CONFIGURE Screen		• DRIVES Screen
	• SYSTEM Screen		• CHILLER Screen
	• DOCUMENTS Screen		• ALARMS Screen
	• DIARY Screen		• DIAGNOSTICS Screen
			• SETUP Screen
			• TRENDS Screen
			• REPORTS Screen

Please note that some of these screens are only available at higher login levels.

To change the screen, touch the relevant section/tab at the top of the screen being displayed

A description of all the functions of each screen is in various sections of this manual.

1.8 Pneumatic System

Air is supplied to various parts of the machine from the central air regulator located on the booster group frame. There is an isolation valve and regulator located on this panel.

The specification for the pressure and volume requirements can be found on the pneumatic schematic drawing provided.

Air is used for the following functions;

- Operation of the vacuum isolation valves (Fail-safe to Close on loss of power)
- Operation of moving shutter
- Operation of Nip roller mechanism
- Chiller system - auxiliary valves
- Raising and Lowering of the coating shield
- Operation of chiller heat valve.

The air supply is monitored at all times by a pressure switch and any faults are recorded by the computer and displayed upon the alarm screen.

	Caution !
	**Ensure that if the air supply is clean and dry. Dirty air will cause problems in the pneumatic systems**

1.9 Water Systems

Water is provided to the machine as a cooling medium. It is used to cool the following;

- Source busbar system
- Diffusion pumps
- Vacuum pumps
- Source enclosure
- Evaporator contact post and blocks
- Drum Shields and frame
- Chiller Condenser

The specification for the water temperature and flow requirements can be found upon the water schematic drawing supplied.

Pressure sensors are provided on all the water manifolds to monitor the water supply to the machine. These sensors are monitored by the computer for low pressure and loss of pressure differential across the feed and return lines. In either situation, an alarm will be indicated.

	**Caution!** **Overriding of these sensors will invalidate any warranty agreement**

A strainer is fitted in the supply line to the machine to ensure that debris does not enter cooling pipes. This should be cleaned on a regular basis and should not be left out of the circuit at any time.

	**Warning !** **Do not remove the element from the strainer as this will invalidate any warranty agreement** **Do not install valves in the return lines of the water circuit as this could lead to accidental damage to water cooled items**

1.10 Process Chiller Unit

A free standing chiller unit is located on the winding cart for the purpose of supplying a coolant to the machine's cooled process rollers. The coolant, a glycol and water mixture are used to chill the process drum and roller to a temperature set between + 5 and - 15 degrees C.

The chiller and associated valving operation is controlled from the chiller plc system and interfaced to the main plc.

It is interlocked to change from the heat mode into the cool mode when the pumping system is in the vacuum mode. The mode is changed to heating to allow the drum to be heated back to ambient before venting the machine. If the process drum and roller are not heated up, condensation will occur.

The chiller is supplied with process water to allow its operation. Pressure sensors are used to monitor the water supply.

1.11 Plasma Treater Unit

A plasma treater unit has been incorporated into this machine to allow treatment of the surface of the substrate, while it passes through the unit.

This is achieved by creating a gaseous atmosphere and applying a high voltage, which in turn creates a plasma.

The plasma treater operates as an isolated plasma zone within the vacuum chamber. The plasma is generated and maintained by a medium frequency AC power supply.

The plasma treater is supplied as a robust ergonomically designed assembly to provide optimum operating conditions.

Figure 1.11 - 1 Photograph of Plasma unit in Vacuum chamber

SECTION 2 OPERATION OF THE MACHINE WINDING SYSTEM

The winding mechanism’s function is to wind material from the unwind roll to the rewind roll and allowing the coating processes to take place. The system is designed to process the material without causing any deformations. If faults are present in the parent base roll, it must be recognised that the winding mechanism cannot remove all defects.

2.1 Loading and Unloading of Rolls of Substrate

Warning !

It is essential that any operator that is to load or unload the machine has received training in the operation of the loading device being used.

Care must be taken to obtain truly vertical lifts of the rolls from the chucks in order to prevent damage to the machine by the rolls swinging, if an overhead crane is being used.

Figure 2.1 - 1 Photograph of Roll Being Removed From Machine

In order for the deposition monitor system and spreader rollers to function correctly it is essential that the unwind roll is placed in the centre of the winding mechanism. The distance 'X-X' must be the same on both sides of the roll.

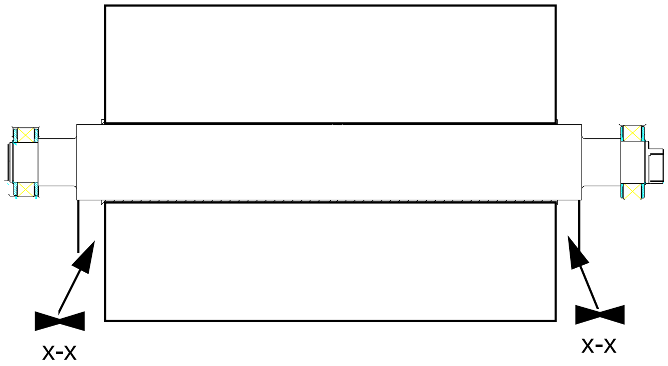


Figure 2.1 - 2 Roll Location On Shaft

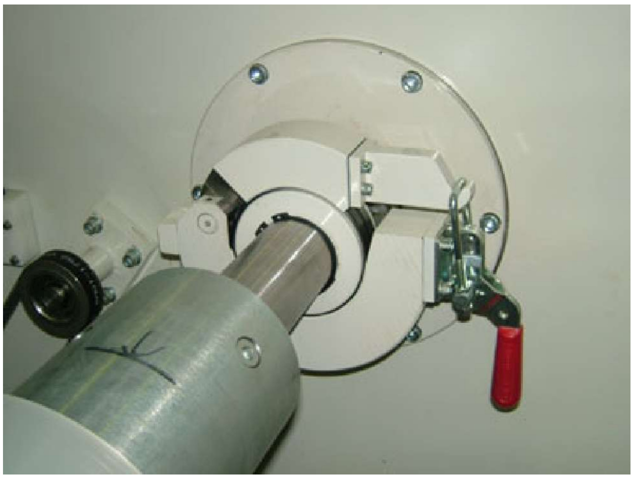
The machine is fitted with a "triangle" shaped drive chuck on the end of each reel shaft (arbour), which locates in a receptacle section on the driven part of the chuck assembly. It is essential that the open section of the assembly is upwards to allow the removal of the reel shaft.

Figure 2.1 - 3 Photograph Showing Triangular Section of Driven Chuck

2.2 Operation of Roll Chucking System

The rolls of substrate are supported on reel shafts (arbours) which have in-built bearings. These bearings are located in bearing supports that extend from the winding mechanism side-plates.

When the reel shafts are lowered into the bearing holders, a swing over clamp is engaged automatically.



The mechanical clamp is locking into place by engaging the toggle clamps provided.

Figure 2.2 - 1 Reel Shaft Located In Chucks

Warning !

Ensure that the rolls of material are supported by a sling and hoist before un-chucking the machine interlocks are incorporated within the machine to provide a safe working environment.

Under no circumstances, override any interlock, either mechanically or electrically

Figure 2.2 - 2 End View of Roll Chuck in an Open Position.

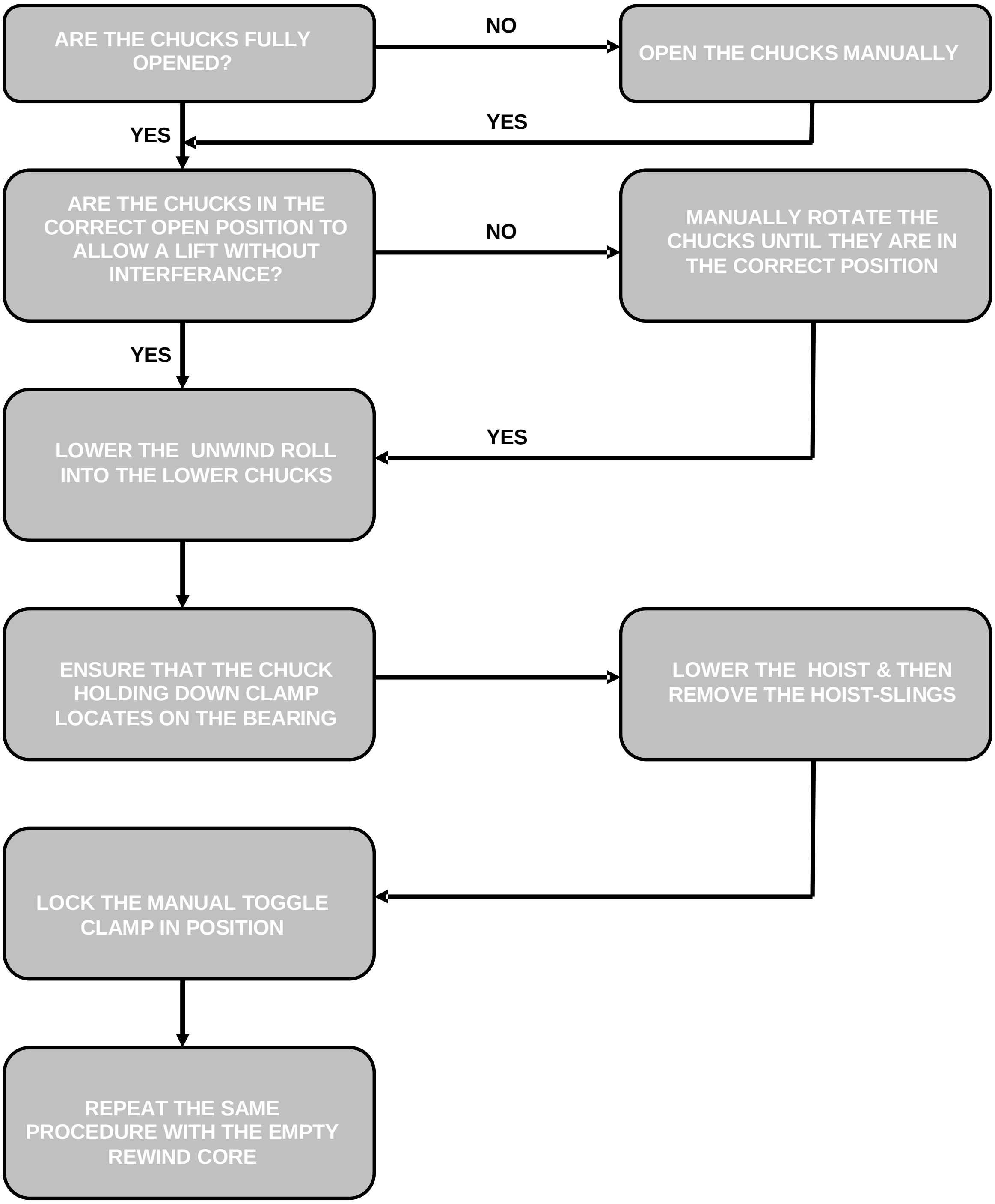
A flow chart for the chucking operation is shown in chart 2-1

Warning !

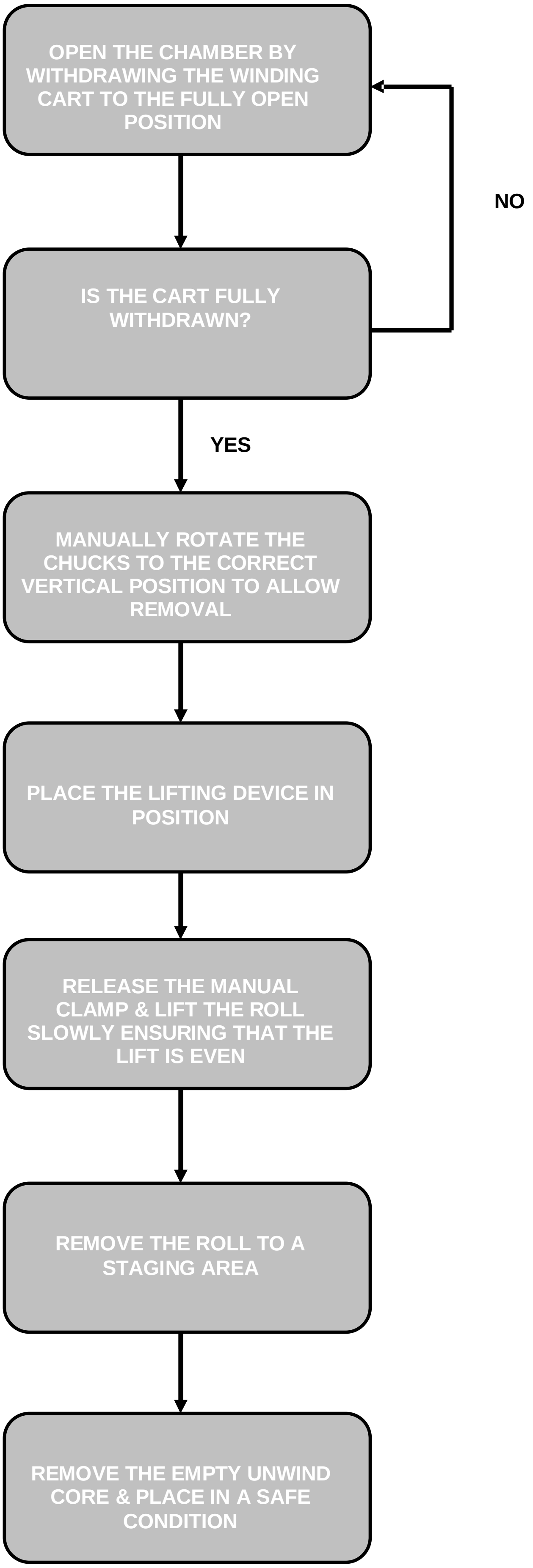
Before attempting to lift the rolls out of the chucks,

Ensure that the clamps have been opened and the driven end is in the correct vertical position for removal.

Figure 2.2 - 3 Photograph of Roll Being Removed From Machine Using Overhead Crane



Flow Chart 2.2 - 1 Chucking Flow Chart



Flow Chart 2.2 - 2 Unchucking Flow Chart

2.3 Reel Shafts

It is normal that 152mm shafts are provided with the machine. The shafts are manufactured to fit inside a core with a minimum inside diameter of 154mm.

Bearings are fitted to the shafts to allow the shafts to be supported from the machine side-plates.

The shafts are expanded into the paper cores by mechanical lugs that engage into the core.

They should provide a long operating life, if used as advised.

On machines with a maximum width of 1650mm, 67mm shafts maybe used. However there will be a speed limitation at small diameters

	Warning !
	**Do not operate the machine outside the limits given in the specification. The minimum width should not be exceeded, as bending moments on the reel shafts**

2.4 Operation of the Mechanical Reel Shafts

It is recommended that the paper core wall thickness should be a minimum of 17mm thick to stop potential winding issues with deformation.

The roll or core (shell) should be placed centrally on the reel shaft using a loading and cantering device if available. To check that it is central, a quick measurement from the edge of the reel to the inside of the bearings will suffice.

To engage the reel shaft, insert the T-wrench supplied into the end of the shaft and turn it clockwise until the core is locked to the shaft (Approx. 100 NM Torque). The rotation on the end of the shaft results in the extending of the locking segments from within the shaft.

Alternatively an air wrench can be used to tighten the shaft within the roll.

Note: If the reel shaft is not tightened sufficiently, slippage between the core and shaft may take place. This can lead to telescoping of rolls.

To disengage the reel shaft from the core, turn the T wrench in an anti-clockwise direction, until the segments are retracted.

Figure 2.4 - 1 Expanding A Mechanical Shaft Within A Roll

	**Caution !** **Do not use excessive tensioning torque on the reel shaft as mechanical damage can take place.**

2.5 Threading the Machine

Reference is to be made to the machine web path diagram to ensure that the correct web path is followed through the machine.

Before any rolls are placed in the winding mechanism, it is suggested that a "leader" tape be threaded through the machine, to assist the threading procedure.

However it is normal practice to attach the new unwind core leading edge roll onto to a "tail" from the previous roll. This will assist in the threading procedure.

It is normal practice to cut the substrate, on the unwind roll, and therefore provide a lead edge to ease its path through the winding mechanism.

	Warning !
	**Ensure that the movement isolator is in the off position and locked out before entering the winding mechanism area.** **Under no circumstances, override any interlock, either mechanically or electrically**

Thin double-sided self adhesive tape must be used on all splices and to fix the web to the rewind core, taking care to avoid any creasing of the substrate.

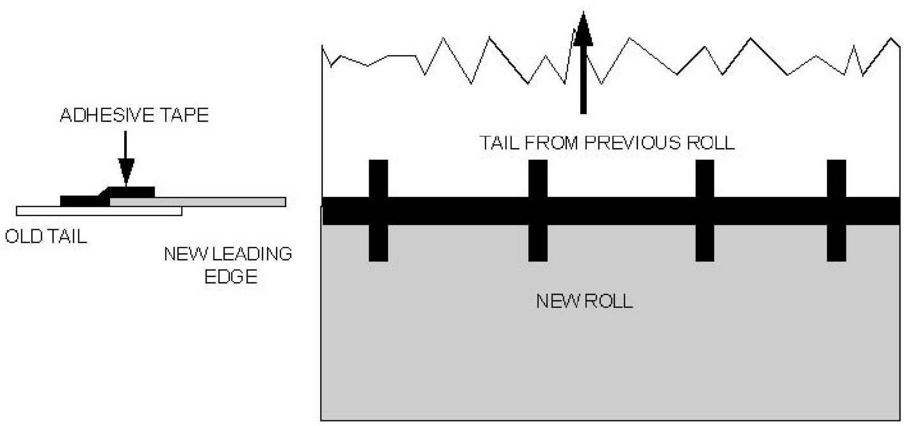


Figure 2.5 - 1 Examples of Joining of the Substrate At The Unwind Roll

Any creases located in the substrate at splices or when attaching it to the core will affect the quality of the winding.

Substrate Details

When threading is completed, the following measurements are required by the machine drive system to allow normal operation:-

- Substrate Thickness
- Substrate Width

This allows the operator to input the initial start conditions of the drive system in the ROLL SET-UP screen. Insert this information into the computer as described later in this manual

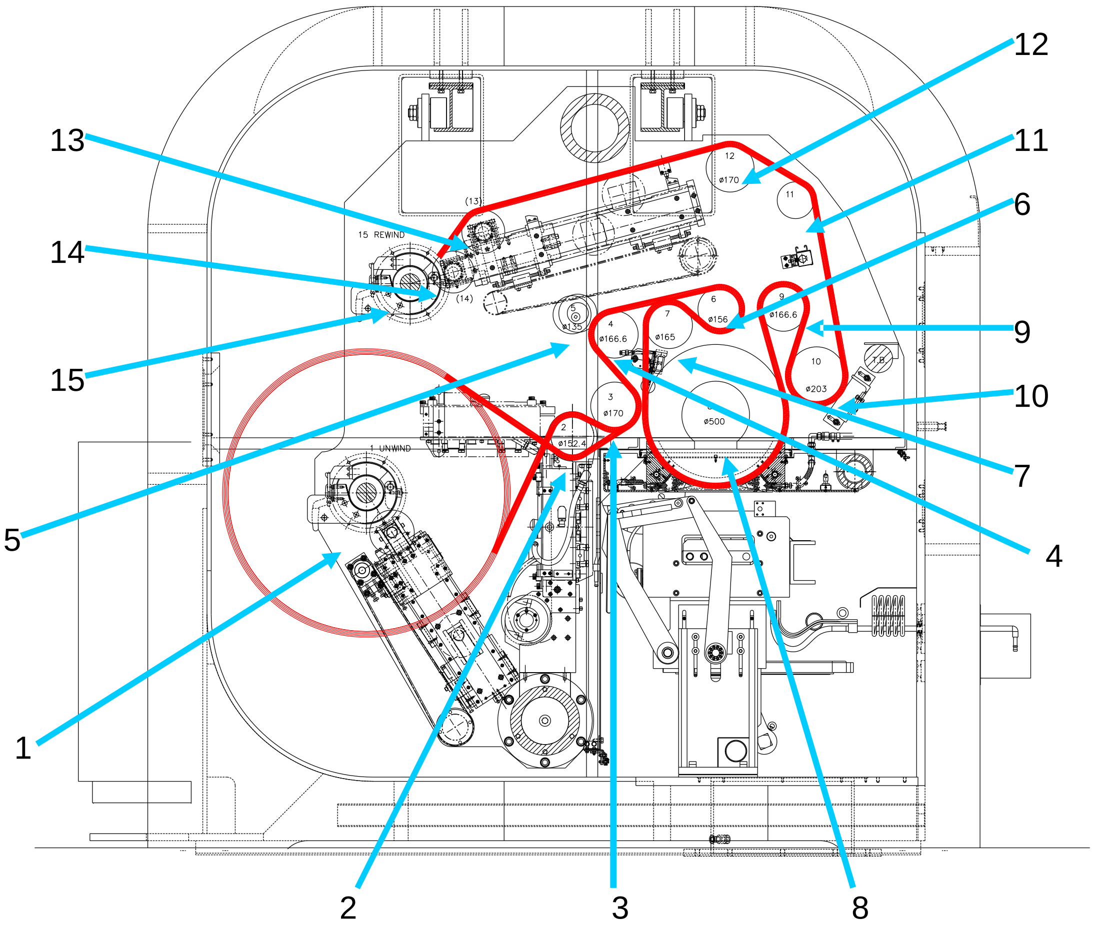


Figure 2.5 - 2 Web Path Drawing

Roller Number	Description
1	Unwind Reel – Driven
2	Special Spreader Roller
3	Load Cell Roller
4	Chromed Roller – Driven From Drum
5	Nip Roller
6	Chromed Roller – Tendency Driven
7	Special Spreader Roller
8	Water Cooled Drum – Driven
9	Water Cooled Roller – Driven
10	Cork Roller – Driven
11	Mount Hope Spreader Roller
12	Load Cell Roller
13	Special Spreader Roller
14	Idler Roller
15	Rewind Reel - Driven

Table 2.5 - 1 Rollers

Note: Ensure that all adhesive tape used for threading the machine is removed before running the drive system. Any adhesive tape remaining in the winding mechanism can lead to web breaks etc.

2.6 Direction Selection of Winder Drives

An under-wind/overwind facility is provided to allow the unwind and rewind rolls or cores to rotate in either a clockwise or counter-clockwise direction. This is provided to allow the coating of the inside or outside surfaces of the roll and the ability to wind the finished product with the metallized coat on the inside or outside of the roll.

The selection of the winding modes is made in the computer in the ROLL SET-UP screen. It can be also modified in the individual pop up screens in the DRIVES screen.

NOTE: To change this selection, the drive system must be in a powered down state (that is, tension off).

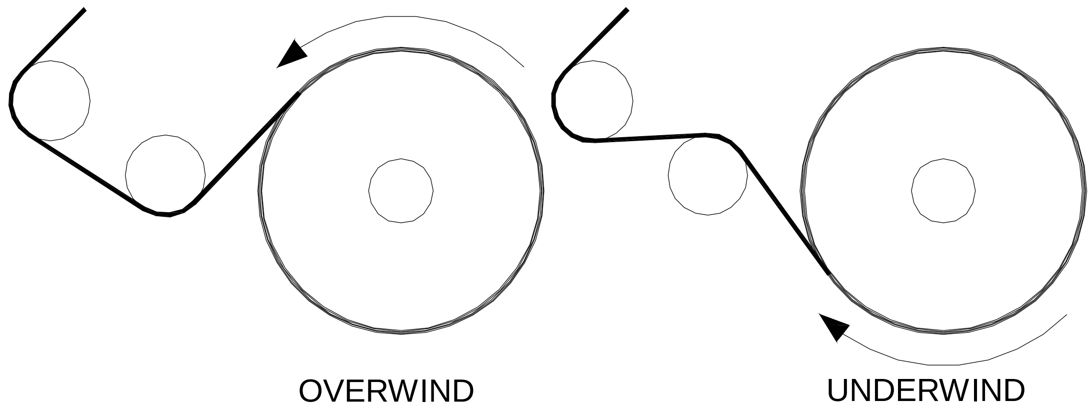


Figure 2.6 - 1 Diagram Showing Underwind/Overwind Web Path From the Unwind Roll

2.7 How to Measure the Substrate Roll Diameters

The computer requires the roll and core diameters to determine the initial speed that the winders need to rotate.

This is achieved by the drive system driving in the layarms into the roll and core.

This is carried out by pressing the Get Diameter button on the drives screen.

This is normally hid until the layarm home position action has taken place.

Figure 2.7 - 1 Get Diameter Button on Drives Screen.

After pressing this button, a further screen will be shown asking if it is safe to carry out this function.

	Warning !
	**Ensure the area is clear before pressing the YES button.**

The lay arms will then drive into the core and roll giving the diameter feedback back to the drive system.

	Note!
	**The machine will not allow tension to be applied until this function has been carried out**

2.8 Setting of Running Tensions

The machine winding system is designed such that the tension applied to the substrate at the unwind, coating and the rewind sections of the winding system can all be at different set-points.

The value of this web tension set-point depends on the type, thickness and width of the substrate, the physical condition of the unwind reel and the required condition of the rewind reel.

It is suggested that the starting values listed below are used initially as starting tension set-points. Experimentation with the substrate will lead to normal operating conditions to be determined.

Typical starting values are:

Material	gms/ linear mm of web width	oz/ linear inch of web width	Newtons/ linear 100mm of web width
12 micron Polyester	9-11.25	8-10	90-113
20 micron BOPP	10.1-12.3	9-11	102-125
40 gsm Paper	27	24	272
80 gsm Paper	36-45	32-40	363-454
12 micron Nylon	. 9-11.25	8-10	90-113

Table 2.8 - 1 Tension Guidance Levels

Once the values have been calculated, the required set-point values can be entered into the SETUP screen.

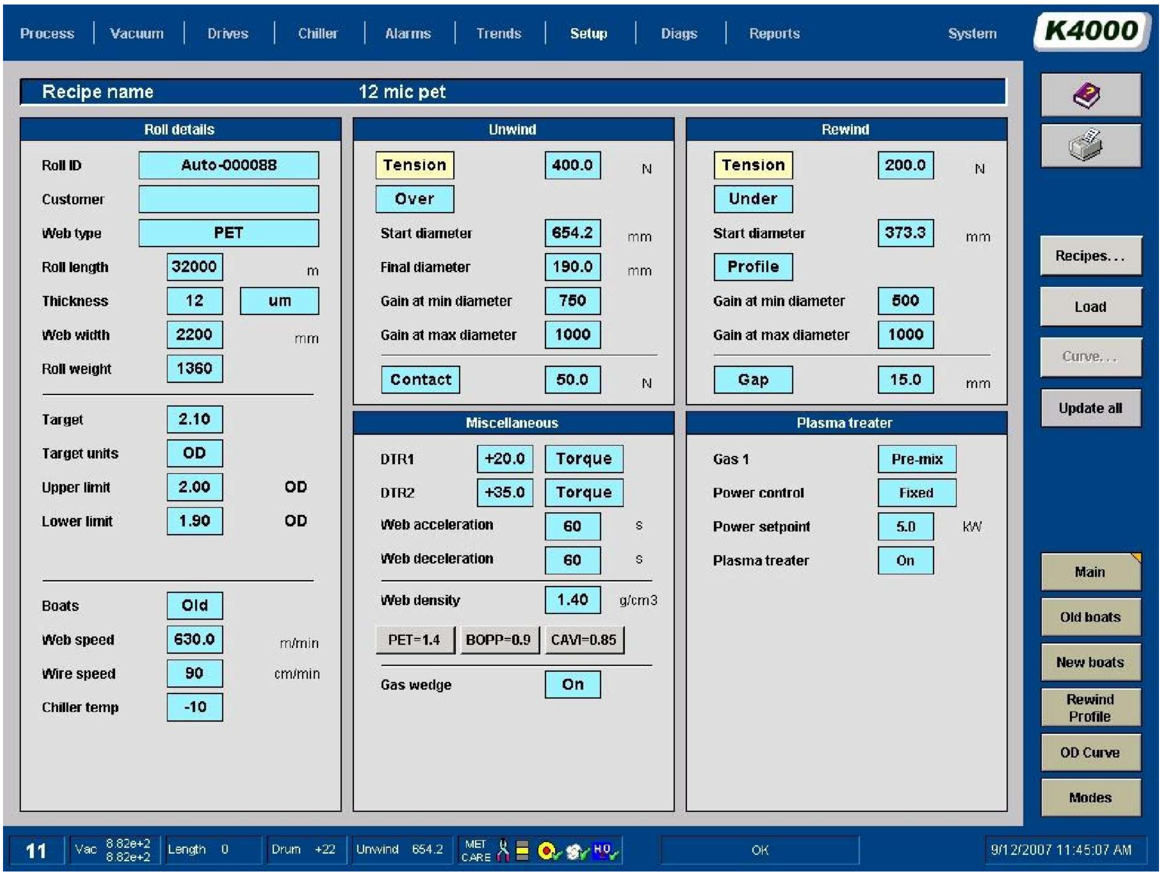


Figure 2.8 - 1 SETUP screen

The set tension and actual tension levels will be shown on the DRIVES screen. The operator can adjust the set-point while the coating process takes place from the DRIVES screen.

2.9 Process Roll Set-up Recipes (Operator Access Level)

To assist the user of the equipment, individual recipes can be stored within the computer.

Also refer to section 6 of this manual when a plasma treater is installed.

These RECIPE files will advise the user of the following running conditions;

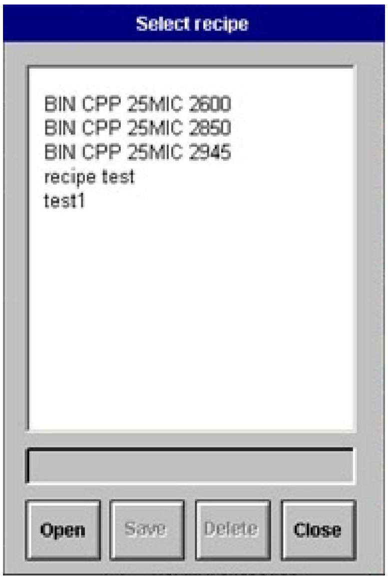
Parameter	Description
Roll ID	The roll identification reference can be used to find an archived report at a later date. As the recipe is loaded, the user will be questioned if to use a new reference or an automatically created code.
Customer	The customer name can be used to find an archived report at a later date.
Web Type	Information for reports
Roll Length	Information for reports
Thickness	Important that the drive system has this information to allow automatic diameter stopping.
Web Width	Important information for the machine to select the correct number of evaporators and aluminium wire feeders.
Roll Weight	Information for reports
Target	Aluminium deposition target
Target Units	Mode of measurement - Ohms or Optical density
Upper Limit	Declares the upper display limit on the deposition
Lower Limit	Declares the lower display limit on the deposition
Uncoated web transmission	Clear Tolerance (Used with optical monitor system)
Boats	Selects ‘Old’ or ‘New’ Evaporators
Web Speed	Sets required Line Speed
Wire Speed	Required aluminium wire feeder speed
Chiller Temp	Sets Chiller Unit Process Temperature
Unwind Tension	Sets Tension set point
Unwind Roll Direction	Sets roll winding direction
Unwind Start Diameter	Important information for the drive system
Unwind Final Diameter	Important information for the drive system
Unwind Gain at minimum diameter	Sets the drive response to changes in tension at the roll minimum diameter
Unwind Gain at maximum diameter	Sets the drive response to changes in tension at the roll maximum diameter
Unwind Layarm - mode of operation	Selects the mode of operation.
DTR1 Control Mode	Selects the mode of operation
DTR1 Set point	Set point value entered here
DTR2 Control Mode	Selects the mode of operation
DTR2 Set point	Set point value entered here
Web Acceleration	Set point value of time machine takes to accelerate to “full” speed
Web Deceleration	Set point value of time machine takes to decelerate to “full” speed
Web Density	Used by drive system. Select from default value buttons, or enter value found on material data sheet
Gas Wedge	Select ‘ON’ to allow usage
Rewind Spreader Position (Option)	Sets the position of the spreader roller.
Rewind Tension	Sets Tension set point
Rewind Roll Direction	Sets roll winding direction
Rewind Start Diameter	Important information for the drive system
Rewind Taper Profile Mode Selection	Important information for the drive system
Rewind Gain at minimum diameter	Sets the drive response to changes in tension at the roll minimum diameter

Parameter	Description
Rewind Gain at maximum diameter	Sets the drive response to changes in tension at the roll maximum diameter
Rewind Layarm - mode of operation	Selects the mode of operation.
Plasma – Gas 1	Select the type of gas being used.
Plasma – Power Control	Power control either fixed set-point or ramp with speed
Plasma – Power Set point	Power value set-point
Plasma – Plasma Treater	Select 'ON' to allow usage

Table 2.9 - 1 Roll Set Up parameter Table

The user will still have to place certain information in the relevant screens about roll widths and diameters, etc.

Choosing an existing RECIPE (Operator Access Level)



There is a facility within the system to use an existing RECIPE that has been stored in the RECIPE database.

Figure 2.9 - 1 Selection of Recipe Files Using The OPEN Pop Up Screen

Open Button

To access the list of recipes already stored in the database, the OPEN button should be touched.

A list of existing RECIPE files will appear in a pop up screen. The RECIPE required can be chosen by moving the highlighted cursor up or down the list until it is highlighting the RECIPE required. The ENTER button is then touched to load the recipe into the SETUP screen.

Save Button (Supervisor Level Only)

To allow the user to store RECIPES under a particular name, the SAVE button has been provided.

Figure 2.9 - 2 Screen Keyboard

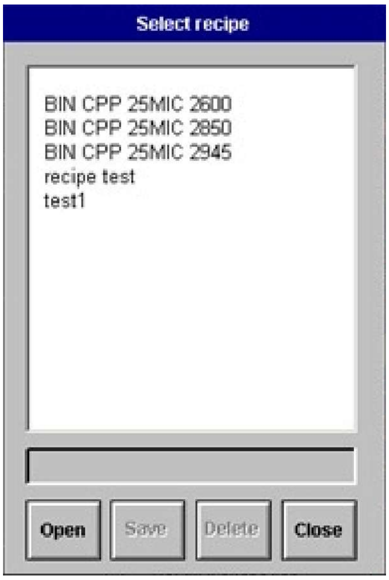
To enter a name, touch the screen to the right of the RECIPE NAME on the SETUP screen. A pop-up keyboard will appear to allow the name to be typed. After completion, touch the ENTER button on the keyboard.

The name will then appear next to the RECIPE NAME segment on the screen.

After checking all the details, the SAVE button should be touched.

Deleting RECIPE Files (Supervisor Level Only)

To remove a RECIPE file, the DELETE FILE pop up window should be selected from the SET-UP screen. The file should be selected placing the cursor on the file to be deleted. The following screen will be displayed.



Touch or select the ENTER button on the screen to remove the file

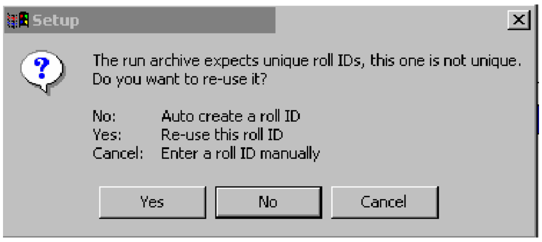
Figure 2.9 - 3 Removing RECIPE Files

Downloading of SETUP RECIPE to Control Program.

Once the user has inputted the relevant information, the recipe can then be downloaded to the control program.

Touching the button on the screen carries this out.

Figure 2.9 - 4 LOAD RECIPES Section Of SETUP Screen



When attempting to load the recipe, a pop up message screen will question the user as below. A selection has to be made.

The roll ID is very important when using the reporting function.

Figure 2.9 - 5 LOAD Pop up Screen

2.10 Rewind Tension Level Set point - Taper or Profile Control

The operator has the ability to select either taper tension or profile tension control for this machine

When To Use Taper Tension?

For the majority of plastic web materials running in vacuum constant tension control is adequate. However, with certain materials, such as paper, it has been found that the tension level is required to increase or decrease to give a good rewind reel hardness profile.

If there is sideways slipping of the substrate at the rewind roll as it increases in diameter, the set tension should be increased. If the taper profile curve is plotted correctly, the drive system will increase the tension sufficiently to prevent the slippage taking place.

When processing paper, it is normal to decrease the tension set point as the roll diameter increases. If it is not used, a "corkscrew" effect can take place where the roll becomes tighter, the larger that it becomes. The end result can be that the roll will telescope side-wards.

What is Taper?

The term taper is described as a "narrowing of a solid item along its length".

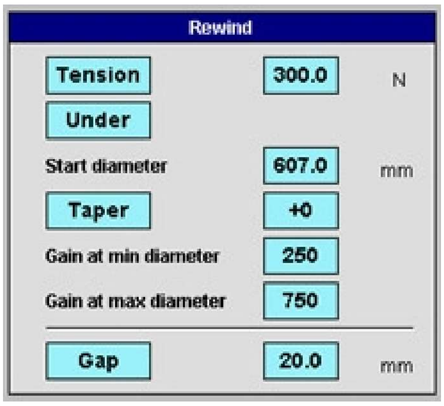
Whilst the design of the winding system is such that set tensions are maintained from start to finish of a roll. There are instances, when it is necessary to have a controlled change of tension as the build up of the substrate on the rewind proceeds.

What is Tension Profile?

The term profile relates to a tension set point value curve that can be created within the computer. The set point will follow this curve as the reel diameter increases.

Setting Taper Tension Set point

On the SETUP screen, the TAPER mode can be selected to either “Taper” or “Profile”



When this mode of operation is selected, the rewind set-point tension will increase or decrease by the value indicated on the screen in a linear action as the roll increases in diameter from the core to the full diameter

Figure 2.10 - 1 TAPER Selection Section of SETUP Screen

- Example with a –20% taper and a starting set point of 100 newtons, the set point at 1000mm will be 120 newtons.
- Example with a +20% taper and a starting set point of 100 newtons, the set point at 1000mm will be 80 newtons

When to Use the Rewind Tension Profile Curve?

There are situations when it has been found that increases or decreases at certain diameters are required at the rewind roll to allow formation of a good wind up.

This function allows a curve to be set up in such a method that gives this flexibility.

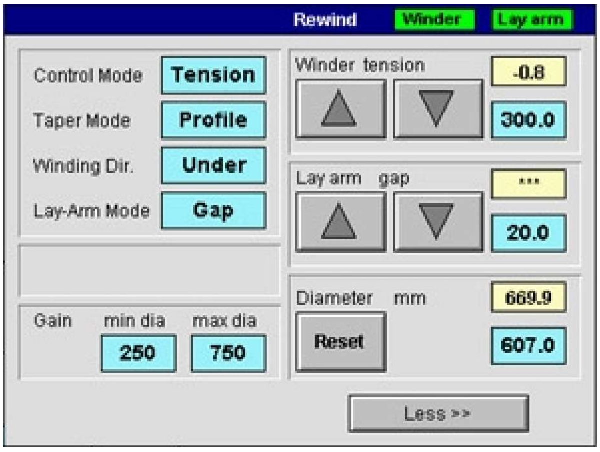


Figure 2.10 - 2 Profile Selection Section of DRIVES Screen

Setting Rewind Tension Profile Curve

On the DRIVES screen, the TAPER mode can be selected to “Profile” curve

The rewind PROFILE TENSION curve is accessed from the DRIVES screen by touching the button shown.

Figure 2.10 - 3 Selection Buttons to view Tension Profile Curve

The following screen will be displayed.

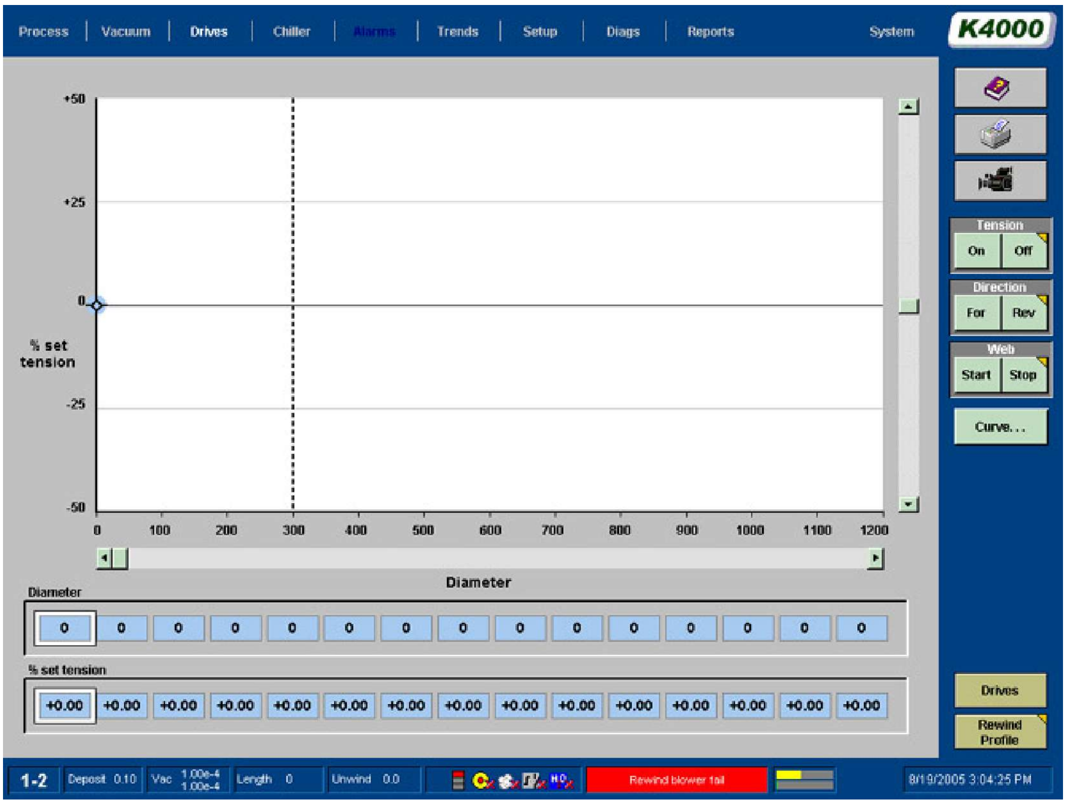


Figure 2.10 - 4 Tension Profile Screen

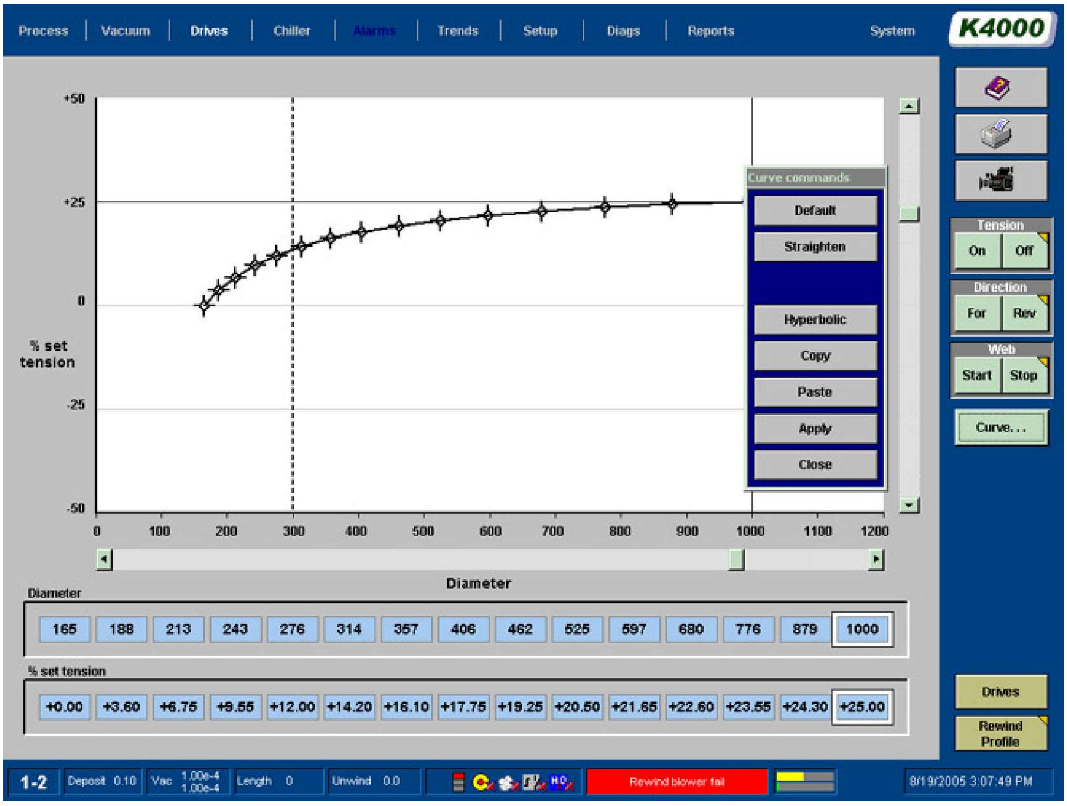


Figure 2.10 - 5 Tension Profile Screen Showing Curved Profile

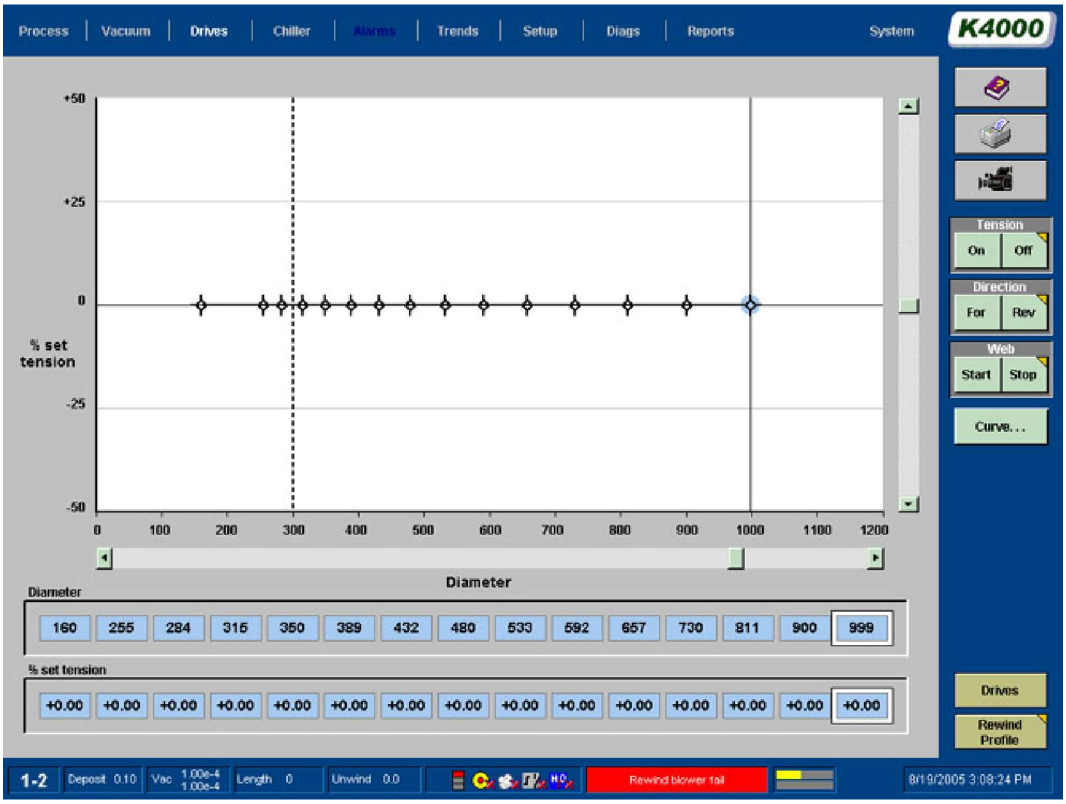


Figure 2.10 - 6 Tension Profile Screen Showing Straightened Profile

Selection will insert the default code as figure 2.10 - 6

Will plot a straight line between first and last points

Will produce a hyperbolic curve between first and last points

Copy the curve to use in another recipe

Paste the copy of curve, selected by the ‘copy’ button.

Apply curve to the recipe. Without applying, the curve is not ‘set’

Figure 2.10 - 7 Tension Profile Curve Commands

The X. Axis is scaled between the minimum and maximum diameters for the rewind roll. The Y. Axis is scaled between -50 and +50 per cent of the actual rewind tension set point, as read from the SETUP screen

There are 7 points on the curve. The first and last are fixed on the x-axis at the core and maximum roll diameters. The other 15 points can be moved on the screen to create a profile curve.

The reference points can be moved either by selecting them with the user's finger and moving them or by a “computer mouse”.

If there is no requirement to increase or decrease the tension throughout the coating cycle, the profile should be a straight line at 0 %.

Once the curve has been completed, the user can return back to the SETUP screen by pressing the SETUP button.

2.11 End of Roll Sensing (Auto Slowdown)

A function of the drive system is to detect when the unwind or rewind rolls are reaching their minimum or maximum diameters and then stop the drive system.

The function is designed to trigger the deceleration of the drive line speed to zero and to stop if the unwind is running off the core, and rewind exceeding the maximum roll sizes. Normally it is initialised by the material on the unwind approaching the core. However at the same time, there is a requirement to leave a minimum amount of material on the unwind core.

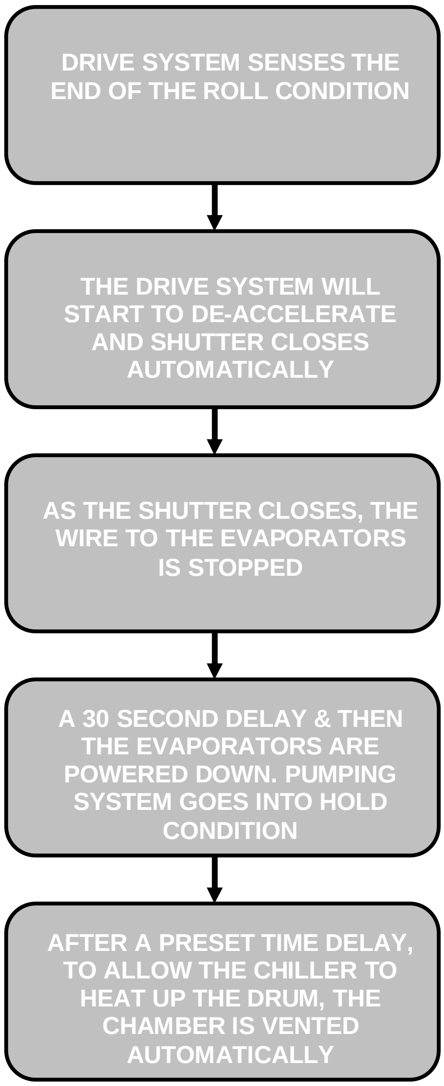
The drive system calculates the diameter at which the unwind roll should start to slow down to finish at the final unwind roll diameter, based upon the following inputted information.

- Line Speed (Actual feedback)
- Required diameter of material left on the core

The information is placed into the ROLL SETUP screen, by the user, and downloaded to the drive system.

The machine will make alterations for speed changes during the coating cycle.

When auto slowdown is used, the machine will go through the sequence shown in the 'auto slowdown' sequence flowchart, to vent the chamber up to atmosphere and then wait for the operator to open the machine.



Flow Chart 2.11 - 1 Auto slowdown Sequence Flow Chart

2.12 Monitoring Running Conditions

The DRIVE screen will indicate to the user, what is occurring within the drive system at all times.

The following list of feedback signals and conditions are monitored;

- Unwind Tension
- Rewind Tension
- Line Speed
- Layarm Gap or Pressure
- Drive unit conditions - Healthy
- Length Counter
- Roll Diameters
- Spreader Position
- Gain Settings

Setting/Adjusting Parameters:

The following parameters can be adjusted on line, using the arrow sections on the screen;

Figure 2.12 - 1 DRIVE Screen Adjustments

- Line Speed
- Tensions
- Draw Ratio's
- Current/Torque Limits
- Gain Settings

There are sets of arrow, one to raise the parameter, and one to lower it.

As the finger is placed upon the arrow, the parameter will increase or decrease within the minimum and maximum values. If the finger is left upon the arrow and the parameter starts to change, the rate of change will increase, as long as the finger is left upon the screen.

Touching the screen above it and entering a new value via the keyboard can also alter the value of the tension set point.

Monitoring Set-point/Actual Operating Parameters

This display will generally show the set parameter below the line and the actual parameter above the central horizontal line.

Figure 2.12 - 2 DRIVE Screen Adjustments

The upper black numbers indicates the tension set point. While the lower blue section indicates the tension feedback value. There will be numerical information of both the set and actual parameters at the right-hand side of this section of the screen.

Figure 2.12 - 3 Drive Status Message Box on DRIVES Screen

In the case of an "Emergency Stop" or "Web Break" situation, the E-Stop key switch on the desk has to be turned to the RESET position, and the drives reset button on the DRIVES screen has to be touched.

Monitoring Unwind Diameter

The unwind section of the screen allows the user to monitor the performance of the unwind roll's diameter during the coating cycle.

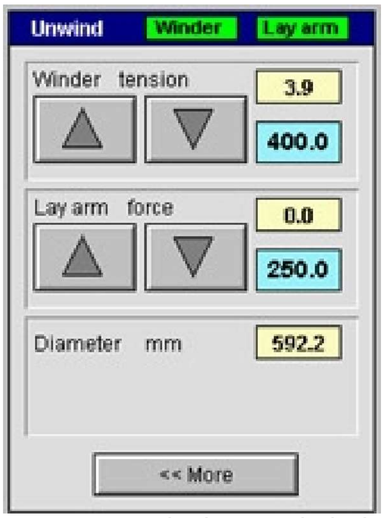


Figure 2.12 - 4 Monitoring Unwind Diameter

The following parameters are shown;-

- Tension Set-point
- Actual feedback Tension
- Actual diameter.

As the unwind roll diameter becomes smaller the Auto slowdown sequence will take place. Pressing the MORE button will bring up a further section of the pop up screen.

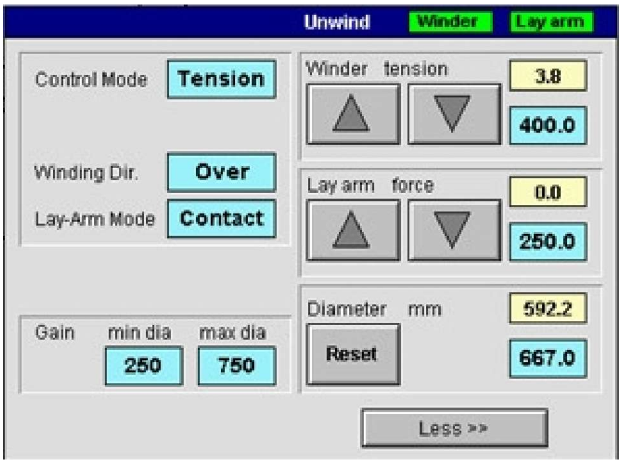


Figure 2.12 - 5 Expanded Unwind Monitor pop up Screen

The further functions are

- Gain Settings
- Under/over wind direction selection
- Diameter Change
- Layarm Control Mode

Footer Display

The unwind diameter is displayed at all times in the screen footer

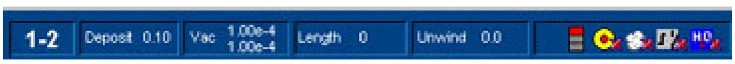


Figure 2.12 - 6 Footer showing Unwind Diameter

2.13 Gain Settings - Rewind/Unwind

The drive gain setting value that can be entered in the unwind and rewind pop-up boxes is to allow the user a certain degree of control of the response of these drive units.

Figure 2.13 - 1 Gain Settings Section of SETUP Screen

There are two values, the left hand one for the gain setting at core and the right value for the maximum roll setting.

Within the computer, there is a look up table that increases the responsiveness of the drive control at the diameter increases.

The higher the number that is entered into the DRIVE GAIN value, the more responsive the drives will become.

Do not set the levels to a value where the drive becomes unstable.

Monitoring for responses.

The historical display available from the TRENDS screen can be used to monitor the performance of the gain settings.

Modifications can be made from the pop up box shown below and the response be monitored.

Figure 2.13 - 2 Rewind Control Pop Up Box

Generally, higher values should be used materials such as CPP, LPDE, and smaller values for BOPP, PET and Paper

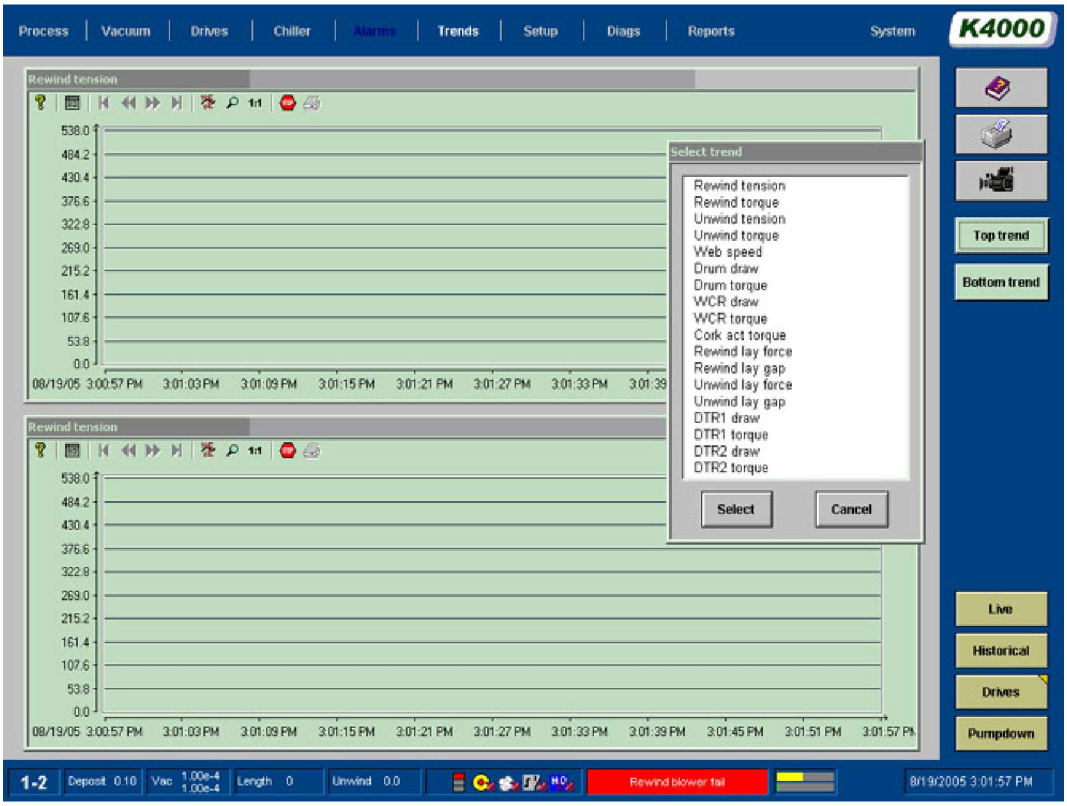


Figure 2.13 - 3 TRENDS Screen – Drives

2.14 Starting the Drive System & Selecting Modes of Operation

The drive system has a number of programmed modes of operation that are selected by the various buttons on the DRIVES screen.

There are keys for the following functions that are active on the DRIVES Screen

- Tension On/Off – applies tension to the web
- Direction Forward/Reverse
- Web Start/Stop –starts to move web at the SET SPEED if all the relevant interlocks are made.
- Nip Roller Engagement On/Off (see section 2.31)
- Inch Drive System (see section 2.18)
- Reset Drives (see section 2.17)
- Reset Lay-arms (Only appears when arm(s) need to be reset.
- Reset Length – resets length counter

The start function is not available unless the winding cart is driven fully into the chamber and tension mode has been activated. When the buttons are activated, the small triangle will move within the button.

Figure 2.14 - 1 Mode Buttons

2.15 Web Break Detection

If there is a rip/break in the web/substrate, the machine has been designed to stop the drive system. If not, the coating drum will become coated and possible wrap around the web on rollers can occur.

The drive system will perform a "Fast Stop" procedure if it detects a web break.

To determine if a web break has occurred, the drive systems monitors both the rewind and unwind tension feedback levels.

If one of these levels falls below a preset low level (normally 3% full tension range), it triggers the web break detect circuit.

The drive system will de-accelerate in a pre-designed time between 10 and 20 seconds to a stop. After this time period, the drive system is powered down.

2.16 E - Stop Function

In an emergency situation the operator should press the "E - Stop" push-button.

When activated, the power to the drive system and other equipment will be removed.

At the same time, all other power circuits will be de-energised.

The drive rollers and rolls will "coast" to a standstill condition if they are still rotating.

The MMI will indicate that a "E-Stop" has occurred on the Alarm page

Resetting After an “E-Stop” Function

Warning !

Ensure that all sections and areas of the machine are clear of personnel, before resetting.

After ensuring that the machine is safe to be reset after a “E-Stop” condition has occurred, the circuit can be reset.

The flashing “RESET” button has to be pressed to carry out the reset.

Machine Start Up

On powering up the machine, the safety circuit needs to check that all components are present in the correct condition. The “E-Stop” button has to be pressed before resetting the system.

2.17 Drive Reset Function

If the E-Stop function has been activated, the machines ‘E-Stop’ and ‘Guard-Stop’ reset pushbuttons have to be pressed.

Also the RESET DRIVES button on the drive's screen has to be touched to allow normal operation.

Figure 2.17 - 1 Reset Drives Button

This will place the drive system in a ‘ready state’.

2.18 Drive Inch Function

The drive system can be “Inched” (Jogged) slowly by pressing the INCH push-button at the rear of the chamber at the loading area.

There is also a button on the DRIVE screen to allow the same function.

The direction of the movement is selected using the DIRECTION switch on the DRIVE screen

Before the rollers will move, there is a 5 second delay while the warning lights flash and siren operate. These warning indicators will stay activated during this operation.

To comply with the latest recommended safety requirements (PrEN 1010:1993), the control system is programmed to allow the following sequence;

Warning !

Ensure that all relevant sections and areas of the machine are clear of personnel, before continuing with this operation.

- After initial holding down of the push button for a minimum time period of 3 seconds, release the push-button.
- Wait for one second and then press the push button again.
- The rollers will turn in the direction selected
- After the push button has been disabled, the siren and warning lights will be activated for a further 5 seconds.

2.19 Description of Draw Control Function

A mode of operation available in for controlling the DTR roller section of the drive system is described as "Draw Control".

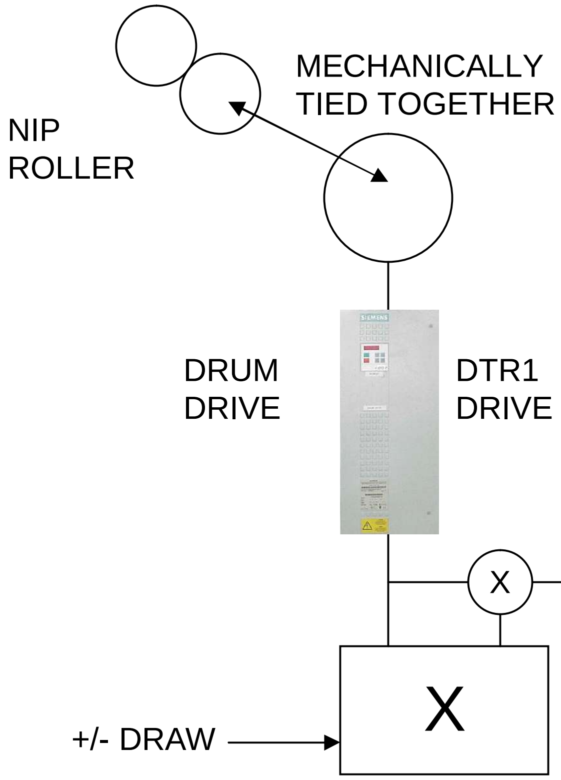


Figure 2.19 - 1 Draw Control

In this machine, the drum is the master reference.

This mode of operation compares the speed difference between the drum, the master reference, and the DTR roller. The draw value is said to be zero when both the reference drum, and the roller being monitored are rotating at the same speed.

The speed of the DTR roller can be adjusted to rotate faster or slower than the master reference, by adjusting the draw value. For example, if the draw value is changed to 0.5% on the interface unit, the DTR roller will rotate at a speed 0.5% faster than the master reference, drum. If the value is set to 0.95%, the speed will be slower by 0.5% of the master reference, drum.

It is usual to operate rollers after the master reference, slightly faster. This allows the substrate to be held tight across the surface of the drum, to provide the best conditions for cooling of the substrate.

This mode of operation is described as a closed loop system.

2.20 Description of Torque Control Function

A mode of operation available in controlling the DTR roller section of the drive system is described as "Torque Control".

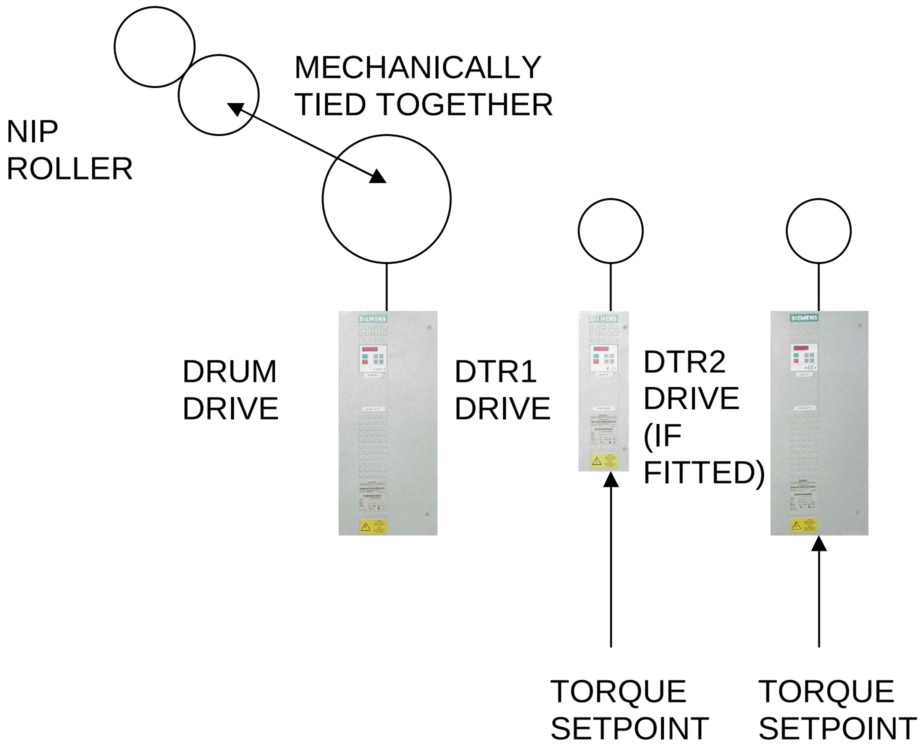


Figure 2.20 - 1 Torque Control

In this machine, the Drum is the master reference and the DTR roller can be operated in a "Over speed Mode";

This mode of operation allows over speeding of the above DTR roller difference in reference to the drum, the master reference.

The torque required by the DTR roller can be limited in this mode of operation. With the torque value set to zero, the drive motor will not attempt to drive the relevant roller. In fact the roller will either slip under the substrate or be "pulled" around by the roller.

The speed of the DTR roller can be adjusted to rotate faster than the master reference, by adjusting the torque value. For example, if the torque value is changed to 5 % on the DRIVES screen, the relevant roller will rotate at a speed faster than the master reference, drum. It is normal to monitor the substrate while running and adjust the value to ensure that the substrate is flat on the relevant roller.

	Note!
	**If the value of the torque is increased to a level that will over-speed the roller to a point where the substrate will slip over the roller. This can be seen by the current taken by the relevant motor to decrease at this point.**

This mode of operation is described as an open loop system.

2.21 Rewind Lay arm Roller Assembly

To ensure that the substrate is supported until it is wound onto the rewind roll, a layarm roller assembly is used.

The rewind layarm assembly incorporates two rollers mounted on a bracket. This bracket is mounted on a slide rail, which moves in a parallel path as the rewind roll increases in diameter.

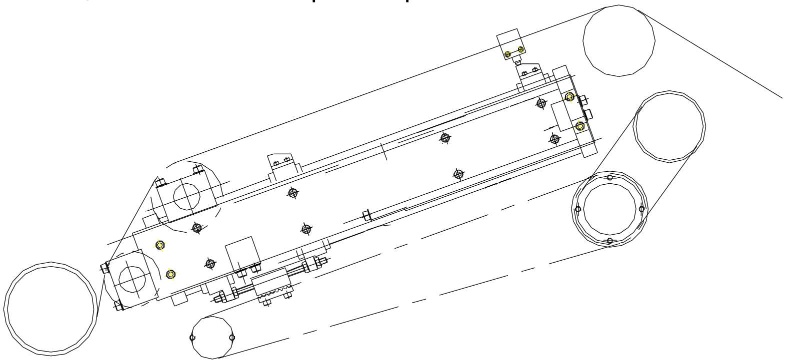


Figure 2.21 - 1 Rewind Layarm Arrangement

In the GAP mode of operation, a small gap is maintained as the roll increases in diameter.

Alternatively, the system can be operated in the CONTACT mode, where the last roller before the rewind roll presses on the rewind roll.

Note: If excess pressure is applied, the substrate can be damaged.

Setting the Operating Mode

The mode of operation is set in the ROLL SETUP screen. It can also be changed from the LAYARM pop up box on the DRIVES screen.

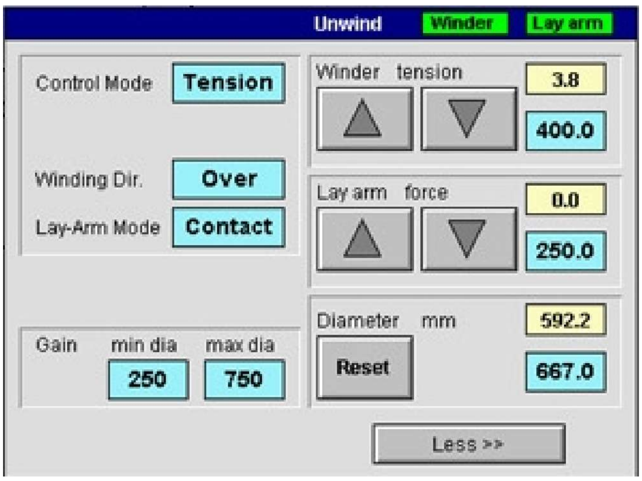


Figure 2.21 - 2 LAYARM Section Of Screen

Touching the top section of the block in the box changes the mode of operation.

The amount of contact pressure applied or gap required is set on DRIVES screen. These values can be changed while the coating cycle takes place, but it is advisable not to change between the modes of operation.

Theory of Operation

The drive system generates calculated diameters by comparing the feedback signals of the drum and winder motors. A secondary function of the calculated rewind diameter is to control the position of the layarm in the layoff mode.

The arm assembly is moved up or down the slide rails by a motor. The motor that is located on the outside of the chamber drives a shaft inside the chamber, via a vacuum lead-through.

The motor is an ac vector drive system, which is very accurate in its operation.

The encoder on the motor generates a feedback signal.

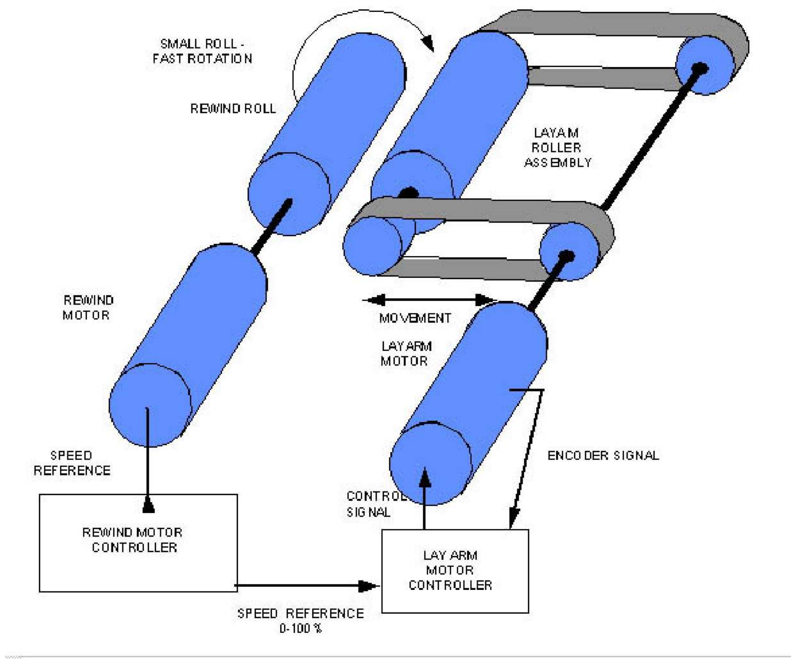


Figure 2.21 - 3 Layarm Theory

GAP Mode

The motor is driven by a signal from its controller to match the calculated diameter and the feedback signal. With a set gap of 0, the layoff roller should touch the rewind roll as the diameter changes. In practice, it is normal to have a gap of about 15mm set from the drive interface unit.

CONTACT Mode

In this mode, the layarm roller is in contact with the rewind roll of substrate. The contact pressure can be varied and the arm will move at a constant pressure as the roll increases or decreases in diameter.

In the CONTACT mode, the current supplied to the motor is reduced or increased to allow the arm to move down the slides. The maximum pressure that can be applied is the weight of the roller assembly, if the assembly is sloping down as it approaches the roll.

In the case where the assembly has to move up the slide rails, the maximum pressure is set up during the drive commissioning.

The CONTACT pressure set point is controlled by the user from the DRIVES screen, which indicates the pressure set point.

The drive system takes into account tension and diameter changes and keeps the pressure constant.

Movement by Operator

With the chamber open, the roller assembly can be adjusted by the operator from the rear control station with the “Raise" and "Lower" push buttons.

This function is only available with the chamber fully open.

Limitation of Movement

The upper and lower limits of movement are stopped by mechanical stops at either end of the slide rail. A limit switch also monitors the fully raised position.

Modes of Layarm Motors during various periods in operating cycle

At certain times in the operating cycle of the machine, the drive system will override the operator’s input in the control of the layarms. These situations are described below; -

- **Winding cart open, tension applied and the arm is in the FULLY OUT (Parked) positions**

The arm should not move from this position unless the user moves them with the local push buttons. These push buttons are disabled by the activation of the drive interlock (winding cart inside the chamber).

- **Winding cart open, tension applied, arm fully out (parked) position, inching of drive system**

The arm should not move from this position unless the user moves them with the local push buttons. These push buttons are disabled by the activation of the drive interlock (winding cart inside the chamber).

If the arm has been moved from its parked position and the drive system is inched, the arm will move to the roll and operate in the selected operational mode, that is, CONTACT or GAP..

- **Winding cart inside the chamber, tension applied and the drive interlock activated**

The layarm will move to the rewind roll. There is a timer that allows the roller to be driven down. It will then apply pressure to the roll of substrate to stop possible movement of the upper layers.

- **Winding cart inside the chamber, tension applied, the drive interlock activated and the line is started**

When the line start signal is received, the mode of operation should change (if required) to that selected by the user

- **Winding cart moving out of chamber, tension applied, drive interlock not activated**

As the winding cart starts to move out of the chamber, the layarm assembly will be driven back to its parked position. The limit of this travel is stopped by localized switch inside the chamber.

2.22 Unwind Layarm (Gap) Roller

To stop the substrate sliding, feathering or telescoping at the unwind position, a layon assembly is provided.

The unwind layon roller assembly consists of a roller that slides upwards to the unwind roll on the slide assembly.

The principle of operation and control is the same as the rewind layarm.

The major difference is that this assembly consists of a single roller, which is pivoted to allow for differences in the roll circumference.

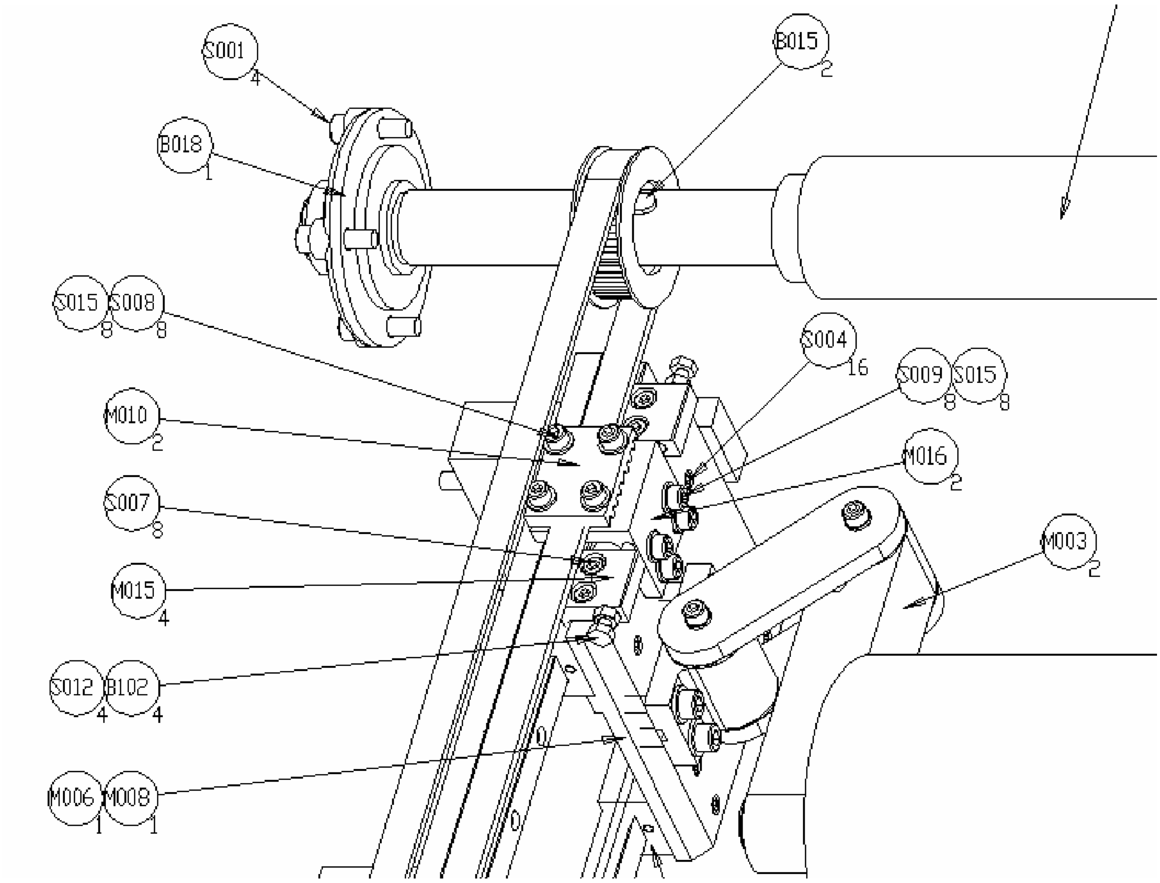


Figure 2.22 - 1 Drawing of Unwind Layarm Pivot assembly Roller Alignment

2.23 Process Driven Rollers

Pre-Drum Driven Roller

To allow a contact nip roller to be used in the winding mechanism, a roller is driven by the drum at the same surface speed.

The drum drives the roller via a timing belt located within the tendency belt enclosure.

The surfaces of the rollers are manufactured to a fine tolerance with an upper surface of chrome. Care should be made not to damage them.

Cooled Process Drum

A cooling drum is provided to allow the removal of heat from the substrate as it is in contact with it.

The heat is generated from the aluminium condensing upon the substrate and radiant heat generated by the coating sources.

It is designed and manufactured to ensure an equal cooling performance across its full width.

The cooling medium is supplied to the process drum from the process chiller system at a preset temperature.

The surface of the drum is manufactured to a fine tolerance with an upper surface of chrome. Care should be taken not to damage the drum, as this will cause potential problems with overheating of the substrate.

Also refer to the section on the gas wedge and its operation.

Cooled Process Roller (DTR1)

A cooled driven roller is provided to allow the removal of heat from web as it is in contact with it.

The cooling medium supplied to the roller from the process chiller at a preset temperature.

The roller is manufactured to a fine tolerance with an upper surface of chrome. Care should be taken so as not to damage it

Ebonite Roller (DTR2)

The driven roller is constructed from a non-metallic material called ebonite.

This roller is the backing roller for the monitor system and should not be replaced by a roller of a different material.

This roller has been covered with a cork tape to ensure good contact between the substrate and the roller surface.

If it is damaged or badly contaminated, the covering will need to be cleaned or replaced.

2.24 Load Cell Rollers

A number of lightweight rollers are supplied and are mounted between sets of load cell transducers. Theses are normally located in the following areas;

- Unwind Roll
- Rewind Roll

The rollers rotate on bearings located in the load cell transducers

The purpose of this combination of rollers and transducers is to register the amount of tension being applied to the substrate across its width. This information is then transmitted to the drive, PLC and sub-systems for control and monitoring.

The tension levels are displayed upon the DRIVES and PROCESS screens.

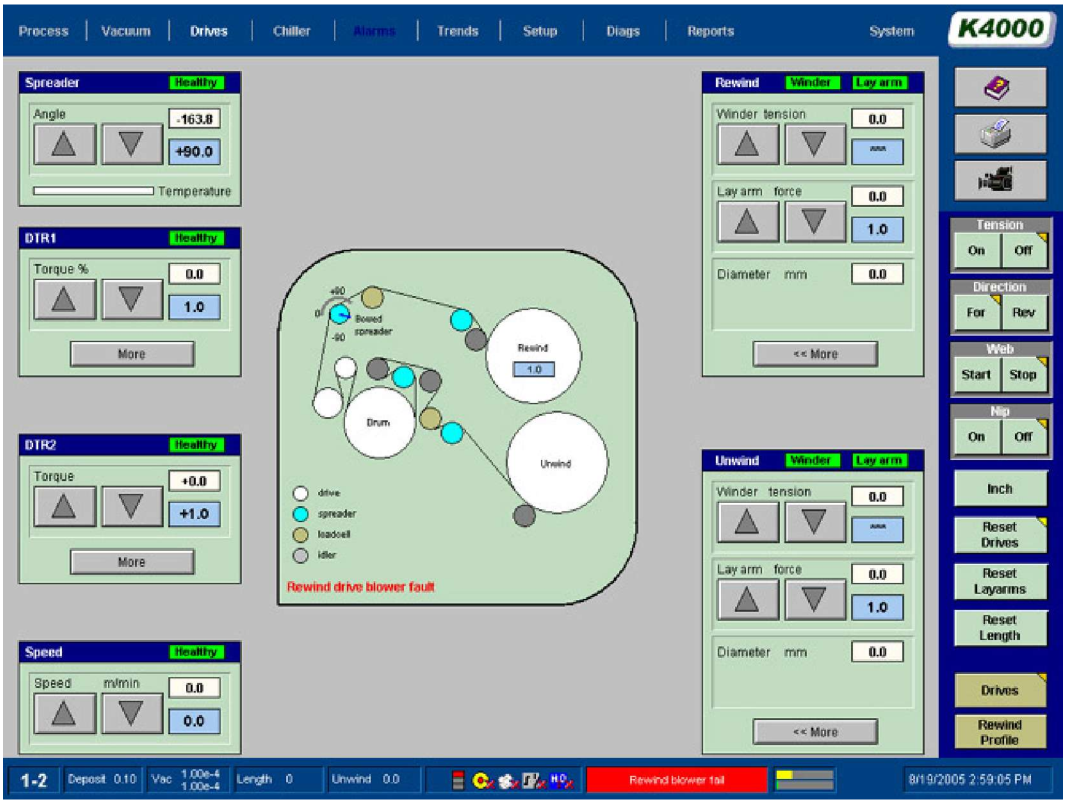


Figure 2.24 - 1 DRIVES Screen

2.25 Straight Cut Spreader Roller

A number of these rollers are located in the machine to spread the substrate. The substrate presses on the roller fingers, which in turn move sideways and spread the film.

Figure 2.25 - 1 Photograph of Spreader Roller

The roller's exclusive design is increasingly flexible from the center to the edges, providing maximum wrinkle removal action without distorting or causing uneven tension throughout the web.

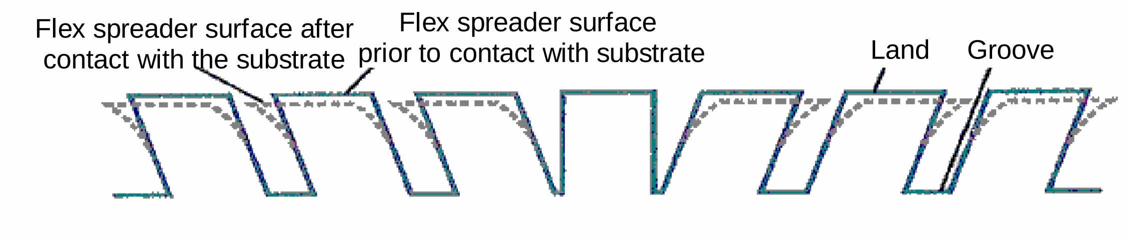


Figure 2.25 - 2 Straight Cut Spreader Roller Principle

Profile Depth

From the start of the groove, the profile becomes deeper and more angled starting at 6mm ang increasing to 30mm at the edges.

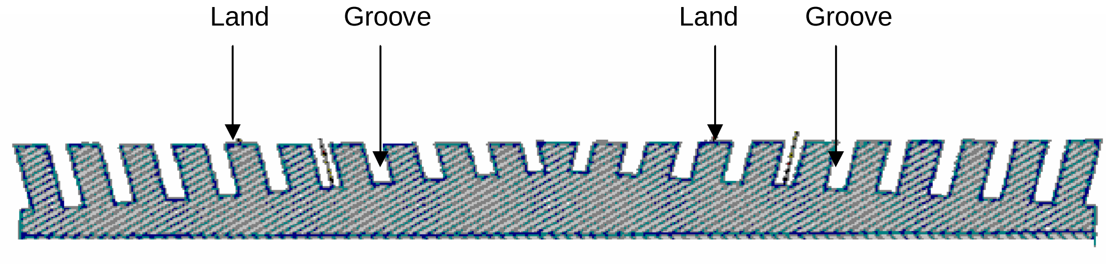


Figure 2.25 - 3 Straight Cut Spreader Roller Groove Profile

This roller will become worn over a long period of time. This is noticeable when moving out to a wider width substrate than has been run previously. If there is problem, it will normally show up as a wrinkling problem.

2.26 Bow Type Spreader Roller

This type of roller is also sometimes called a 'Banana Roller'

A bow type roller may be located on the winding mechanism to spread the substrate before the rewind roll.

The amount of bow in the roller is fixed.

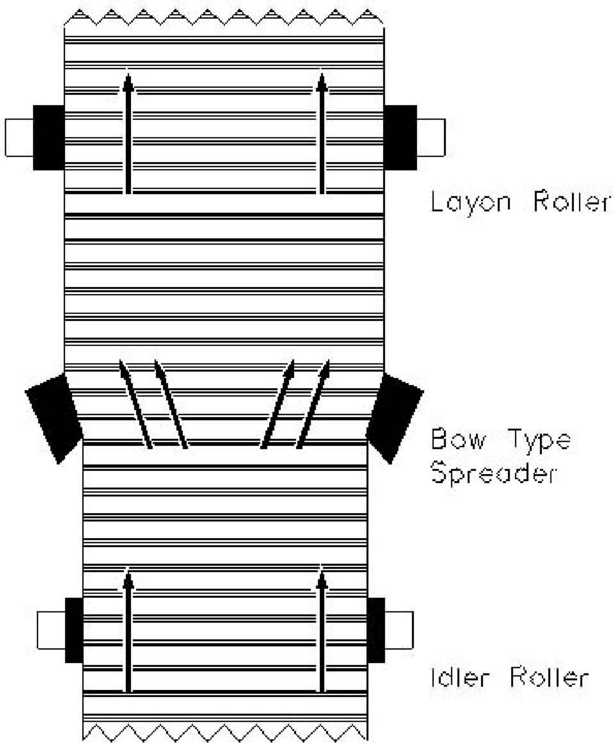


Figure 2.26 - 1 Operation Of Bow Type Spreader Roller

Position of Apex

The apex of the bow can be adjusted at the roller. The bow should be moved depending upon the web path used. Generally, once the test engineer has set it, it will not require any further adjustment. Some machines have a remotely adjustable apex, which can be adjusted whilst the machine is closed.

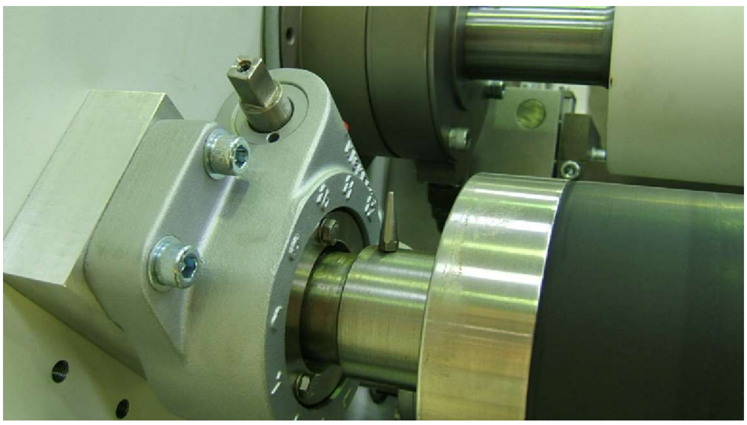
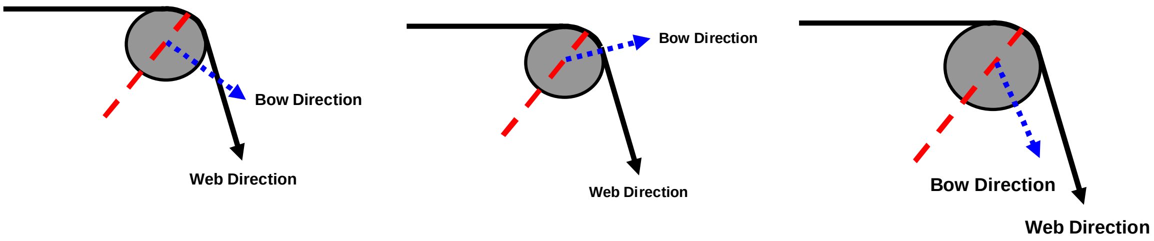


Figure 2.26 - 2 Photograph Showing Kickert Spreader Roller Apex Indicator

See the vendor manual for information on the operation and positioning of the bow in relationship to the substrate.



Normal Bow Direction

Bow Position to Tighten Edges of Substrate

Bow Position to Tighten Middle of Substrate

Figure 2.26 - 3 Bow Position Affects of Spreader Roller

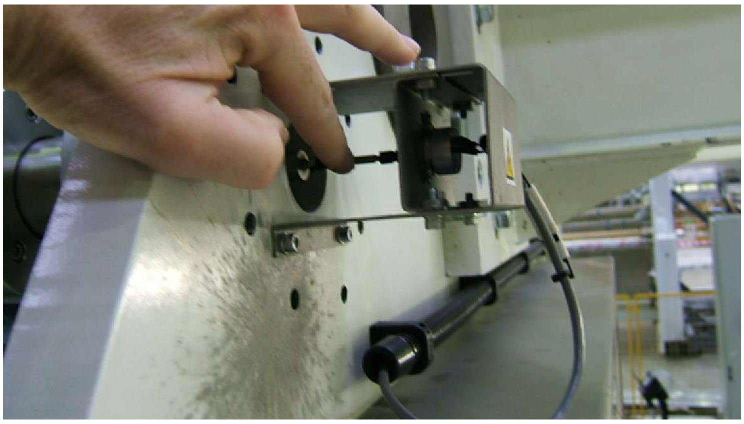


Figure 2.26 - 4 Photograph of Spreader Feedback Potentiometer and Coupling

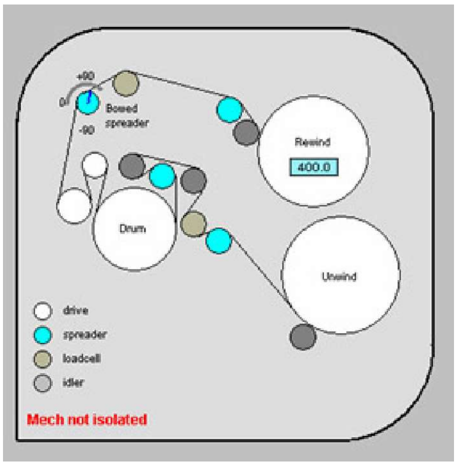


Figure 2.26 - 5 Drive Screen Indicating Spreader Roller feedback.

2.27 Nip Roller

The nip roller is a rubber-covered roller that is used to nip the substrate between itself and a pre-drum driven roller.

The nip roller is moved away from the driven roller by a pneumatic actuator via a mechanical linkage mechanism.

The cylinder is located on the outside of the chamber. The control of the cylinder operation is by a button on the DRIVES screen..

The machine is designed to operate with the nip roller pressed against the driven roller.

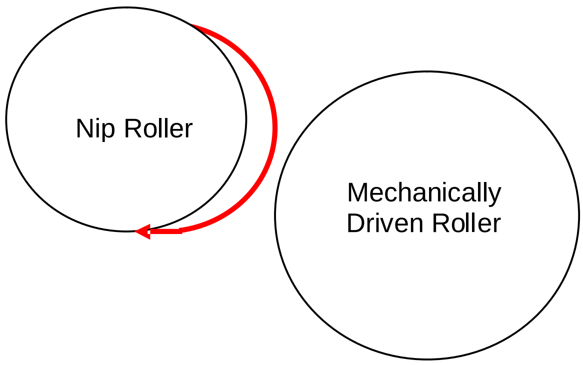


Figure 2.27 - 1 Sketch of Nip Roller Relationship

This is to ensure a tension isolation point between the unwind roll and the rewind sections of the winding system.

When a "web break" condition occurs, the nip roller is withdrawn from the driven roller to stop possible damage to it.

Operating Pressure

The nip roller is operated by pneumatic actuator, which is regulated by a local pressure regulator.

This air supply should be set to 2 bar. Excess pressure will damage the rubber on the roller.

<table>
<tr><td rowspan="2"></td><td>Caution !</td></tr>
<tr><td>Do not apply more than 2 bar pneumatic pressure or excessive contact the roller surface.

If a padlock has been applied to the regulator, do not remove it.</td></tr>
</table>

Loss of Nip Action

If the nip roller is retracted from the driven roller, slippage can take place around the drum. This can lead to incorrect diameter calculations by the drive system, due to slippage of the substrate over the drum.

If the nip roller is removed when the tension difference between the unwind and rewind sections of the machine are more than 2:1, it is usual that the substrate will creep in the direction of the higher tension.

Creases at Nip Roller

There are occasions, due to poor quality substrate or badly wound unwind rolls, and it may not be possible to use the nip roller due to creasing of the substrate.

Careful balancing of the tensions throughout the machine should enable the substrate to be processed successfully in these cases.

Testing Nip Roller Contact

The method is described in the maintenance manual.

2.28 Gas Wedge System

It has been found that the addition of a system allowing a small amount of dry air to be placed between the substrate and the process drum when metallizing allows better conduction of heat away from the substrate.

Recorded Benefits are

- Improves film cooling
- Little or no temperature rise at the rewind
- Improved convertibility
- Removes 'tramlines'

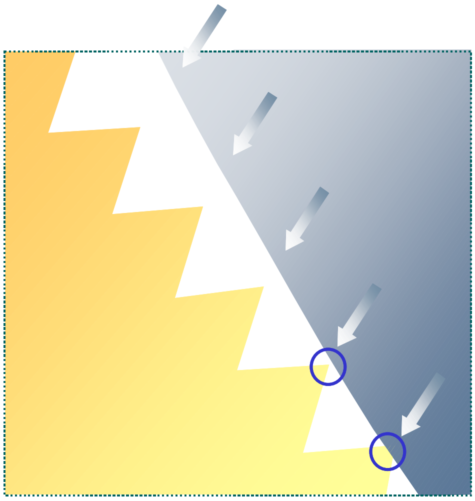


Figure 2.28 - 1 Effective cooling at points of contact only

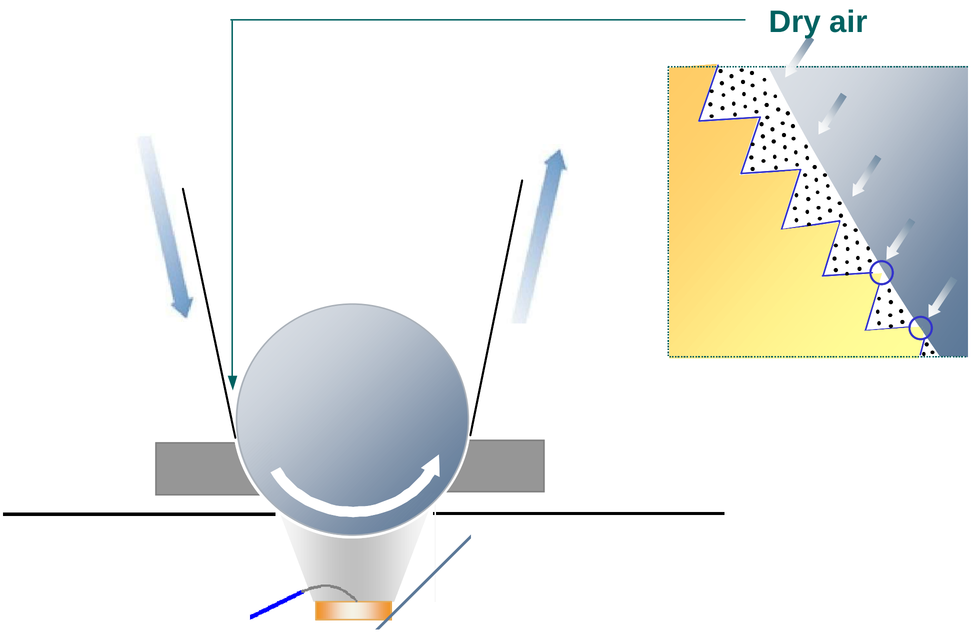


Figure 2.28 - 2 Effective cooling at all points along the substrate's length

Why is the device called a Gas Wedge?

The air (gas) is fed into a specially designed distribution device that allows equal gas flow across the full width of the process drum.

Viewing the Gas Wedge Position.

The gas wedge can be seen from the rear of the machine in location. There is also a hole placed in the side plate to allow the user to check that the device is not in contact with the substrate.

Adjusting Gas Flow

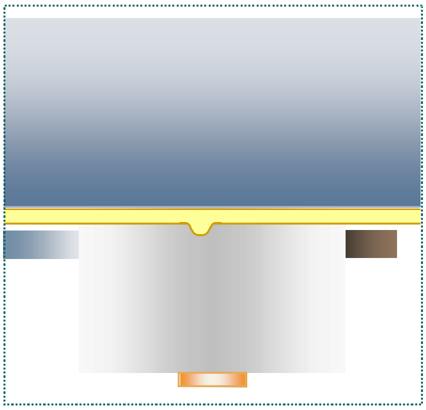
The amount of air being fed to the system is controlled from a manual flow regulator on the winding cart, close to the operator position.

If the flow is allowed to be too high, there will be a vacuum breakdown in the evaporation zone. Normal operating level is approximately 2 litres/sec.

Figure 2.28 - 3 Operator Adjusting Gas Flow To Gas Wedge

Removing Tramlines From The Substrate.

The word “Tramline” described as phenomena when the substrate is not flat as it passes through the evaporation zone.



The tramline can be formed because of a number of reasons.

- Expansion of the substrate as the aluminium contacts the surface.
- High co-efficient of friction between the drum surface and the substrate.

Figure 2.28 - 4 Drawing Indicating a Tramline Being Formed

The outcome is a visible defect as shown below

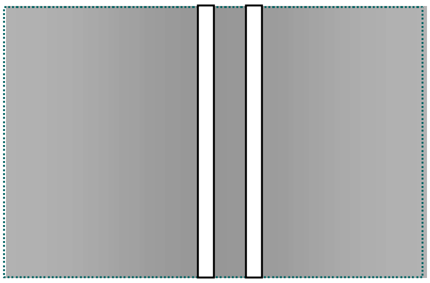


Figure 2.28 - 5 Drawing of a Visible Tramline Defect

The injection of air between the substrate and the drum allows the substrate to be processed and;

- Reduces expansion by improved film cooling.
- Acts as a ‘cushion’ to reduce the co-efficient of friction
- Prevents formation of heat wrinkle

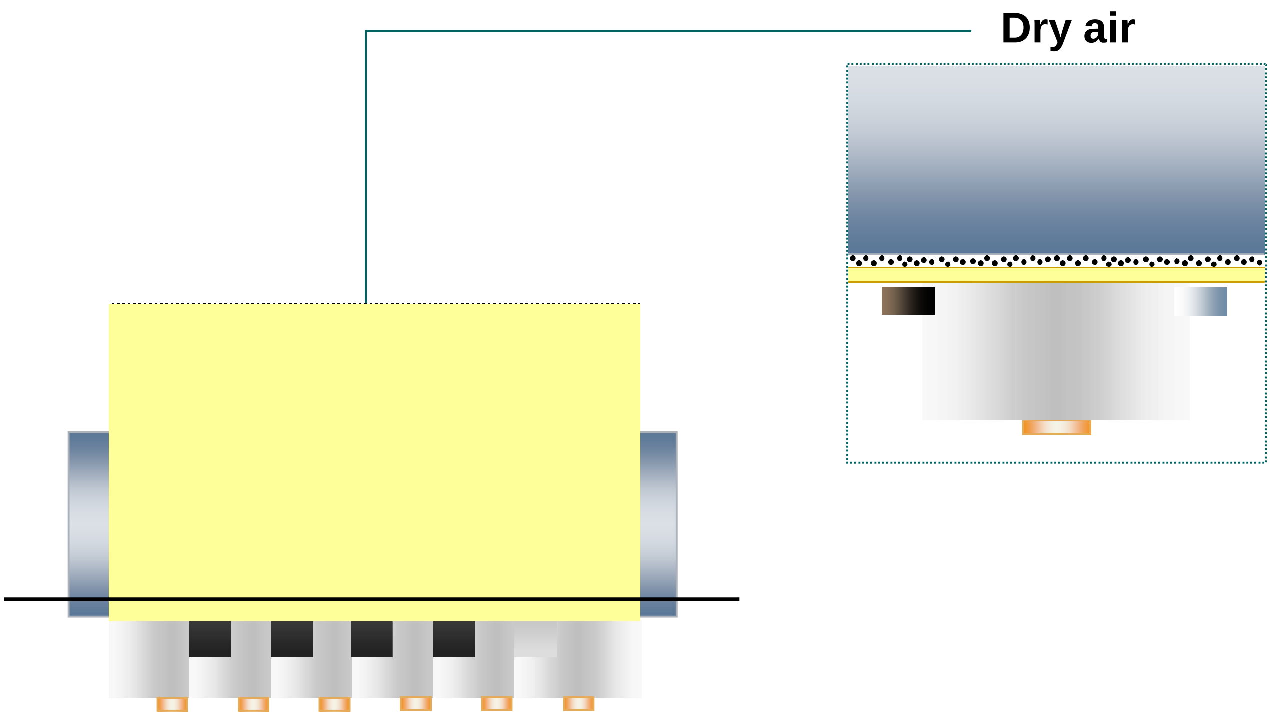


Figure 2.28 - 6 Drawing Showing Gas Wedge in Action

It has been found that on thinner materials, for example 20 micron CPP. Excessive gas can actually cause tramlines, so care should be taken when setting the gas flow level.

2.29 Web Threading System (Option)

A device is provided in the machine to allow the user to automatically web the substrate through the winding mechanism.

To allow this to be used the following situations must be correct.

- The winding mechanism is fully retracted from the chamber.
- The drum coating shield is in the fully lowered position

There is a nylon attachment strip located between two chains. The chains run on sprockets through the winding mechanism and provide a path around the rollers.

The attachment strip is normally parked in a position where it will not interfere with other operational aspects of the machine. It is driven to this location and is automatically stopped by a position switch.

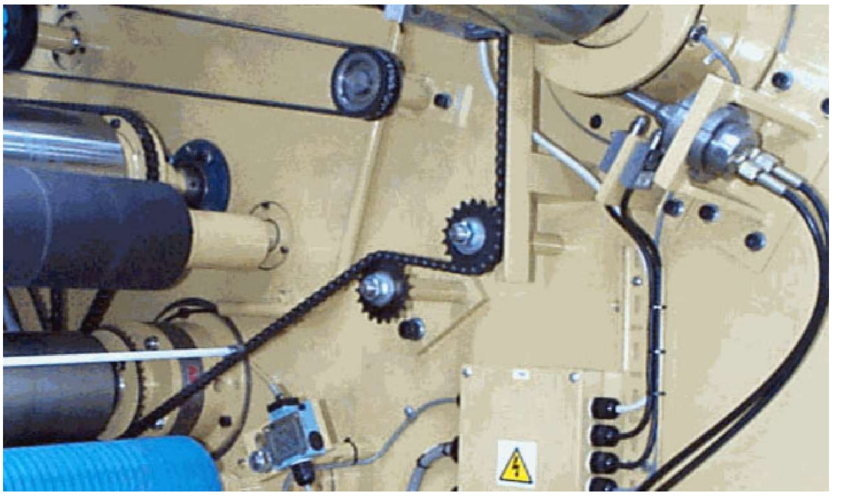


Figure 2.29 - 1 Photo of Threading System

SECTION 3 MACHINE EVAPORATION SOURCES AND PROCESSES

3.1 Aluminium Evaporation Source

Evaporator or 'boat' is the name given to the ceramic bar that is heated by an electrical power source.

They are heated until its surface temperature is approximately 1650 °C. At this temperature aluminum wire, when fed onto them will melt, and then vaporize onto the passing substrate.

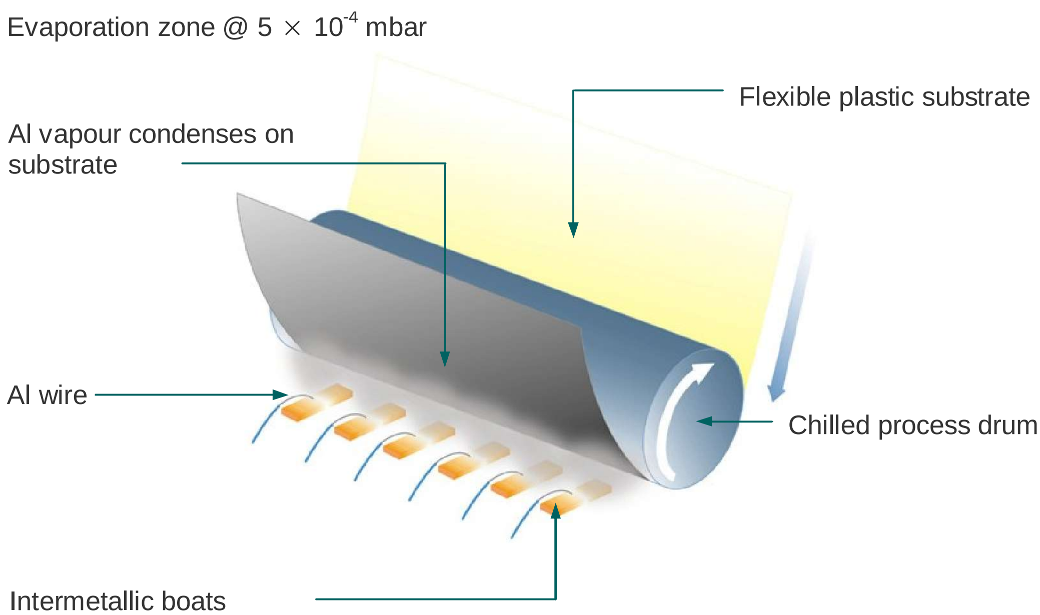


Figure 3.1 - 1 Evaporation Process Principle

The boats are located within the chamber and are in a stationary position, while metallizing takes place.

Overview of Evaporation Area

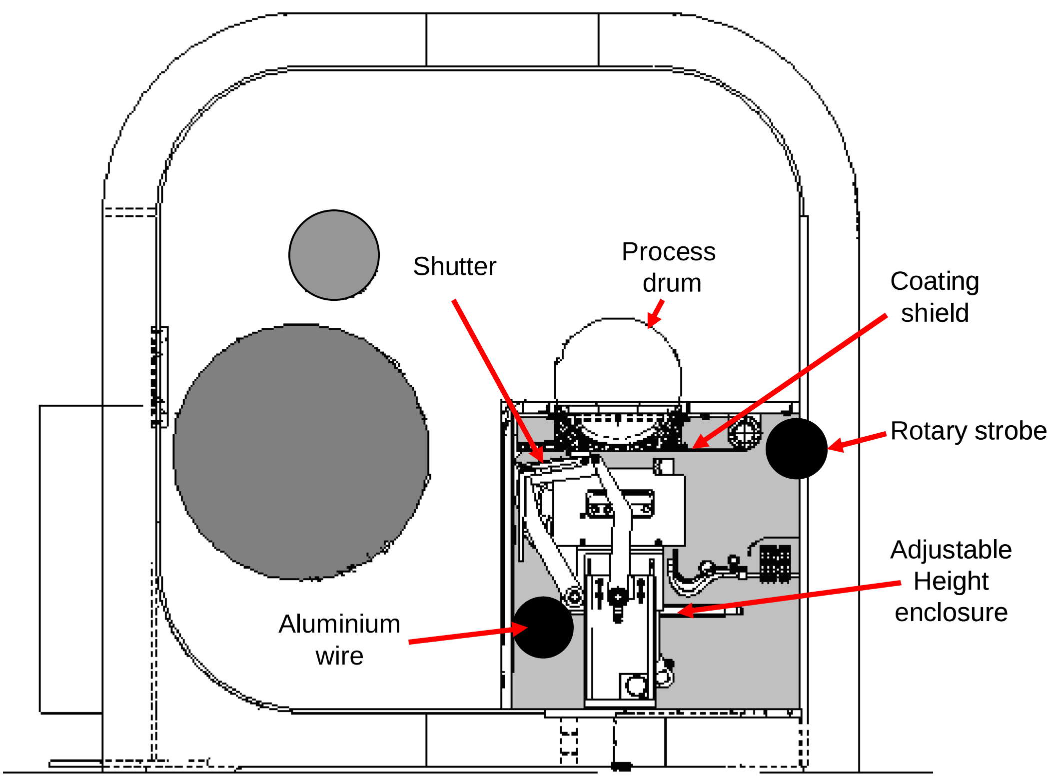


Figure 3.1 - 2 Cross Sectional view of Evaporation Source Area

3.2 Fitting the Evaporators

Each evaporator is supplied with the electrical power necessary to heat it via two copper terminal blocks, located at either end of the evaporator.

In order to ensure good electrical contact and to allow the evaporator to expand when heated, one of the terminal blocks is spring-loaded against the end of the evaporator, while the other is stationary.

It is essential that for electrical contact to take place between the surfaces of the evaporator and the terminal blocks, the blocks are clean.

The evaporator posts at the spring-loaded side of the evaporator are mounted lower than those at the fixed side. They are manufactured this way to allow the evaporator to slope away from where the wire is fed upon it, i.e.; one degree fall.

After a period of time, erosion may take place in these contact blocks. They are a wear part and can be replaced using high conductive/high density copper. The blocks are bolted onto the copper posts using stainless steel bolts. These bolts should be checked periodically for tightness.

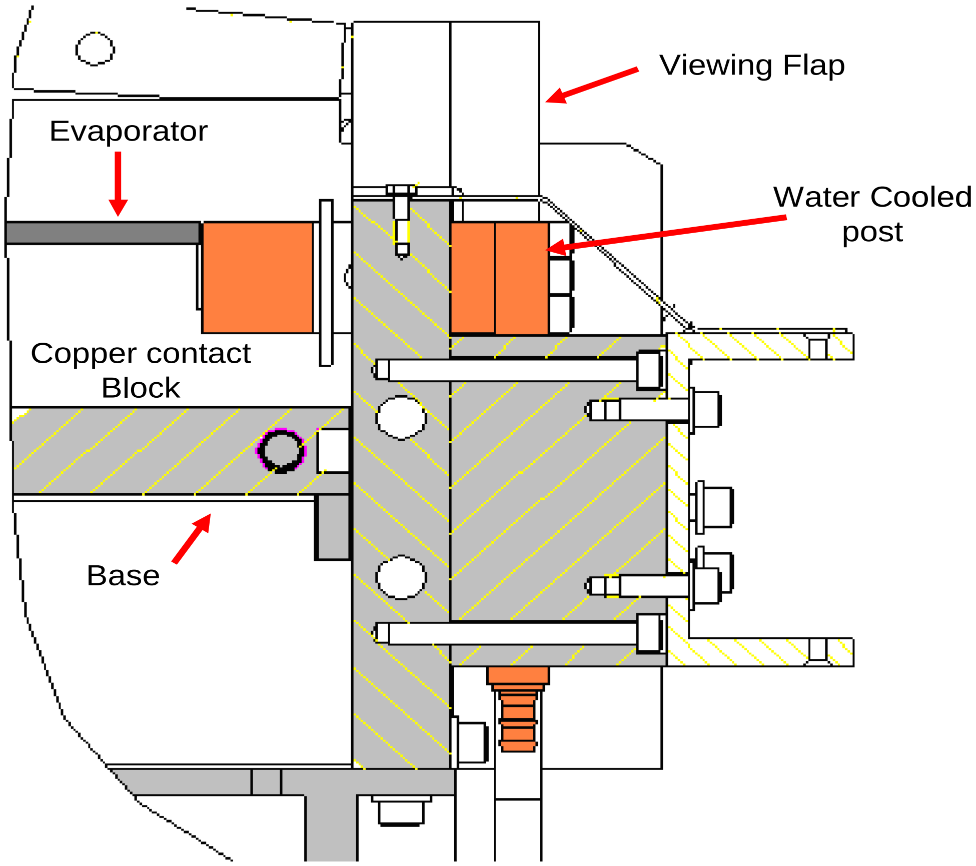
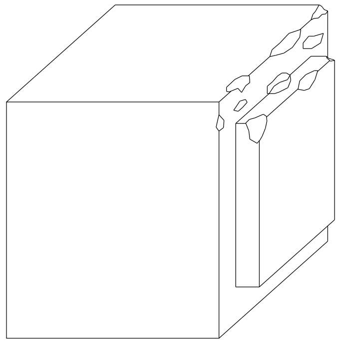


Figure 3.2 - 1 Copper Contact Blocks

The wear of the contact blocks can be reduced by using the correct cleaning and operating procedures. The temperature of the cooling water should kept within the required specification, i.e. below 28 °C



Damage can take place if the cooling water to the source is sufficient, but at the same time, if the water is too cold, poor contact can take place.

Figure 3.2 - 2 Sketch Showing Example of Damage to a Copper Contact Block.

Electrical contact between the evaporator and the blocks is assisted by the use of a graphite impregnated tape between the end of the evaporator and the terminal block.

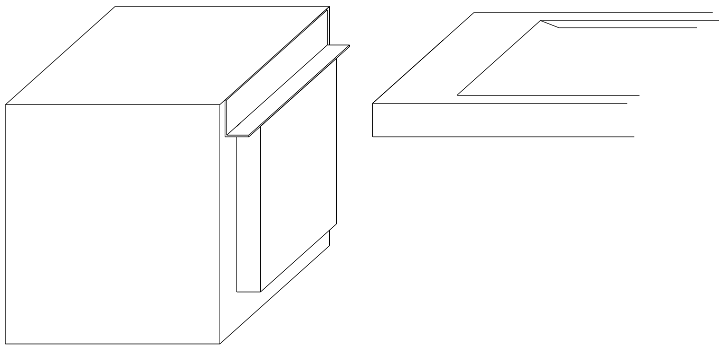


Figure 3.2 - 3 Fitting Of Graphite Between The Evaporator And Copper Block

When fitting an evaporator, the spring loaded block must be pulled or pushed back, the evaporator placed between the blocks and a single layer of graphite tape inserted at each end of the evaporator.

After the terminal post is released trim the graphite tape so that it does not protrude beyond the evaporator surface.

Check also, that the evaporator is pressed down onto the horizontal ledges of the terminal blocks.

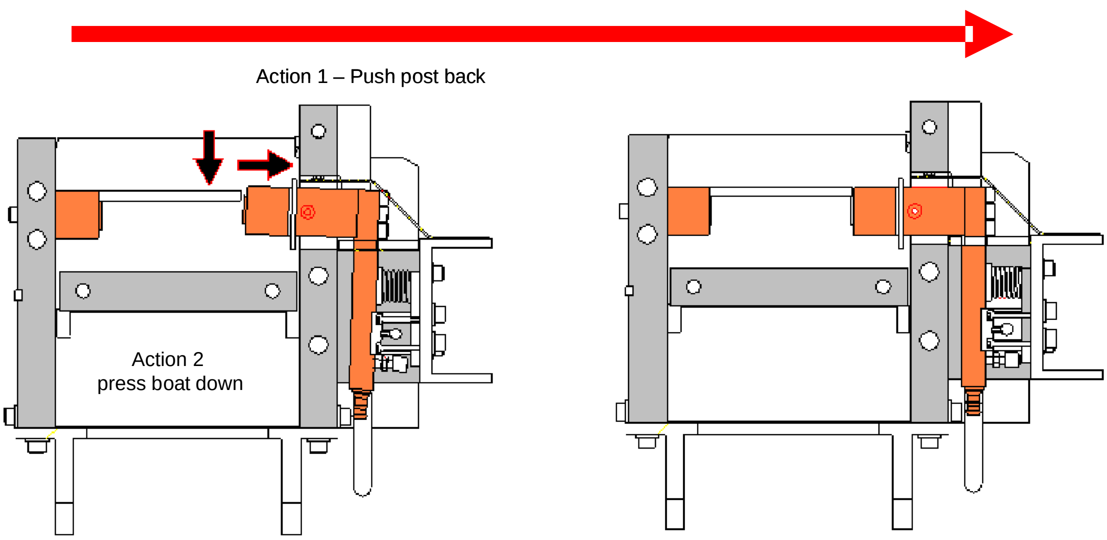


Figure 3.2 - 4 Source Spring Loaded Block Retracted & Then Clamped Between Contact Blocks

3.3 Care of Evaporators

Evaporators give the best working life when they are cared for correctly. This has been proved over many years of metallizing experience.

Evaporators are delivered to the customers in sealed packages to keep them moisture free. It is important to keep the evaporators in their packaging until the time of usage.

It is essential that the evaporators are heated up slowly the first time that they are used.

Supplying power to them quickly will cause them to heat up too quickly and they are liable to suffer from thermal shock. This in turn will shorten their working life.

This machine provides a method of heating up the evaporators in a manner recommended by all the evaporator manufacturers. The rate at which the evaporators will heat up is selected on the SETUP screen. A pop-up screen allows the user to select the heating ramp curve.

	**Note!** **The user has a choice of selecting OLD BOAT or NEW BOAT and they should always select the NEW BOAT setting, when new evaporators are fitted.**

Selecting the NEW BOAT facility conditions the evaporator by heating it slowly to approximately 500 °C to allow absorbed gases to be driven off. Above 500 °C the rate of heating ramps up at a quicker rate automatically.

Subsequent heating of the evaporators can be carried out with the faster heat up ramp given when OLD BOAT is selected. After the initial selection of NEW BOAT, the system will normally switch back to OLD BOAT.

A time period of approximately 12 minutes is normally used for the NEW BOAT ramp compared with 4 minutes for the OLD EVAPORATOR ramp. This time base and ramp rate can be changed by the user.

Both the OLD BOAT and the NEW BOAT ramp curves are adjustable in the OLD/NEW BOAT pop-up screen from the SETUP screen.

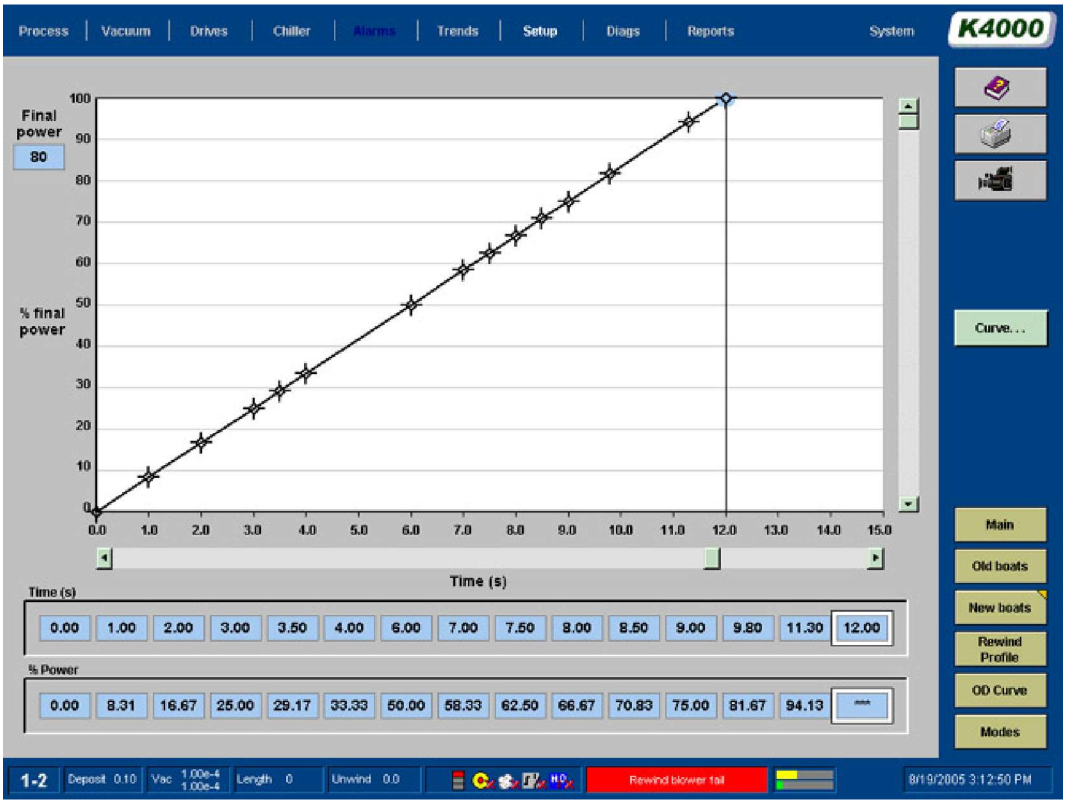


Figure 3.3 - 1 New Boat Curve Screen

Figure 3.3 - 2 Old Boat Curve Screen

3.4 Evaporator Power Supply and Temperature Control

To allow the evaporators to be heated up with a low voltage and high current supply, step down transformers are used.

The connector that passes through the chamber wall is a thick walled copper tube that is cooled by water.

It is connected to a solid busbar, which in turn connects to the transformer.

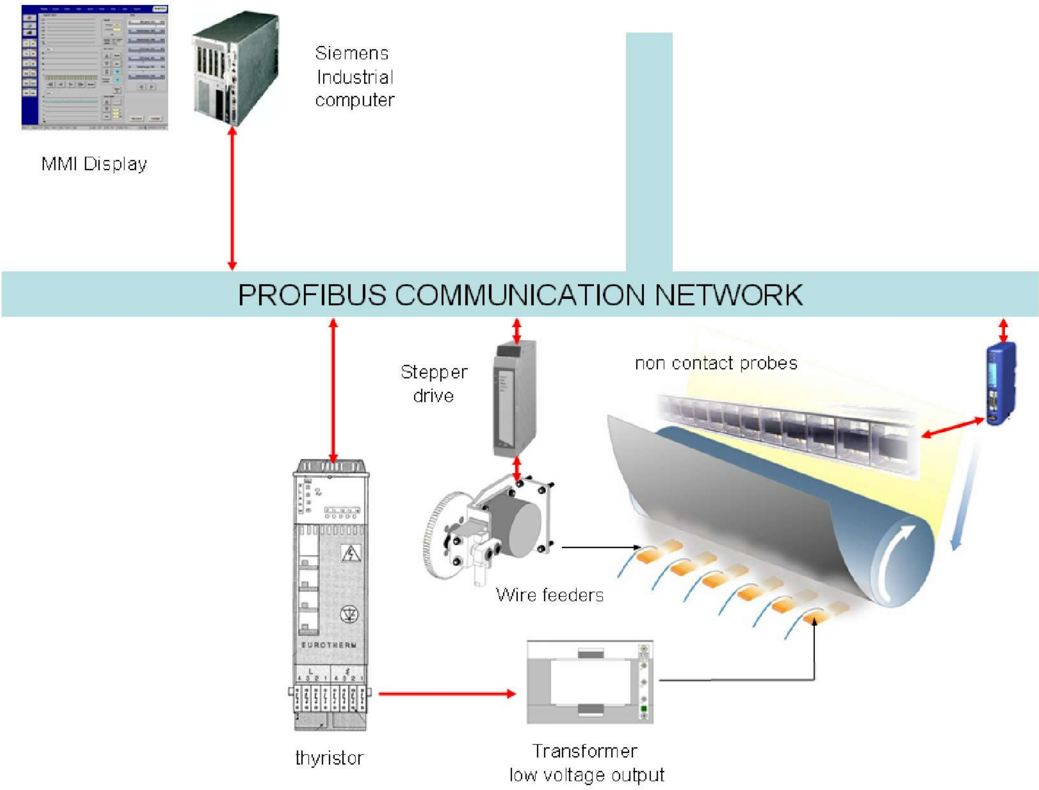


Figure 3.4 - 1 Control Schematic Drawing

Each evaporator power supply is controlled from the computer, via the thyristor, in either an automatic or manual mode. The value of the power being supplied to the evaporators can be read from a bar graph display on the PROCESS screen

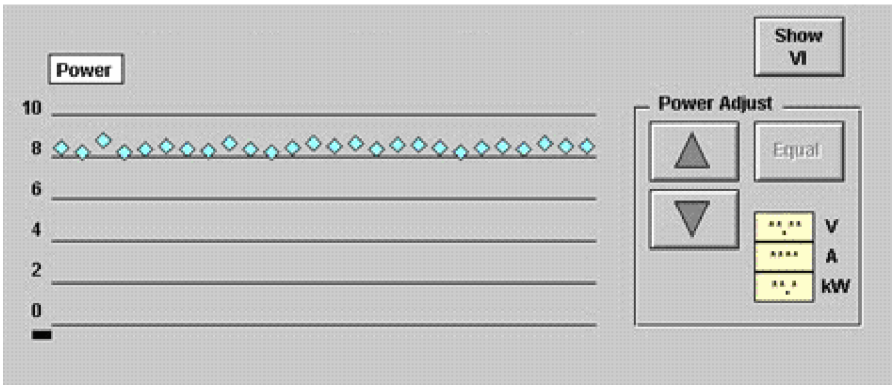


Figure 3.4 - 2 PROCESS Screen - Power Control Section

The indicators shown on the screen indicate the power setpoint as a percentage value between 0-100%.

The voltage and current can be displayed on a pop-up-screen by pressing the SHOW KW button on the PROCESS screen.

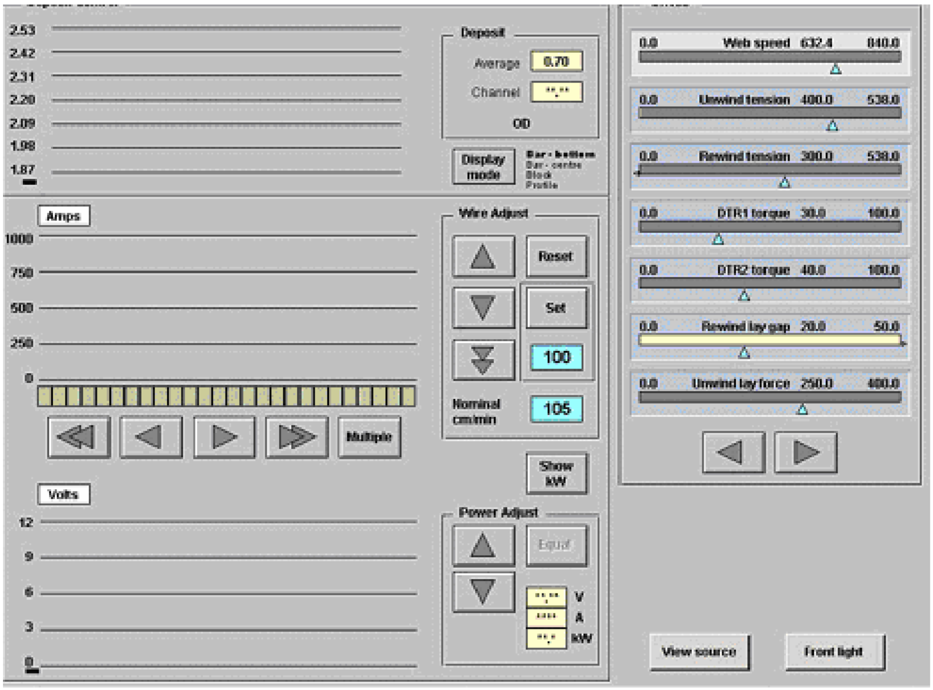


Figure 3.4 - 3 PROCESS Screen Displaying Volts and Amps Section

The power supplied to the evaporators is ramped from the computer after the source's "BOATS On" button is touched on the PROCESS screen.

As this circuit is enabled, the computer will switch from a STOP to a WARM UP mode.

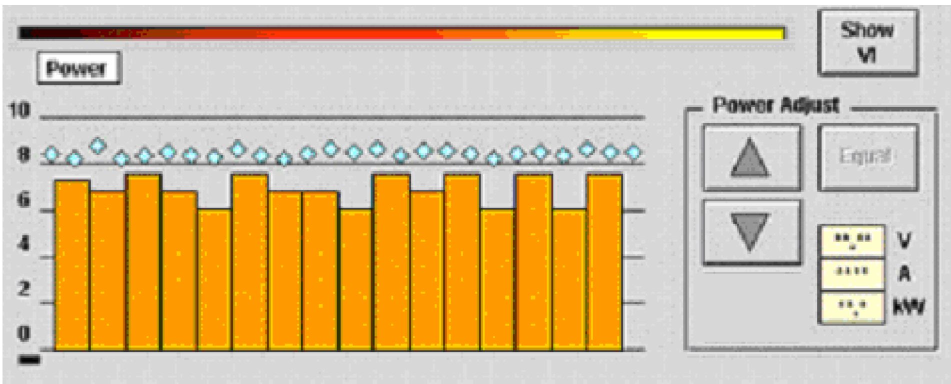


Figure 3.4 - 4 PROCESS Screen - Power Warm Up/Ramp Mode

The computer will supply signals to the thyristors to increase their power levels to a predetermined level. This level is set by the computer based upon the "NOMINAL WIRE SPEED" value setting in the SETUP screen.

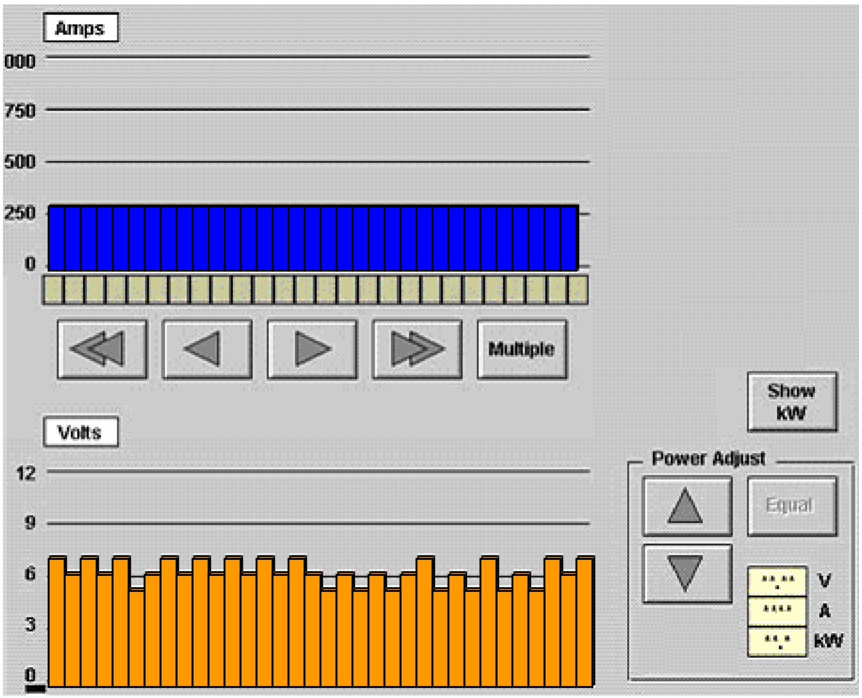


Figure 3.4 - 5 Volts and Amps Screen During Ramp Period

If a evaporator becomes loose and there is no contact between the copper contact blocks and evaporator, no current will be drawn. This will be observed as a high voltage with no current.

It can be seen that the current levels being drawn by the evaporators can be at different levels. The reason for this is that they all heat up at different rates due to their different resistances.

The mode of operation of the power and wire feed outputs is selected at this stage by touching the control buttons.

BOATS OFF — This removes ramp signals from the thyristor units and therefore the power to the evaporators.

BOATS ON — This button energizes the main source contactor and starts the ramp signal to the thyristors.

WIRE OFF — This removes the control signal and the power supply to the stepper controllers.

WIRE ON — This button energizes the stepper controllers and allows control signals to be applied as requested by the computer.

MAN. CONTROL — This mode allows manual operation, changes of the wire feed rates and power levels.

AUTO CONTROL — This mode will place the controller in an automatic mode of operation.This can only occur when all the deposition readings are within 50% of the set-point.

Figure 3.4 - 6 Indication Of Operating Modes Of Deposition Control System

Note: Do not select the BOATS OFF mode on the screen while the substrate is being coated. Selecting this mode will switch the power supply to the evaporators off (control signal) and stop the wire feeder's feeding wire.

If this occurs accidentally, stop the substrate by selecting the WEB STOP button.

Then the evaporator power on, to allow them to clean themselves of any remaining aluminium, by vaporization for 5 minutes.

Operational Mode Changeover - Warm Up to Auto modes

The operational status of the deposition control will be indicated on the PROCESS screen on the Auto button..

Figure 3.4 - 7 Deposition Mode Display Auto Button

The system is designed to only switch into an automatic mode when all the feedback readings are within a set error percentage of the target. This parameter is set up in the software and is normally 50% of the set-point.

If the deposit is initially too light, the controller will not switch into the AUTO mode, even though it has been selected. The line speed must be reduced slightly or the base wire feed rate increased to overcome this problem.

Before the AUTO control is selected, the button will have a yellow triangle on the MAN section of the button. To select AUTO mode, touch that section of the screen. The yellow triangle will move from the MAN to AUTO section.

Once the feedback signals are within the 50% level, the controller will attempt to bring the error to 0% throughout the coating cycle. Until this time, the yellow triangle will be shown intermittently.

If a feedback signal goes outside the 50% range, the controller will "freeze" the wire feed rates constant at that time, until the out of range channel returns below the 30% level. The yellow triangle on the AUTO section of the button will be displayed intermittently.

Reasons that the feedback signal will go beyond the normal range are described in section 8 of this manual. There is also an explanation of what procedures should be taken to overcome the issue.

Cleaning Flooded Evaporators

In the case of this type of evaporator, all the remnants of any aluminium have to be removed from them before venting the chamber. Therefore the machine software has been programmed to allow the evaporators to be heated for a 30 second time period after the slowdown sequence has been instigated.

Another factor that is taken into this calculation is the power settings that were being used at the end of the previous coating cycle. The computer assumes that these power levels were optimum for the coating uniformity. Therefore the power levels may differ between evaporators.

Operator - Evaporator Power Level Adjustments

The operator has the ability to increase or decrease individual evaporator power levels from the PROCESS screen.

To move the black arrow cursor under the channel that the power level is to be adjusted, touch the screen on the arrow. Alternatively, the arrows at the top of the power section can be used.

Use the "up" or "down" arrows to adjust the power levels. It can be noticed that the power will fall slightly after the channel has increased. This is a design feature to ensure that rapid substantial increases are not made by the operator.

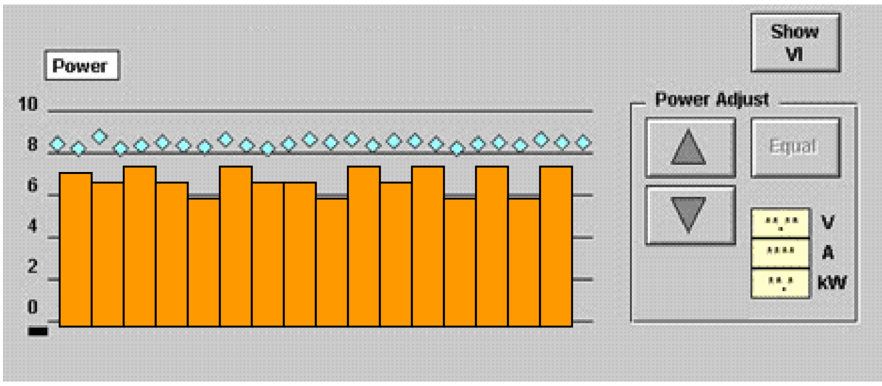


Figure 3.4 - 8 Indication Of Power Levels On PROCESS Screen

To raise or lower all the power levels, the SELECT ALL button, should be touched and all the selected evaporator power supplies will be selected.

Selecting Multiple Channels

There is a function designed into the control of the machine to allow multiple selection or individual selection of channels.

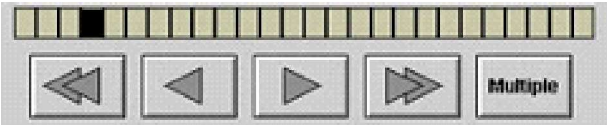


Figure 3.4 - 9 Channel Selection Bar Showing selection of a single Wire Feed Output

The selected channel is shown by the single cursor, which can be moved using the fast and slow movement buttons.

Selecting the MULTIPLE button, will change the buttons.

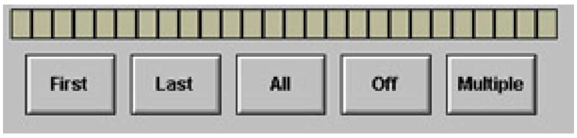
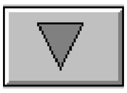


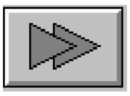
Figure 3.4 - 10 Channel Selection Bar Showing the Multiple Selection Wire Feed Output

Figure 3.4 - 11 Channel Selection Bar Showing selection of All the Wire Feed Outputs

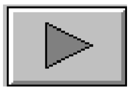


This button will allow the selected channels output values to be raised or lowered.

This button allows multiple output channels to be selected, and when pressed changes the display.



This button will move the cursor from one step to another at high rate.

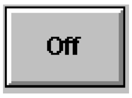


This button will move the cursor one step at a time.

This button when pressed will select the first of a multiple of output channels.

This button when pressed will select the last of a multiple of output channels. These channels will then be controlled by the increase/decrease buttons, etc.

This button selects all the output channels.



This button deselects the multiple function.

Figure 3.4 - 12 PROCESS Screen Multiple Control Buttons

If an individual channel has been selected in normal display and then the FIRST button is touched, and then the MULIPLE screen is selected, a marker will be placed on the first channel. If another channel is selected in the normal view and the LAST button selected in the MULIPLE view, all the channels between the first and last selected will be used.

Equal (EQUAL) Function Button

To allow the user to adjust all the power set-points to the same level, an equalize (EQUAL) button is provided. The cursor (arrow) should be placed above the evaporator on the power section of the screen. This should be the evaporator that the power supply is giving the correct amount of power and optimum conditions. When the equalize button is touched, the other evaporator power supply set-points will alter to become even with the one selected.

How Many Evaporators Do You Use?

The number of evaporators used is dependent upon the width of substrate being processed. When the coating width is placed in the SET UP screen and downloaded, the computer will automatically select the number required. The user is informed of these requirements on the PROCESS screen by the number of channels selected.

Using Partly Used Evaporators

When coating substrate less than the full width of the machine, some evaporators are not used fully. When the evaporator set is taken out of the machine, these evaporators can be stored in a dry location and used to replace any evaporators that fail during normal cycles.

It is normal to leave evaporators in all positions. This stops the evaporator contact blocks from being coated by aluminium.

3.5 Evaporator Normal Running Mode

The evaporators normally run at a vacuum level below 5x10-4 (mbar) and are therefore interlocked not to switch on unless the high vacuum valves are open (i.e. VV7)

The normal operating cycle, including heating up, etc.; can be understood better from figure below, which shows the normal cycle in a graph format. This figure also shows the interaction of the evaporator power levels.

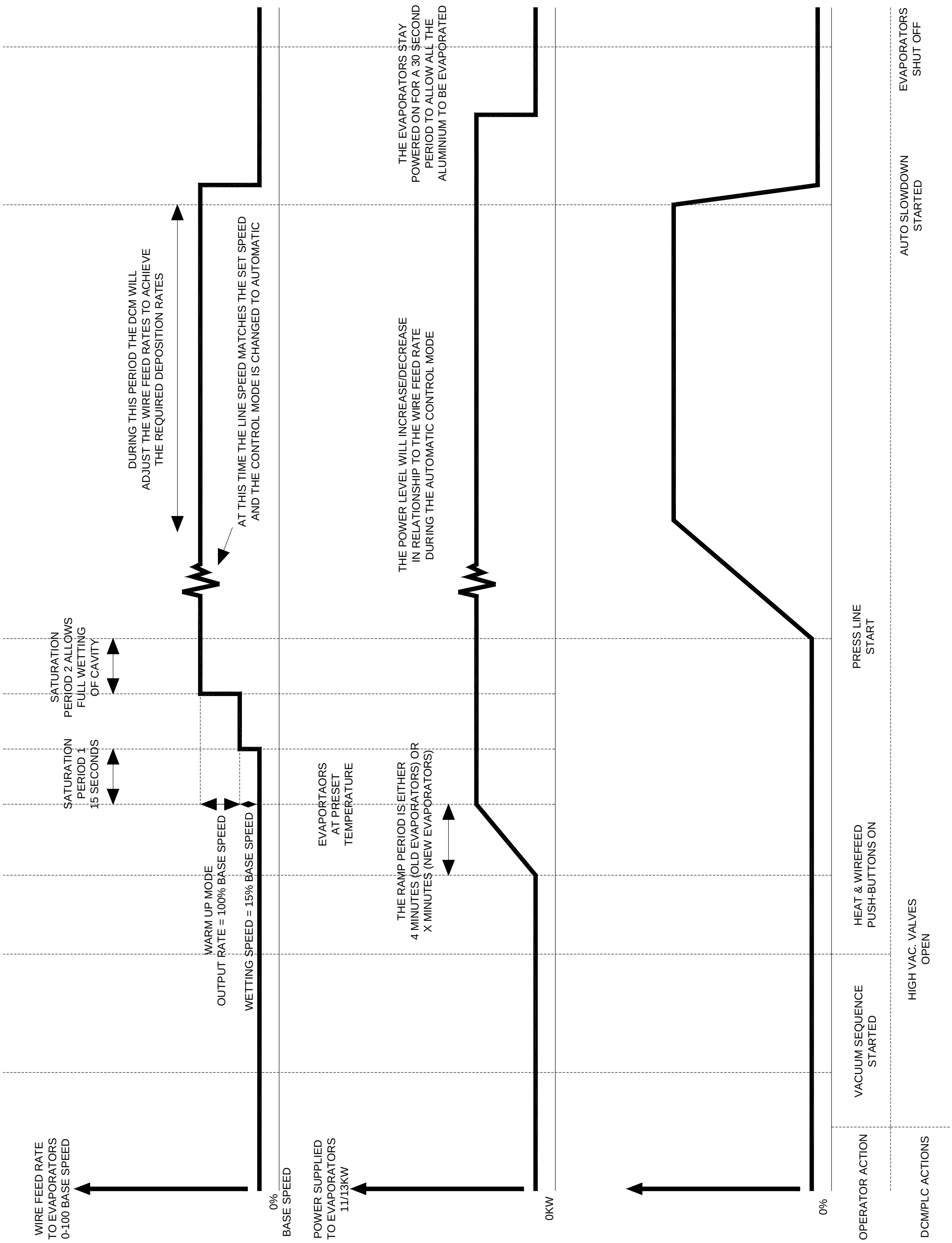


Figure 3.5 - 1 Normal Heat-Up Cycle Of Evaporator,s

At the end of the ramp and saturate time period, the wire will automatically be fed onto the evaporator for a fixed time period at a limited rate.

When the evaporators are at the correct temperature, they will melt the wire as it is fed onto them.

This will not be the case if they are too hot or too cold

The temperature of the evaporators should be adjusted manually to allow the aluminum to "flood out" the whole of the cavity, see figure. If the cavity is overfilled, "spitting" can occur and the end contact blocks will erode quickly. If under filled, there is more chance of poor uniformity across the web, through "flash evaporation", where the metallic coating looks patchy.

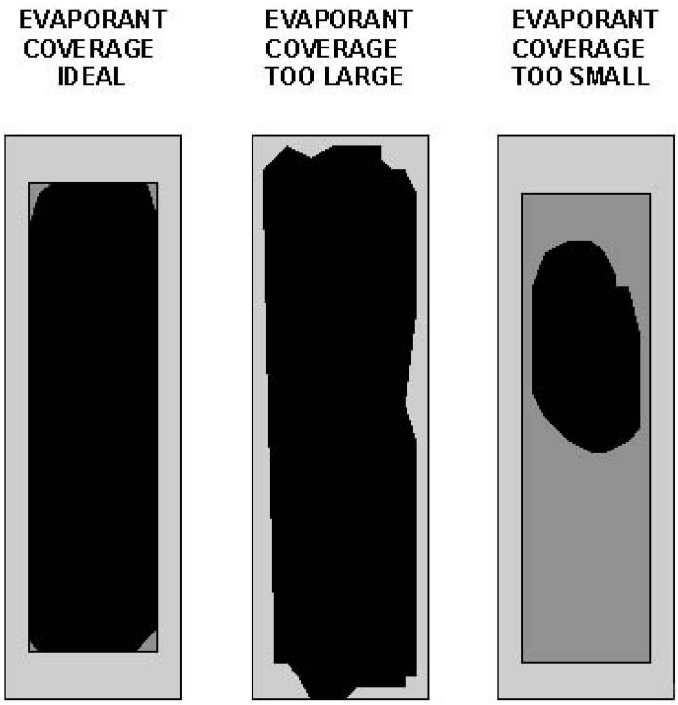


Figure 3.5 - 2 Evaporator Cavity Usage

Cavity coverage can be seen through the source viewing window.

Evaporator - Applying the Correct Power Levels to New evaporators

If the power level supplied to evaporators initially is too much, the temperature can rise above the recommended operating temperature. This leads to shorter evaporator life.

If the power level is too low and therefore the temperature, when wire is applied, it does not melt and evaporate. This leads to flooding problems.

A single piece of aluminium wire can be placed on the new evaporators when the machine is open.

When the power is applied to the evaporators, the wires will melt and then evaporate if the temperature is correct. This is a method on ensuring that the evaporators are at the correct temperature before energising the aluminium wire-feeders.

Evaporator - End of Useful Life

The normal operating life of the evaporators should be in the 12 - 15 hour range. It will be noticed, as evaporator’s age, various parameters will change:

- The working voltage and current levels increase.
- The evaporation rate decreases.
- Parts of the evaporators appear to be hotter than others.
- The uniformity control becomes more erratic and the controller struggles to control the deposition.

Usually, it is found that when the evaporator is at the end of its useful life, the current and voltage are at maximum values. Also the evaporation rate has fallen and therefore uniformity cannot be kept constant. It can be noticed that the surface of the evaporator is sometimes grooved longitudinal where the wire has come in contact with the evaporators.

	Note!
	The options to change individual evaporators or the full set is dependent on the usage of full width coating (i.e. outer evaporators may have run only a few hours).

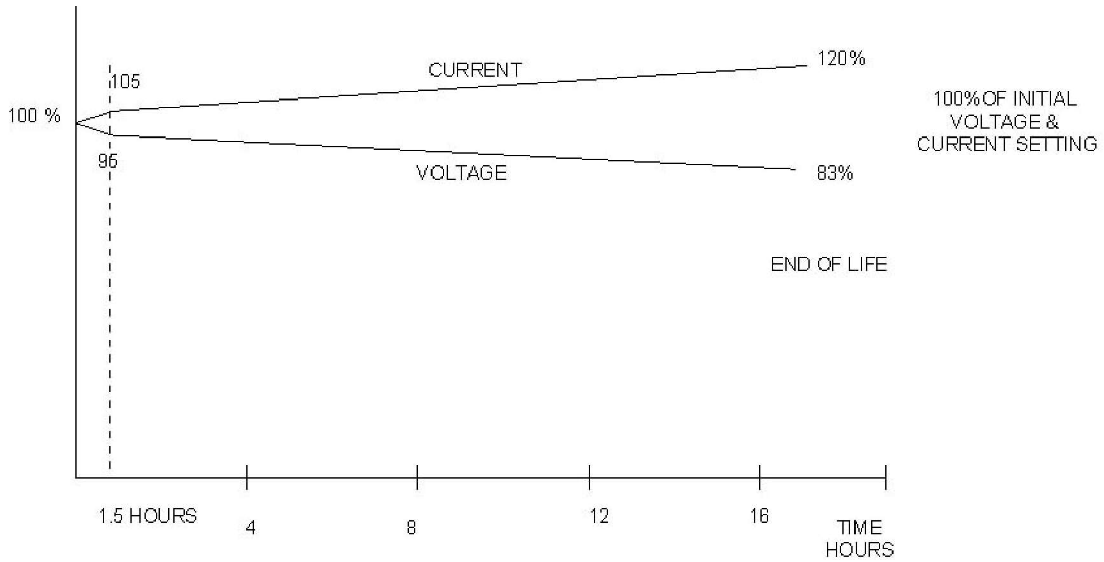


Figure 3.5 - 3 Normal Current And Voltage Parameters Throughout Evaporator Life

Prolonging Evaporator Life

If you change a number or all the evaporators, attempt to replace them with evaporators that have a similar resistivity value

To prolong evaporator life, the following rules should be followed:

- Always use the NEW EVAPORATOR warm up facility.
- Do not supply the evaporators with more than 9.5 volts on their first warm up, i.e. reduce the BASE WIRE FEED speed for the first run.
- Keep the evaporator cavities full at all times.
- Keep the evaporator's temperature at or below the optimum temperature of 1450 C.
- Ensure that the contact between the contact blocks and evaporators is parallel.
- Do not run excessive wire feed rates.
- Clean the oxide deposits from the cavity surfaces between coating cycles using a scraper.
- Do not allow the evaporators to be painted with release paint.
- Do not vent the chamber while the evaporators are still hot.

3.6 Cleaning of Evaporator Surfaces

Oxide Contamination

To prolong the life of the evaporators, the surface area that the evaporant covers must be kept to a maximum. Due to the fact that there are some contaminants in the aluminium wire, which do not vaporize, they are not evaporated and are left behind on the evaporator surface.

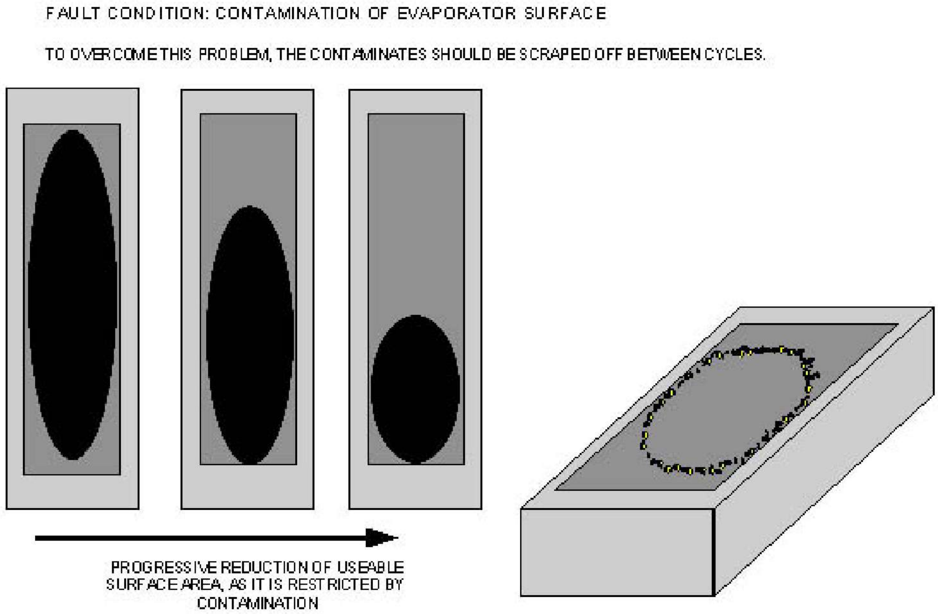


Figure 3.6 - 1 Contamination of Evaporator Surface

It has been noted that as the surface area becomes smaller, the deposition uniformity becomes poorer.

Figure 3.6 - 2 Photograph of a new evaporator Surface

Figure 3.6 - 3 Photograph of an evaporator after one coating cycle.

The surface of the evaporator should be cleaned between cycles using the edge of an old evaporator.

Figure 3.6 - 4 Photograph of operator cleaning the surface of the evaporator using an old Evaporator.

All the oxide deposits should be cleaned off to allow the full surface area to be used. Use a vacuum cleaner to remove any remaining dust.

Figure 3.6 - 5 Photograph of used evaporator after being cleaned.

Figure 3.6 - 6 Photograph of used evaporator that has not been cleaned.

3.7 Wire Feed System

Aluminium wire is pulled off reels, mounted below the source box, by wire feed units and fed onto the evaporators through guide tubes. The wire is wound upon plastic reels and is supplied in standard sizes, from a number of manufactures.

The wire feeders are driven by stepper motors located below the source box. The stepper's motors are controlled from the computer via a communication link to the wire feeder motor controllers.

The wire feeders are mounted upon insulated pads that insulate the evaporator power from shorting to earth.

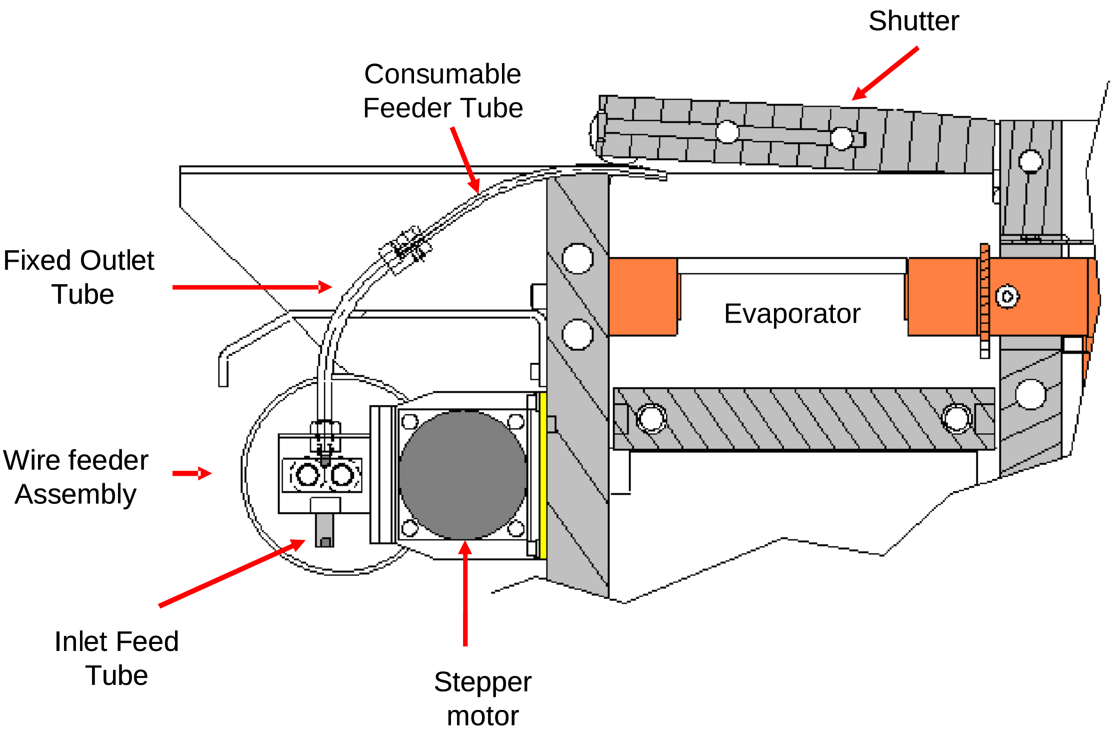
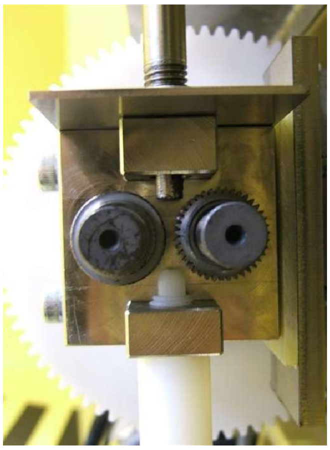


Figure 3.7 - 1 Drawing of Wire Feeder Assembly Overview

There are two guide tubes used on the wire feeders, inlet and outlet. The inlet tube is used to deliver the wire straight into the feed gear-wheels. It is important that this tube is located into the wire feeder fully and tightened into place.



The outlet tube is a two piece assembly, one piece is fixed and the other is a wear part that passes through the side of the shielding.

The fixed section should be rotated into the feeder until the feeder will not rotate. At this time, rotate the tube anti-clockwise, until the outlet points towards the evaporator. Tighten up the locking nut at this time.

Figure 3.7 - 2 Photograph Showing the Correct Setting of Wire Feed Guide inlet and outlet tubes.

When fitting new reels of aluminium check that the free end of the wire is not trapped under any of the top layers of the wire.

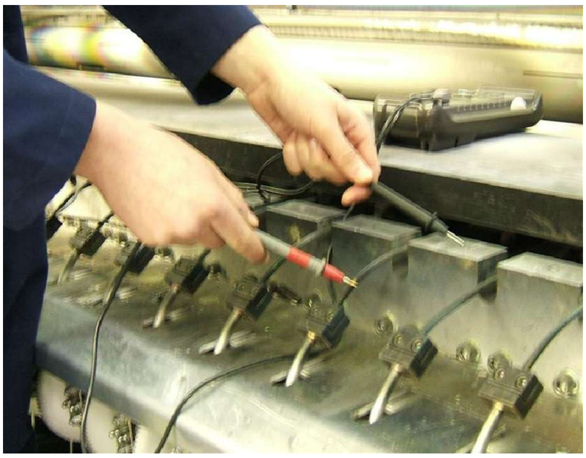
The spools of wire are located on free running holders sat on a carriage located in the bottom of the chamber.

Figure 3.7 - 3 Photograph of Aluminium Wire Being Drawn of Spool

	**Note!**
	When handling the wire spools, do not allow the wire to uncoil as this can cause tapping of the wire, under loose layers.

Whenever the source area is cleaned or the wire feed tubes adjusted check that there is no electrical connection between any wire feed and the trough. This can be done visually but the use of an Ohm meter is preferred.

If any of the wire feeds have a connection to ground (e.g. the trough, this will cause "spitting" of aluminium particles from the evaporator onto the web. These can be described as small "flashes as seen in flash photography" at the point where the wire touches the evaporator.



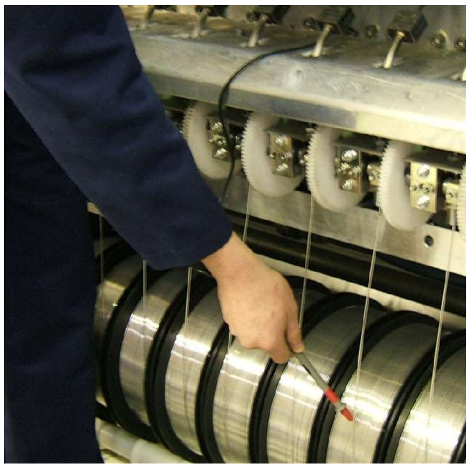


Figure 3.7 - 4 Photographs of Operator Checking for Potential Shorts on the Wire-feeders

Wire Feed Tubes - Consumable

The top section of the wire feed tube, closest to the evaporators is classified as a consumable (wear) part.

It will be normal that this part will have to be cleaned regularly as described in section 8 and replaced as required.

The tube should be painted with the release paint to ease the removal of the build up of aluminium from it.

Note!

If the wire feeder tubes are not kept clean, blockages can occur that will affect the uniformity of the deposition.

The clamping section of the wire feeder should be covered in aluminium foil to protect it from aluminium oxide build up.

Figure 3.7 - 5 Photograph of Wire Feeder Clamp Section With and Without Protection.

The consumable tube is clamped in a holder at the end of the fixed section as shown in the figure below. The tube should be clamped square to stop any possible friction on the aluminium wire. Ensure that both ends of the tube are free from debris and burrs that may affect the performance of the feeder.

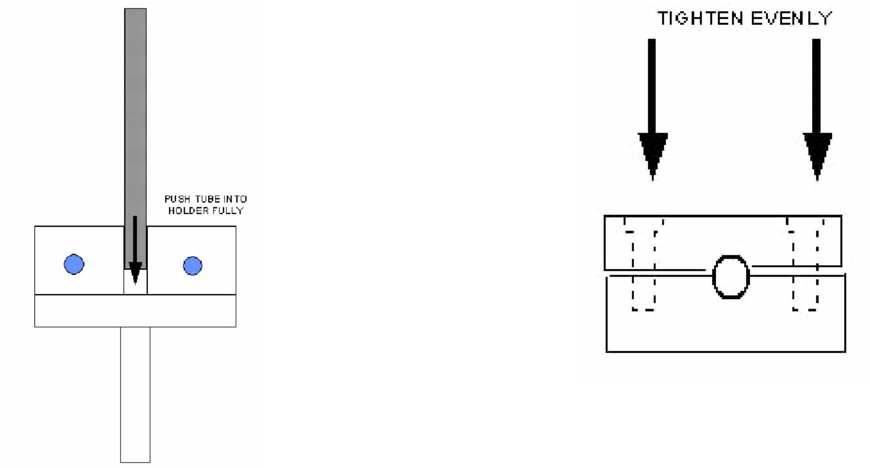


Figure 3.7 - 6 Inserting Wire Feed Tube into Fixed Section

The relationships between guide tube and evaporator and the position where the wire makes contact with the evaporator is extremely important.

It is important to feed the wire on to the evaporators at a point between 1/3 and 1/2 from the wire feed end to have good cavity use. For high feed rates, above 70 cm/min., it is suggested that the wire be fed to the middle of the evaporator.

It is essential that the relationships shown in the figure are followed to achieve the best performance from the evaporation system.

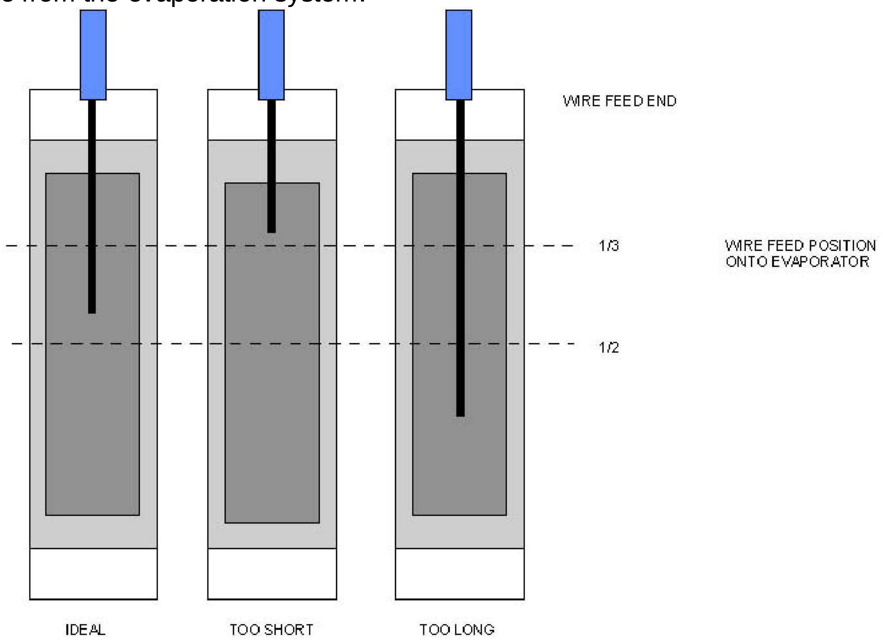


Figure 3.7 - 7 Correct Setting Of Wire Feed Guide System

3.8 Wire Feeder Control

As previously described, the aluminium wire is supplied to the evaporators from a spool located below the evaporators.

The wire feeders are controlled from the computer and feed the wire in a controlled manner up to a maximum rate of 350 cm/min. However, it is recommended that a maximum rate of 120 cm/min (at approx 100%) should be used to ensure the minimum amount of coating issues and a longer evaporator life.

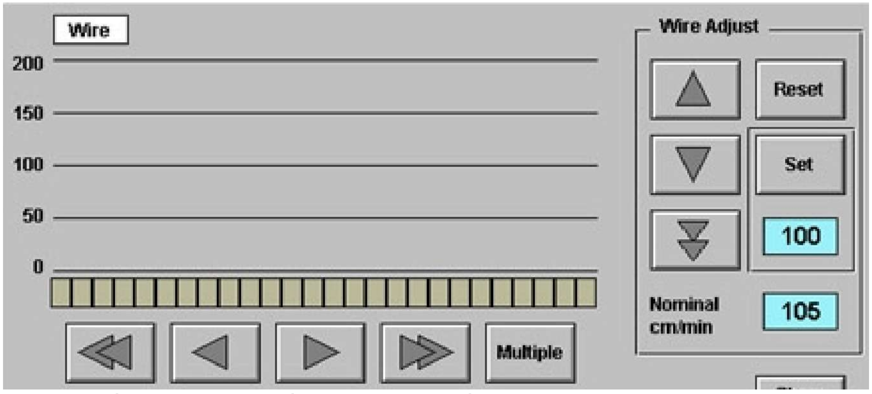


Figure 3.8 - 1 Wire Output Section On PROCESS Screen

The amount of wire that is fed by the feeders will be a percentage of the base wire speed that is set by the operator. This base speed can be set from wire feed speed raise or lower sections on the screen shown above.

The stepper motors are supplied with drive pulses from the drive controllers located in the relevant electrical panel. The controllers are fed from a 48 volt dc power supply when the wire feeder ‘ON’ button is selected.

A control signal is supplied from the computer when the control mode is changed from STOP to MAN or AUTO modes. The normal operation sequence can be better understood from looking at figure 3.8 – 2.

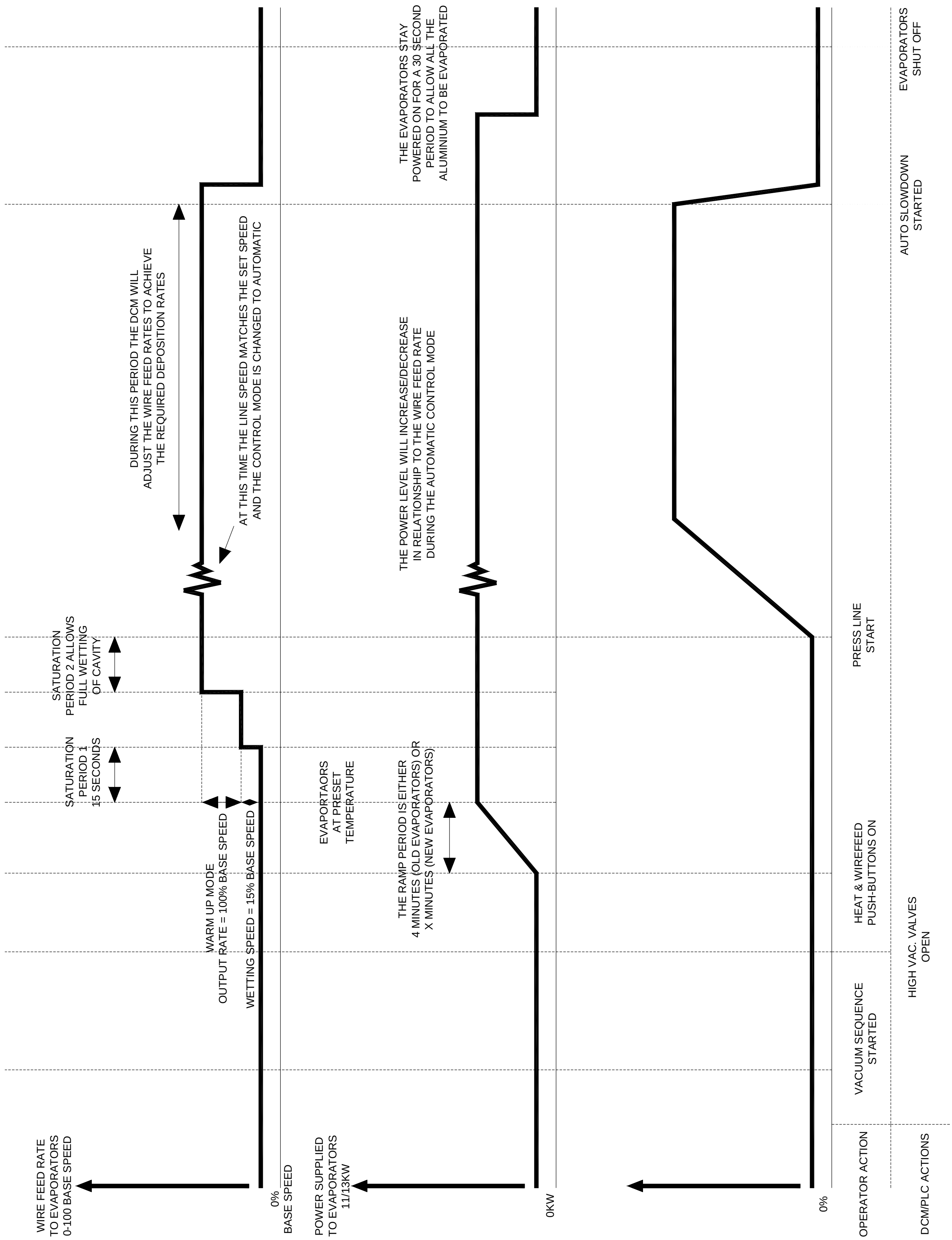


Figure 3.8 - 2 Normal Operation Of Wire Feeders And Evaporators

The wire feeders can only be operated normally with the chamber closed and the evaporators powered up to their operating temperature.

The wire feeders are operated with the cart fully retracted, for calibration purpose.

Operation of Wire Feeders with the Winding Cart Fully Retracted

The computer will only allow the required preset wire-feeders to run when carrying out this operation.

To operate all the wire feeders all together, carry out the following procedure;

1. Set a base speed on the PROCESS screen.
2. Touch the WIRE FEED ON button to drive all of them at 100% of their set base speed.

The individual wire feeders can be run in the above procedure, by reducing the individual outputs to zero, except the one being tested.

This is carried out by moving the cursor to below the channel in question and using the keyboard "up or down" cursors to raise lower the output.

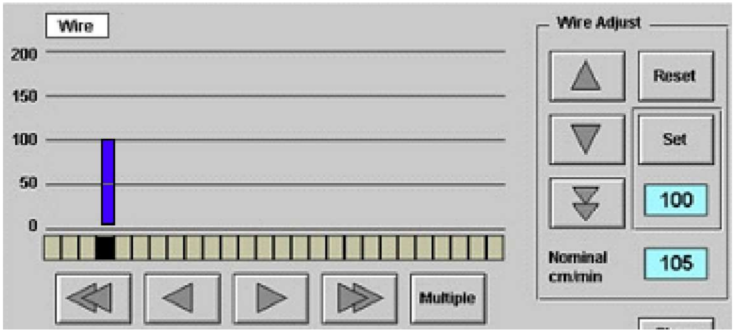
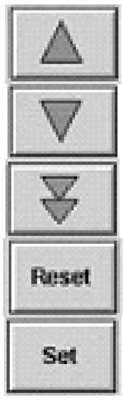


Figure 3.8 - 3 Individual Control of Wire Feeders- Output Screen



Increases the selected channel value.

Lowers the selected channel value

Sets the selected channel to 0% Output

Resets the selected channel to 100% output

Sets the selected channel output to the percentage value in the box displayed below.

Figure 3.8 - 4 Description of Wire Feed Control Buttons.

Selecting Multiple Channels

There is a function designed into the control of the machine to allow multiple selection or individual selection of channels.

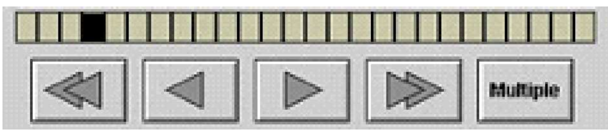


Figure 3.8 - 5 Channel Selection Bar Showing selection of a single Wire Feed Output

The selected channel is shown by the single cursor, which can be moved using the fast and slow movement buttons.

Selecting the MULTIPLE button, will change the buttons.

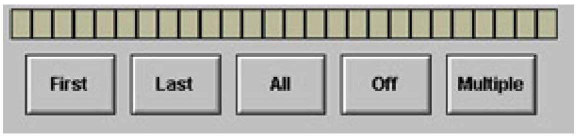


Figure 3.8 - 6 Channel Selection Bar Showing the Multiple Selection Wire Feed Output

Figure 3.8 - 7 Channel Selection Bar Showing selection of All the Wire Feed Outputs

Selecting the ALL Button will select all the outputs and highlight all the channels.

If an individual channel has been selected in normal display and then the FIRST button is touched, and then the MULIPLE screen is selected, a marker will be placed on the first channel. If another channel is selected in the normal view and the LAST button selected in the MULIPLE view, all the channels between the first and last selected will be used.

Operation Under Normal Production Conditions

Ensure that the SETUP screen has had the correct conditions loaded within them, i.e.

- Substrate Width
- Nominal Wire Speed

The WIRE FEED ON button should be touched at a similar time to the BOATS ON button.

When the evaporators have been heated up to a level that they will evaporate the aluminium being fed to them, the wire will be fed automatically at a reduced speed. After a time delay, the wire will be fed instantaneous at the nominal speed set-point. To stop the wire, touch the "Wire Feed Off" button

The user can modify the wire feed rates using a combination of the "Nominal cm/min Feed Rate" and the wire feed "Output" percentages. However, once the controller is operating in the AUTO mode, it will make corrections to the "Output" percentages.

At the end of the cycle, the auto-slowdown function in the drive system will shut the wire off when the shutter is closed.

Alternatively, the wire can be stopped by touching the BOATS OFF button on the PROCESS screen

Operation under New Evaporators Conditions

As already mentioned in the chapters about the evaporators, problems may exist when feeding wire onto them.

If the wire does not melt and evaporate on the evaporators, the power level to them all can be increased. Alternatively the wire rate can be turned down.

If there is one evaporator that is giving problems, remove the wire feed outputs from all the evaporators except that one. Adjust the power rate until the cavity is being filled as described previously. Then bring all the wire feed rates back up to the normal operating level.

3.9 Use of Release Agent

The collection efficiency of the evaporated aluminium onto the passing substrate is approximately 70%. The other 30 % will be evaporated and collected on the shielding around the boats.

In order to prevent evaporated aluminium adhering to the source box, surrounding collection shields and shutter, the surfaces must be painted with a release agent.

The recommended release agent is Boron Nitride paint or Aquadag Graphite Paint diluted with water.

Careful and adequate application of the paint will be repaid by easy cleaning of the machine.

In places such as the underside of the shutter, where some adhesion of the aluminium is required, the paint should be applied more sparingly than on the inside of the source box.

Avoid painting the electrical contact surfaces and any evaporators that may be in the source box.

The specification of the boron nitride can be found in the consumable section.

	Caution !
	**Do not use hammers or chisels to remove aluminium from the machine's collection surfaces.**

3.10 Visual Inspection of Coating Source Area Before Closing the Chamber

The evaporators and immediate area must be checked before closing the chamber. The coating shield , viewing flap and shutter must be also checked and cleaned for loose aluminium coatings.

This includes;

- Checking that there is enough aluminium wire left on the feed spools. (section 3.11)
- Cleaning the evaporator surface (section 3.6)
- Checking the wire feeder tubes for no potential blockages (section 3.7)
- Checking that no potential shorts can take place between the wire feeders/tubes and the surrounding steelwork. (section 3.7)
- Checking that no loose debris is hanging of the shutter (section 8)
- Checking that there is no loose aluminium debris that can cause shorts to the evaporators. (section 8).

3.11 Checking Aluminium Wire Spools for Low Wire Levels

The aluminium wire is usually supplied by a commercial supplier on plastic spools.

The amount of wire is normally measured in weight. This is normally between 6.5 and 10.5 KG.

To enable the user to visually see how much wire is still on a reel, a slot has been designed into the plastic spools.

On a standard spool, as the wire initially gets to point where the slot is no longer full, there is approximately enough wire for one process cycle (<40 minutes) left. With this knowledge and knowing the speed that the feeders are being operated at, the user can decide if to or not to change the spool before the wire runs out.

Figure 3.11 - 1 Photograph Of Full Wire Spool

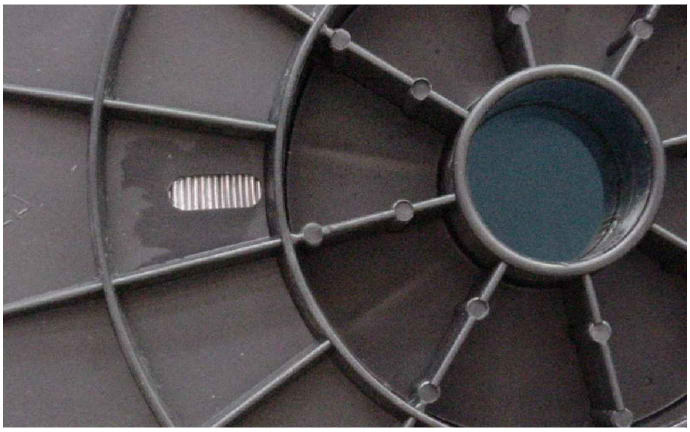


Figure 3.11 - 2 Photograph of Opening in Spool Side Used For Checking How Much Wire is remaining

3.12 Pre-coating Plasma Treatment

Web based substrates used for vapour deposition, often require surface preparation or chemical modification to enhance the adhesion of the deposition layer to the web surface or perhaps a smooth surface. With today's improved understanding of the impact of film surface chemistry there may also be good reasons to chemically modify web surfaces to enhance moisture, gas barrier and adhesion.

Today there are many web substrates available for vapour coating with a wide range of surface chemistries and topographies from paper, non-woven polymers, cellophane and synthetic polymer films. These substrates cover a wide range of chemical functionalities, which will influence what surface treatments if any, will be necessary to prepare them for coating.

In preparation for depositing a coating, the web can be surface modified, in the vacuum chamber, prior to, the coating being deposited.

One common method used is plasma treatment of the substrate.

What is a Plasma ?

Plasma is often described as the "Fourth State of Matter," the other three being solid, liquid and gas.

A plasma is a distinct state of matter containing a significant number of electrically charged particles, a number sufficient to affect its electrical properties and behavior.

In an ordinary gas, each atom contains an equal number of positive and negative charges; the positive charges in the nucleus are surrounded by an equal number of negatively charged electrons, and each atom is electrically "neutral."

A gas becomes a plasma when the addition of heat or other energy causes a significant number of atoms to release some or all of their electrons. The remaining parts of those atoms are left with a positive charge, and the detached negative electrons are free to move about. Those atoms and the resulting electrically charged gas are said to be "ionized." When enough atoms are ionized to significantly affect the electrical characteristics of the gas, it is a plasma.

In many plasma coating applications positive ions are generated by collisions between neutral particles and energetic electrons.

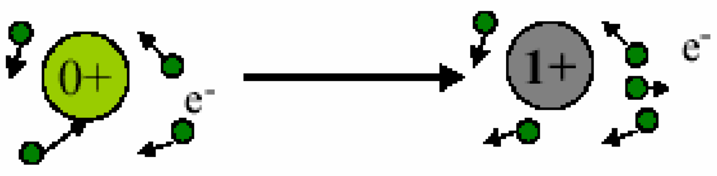


Figure 3.12 - 1 Drawing Demonstrating the Effect of Energy on a Atom

By using a magnetic field, the intensity and therefore effect can be improved.

The construction of plasma Treaters on General Vacuum Equipment machines has optomised the performance of the plasma treatment of the substrates.

In-Chamber Plasma Treatment

There are several principle plasma generation technologies in use. General Vacuum Equipment has supplied a system with an AC magnetron source that can be used with reactive gas blends.

The essential features of all the plasma generator is that it can supply high energy electrons in sufficient quantities to generate heavy ions and additional electrons from the gas blend, for bombardment of a substrate surface. The ions and electrons are constrained in a ring shaped magnetic field above the generator and the substrate is passed though the plasma.

SECTION 4 VACUUM PUMPING SYSTEM

4.1 Vacuum Pumping System Mimic Display Screen

The vacuum pumping system is controlled from the VACUUM screen. Select the VACUUM tab at the top of any screen and the VACUUM screen will be displayed. The screen has a number of buttons that control the operating mode of the vacuum pumping system.

Button	Mode of Operation
Vacuum	Places the vacuum pumping system into a pumping condition to evacuate the chamber
Hold	Places the vacuum pumping system in a standby condition
Vent	Vents the chamber to atmosphere
Pumps	Starts all the required pumps from a non energised condition
Shutdown	Places the vacuum pumping system in a controlled shutdown sequence
Auto	When highlighted, the system is in the AUTO mode
Manual	When highlighted, the system is in the MANUAL mode – not available in ‘operator access level’

Table 4.1 - 1 Vacuum Screen Operating Buttons

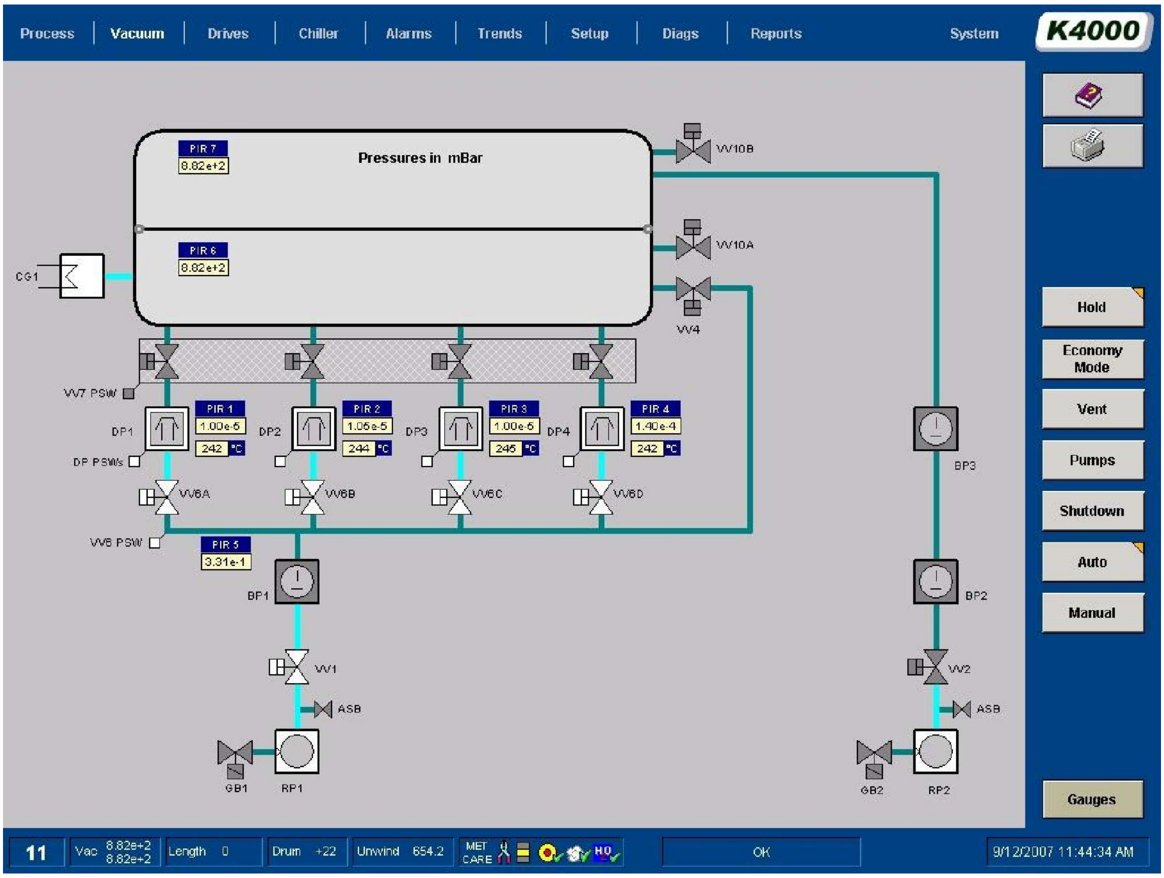


Figure 4.1 - 1 Vacuum Mimic Screen

The pumps and valves are colour coded to show their operating condition;

- Grey - Closed or Off
- White - Open or Running
- Red - Neither Open or Closed, but in a ‘fault’ condition

In the ‘maintenance or supervisor’ level, manual control can be made of the system.

The touch screen will allow operation by a pointer will be indicated on the screen.

A mouse has been supplied to allow the user the facility to start individual pumps and open valves safely when in the manual mode.

If a red box is shown around a pump, it is an indication that there is a problem with the pump. This problem has been sensed by the machine computer.

4.2 Vacuum Pumping System Control

As previously stated, the vacuum pumps and valves are controlled from the VACUUM screen. Full protection and interlocking of the pumping-group are provided by software. It is used together with interfaces on the valves and pressure monitors located at critical parts of the system.

The AUTO and MANUAL buttons can be used in the supervisory mode to change the normal operation between the auto and manual modes. Indication of the mode will be highlighted with an orange triangle I the top right hand corner of the button.

Figure 4.2 - 1 AUTO and MANUAL buttons on VACUUM screen

Normal operation of the machine is in the operator mode, not the supervisor mode. If the system has been operating in the supervisory mode for set up or maintenance reasons, ensure that is placed back in the normal operating mode.

Manual Operation – Supervisory Access Level

Valves and pumps can be operated directly from the VACUUM screen. This facility is useful during testing and maintenance procedures. Note that all the safety interlocks remain in operation so that the system will not allow any unsafe operation, e.g. the chamber cannot be evacuated unless the winding cart is fully engaged within the chamber.

Automatic Operation – Operator Access Level

When switched to the 'Auto', control of individual valves and pumps are not permitted. This is the normal operating mode of the pump group.

In the 'Automatic' mode, operation of the 'vacuum' button starts the sequence of starting the pumps and opening valves. This gives the fastest possible evacuation of the chamber to metallizing conditions, under safe conditions.

An alternative push-button located on the operator station is normally used in conjunction with the enable button to start the sequence and drive the traverse motor for a few seconds.

Operational modes

"Vacuum"

When this mode of operation is selected, the air contained within the chamber will be evacuated by the pumping system. The level of vacuum reached will be generally lower than 8 x10-4 mbar in the evaporation zone.

Interlocks are used to ensure that this operation only occurs in a safe manner.

“Hold"

This mode is selected to place the pumping system in a standby situation. The pressure level in the chamber will not change, when this mode is selected.

"Vent"

When this mode is selected, valves (vent) will be energized to allow the air surrounding the chamber back into the chamber. The pressure levels inside and outside the chamber will equalize.

Interlocks are used to ensure that this operation only occurs in a safe manner.

Diffusion Pump – Hot Condition

Note: When the diffusion pump(s) are not at the operating temperature, the diffusion pump timer will indicate the time left until the pump is ready. Until then, the vacuum sequence cannot be started.

Normally the temperature of the diffusion pumps is not measured and displayed, but a timer allows them to heat up over a set time period of 40 minutes. On some machines a temperature indicator is supplied as an option.

Note: If the power to the machine is lost, the timer will reset to 40 minutes and the delay will start again.

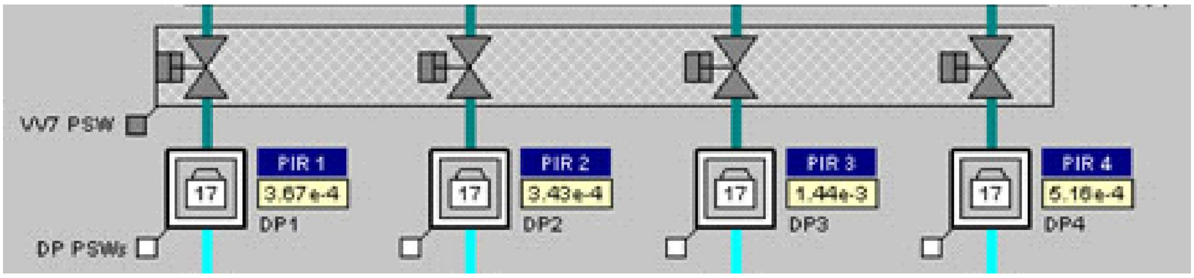


Figure 4.2 - 2 Diffusion Pump Timer Showing Elapsed Time to Fully Heated Condition

Diffusion Pump – Temperature Sensors -Option

An option that is sometimes installed on a machine is temperature indicators for diffusion pumps. This option allows the user to monitor the pump performance.

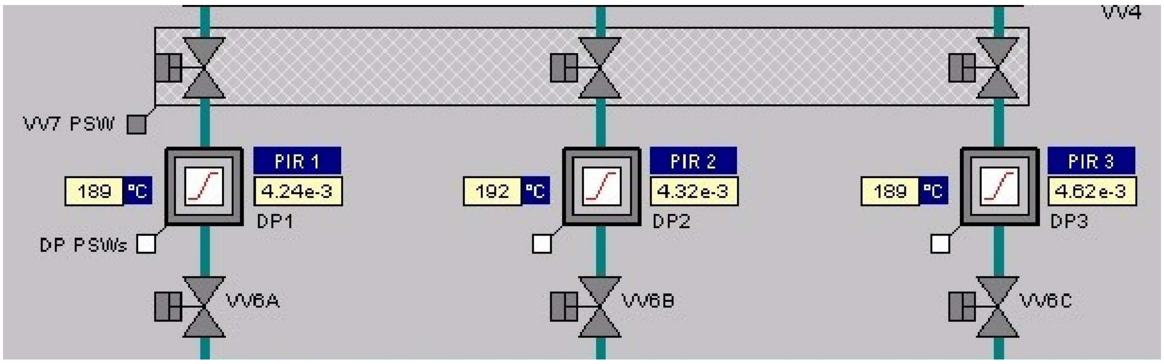


Figure 4.2 - 3 Vacuum Screen showing Diffusion Pump Temperature Feedback

Lauda Cryogenic Unit Control – Option

A different type of cryogenic unit is sometimes fitted on machines. It is manufactured by a company called ‘Lauda’ and is referred to as a ‘Lauda’ unit.

This unit is remotely controlled through the vacuum screen.

With the vacuum system in the ‘automatic’ mode of operation, the unit will be controlled automatically.

When the vacuum system is placed in a ‘manual’ mode, extra interface buttons are shown to allow manual operation of the ‘Lauda’ unit from this screen.

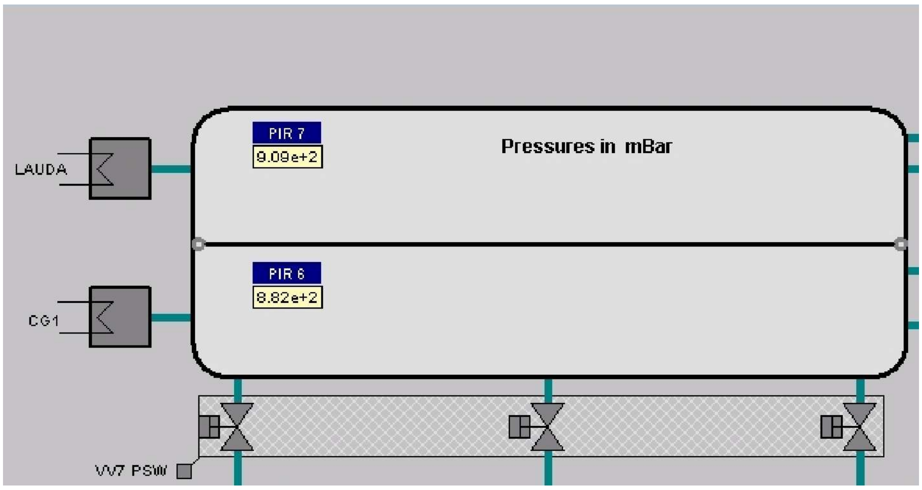


Figure 4.2 - 4 Vacuum Screen – Lauda Option – Automatic Control

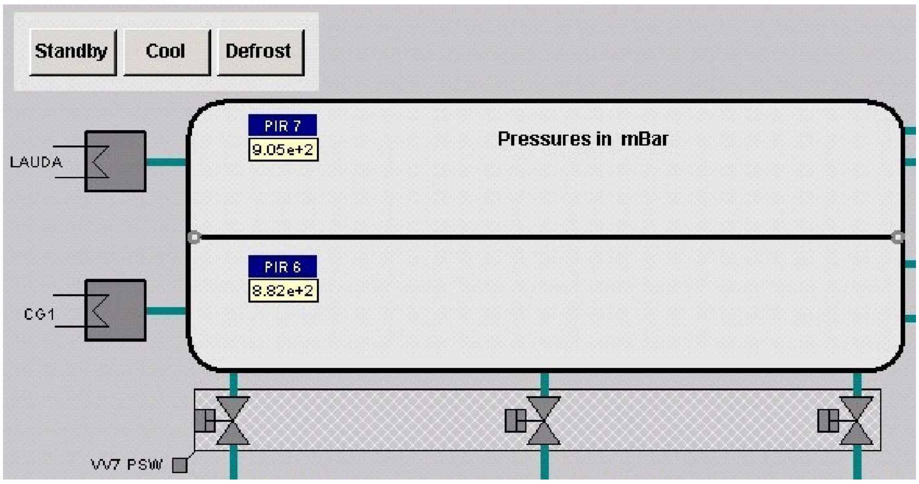


Figure 4.2 - 5 Vacuum Screen – Lauda Option – Manual Control

4.3 Vacuum Pumping System Start-Up Sequences

Utility Requirements

The vacuum pumping system requires compressed air at a minimum pressure of 5.5 bar (approx. 80 psi) and cooling water at a minimum pressure of 4 bar (approx. 60 psi) at the various cooling water manifolds. When both of these requirements are met the "Services" indicator on the interlock popup screen will be green. (See the Pneumatic and Water Schematic drawings for exact requirements.)

Caution !

Note the process drum chiller unit should have been energized for a minimum time period of four hours before the unit is placed in the cool mode.

Normal Start-Up

With the pumping control system in the "Automatic" mode, the process chiller unit, diffusion pump heaters and the rotary pumps will be started by pressing the PUMPS push-button, as shown in the flowchart below.

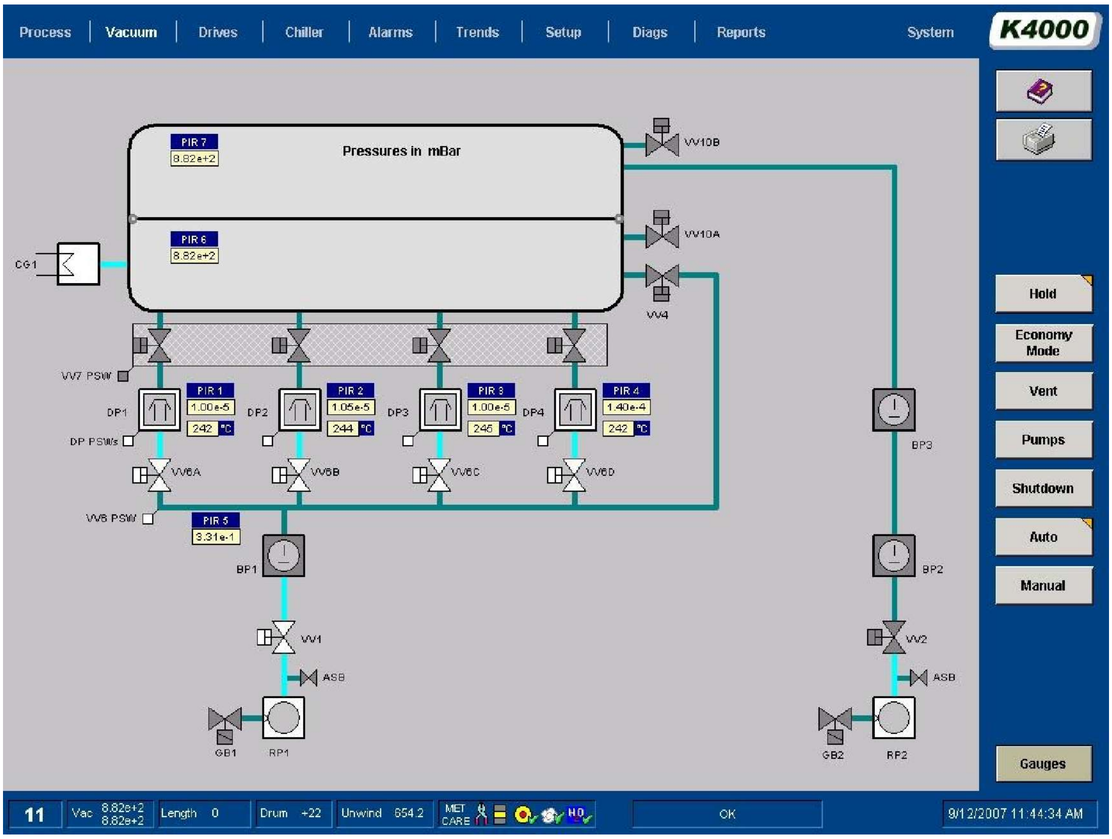
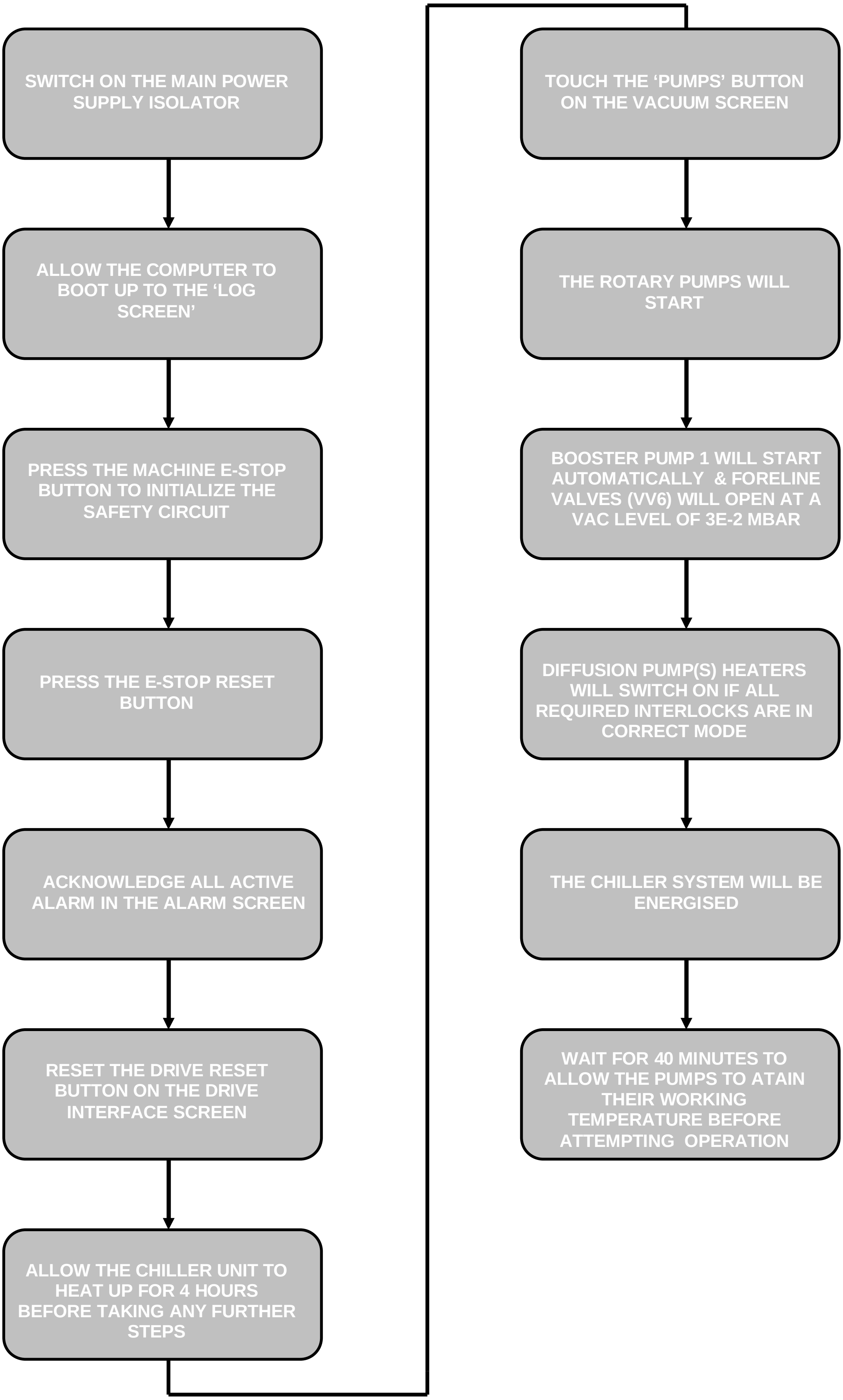


Figure 4.3 - 1 Sample Vacuum Mimic

The timer displayed on the diffusion pump will "count down" as the diffusion pump is heated up.

The Valve VV7 will not open if the timer has not timed out fully.
With the sensor option installed for diffusion pump temperature monitoring, the temperature has to be above 200 degrees centigrade.



Flow Chart 4.3 - 1 Start Up Flow Diagram – PUMPS

Pump Interlock Diagnostic Screen

For each pump, a diagnostic screen is provided to inform the user why the pump will not operate if the icon is touched.   These individual screens can be accessed from the DIAGNOSTICS screen.

There must be a "leg" of green blocks from one side of the screen to the other to allow the equipment to be operated. If the block is red, it signifies that the interlock has not been energised.

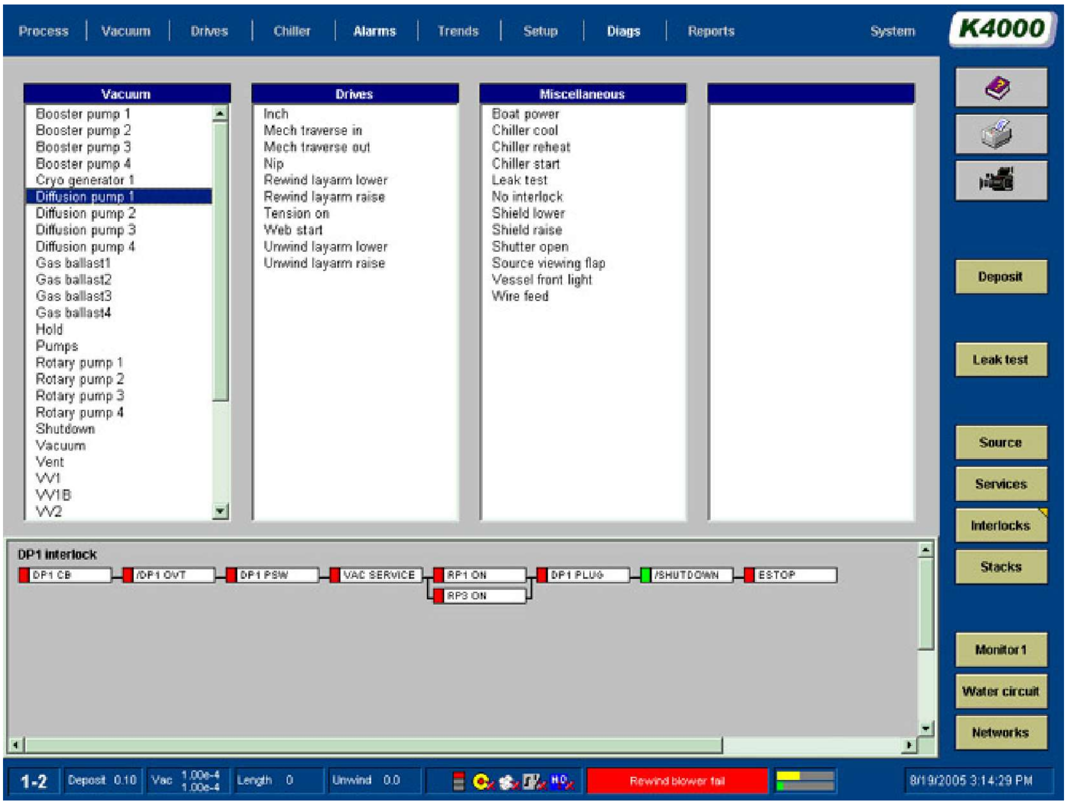


Figure 4.3 - 2 DIAGNOSTICS/INTERLOCK Screen

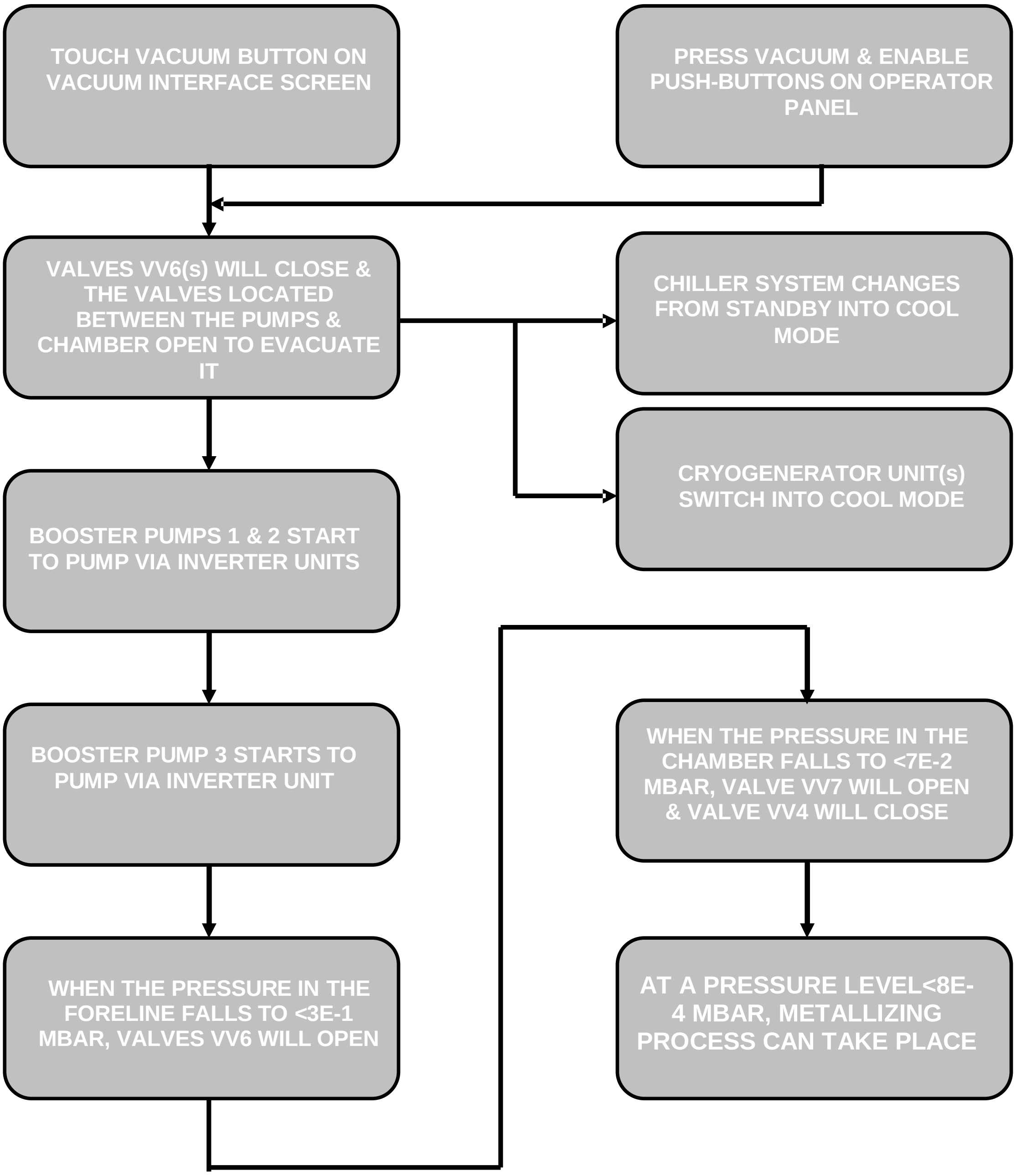
4.4 Pumpdown Sequence

The pumping system is designed to evacuate the chamber in an efficient and safe manner.

Warning!

No adjustments must be made to the sequence without contacting and gaining the permission in writing from General Vacuum Equipment.

The flow diagram for this sequence can be seen below.

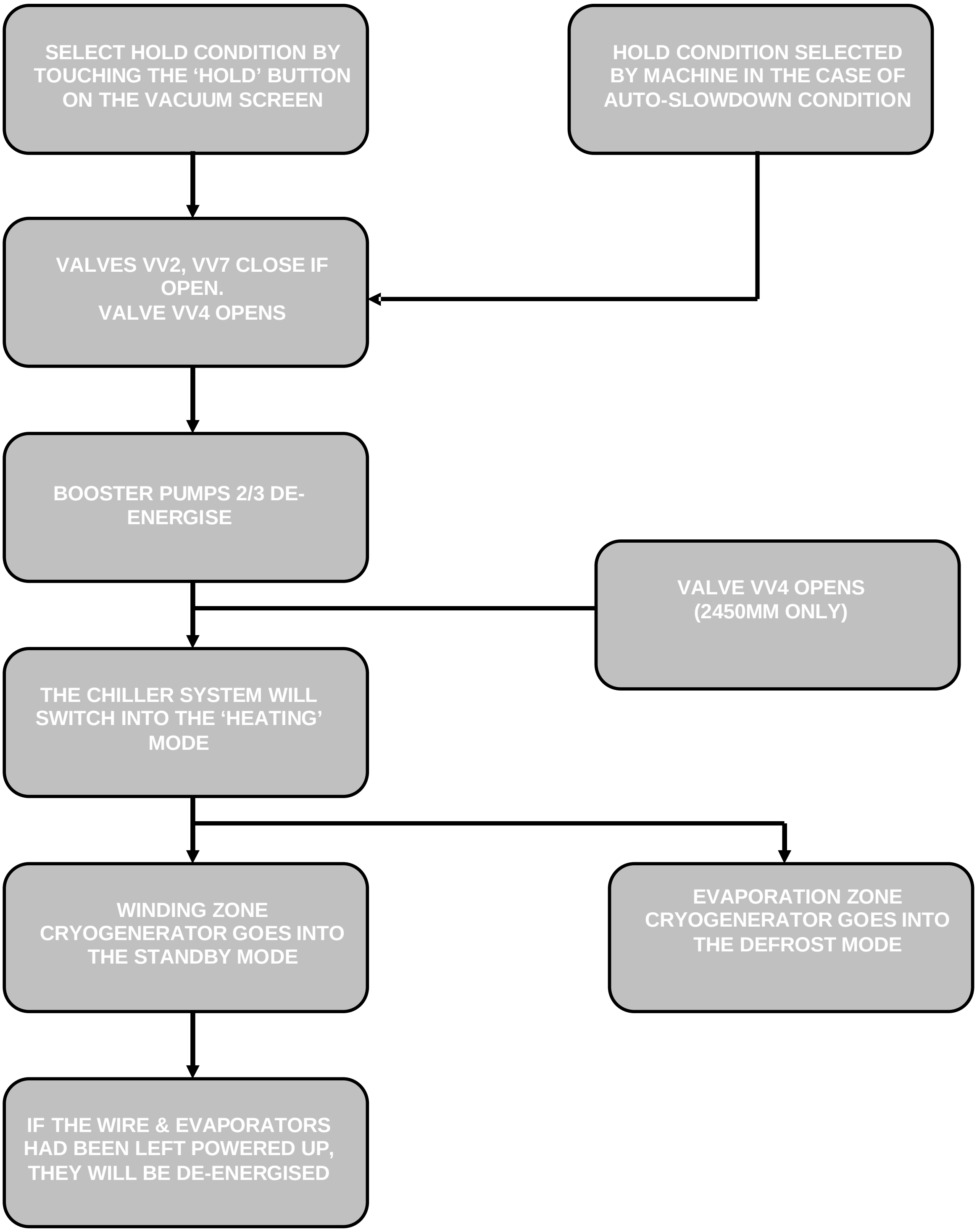


Flow Chart 4.4 - 1 Machine Pumpdown Sequence

4.5 Hold Condition

The HOLD condition is a description given to the vacuum pumping system when the vacuum levels are held constant within the chamber. This pressure may be at atmosphere or at the end of a cycle.

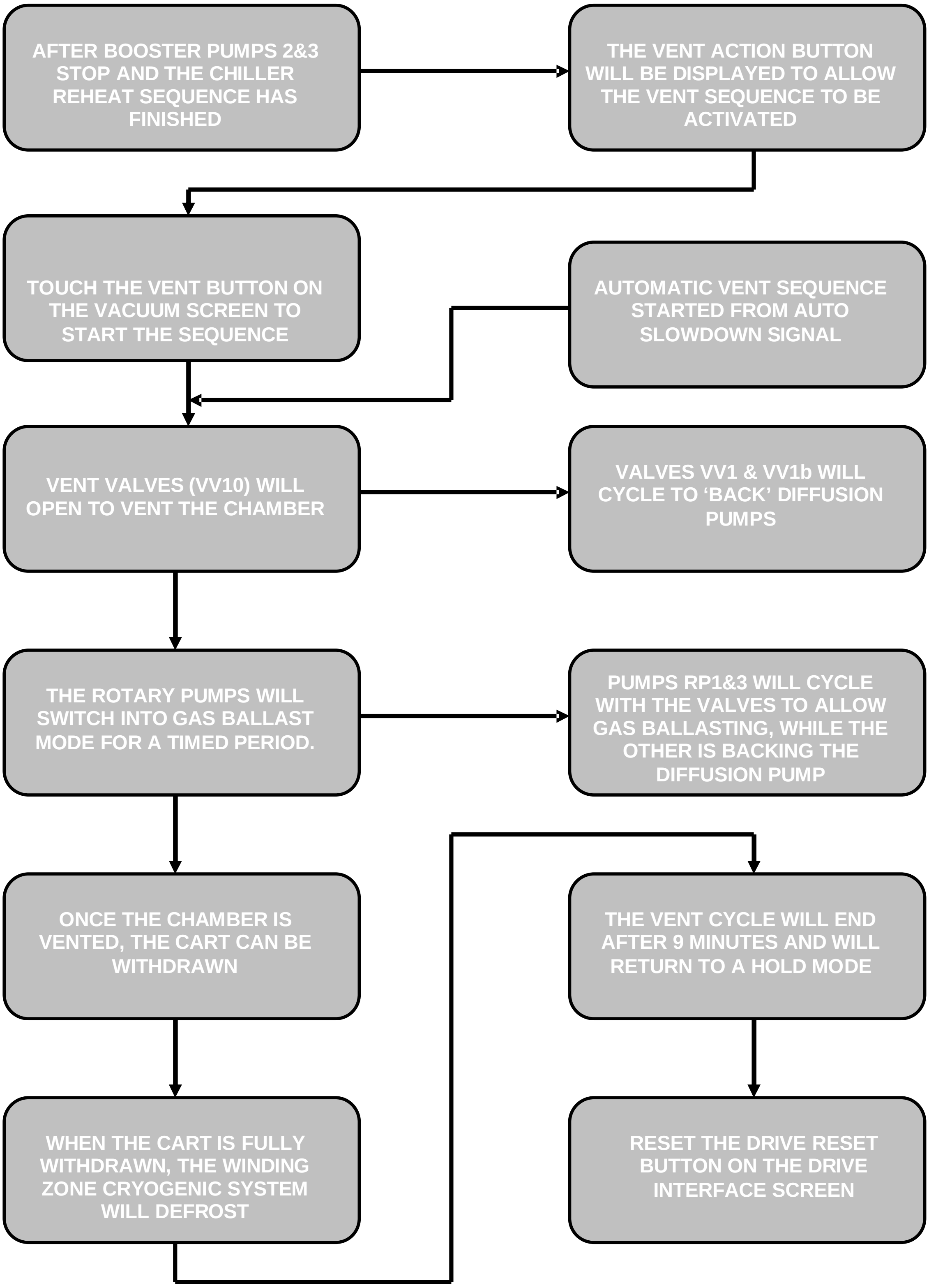
The HOLD condition is used as a buffer condition between changing from VACUUM to VENT for example. The hold mode is indicated by a triangle on the HOLD push-button.



Flow Chart 4.5 - 1 Hold Sequence

4.6 Vent Sequence

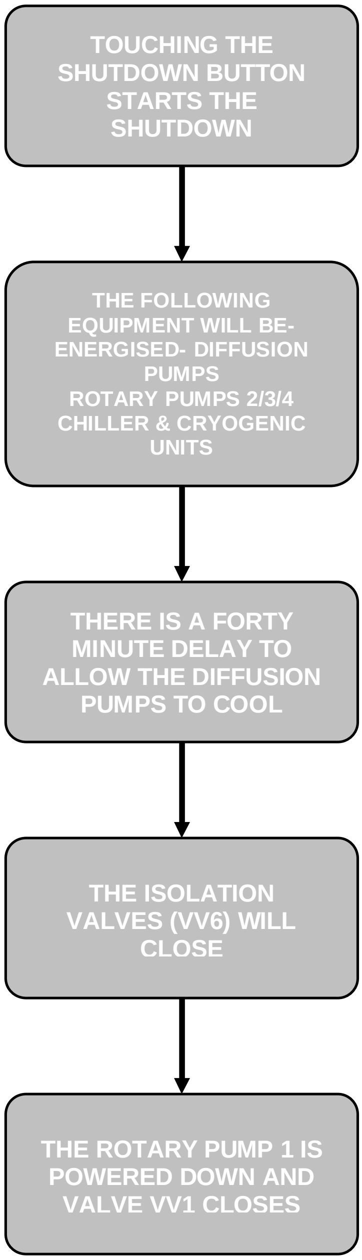
The pumping system will be isolated from the chamber and the air surrounding the machine will be vented into the chamber. The following block diagram explains this sequence;



Flow Chart 4.6 - 1 Vent Sequence

4.7 Shutdown Sequence

The pumping system can be switched off in a correct sequence by pressing the shutdown function key from the Vacuum screen. The flow diagram for this sequence can be seen below;



Flow Chart 4.7 - 1 Shutdown sequence

Partial Evacuation

Before placing the machine in the SHUTDOWN mode.

It is normal to close the machine and partially pump it down before placing the machine in the shutdown mode. This will ensure that the chamber is kept clean. It is suggested that the chamber be pumped down to about 500 mbar, and then placed into the HOLD mode.

4.8 Types of Pumps Incorporated on Vacuum Machines

A vacuum is created by using a number of suction (vacuum) pumps to remove the air out of a chamber and discharges it to the surrounding atmosphere.

Several different types are used, since a single pump type cannot work efficiently over the range of pressures from atmosphere to1E-5 mbar.

Rotary vane and piston pumps have a comparative low pumping speed, but will pump a high pressure differential between input and output, for example 1 mbar inlet to 1000 mbar outlet.

A mechanical booster pump (Roots type blower) has a high pumping speed, but will only work against a small differential pressure between inlet and outlet ports. The pressure range of this type of pump is 10 to 1E-3 mbar

At very low pressures, e.g., below 1E-2 mbar, diffusion pumps are used to provide a very high method of pumping.

In vacuum coaters, where high volumes of water vapour are present, a cryogenic pumping system is used to condense and freeze the gases to coils within the chamber.

Rotary Piston Pumps

The rotary piston pump shown below uses a piston (a hollow cylinder with a hollow tongue attached), which rotates inside a stator by means of the rotating cam within it. The operating cycle, i.e., increase volume, isolate and discharge is similar to the basic concept as described for the rotary vane pump. Pumps with speeds up to 28,000 liters/min.-1 are available.

Mechanical rotary pumps develop their full pumping speed from the atmospheric pressure to about 1 mbar. The pumping speed then decreases to zero as it approaches the ultimate pressure of the pump.

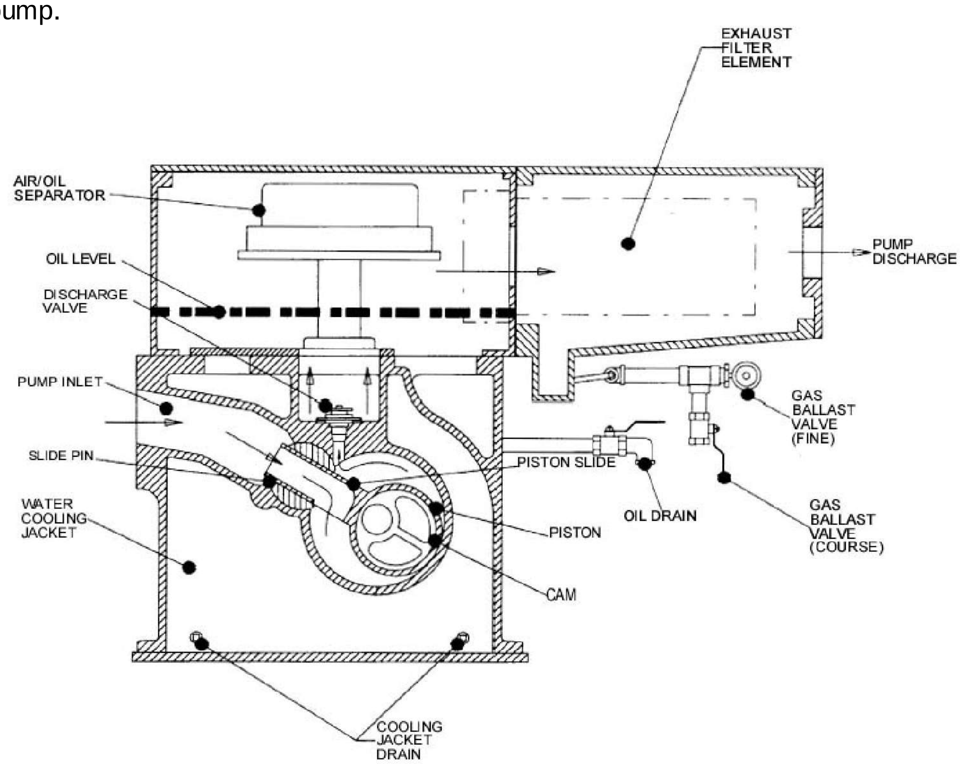


Figure 4.8 - 1 Rotary Piston Pump

These pumps are started from the mimic screen. They should be allowed to warm-up for about 30 minutes before attempting to pump down the vessel and must be left running whilst changing reels, etc.

Rotary Pump Auxiliary Pump System Functions

Gas Ballast Function (Located behind pump guard enclosure)

Any water vapour that is removed from the chamber by the pumping system must pass through the discharge valves in the rotary vane or piston pumps, which is oil sealed. Some of this water condenses within the pump and forms an emulsion with the oil. The oil will change its appearance from a clear (brown) colour to a milky white.

Oil in this condition can lead to reduced pump efficiency, due to the high vapour pressure of water and eventually to damage to the pump.

The gas ballast function allows air to be sucked into the pump through solenoid-operated valves.

The action of the air (gas load) makes the pump work harder and therefore heat the oil temperature to rise and the water is evaporated through the exhaust port.

This admission of air, creates a gas load within the pump that has to be removed by it. This extra work by the pump will increase the pump's oil temperature, which in turn drives the condensed water out of the oil. (The change in the oil condition can usually be seen at the oil-level sight glass where the oil changes from an opaque to a clear appearance as the water is removed.

The gas ballast function is automatically used when the pumping group is in the "Vent" mode. It also can be activated by the operator in the pumping group manual mode of operation.

Air Admittance Valve – located in pipework above pump

When the rotary piston pumps are switched off, a small valve located in the inlet side of the pumps, called the air admittance valve, will open. The purpose of these valves is to allow both sides of the pump to be at an equal pressure level. Without these valves, it is possible that the oil within the pumps can be "sucked" up out of the pumps and into the pipework.

The valves are "normally open" type and are powered up from the plc when the pump contactor auxiliaries are energized.

They should be cleaned periodically, especially when used in dusty conditions.

Air admittance valves operate automatically whatever operation mode, i.e.; manual or automatic.

Oil Mist Eliminators and Filters

The Kinney rotary pumps are fitted with two in-built oil mist filters. Over a period of time, these filters will become saturated with oil and will start to fail. An indication of them failing is the emission of oil mist during the evacuation of the chamber, or when the pumps are in a gas ballast situation.

Atomizers

Atomizers are fitted to the exhaust ports of the pumps to remove the odour of any oil mist that passes through them. The atomizer contains easily replaced "Activated carbon". If details are required, please contact General Vacuum Equipment.

Cooling Water

The pumps are cooled by water supplied from the vacuum group manifold. The water flow is controlled from a thermostatic valve located on the individual pumps. It is important that these valves are not adjusted.

Rotary Vane Pumps

The rotary vane pump uses a piston (a hollow cylinder with a hollow tongue attached), which rotates inside a stator by means of the rotating cam within it. The operating cycle, i.e., increase volume, isolate and discharge is similar to the basic concept as described for the rotary vane pump. Pumps with speeds up to 28,000 liters/min.-1 are available.

Mechanical rotary pumps develop their full pumping speed from the atmospheric pressure to about 1 mbar. The pumping speed then decreases to zero as it approaches the ultimate pressure of the pump.

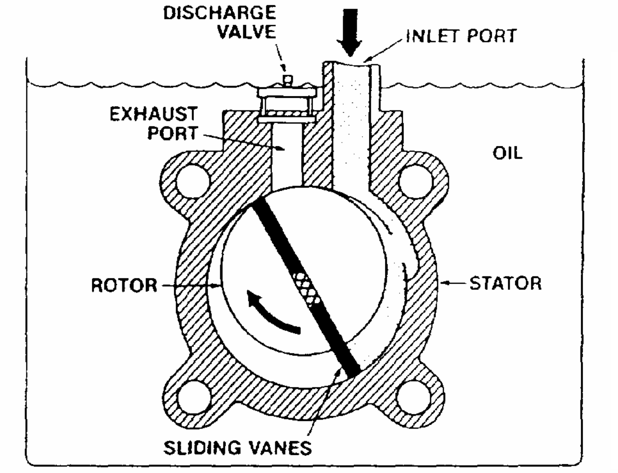
The rotary vane, oil-sealed mechanical pump removes gases by compressing them to a point slightly above atmospheric pressure. It then expels the gases to the outside world. It is used to produce roughing or fore pressures lower than the rotary piston pump.

The pump module is immersed in an oil bath. This oil is purified to remove high-vapour pressure contaminants. The oil serves the following purposes:

1. Cools the pump.
2. Lubricates.
3. Seal against atmospheric pressure.
4. Opens second-stage exhaust valve at low inlet pressures.

Components

An oil-sealed mechanical pump includes a housing, or stator, an offset rotor with spring-loaded vanes, an intake port and an exhaust port equipped with a discharge valve. It may also have a ballast valve.



The pump rotor is driven by a belt-drive mechanism or it may be directly coupled to the drive motor. Most pumps have two stages to produce better vacuum. This view of a mechanical pump shows the inlet and exhaust ports of one stage.

Figure 4.8 - 2 Rotary Oil-Sealed Mechanical Pump Module

How the Pump Works

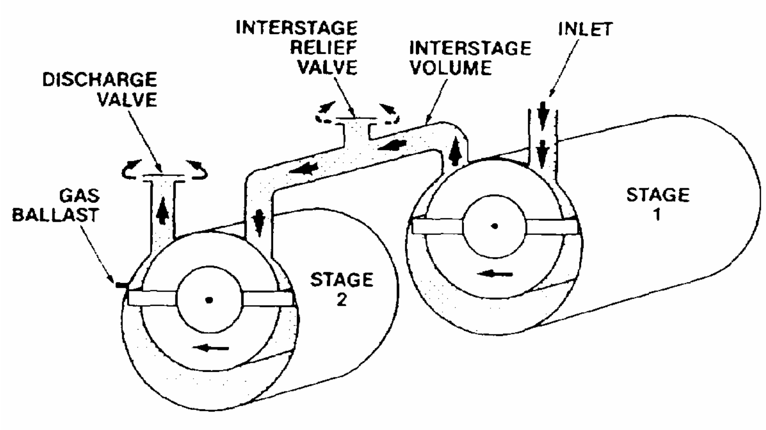


Figure 4.8 - 3 Operation of a Rotary Vane Pump

Gases from the chamber enter the inlet port. They are swept around by vanes, and compressed. Compression builds up pressure to overcome atmospheric pressure. The spring-loaded discharge valve is opened. The air is then expelled to atmosphere. These pumps can remove over 99.9% of the air from the chamber.

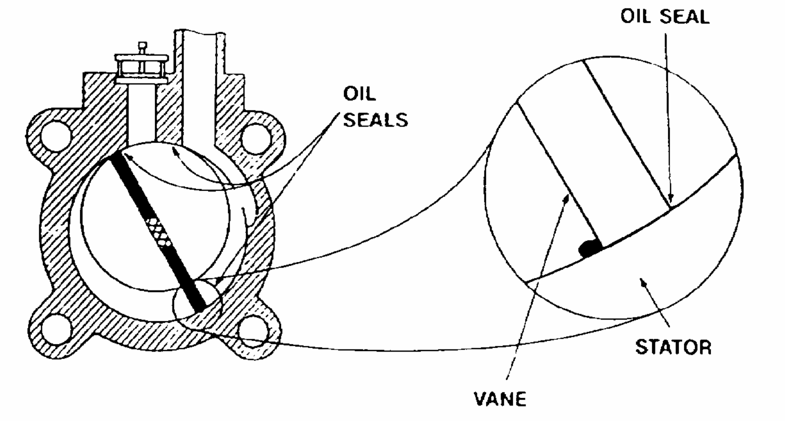


Figure 4.8 - 4 Operation of the Rotary Vane

A thin film of oil makes the final seal in these pumps. Therefore, base or ultimate pressure is partly determined by the vapour pressure of the oil. If the oil becomes loaded with water, or other impurities, these contaminants increase the vapour pressure.

Then it is impossible to reach satisfactory low pressures. The oil is a very important part of your mechanical pump. Dirty pump oil is usually the reason why the pump is not performing well.

These pumps are started from the mimic screen. They should be allowed to warm-up for about 30 minutes before attempting to pump down the vessel and must be left running whilst changing reels, etc.

Rotary Vane Pump Auxiliary Pump System Functions

Gas Ballast Valves (Located within the pump enclosure)

Rotary vane pumps have a 'Gas Ballast' facility that admits air into the pump inlet whilst it is running, but not pumping the vessel (during reel change, etc.). This function takes place automatically as the chamber is vented.

This admission of air creates a gas load within the pump that has to be removed by it.

This extra work by the pump will increase the pump's oil temperature, which in turn drives the condensed water out of the oil. (The change in the oil condition can usually be seen at the oil-level sight glass where the oil changes from an opaque to a clear appearance as the water is removed.

Anti Suck back Valves

This type of Pump is fitted with an anti suck back valve at the pump inlet. This is part of the pump.

This valve is closed when the pump is switched off or at other times. It can be used to allow the pump’s internal pressure to rise up to atmospheric pressure whenever the pump is stopped. By this action oil is prevented from being forced through the pump into the vacuum lines

These valves operate sequentially in the normal operation mode of the machine, i.e.; manual or automatic.

Oil Mist Eliminators and Filters

The rotary vane pumps are fitted with in-built oil mist filters. Over a period of time, these filters will become saturated with oil and will start to fail. An indication of them failing is the emission of oil mist during the evacuation of the chamber, or when the pumps are in a gas ballast situation.

Cooling Water

The pumps are cooled by water supplied from the vacuum group manifold. The water flow is controlled from a thermostatic valve located on the individual pumps. It is important that these valves are not adjusted.

Oil temperature

On this type of pump, the oil temperature is monitored, and if found to be above a predefined level, an alarm will be indicated on the MMI

Oil Level

On this type of pump, the oil level is monitored, and if found to be low, an alarm will be indicated on the MMI

Mechanical Booster Pump

A typical mechanical booster pump utilizes two figures of eight shaped impellers, synchronized by external gears and rotating in opposite directions inside the stator. The gas is trapped between the impellers and the stator wall and transferred from the high vacuum inlet to the discharge side of the pump. The gears are oil lubricated and are isolated from the impellers, so that the impellers run dry. The clearance between the impellers and the stator is generally about 0.005 to 0.025 cm. As a consequence, back-leakage of gas occurs at a rate governed by the pressure difference between input and output and type of gas being pumped. Under normal running conditions, a pressure difference of about ten to one is obtained across the pump.

Pressure Range 10 – 1E-3 mbar

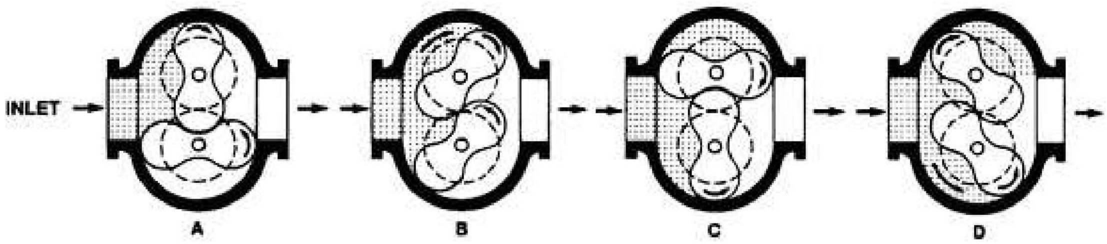


Figure 4.8 - 5 Mechanical Booster Pump

This type of pump can be also described as a "Roots" or "Blower" type pump.

Depending upon the size and type of machine, the numbers of this type of pump will vary.

This type of pump is used as a second roughing stage to bring the vacuum level rapidly down to a level of 5E-2 mbar before the high vacuum valves can be opened.

They consist of two impellers that rotate inside the pump body. The action of the movement is to sweep gases from the inlet side to the outlet side of the pump.

Inverter Drive Control (VFD)

All booster pumps installed on the machine have their speed controlled by either panel or motor mounted variable frequency drives.

The principle of operation allows the pumps to be used earlier in the pumpdown cycle without overloading them.

Diffusion Pumps

Once the roughing pumps have brought the vacuum level inside the chamber to a level below 7E-2 mbar. The high vacuum valves open, which allow the diffusion, pumps to pump the evaporation zone, while the roughing pump are pumping the winding zone.

The level of vacuum that this type of pump will achieve will depend upon the condition of the evaporation zone and the material being coated. This range is normally 1E-3 mbar down to 8E-5 mbar during metallizing.

The basic mode of operation of the pump is as follows: see the figure shown.

The silicon oil located in the boiler (base) of the pump is heated by external banks of heaters. The oil is vaporized as it is heated and rises within the stack. As it reaches the "cold cap" area, it is forced to change his direction and becomes a "vapour jet." This "vapour jet" is in a direction away from the pump inlet port and it expands as it passes from a high pressure to a low-pressure region of the pump. The vapour jet travels at a high velocity and collides with gas molecules of gas from within the pump inlet area. The oil molecules combine with the gas molecules and are forced downwards to the base of the pump.

As the combined molecules are re-heated, they separate and the gas molecules are pumped out of the pump via the foreline pumps. The oil vapour hits the walls of the pump, which are cooled. It condenses and runs down to the base of the pump.

This process is continuous while the heaters are switched on and therefore a constant cycle is maintained.

To improve the efficiency of the pump, most designs use multi-stage jets within the stack. Normally, there are three stages within this stack plus a "cold cap" at the top of the pump.

The "cold cap" is used to stop the "backstreaming" of oil back into the chamber(inlet) port. "Backstreaming" occurs when some of the oil molecules pass out of the stack. Instead of being forced down, they keep on travelling upwards towards the chamber.

The cold cap is usually cooled by cooling water fed from the vacuum water manifold. The relative parameters of the required water supply can be found upon the water schematic supplied. The water returning to the manifold from the base of the pump should not exceed 130 degrees F.

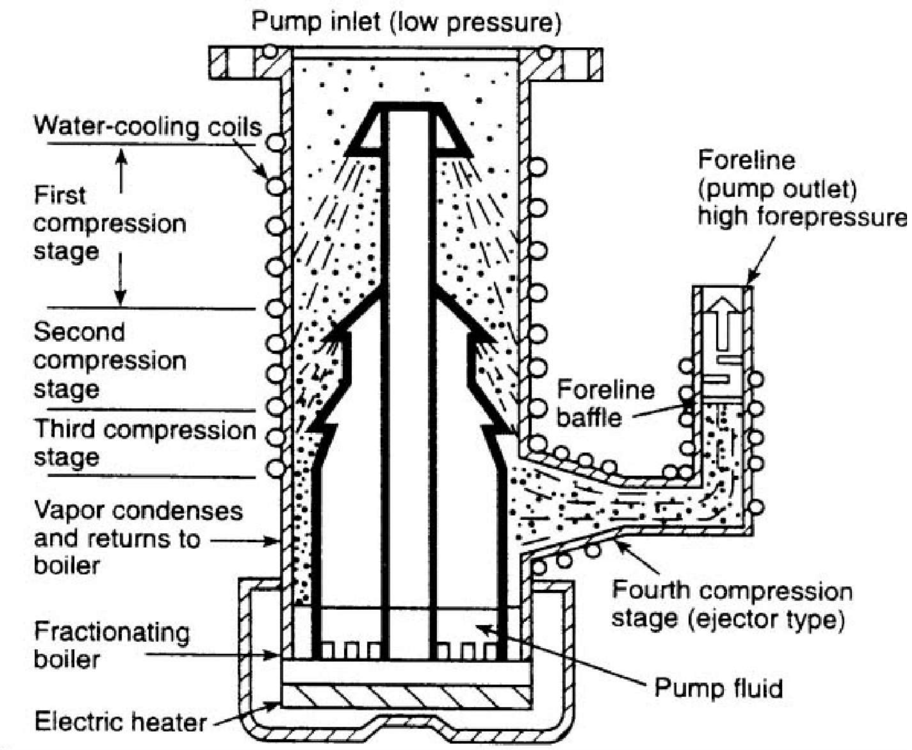


Figure 4.8 - 6 Oil Diffusion Pump

Cryogenic Pumping

A minimum of one cryogenic coil system is built within the chamber to help the other pumps within the system to "pump" the large amounts of water vapour that is given off by the substrate. The coil is supplied by a cryogenerator system.

The refrigerated coil is an integral part of the pumping system and both manual and automatic control are provided.

This coil is cooled to - 130 degrees C in order to collect any water vapour given off the substrate as it is passes through the machine.

The cryogenerator is fully interlocked with the rest of the pumping system.

- While the machine is in the "Hold mode", the cryo generator is in standby.

- While the machine is in the Vacuum Mode", the cryo generator is in the cool mode.

- When the chamber is placed in the “Hold Mode” at the end of a coating cycle, the cryogenerator will change into the standby mode until the chamber is opened.

Evaporation Zone Coil

Then it will defrost for three minutes and then return back to standby. Any water and solvents collected on the coil are melted and pumped away.
Winding Zone Coil

This coil will defrost when the chamber opens. Any water and solvents collected on the coil melt and run out of the chamber via trays and ‘run aways’.

In our application, "cryogenic" simply means "low temperature."

There is water moisture within the machine. The source of the moisture can be from the following sources;-

- Released from the substrate being processed.
- From a water leak
- From high humidity air leak

This causes a high gas load to be created for the pumping system to "pump away."

It is known that from basic physics, that water will vaporize as the surrounding pressure equals the natural vapour pressure level of water. From basic experiments, we know that water boils (turns to vapour) at 100 degrees C and the vapour pressure is 1000 bar , i.e., atmosphere.

At 10 degrees C, the vapour pressure is 13 mbar. Therefore at 13 mbar, the water "boils" at 10 degrees C and at 10-3 mbar, the water will boil at -78 degrees C !!!!!.

The normal working pressure in the evaporation zone is below 10-3 mbar and therefore the water will boil of at any temperature above -78 degrees C. Unless the chamber can be kept at a temperature of this magnitude, pumping of the chamber becomes a major problem.

Obviously, the chamber or the roll of substrate also cannot be kept at such a low temperature and rolls of paper can contain up to 5% moisture by weight.

Once the water has started to vaporize, it creates a enormous volume change;

At 1000 mbar, 1 gram of water produces 1245 cc of water vapour and using the PV=P'V' formula, the volume at the lower pressures can be calculated;

Pressure Level	Volume
1E3 mbar	1245 cc
1E2 mbar	12,450
1E1mbar	124,500
1 mbar	1,245,000
1E-1 mbar	12,450,000
1E-2 mbar	124,500,000
1E-3 mbar	1,245,000,000

Table 4.8 - 1 Water Volume at Various Pressure Levels

Therefore if water is being released at a rate of 1 gram/min. at a pressure of 1x10-3 mbar, the pumping group will have to pump 1,245,000,000 cc/min. (20,750 litre/sec) of water vapour. This is the approximate capacity of a large booster pump.

To be able to handle the extra pumping requirement, multiple extra pumps would have to be added or a smaller number of cryogenic pumping systems can be used. The cryogenic pumping system is simply copper coil that is cooled down to approximately -120 degrees C. At these temperatures, any water released is condensed and frozen onto the coils.

The coils are supplied with a specially super cooled gas from a cryo-generator unit

The Cryo-generator system is a self-contained stand-alone cryo-generator that provides a refrigerant gas to the copper cryo coil at below - 120 degrees C.

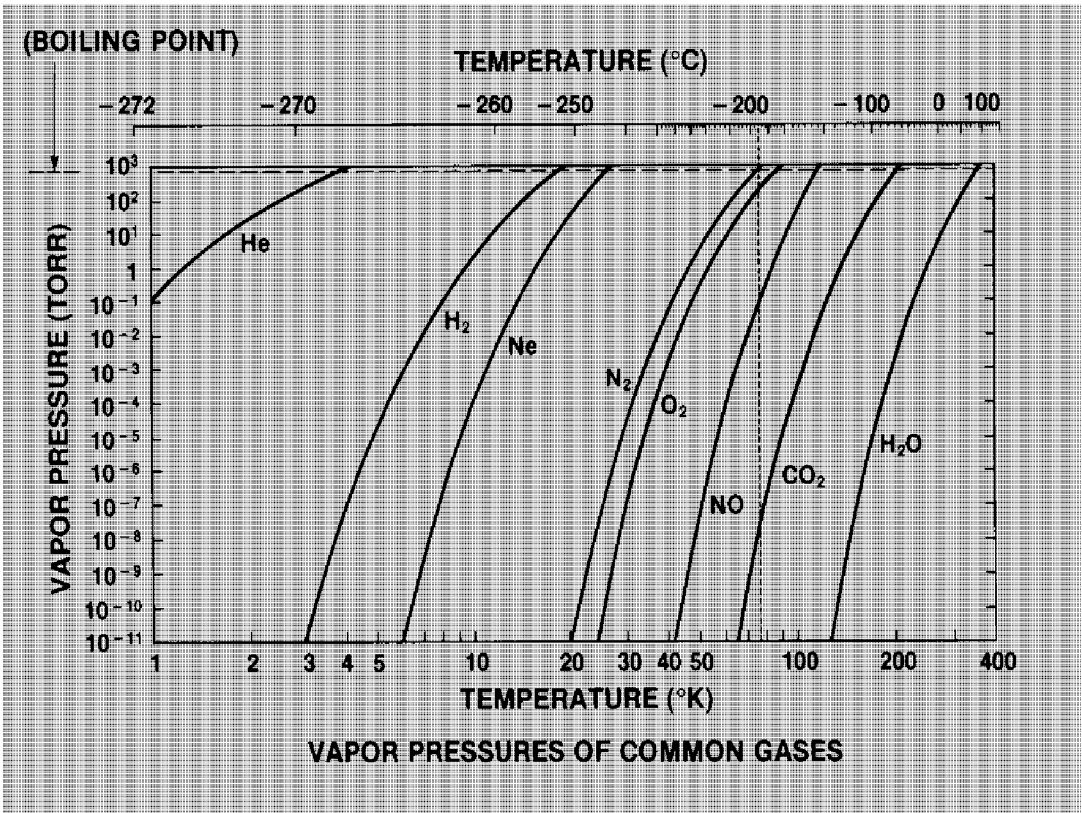


Figure 4.8 - 7 Vapour Pressure Curve of Water V Temperature

4.9 Pipe line Isolation Valves

To allow isolation of the pumping system from the chamber and between sections of the pumping system, isolation valves are used. These "vacuum tight" valves are used for their reliable performance over a number of years service.

The in-line valves located in the pipe work are activated by air driven actuators, which are controlled from the machine computer system.

The pneumatic actuators located either above or on the side of the valves are designed to close the valves in a situation where there is a loss of power to the machine. This is to ensure that the valves return to a safe condition in the case of a major power shutdown situation.

Each valve has a magnetic sensor to indicate the open and close condition of the valve. The condition of the valve is indicated on the VACUUM screen.

- Grey indicates the valve is closed
- White indicates the valve is open

Flashing red indicates that the valve has been commanded to open, but has not reached its fully open position.

The valves are interlocked by the computer to provide a safe efficient pumping cycle of the metallizer.

Vendor manuals are provided in the proprietary binders.

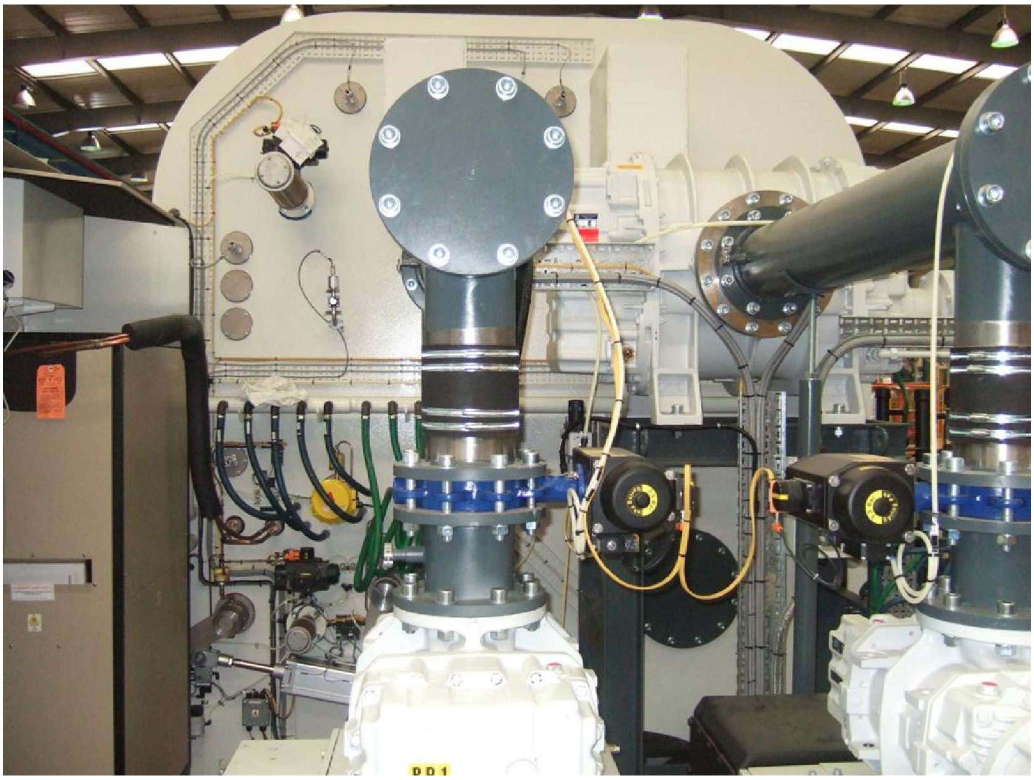


Figure 4.9 - 1 Isolation Valve

4.10 High Vacuum Valves

This type of valve can be described as follows;-

This is the plate valve located above each of the diffusion pump(s).

Figure 4.10 - 1 Photograph of High Vacuum Valve Plate

The valve is activated by a pneumatic cylinder located outside the chamber via a rotary actuator.

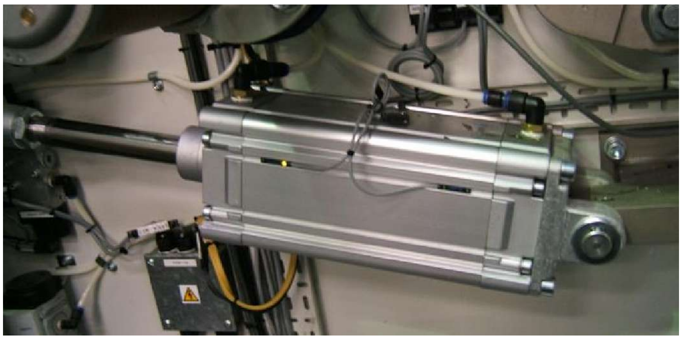


Figure 4.10 - 2 Photograph of High Vacuum Valve Plate Movement Cylinder

Figure 4.10 - 3 Photograph of High Vacuum Valve Plate Rotary Leadthrough

This valve is activated as required by the machine, when the interlocks are in the correct state.

	**Warning !**
	Do not override the interlocks.

The design of the valve allows them to be either fully opened or fully closed. They will close in the case of a shutdown of the machine's power system and leave them in a safe condition.

There are reed (magnetically activated) switches located at either end of the cylinder to indicate if the valve is open or closed.

Flashing red indicates that the valve has been commanded to open or close, but has not reached its fully open position.

4.11 Vent Valves

There are two valves located on the chamber that are used to vent the machine to atmosphere when opened.

Both of these valves have filters attached to stop debris being pulled into the chamber from the surrounding area. It also protects users from trapping their fingers in the valve.

The valves each have open and closed position switches and they are interlocked to the main machine PLC system.

If the valves do not close, the chamber cannot be evacuated.

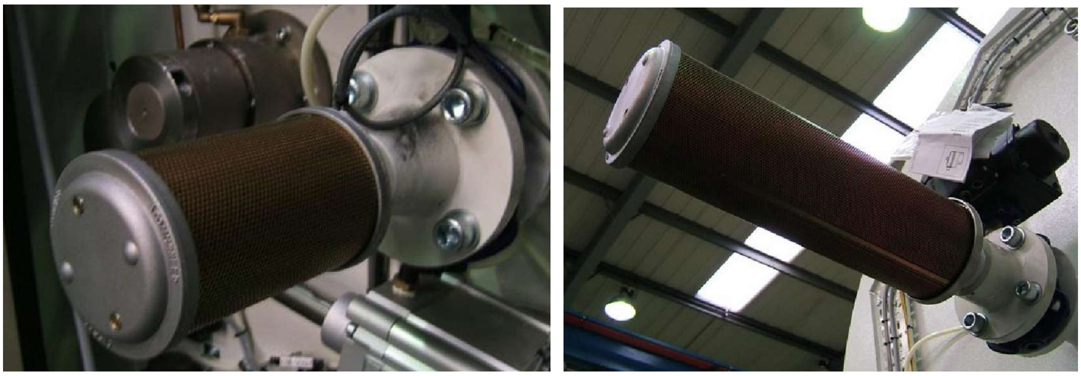


Figure 4.11 - 1 Photograph of Vent Valves

4.12 Anti Vibration Bellows

Steel Bellows

To stop vibration passing from the rotary pumps to the pipework, steel bellows are provided.

Rubber Bellows

To stop vibration passing between the pumping group and the chamber, rubber flexible bellows are used. These isolate any vibration created by the pumps.

To remove them, loosen the large hose clips, slide the bellows back on to one side of the joint area and remove the pipe adapter from the other side.

The gap in the pipework should be parallel and the maximum gap should be no more than 12mm.

4.13 Test Ports - Klein Fittings

To allow test equipment such as vacuum gauges, gas analysers and leak detection equipment to be attached to the machine, test ports are provided.

These ports are supplied to a standard design named Klein. GENERAL VACUUM EQUIPMENT uses the following sizes;

KF16 16mm Inside diameter
KF25 25mm Inside diameter
KF40 40mm Inside diameter

Standard clamping assemblies are used to attach equipment to these ports.

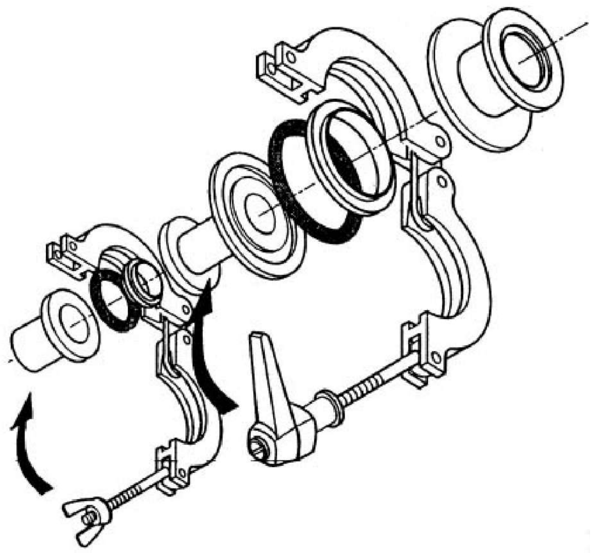


Figure 4.13 - 1 Klein Fitting Arrangement

4.14 Pumping System And Chamber Leak Testing Methods

The process is contingent on the vacuum level within the chamber being in the correct vacuum level, i.e. below 2E-3 mbar.

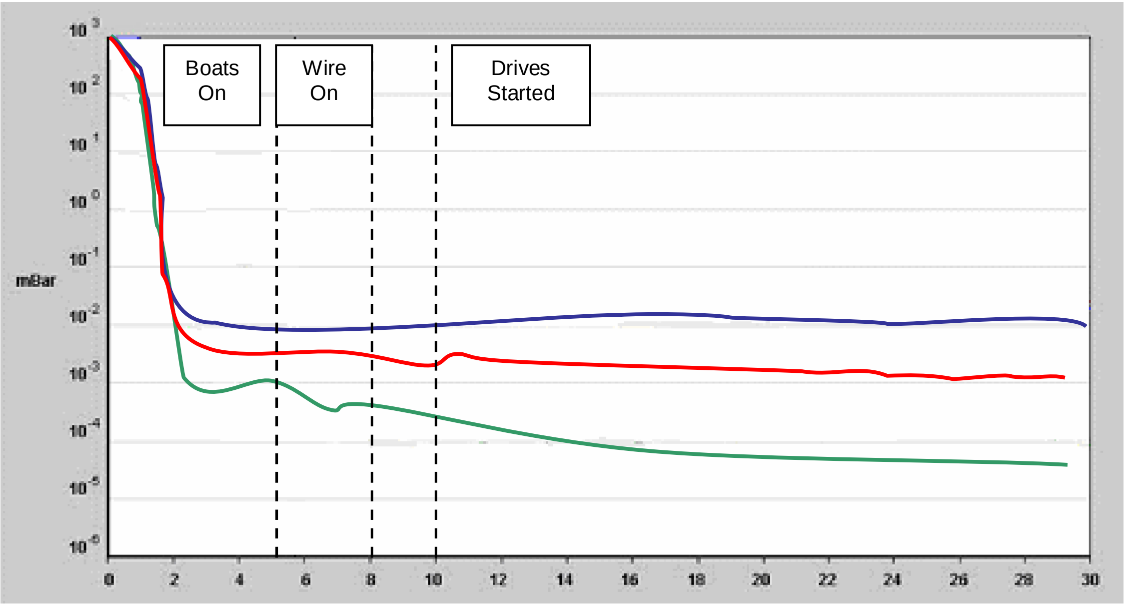


Figure 4.14 - 1 Curve Showing Normal Pumpdown Cycle For The Process

When the required vacuum level cannot be reached, one of the following is normally the reason;

- Air Leak into the chamber
- Internal Water Leak
- Unclean Machine
- Outgassing Substrate
- Pump Failure

Each subject can be explained in a little more detail.

Air Leaks

When there is a suspected leak of air from outside to inside the chamber, the system can be leak checked. The leak detection equipment has to be installed as per figure. 3-19 and the following areas are normally checked;

- All Vacuum Fittings-Klein Fittings
- All Vacuum Gauges
- All Rotary Leadthroughs
- All Isolation Valves
- All Electrical Leadthroughs
- All Water Leadthroughs
- All O-ring Joints

Follow the instructions provided with the leak detector's manual. Leak detection using Helium as a search gas should always start at the top of the chamber, as Helium gas rises.

If a leak is found, repair or replace the defective part.

Normally, it can be assumed that welded joints do not often spring leaks. But if one is suspected, it can be checked using welding inspection dye, which is commercially available.

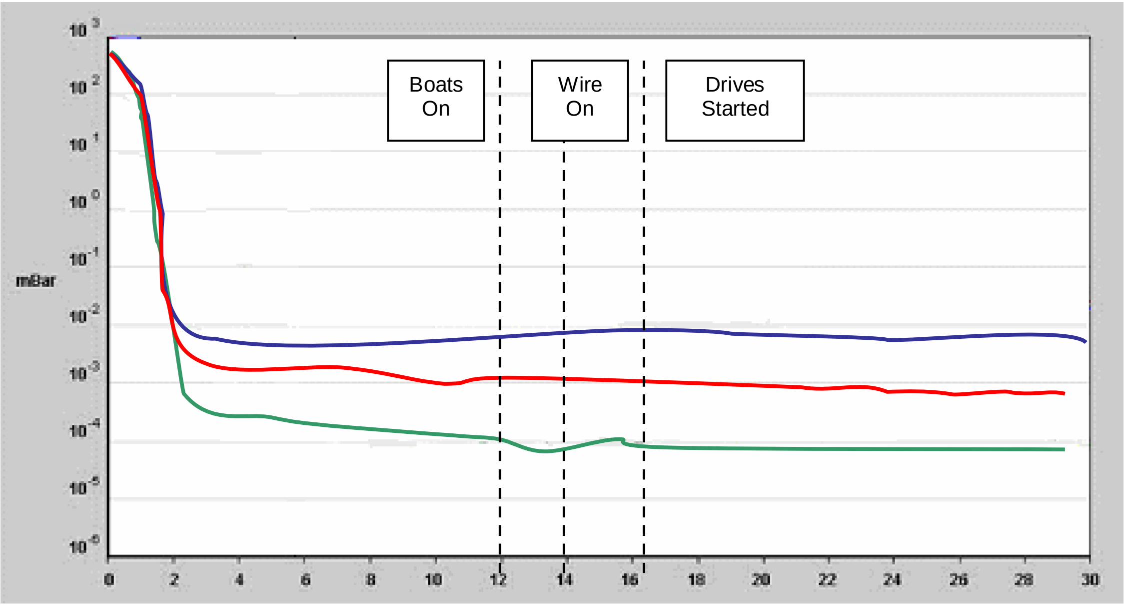


Figure 4.14 - 2 Pumpdown Curve Showing Pumpdown Cycle When There Is An Air Leak Present

Internal Water Leaks

It is necessary to use water within the machine to cool shields etc. These water-cooled sections are normally connected with rubber hoses that can sometimes leak.

The problem of water vapour within the chamber is described in the section on cryogenic pumping systems.

Large water leaks are easy to find, but small leaks can be difficult to find and will generally hold the vacuum level to above 1E-2 mbar.

Close inspection of the cooled area must be made when the chamber is opened for damp areas or small ice deposits, which will indicate leaks.

The reasons for hoses to spring leaks can be overheating, i.e. lack of water and eventual aging.

Replacement of hoses, if necessary must be changed carefully ensuring that there is no possibility of trapment by mechanical assemblies within the chamber.

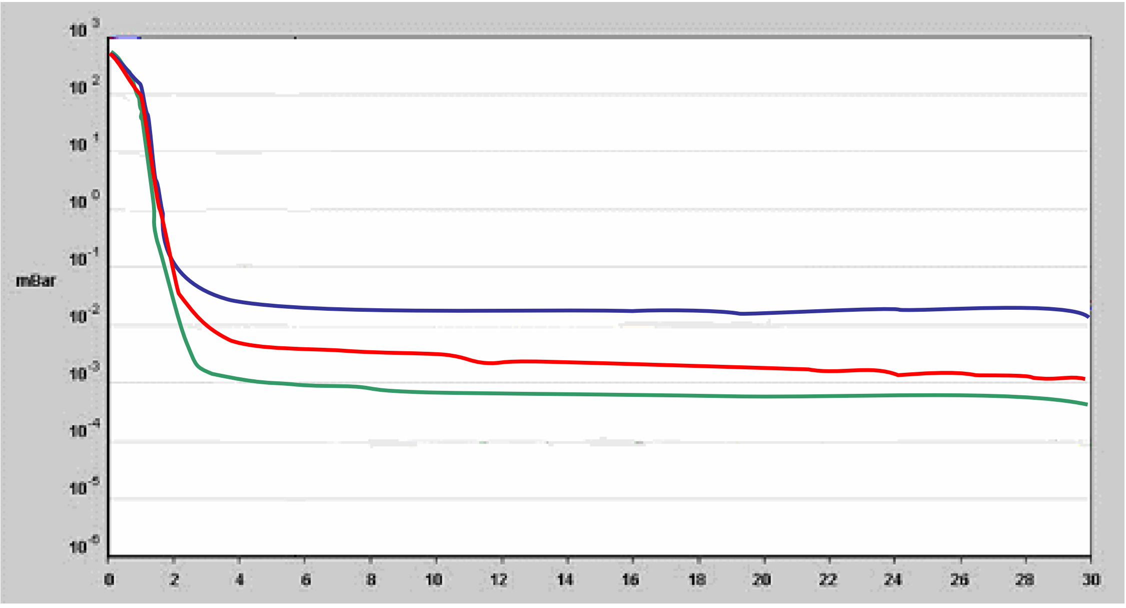


Figure 4.14 - 3 Pumpdown Curve Showing Pumpdown Cycle When There Is A Small Water Leak Present

Unclean Machine

The most common reason that causes vacuum problems is contamination of the machine by aluminium oxide. This is the by-product of the metallizing process and must be removed on a regular basis. Moisture is absorbed by the oxide, which is only released by extended pumping of the machine or when localized heat (from evaporators) drives it out.

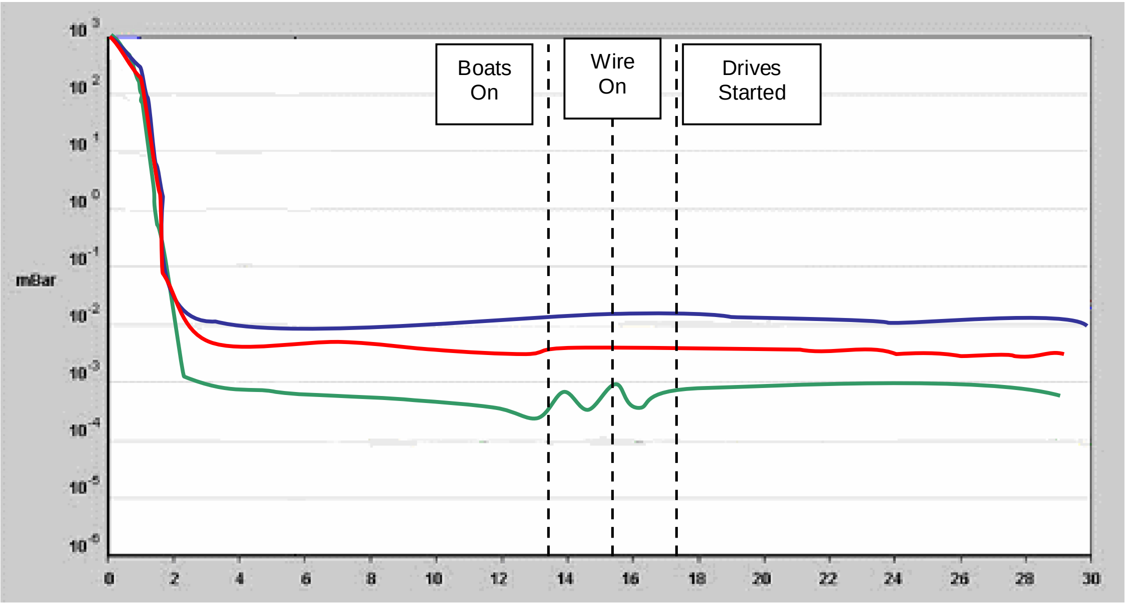


Figure 4.14 - 4 Pumpdown Curve Showing Pumpdown Cycle When The Machine Is Contaminated With Aluminium Oxide.

Outgassing Substrates

Some substrates such as paper release vapour, etc. into the chamber when they are unwound. The release of the vapour presents a large gas load for the pumping system to remove.

Sometimes this gas load is too large for the pumps to remove quickly and the pressure will rise within the chamber. Another indication is the extended pumpdown of the chamber. Machines are designed run certain materials based upon their original specification, i.e. film machines have a smaller pumping group than paper machines.

To test for this problem, pump the machine down twice, once empty, and once with the material in. Compare the results, a outgassing roll will be indicated by a longer pumpdown.

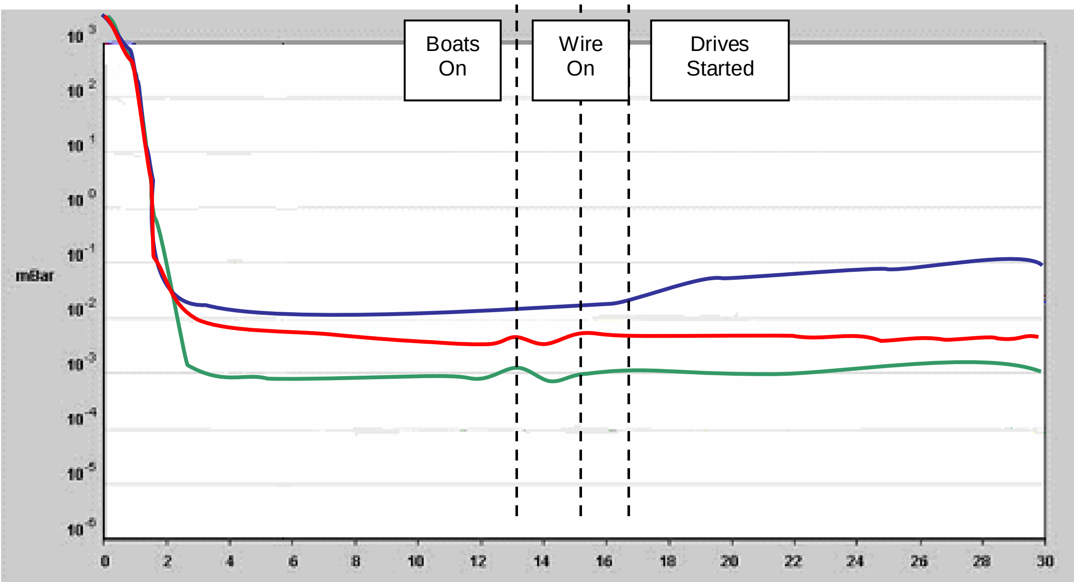


Figure 4.14 - 5 Pumpdown Curve Showing Pumpdown Cycle When The Substrate Is Outgassing

Manual Pressure Rise Test

If a leak detector is not available or the source of the problem cannot be found, a pressure rise test can be carried out. This test is carried out the following sequence;

1. Clean the machine completely from all aluminium oxide coatings
2. Pump the empty chamber for 12 hours without using the Cryo-generator unit
3. Isolate the chamber from the pumping group by closing the isolation hand valve and then placing the pumping group in a "Hold" mode.
4. Record the pressure (vacuum levels) in the chamber using the active pirani gauge located within the chamber. This measurement should be taken every 5 minutes for a period of one hour.
5. Plot the results on log-log plotting paper and compare them with the curves shown. From the plotted graph, the pressure rise of the chamber can be calculated, i.e. the level after 1 hour, minus the level measured at the beginning of the test.

This can be compared with the pressure rise figures given in the machine specification.

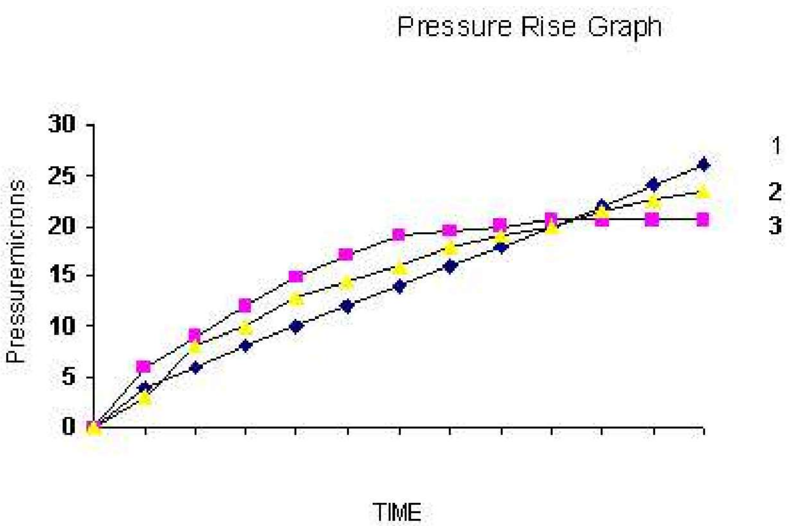


Figure 4.14 - 6 Pressure Rise Curve3 - 1

Key
1. Pressure rise due to leakage
2. Pressure rise due to combination of air leakage and outgassing
3. Pressure rise due to outgassing

The term micron refers to 1E-3 mbar

Therefore if the pressure rose from E-4 to 1.6E-2, this could be termed as 16 microns.

The pressure rise should be less than 30 microns. If the pressure rise is excessive, an air leak is probable and further leak testing will have to take place.

Leak Detection Methods

The following is a guide only and it is suggested that the user of leak detecting equipment follow the instructions supplied with the leak detector.

The leak detection methods described below are designed to pinpoint the air leak by applying a small amount of the search gas to the outside of the evacuated chamber. Monitor the measuring device for a response to the gas.

Vacuum Gauge Method

This method utilizes the varying sensitivity of the ionisation (Active Penning) gauge depending upon the type of gas being used. Penning gauges are calibrated for air/nitrogen by the manufacture and when a helium rich partial atmosphere is present, the reading displayed will fall (lower). If helium is sprayed on a leak, the relatively steady penning reading will be disturbed and the source of the leak identified.

Helium Leak Detection

This is a much more sensitive method of detection and involves connecting the leak detector in the foreline pipework. Follow the leak detector manufactures instructions for setting up the detector and spray gas in the suspected leak area and monitor the detector for a response.

Precautions must be made to avoid excessive flooding of the chamber and surrounding area with the probe gas. The purpose of the test is to determine the exact location of the leak, just not to prove that there is one present. If excessive amounts of the probe gas in the area being tested, the detector will take a long time to "settle" back to its previous level.

Use a fine jet of gas for short periods of time to locate the leak.

Figure 4.14 - 7 Photograph of Helium Leak Detector Attached to Pipe work

RGA (Residual Gas Analyser)

This equipment is used by certain vacuum coating companies to monitor the levels of different gases within the process area. It can be used with a search gas, e.g., helium, to monitor for rises in that gas while leak detecting.

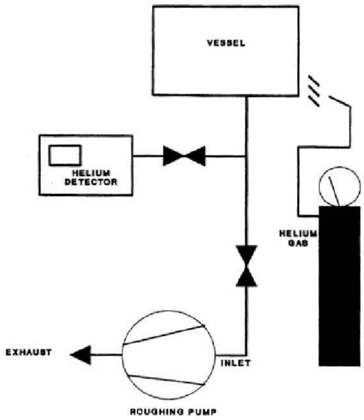


Figure 4.14 - 8 Leak Detector Location

SECTION 5 VACUUM PRESSURE LEVEL MONITORING SYSTEMS

5.1 Vacuum Monitoring Overview

Vacuum systems can be described to operate in different types of vacuum environments.

Description	Pressure Range - mbar
Low Vacuum (Rough)	$1E^{-1}$ mbar
Medium Vacuum	$1E^{-3}$ mbar
High Vacuum	E^{-3} - E^{-8} mbar
Ultra High Vacuum	< E^{-8} mbar

Table 5.1 - 1 Pressure Range

Gauges Used

In machines, the coating process takes place in the high vacuum region and the gauges we use are shown in the table below.

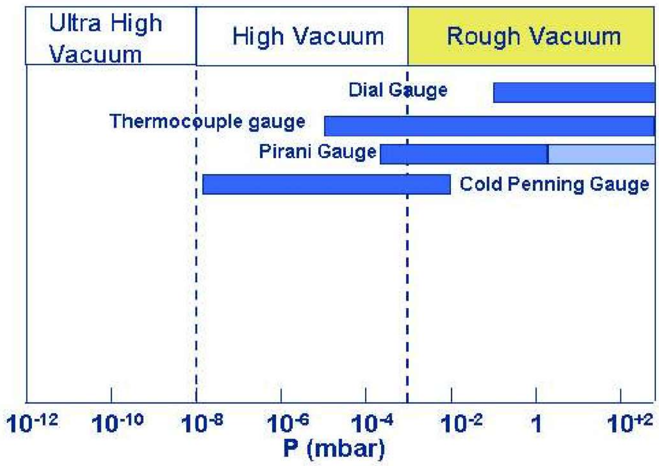


Figure 5.1 - 1 Approximate Vacuum Gauge Range

The vacuum pressure levels for the machine are displayed on the vacuum screen on the interface screen.

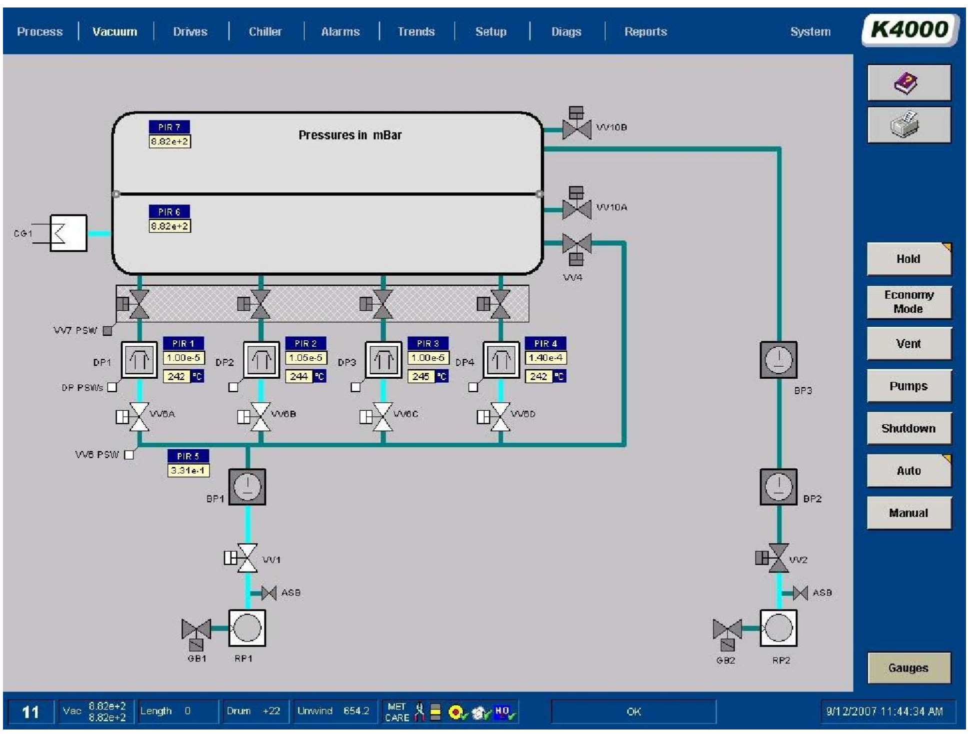


Figure 5.1 - 2 Vacuum Mimic Screen Showing Vacuum Levels

A number of different types of gauges are used to monitor the vacuum levels at various points within the vacuum chamber/system. These pressure levels will be indicated in mbar measurements dependent upon the machine specification.

Labelling for these various points can be seen on the vacuum mimic screen and can be referred to on the vacuum schematic drawing.

Various pressure level settings are set in to a "CONFIGURATION" screen within the interface computer to open and close various isolation valves. Refer to the maintenance manual for further details.

The types of gauges that are used in this system are described in the following sections.

5.2 Active Pirani Gauges

This type of instrument measures the vacuum level between 200 mbar and 1E-3 mbar. They provide information to the monitoring system for interlocking the operation of valves and starting of pumps.

This gauge gives a 1-9 volt dc signal dependent upon the vacuum level, which is transmitted to the PLC input module. The value is shown on the vacuum mimic screen on the interface screen.

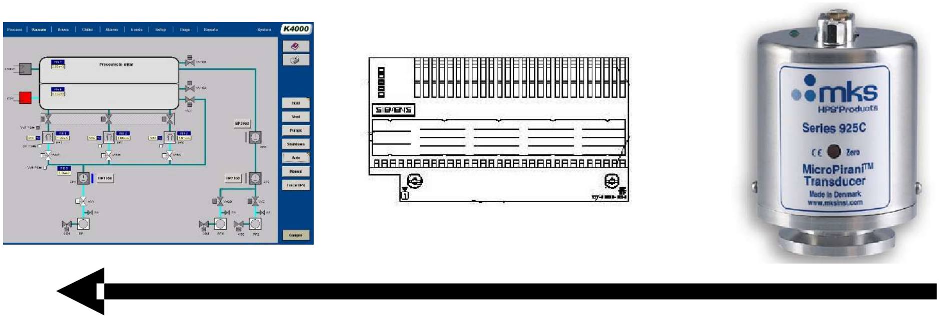


Figure 5.2 - 1 Schematic of Pirani Gauge Signal Monitoring

This type of gauge is located in the following areas of the machine.

- Diffusion pump
- Foreline pipework between the diffusion pump/s and roughing/booster pumps
- Chamber winding zone

	Caution !
	**Do not remove the gauges from their location while they are under vacuum**

Principle of Operation

The gauge is a self-contained vacuum-measuring device with a 1-9 dc volt output. Calibration curves are provided to allow the user to convert the 1-9 volt signal to a relative vacuum level.

The 1-9 volt signal is sent from the gauge to analogue input terminals on a Siemens input rack. The value is converted and then transferred to the computer for displaying the value.

The vacuum levels of the various gauge positions are shown on the VACUUM screen.

The gauges have to be supplied with a stabilized 24 dc volt supply, which is present at all times.

Figure 5.2 - 2 Pirani Gauge

Maintenance

The maintenance (page 34) and calibration (page 32) of the gauge are described in detail within the manufactures manual.

To carry out the calibration at the lower vacuum end, it is suggested that the gauge be placed within the Diffusion pump at it's gauge point. To carry out the calibration, the following procedure should be followed;

1. The pump should have been in operation for at least 1 hour.
2. Close the hand valve between the pump and gauge.
3. Switch the power to the gauges off.
4. Remove the gauge and replace it with the gauge to be calibrated.
5. Switch the power back on to the gauge.
6. Carry out the atmosphere calibration at this point as per the manufactures' instructions.
7. Monitor at the plc input terminals.
8. Open the valve between the gauge and pump.
9. Ensure that the pump is switched back on and leave for 10 minutes
10. Carry out the vacuum calibration as per the manufactures instructions. Monitor the voltage at the plc input terminals.
11. Repeat the steps 1 through 10

5.3 Active Penning Gauge

Normally, only one of these gauges is fitted to the chamber, in the evaporation zone. It is used for measuring vacuum in the range below 1 x 10-3 mbar

It is the gauge that is used most commonly for monitoring the vacuum during the coating of the substrate. The pressure is shown on the vacuum screen on the interface screen.

The gauge is not switched on until the VV7 valve has opened and until that time, its output display on the vacuum scheme will be the same value as the pirani gauge in the evaporation zone.

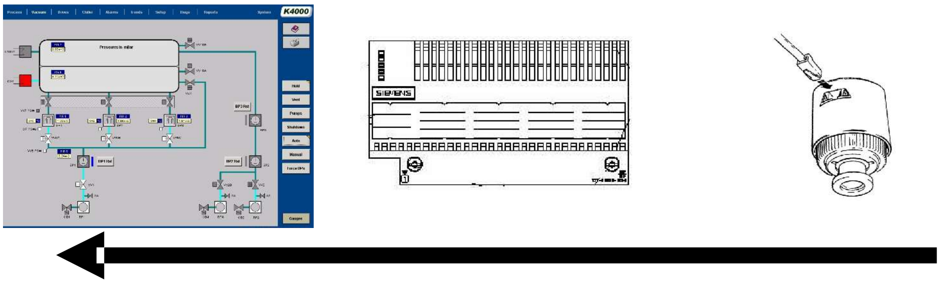


Figure 5.3 - 1 Schematic of Gauge Signal Monitoring

	**Caution !**
	Do not remove the gauges from their location while they are under vacuum

The gauge is a self-contained vacuum-measuring device with a 0-10 dc volt output. Calibration curves are provided to allow the user to convert the 0-10 volt signal to a relative vacuum level.

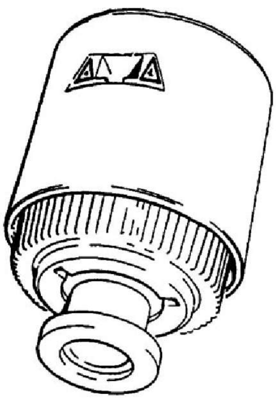


Figure 5.3 - 2 Edwards Active Penning Gauge

The 0-10 volt signal is sent from the gauge to analogue input terminals on a Siemens input rack.

The value is converted and then transferred to the computer for displaying the value.

The vacuum levels of the various gauge positions are shown on the VACUUM screen.

The gauges have to be supplied with a stabilized 24 dc volt supply, which is present at all times. Above a certain pressure, the gauge can be damaged. Therefore, a contact that is switched on when the high vacuum valves open (VV7's) is used.

An internal high voltage supply is supplied across the anode and cathode to produce a glow discharge. Electrons are drawn from the cathode to the anode, but the magnetic field within the gauge deflects them and they follow a complex spiral path.

This is a long path for the electrons to travel and they can collide with gas molecules, ionising them and producing positive ions and extra electrons. The sum of the positive ion current to the cathode and electron current to the anode constitutes the gauge head current that is measured by the internal control circuit.

The current is proportional to the gas pressure until the discharge goes out about 100 mbar. Therefore the indication at atmospheric pressure, is usually false and indicates a very low pressure and this will not change until the pressure falls.

Maintenance

There is very little maintenance that can be carried out upon the gauge and they cannot be calibrated. See the manufacturer's manual for further information.

If the gauge becomes contaminated, the vacuum readings will become erratic. The gauge can be dismantled for cleaning as described in the vendor’s manual.

5.4 Dial Gauge

This mechanical type gauge has a range from atmosphere, $1E^{+3}$ mbar to 1 mbar and is used to monitor the initial evacuation of the chamber.

It is located on the front of the machine above the operator location.

The gauge is provided to allow the user indication of the initial pump down of the chamber.

Figure 5.4 - 1 Dial Gauge

	**Caution !**
	Do not remove the gauges from their location while they are under vacuum

The instrument consists of a sealed capsule mounted inside a vacuum tight case. The gauge is attached to the chamber and is removed by unclamping the klein clamp.

Movement of the capsule by a pressure change is converted into movement of the indicating needle against an mbar dial scale. The pressure indicated is the difference between the chamber and atmospheric pressures. The gauge is barometrically independent.

Maintenance

- Screw Front Type Units

The gauge can be cleaned carefully of dust by dismantling it off the machine. Care must be taken in this operation. Once it is reinstalled, the component should be leak checked.

The manufacturer can only undertake calibration of the instrument.

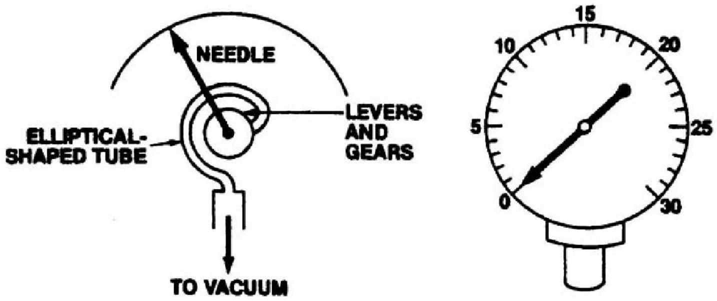


Figure 5.4 - 2 Capsule Dial Gauge

5.5 Thermocouple Gauge

This type of instrument measures the vacuum level between 200 mbar and 1E-3 mbar. They provide information to the monitoring system for interlocking the operation of valves and starting of pumps.

This gauge gives a 2-10 volt dc signal dependent upon the vacuum level, which is transmitted to the PLC input module. The value is shown on the vacuum mimic screen on the soft interface screen.

It is located in the fore-line pipe-work between the diffusion pump/s and roughing/booster.

This type of gauge is used because it is not affected by the zinc sulphide particles in the pressure range in this part of the machine.

Caution !

Do not remove the gauges from their location while they are under vacuum

This type of gauge works on the principle of monitoring the temperature of a heated element. As the number of gas molecules falls, the temperature changes.

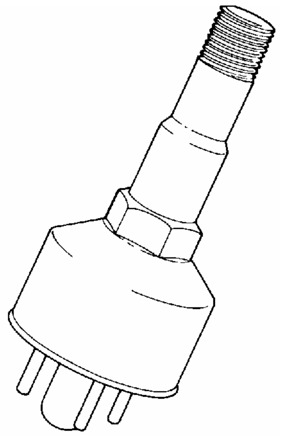


Figure 5.5 - 1 Edwards Thermocouple Gauge

Maintenance

There is very little maintenance that can be carried out upon the gauge and they cannot be calibrated. See the manufacturer's manual for further information.

5.6 Screens Displaying Vacuum Information

The penning and pirani gauges are shown on the vacuum mimic screen, the web path screen and pump down curve screen, as shown below.

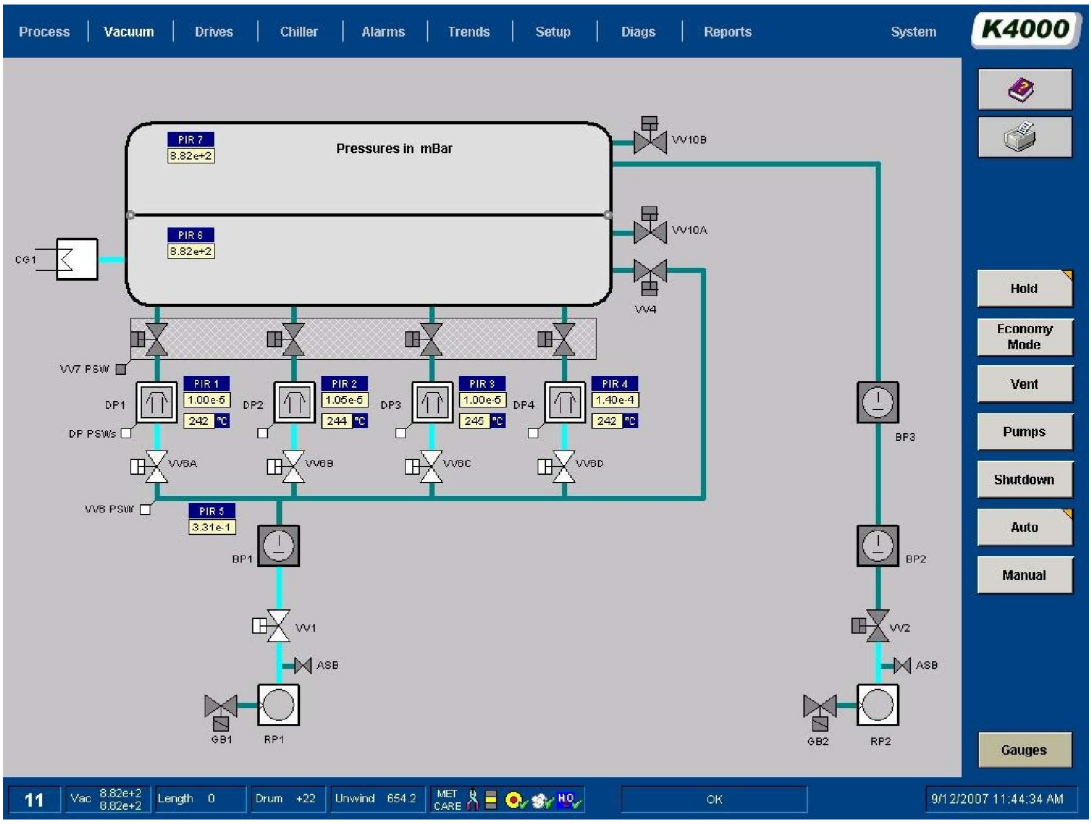


Figure 5.6 - 1 Vacuum Mimic Screen Showing Vacuum Levels

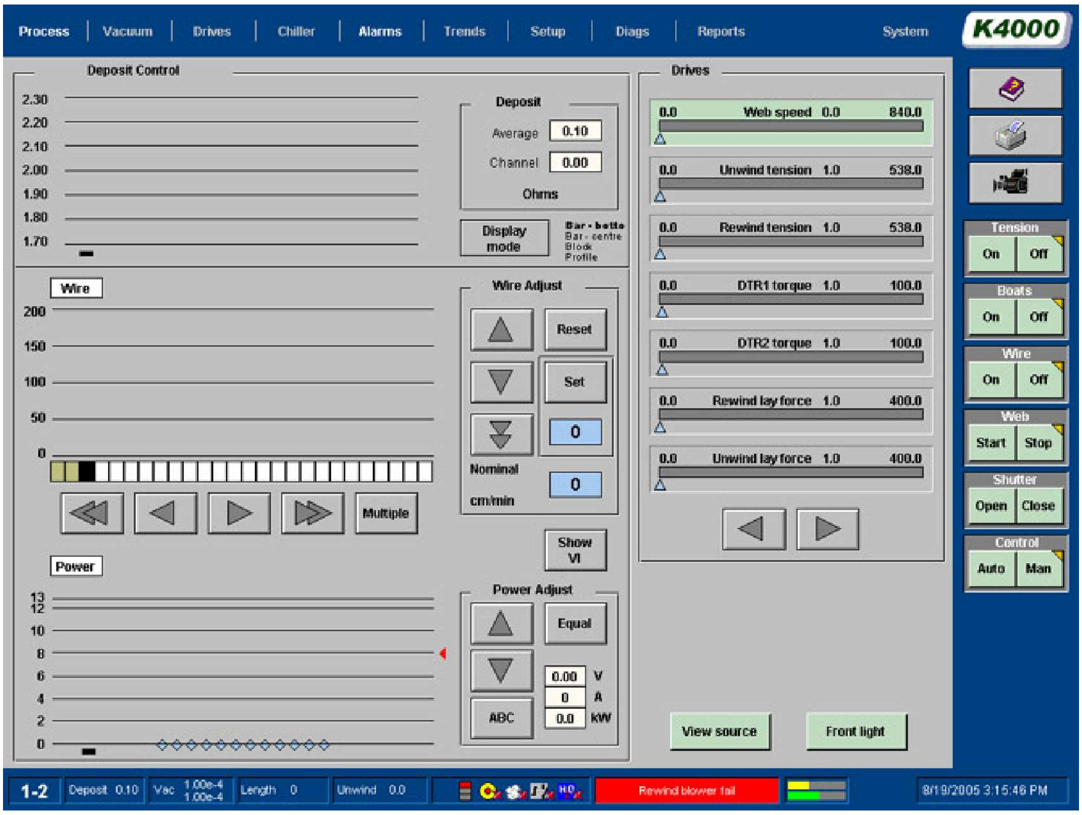


Figure 5.6 - 2 PROCESS Screen Showing Vacuum Levels

The penning gauge vacuum level will only be displayed when the fine valve, VV7's are open.

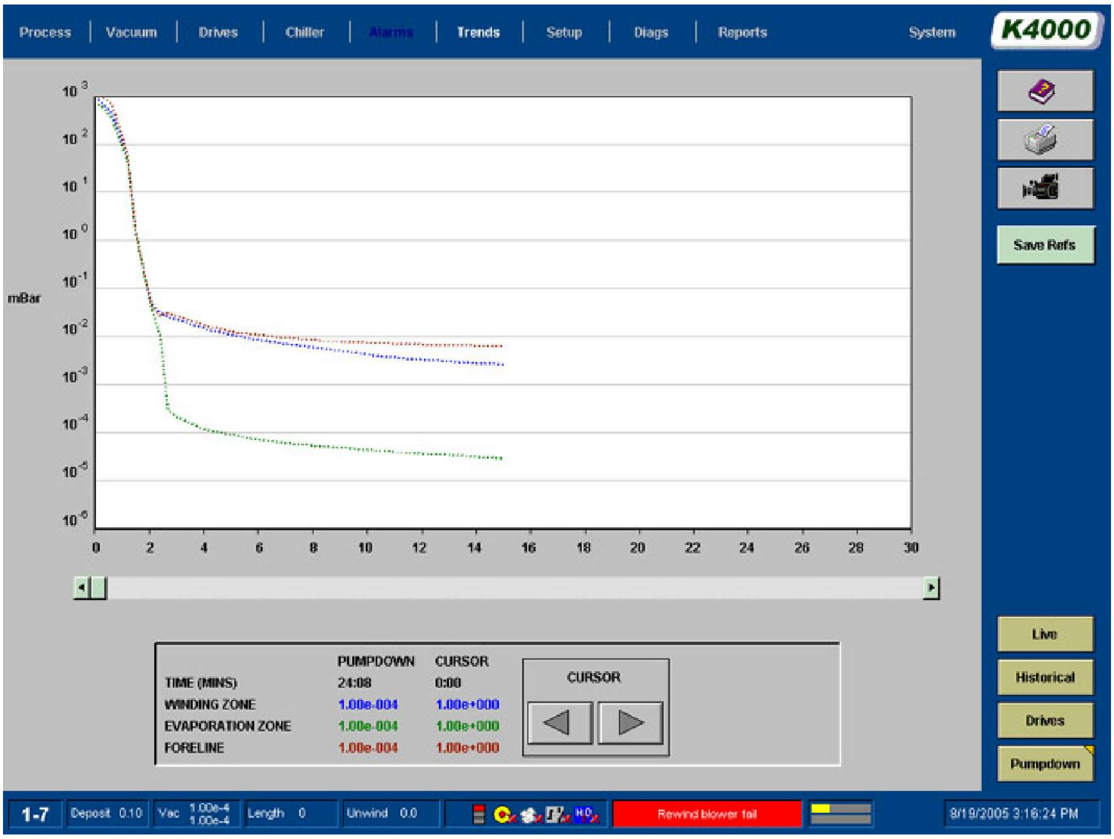


Figure 5.6 - 3 Pump down Curve Showing Vacuum Levels

SECTION 6 OPERATION OF MACHINE AUXILIARY SYSTEMS

6.1 Shutter System Operation - Evaporation Source

An aluminium water-cooled shutter is located above the source enclosure. It protects the substrate from the radiant heat generated by the evaporators whilst they are being heated subsequent to coating taking place/.

The shutter is moved by the action of a pneumatic cylinder being retracted or extended, as the line speed approaches its set level. On deceleration, the shutter closes at a similar rate.

This occurs when the substrate is wound through the machine.

The cylinder is located on the outside the chamber.

If the shutter is opened while the substrate is travelling at a low speed, the substrate can be damaged. It is interlocked to ensure that a minimum web speed is reached before allowing movement to take place. The shutter control system is designed to ensure that the shutter always closes if power is lost to the machine.

To move the winding cart into the chamber, the shutter has to be in its lowered closed position.

An interlocking device is mounted externally for the purpose of monitoring this situation.

<table>
<tr><td rowspan="2"></td><td>Caution !</td></tr>
<tr><td>The shutter must be in the lowered closed position for the winding cart to be moved towards the chamber.</td></tr>
</table>

The shutter's pneumatic control valves are designed to allow safe working upon the shutter and source enclosure or evaporators when the winding cart is retracted from the chamber.

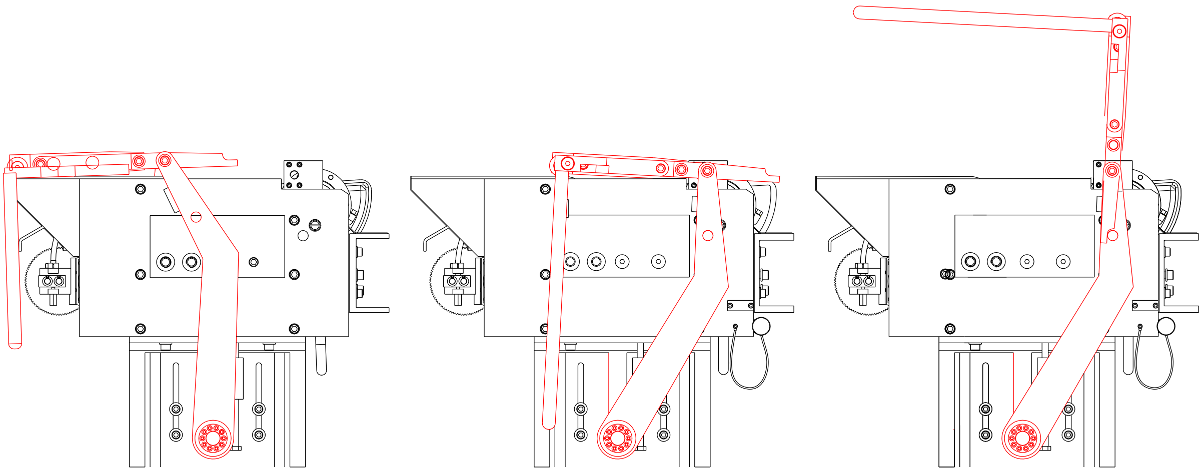


Figure 6.1 - 1 Figure Showing Shutter over the Evaporators and In the Open and Access Positions

A set of valves are fitted that when activated discharge air from both sides of the cylinder. They are activated by the winding cart fully closed limit switch when the winding cart is extracted from the chamber.

Both sides of the cylinder are refilled when the winding cart is fully engaged within the chamber.

The shutter is opened and closed by the touching of the relevant button (section) of the PROCESS screen. Alternatively, the operation can be selected to open automatically from the 'SETUP MODES' screen.

Mode switches

	Manual		Auto	
Chiller		Switching between heat, cool, and standby is done manually	✓	Cooling starts with pumpdown, and reheat starts in Hold
Vacuum		Valves and pumps are sequenced manually	✓	Valves and pumps are sequenced automatically
Web stop		The winders stop when the Web Stop button is pressed	✓	Coating stops when the roll diameter reaches the auto-stop diameter
Shutter		The shutter may be opened at any speed above the minimum limit	✓	The shutter opens and closes as the web speed reaches the speed limits
Wire ramp		The wire speed is constant during web acceleration	✓	The wire speed is ramped up and down in proportion to the web speed
Wire feed on	✓	Wire feed is switched on manually		Wire feed is switched on at the end of warmup
Boats on	✓	The boats are switched on manually		The boats switch on when the pressure is low enough

Figure 6.1 - 2 Mode Section of Setup Screen

Opening of the Shutter (Moving Shield)

Manual Operation

This shutter control allows it to be operated to fully closed or fully open position. It has a "fail-to-safe" action so that it closes to protect the web if there is a power failure, etc.

Once the substrate is running at, approximately above 2 Meters/sec the shutter can be opened by the "SHUTTER OPEN" section of the "PROCESS" screen.

Figure 6.1 - 3 Push Buttons On PROCESS Screen

Automatic Operation

The automatic mode of operation is selected in the "SETUP MODE" screen. It will allow the shutter will be withdrawn automatically as the line speed increases.

It has a "fail-to-safe" action so that it closes to protect the web if there is a power failure, etc.

Once the web is running at, approximately, 2 Meters/sec the shutter will be opened automatically as the line speed increases.

Closing of the Shutter

As the winding drive system decelerates at the end of the coating cycle, the shutter is closed. It will close as described in one of the following paragraphs.

- By watching the unwind roll approach the core, the winding mechanism can be stopped before the substrate is pulled off the core by the operator.

- Touching the “WEB STOP” section on the “PROCESS” screen can stop the substrate. This action will also close the shutter.

Note!

If the shutter is open at low web speeds then the heat from the evaporators will damage the web.

- The shutter will also close when the Auto slowdown sequence is started by the drive system. This occurs once the line speed falls below 2 meters/second

- The shutter can be closed mid-coating cycle by touching the "SHUTTER CLOSE" section of the process screen. This operation can only occur if the manual mode of operation has been selected on the "SETUP MODES" screen.

Warning !

Do not disable the cylinder isolation spool valves, as this could lead to entrapment and injury by the shutter operating.

Opening of Shutter for Source Maintenance

The shutter can be pushed open to allow access to the evaporators. There will be air in the external cylinder and this will restrict the movement initially. As the air is exhausted by the cylinder being moved, this movement will become easier.

Alternatively the shutter can be rotated through 90 degrees using the tool provided. The shutter locates on stop blocks to hold it in position. The photograph below shows this action.

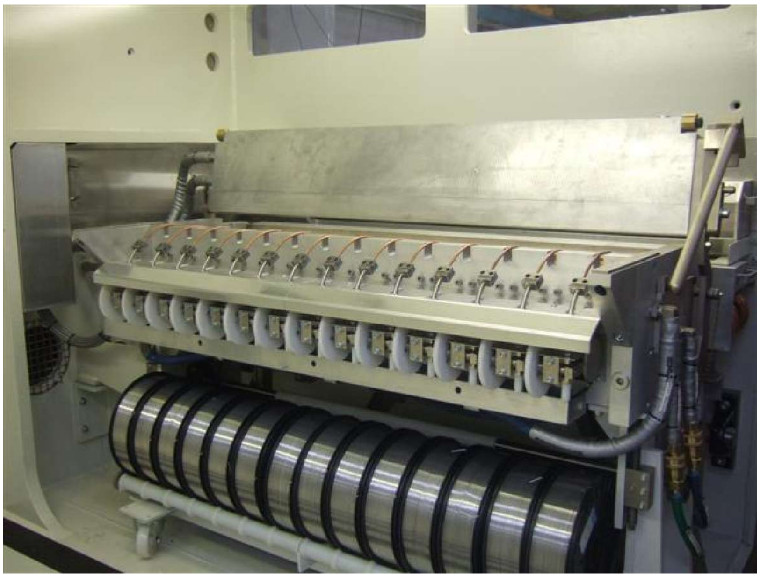


Figure 6.1 - 4 Photograph Showing Evaporation Source Shutter in Open Position

Figure 6.1 - 5 Photograph Showing Operator Rotating the Evaporation Source Shutter

Figure 6.1 - 6 Photograph of Evaporation Source Shutter Locking Pin in Position

6.2 Winding Cart Linear Drive System

The winding cart is moved by pressing the ‘OPEN’ and ‘ENABLE’ push-buttons located on the operator desk.

Figure 6.2 - 1 Photograph of Operator Interface Panel

Before the cart will move, there is a 5 second delay while the warning lights flash and siren operates. These warning indicators will stay activated during this operation.

To comply with the latest recommended safety requirements the control system is programmed to allow the following sequence;

- After initial holding down of the pushbuttons for a warning lights will be activated for a further 5 seconds.

The cart will slow down in the required direction just before reaching the full movement position.

An inverter controlled drive motor is used to move the winding cart in and out of the chamber. It is located on the winding cart under the floor plates.

	Warning !
	**Check that there is no obstructions behind the cart before operating the traverse drive**

Closing the cart is carried out using the same sequence.

The winding cart will not move towards the chamber unless the following interlocks are active or have been activated;

- Shutter Closed and Lowered.
- Coating shield raised and locked in position
- Mechanism water is OK

- Traverse enable push-button is first initially pressed (rear of the machine)

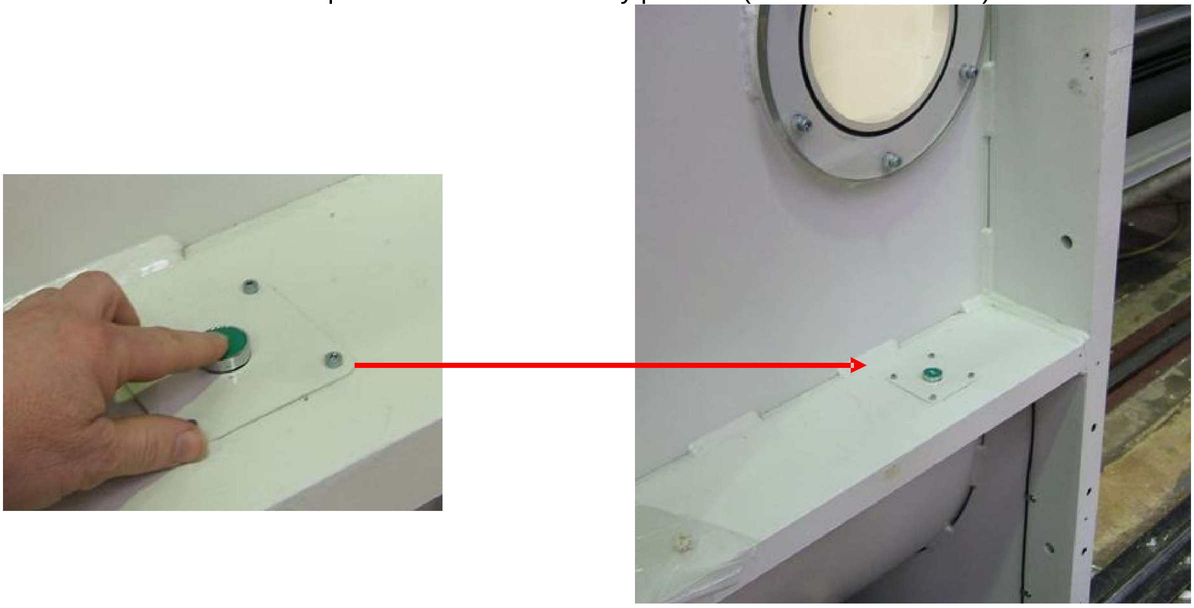


Figure 6.2 - 2 Photograph of Traverse Enable Push-button at Rear of Machine

If one of these parameters is correct indication is given by the lack of any siren or flashing lights on the pressing of the relevant push buttons. The interlocks can also be checked by selecting 'TRAVERSE IN or OUT' interlock logic, from the diagnostic screen.

Limit switches are located below of the cart to indicate the open or closed positions of the winding cart. There are two further switches to change over the speed of movement as the cart approaches the fully open or closed positions. Striker plates located below the cart on the tracks activate the switches.

Figure 6.2 - 3 Photograph of Winding Cart Location Indicating Limit switches

	Warning !
	The movement of the winding cart should be isolated when working inside the chamber by locking out the movement lockout isolator

Locking Out the Traverse Motor

To lockout the traverse movement lockout isolator, turn it through 90 degrees and place a padlock on the switch. This action also disables the winding mechanism drive system

Figure 6.2 - 4 Locking Out Of Movement Isolator

	**Warning !**
	Keep hands clear of the opening between the winding cart and chamber during the closing period

6.3 Evaporation - Winding Zone Separation

The chamber is split into two vacuum zones, and a zone sealing system separates them. Part of this sealing system uses strips of flexible shaped neoprene arranged along the length of the winding mechanism. The seals should be inspected for damage and lubricated as described in the preventative maintenance schedule.

Damage to these seals will hinder the effective action of the pumping system.

Other components built into the winding mechanism, such as low conductance seals, around the drum are used to assist in the separation of the zones.

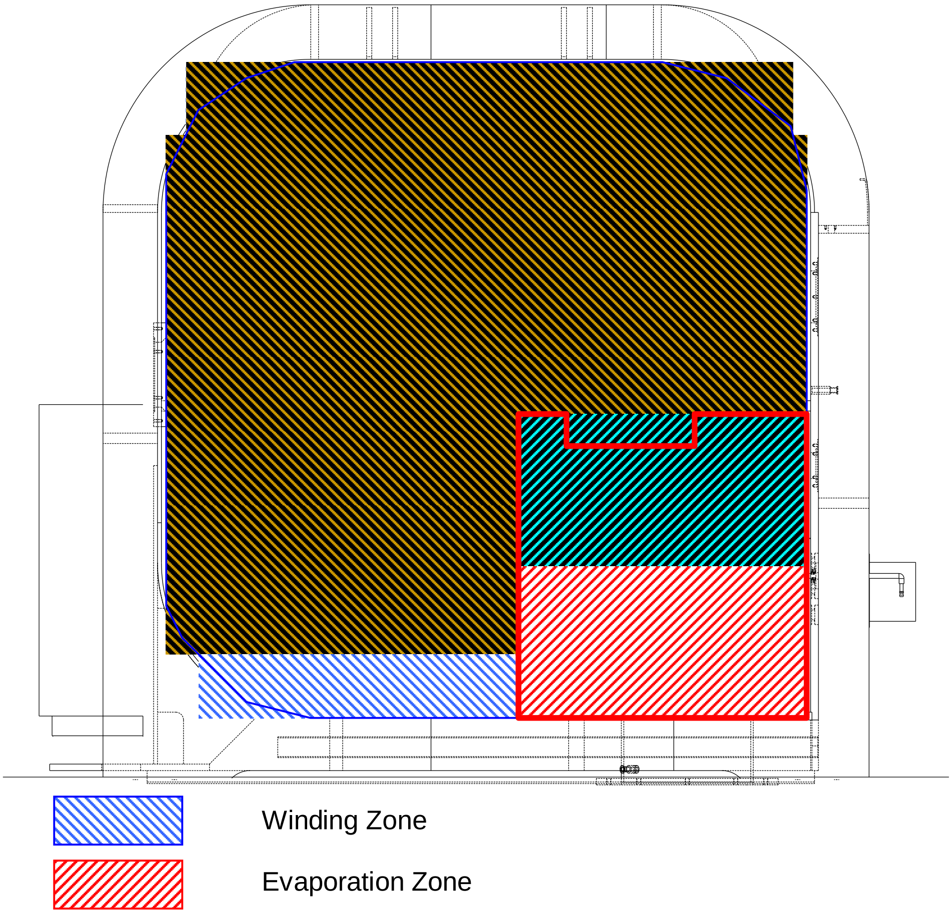


Figure 6.3 - 1 Drawing Showing Cross Section of Chamber indicating the Different Zones

Figure 6.3 - 2 Photograph of Flexible Seals

The assembly around the drum can be lowered to allow cleaning and removal of debris from this area.

The coating shield has to be fully raised to allow the winding cart to be engaged into the machine.

This is achieved by using the RAISE and LOWER push buttons located on the local electrical panel.

Figure 6.3 - 3 Photograph Showing Location of Coating Shield Raise Lower Push-buttons

6.4 Shielding Of Stray Evaporation

Removable steel collection shields are provided to keep the machine clean and they should be fitted at all times when the machine is running.

These shields are removable for cleaning away from the chamber.

To help in cleaning of these shields, they should be painted with release paint.

Figure 6.4 - 1 Operator removing Evaporation Collection Shield - Source

	Caution !
	**Ensure that shields are placed back in the machine following cleaning.**

Figure 6.4 - 2 Operator removing Evaporation Collection Shield – End

Figure 6.4 - 3 Operator removing Evaporation Collection Shield - Wall

6.5 Man- Machine Interface (MMI) Touch Screen

This is the term given to the screen that the operator uses to control the machine.

Boot Screen

After switching on the main isolator, the computer will power up and display the BOOT screen.

Figure 6.5 - 1 Machine Boot Screen

This screen has to be activated by touching the surface of the touch screen or alternatively using keyboard or mouse.

If the computer does not boot up to this screen, it can be suspected that the computer hard drive may have been corrupted and needs to be changed.

6.6 Man-Machine-Interface (MMI) Operational Levels

The MMI software has been compiled from industry standard software packages. To ensure that the possibility of corruption to the operation program is minimised, access to certain software programs is limited by the use of passwords.

'Login In' Screen

The computer has been set up to allow monitoring of usage by different operators and for production purposes, shift patterns.

On start up and following the 'initial boot screen', a SYSTEM screen will be shown and the user must touch the flashing LOGIN button to allow logging on.

At this time of this login sequence, the user will be asked to enter their login name and password.

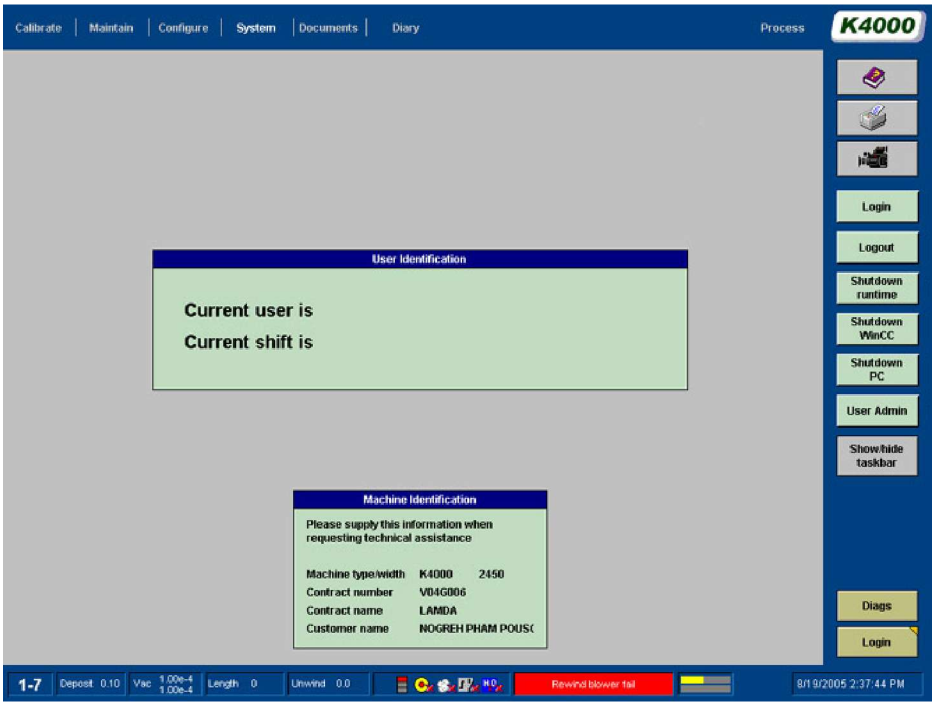


Figure 6.6 - 1 System Screen

All other normally used screens are accessible from this screen. There are some screens that are infrequently used and these are only available via the specific screens that they are attached.

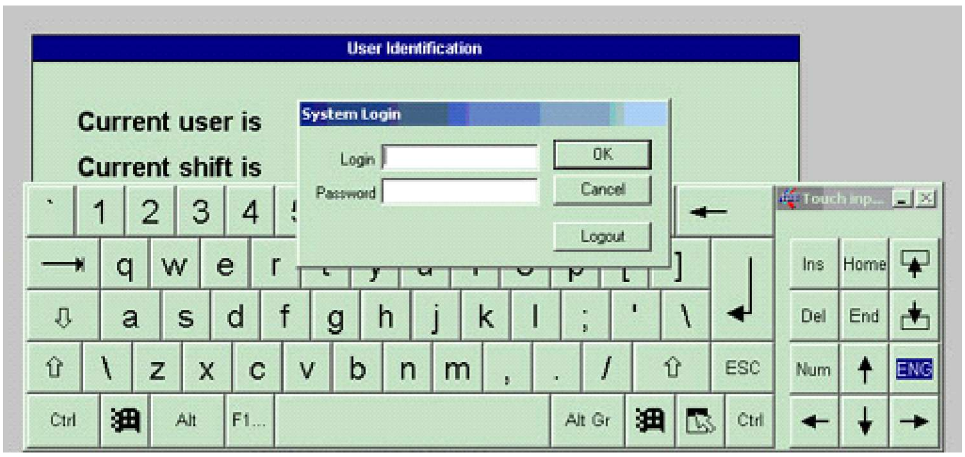


Figure 6.6 - 2 User Log In Screen

Use the keyboard to enter the user's login name and password.

The standard 'factory set' passwords for operators are.

Login name:	Login Password:
Operator 1	Operator 1
Operator 2	Operator 2
Operator 3	Operator 3
Operator 4	Operator 4

From this screen the operator has a choice of the following can choose the following directions to access other screens:-

SYSTEM	***Screens***
	• CALIBRATE Screen
	• MAINTAIN Screen
	• CONFIGURE Screen
	• SYSTEM Screen
	• DOCUMENTS Screen
	• DIARY Screen

PROCESS	***Screens***
	• PROCESS Screen
	• VACUUM Screen
	• DRIVES Screen
	• CHILLER Screen
	• ALARMS Screen
	• DIAGNOSTICS Screen
	• SETUP Screen
	• TRENDS Screen
	• REPORTS Screen

Please note that some of these screens are only available at higher login levels.

To change the screen, touch the relevant section/tab at the top of the screen being displayed

A description of all the functions of each screen is in various sections of this manual.

6.7 Computer MMI Generated Alarms

The computer will report to the user, alarm conditions that occur with the machine hardware and services supplied to the machine.

An alarm condition is relayed to the machine user by two visual indicators. The ALARM tab at the top of the screen will flash and also there is a colour indicator at the bottom of the screen as shown below.

Figure 6.7 - 1 Alarm Indicator

The colour codes indicate the following:

- Green Cleared Fault
- Yellow Acknowledged Fault Present
- Red New Alarm

Pressing the ALARM tab at the top of the screen will bring up the following screen

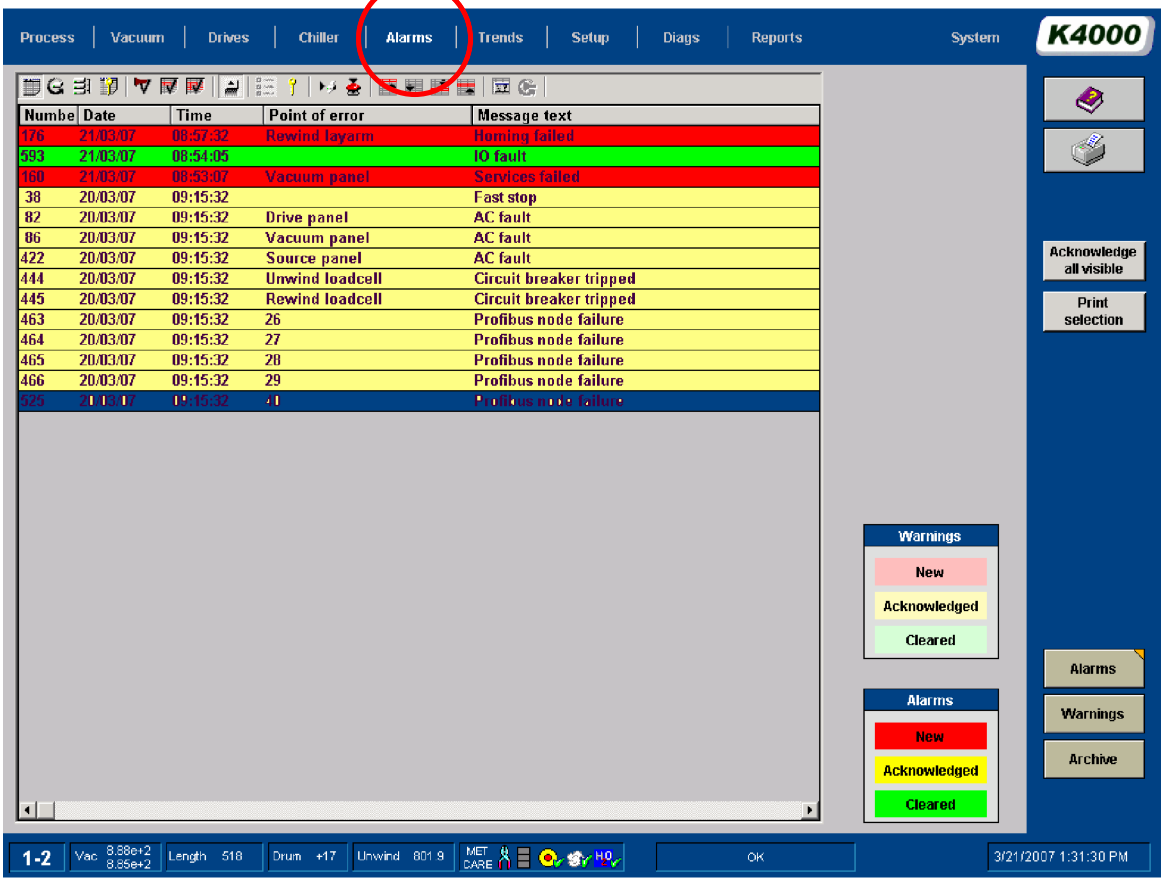


Figure 6.7 - 2 Alarm Screen

Messages displayed in this box indicate the time and date that they occurred and the type of failure.

Alarms that have been acknowledged will be highlighted a different colour from their original.

Only the alarms that can be viewed can be acknowledged. Once the fault has been overcome, the message will be removed after acknowledgement.

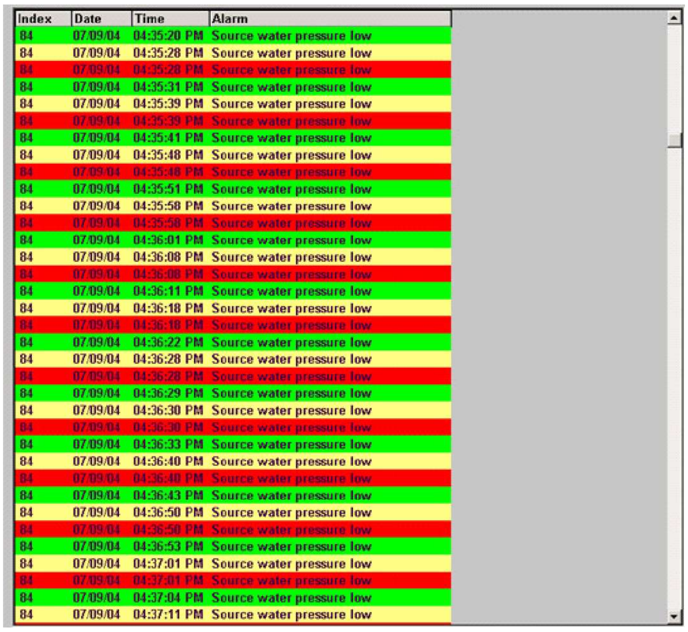
Alarm Screen Options:

Acknowledge all visible	Pressing this button on the screen will acknowledge all those visible on the screen. It will be recorded by the computer that this acknowledgement has taken place.
Print selection	This button can be pressed to print out an individual alarm message.
Alarms	Touching this button will display the current alarms
Archive	Touching this button will display the historic alarms
Warnings	Touching this button will display the current warnings

Figure 6.7 - 3 Alarm Options Table

There are other functions, but these are not used. These are standard WINCC functions.

If the Archive button is then highlighted, the following screen or similar will be displayed.



Index	Date	Time	Alarm
84	07/09/04	04:35:20 PM	Source water pressure low
84	07/09/04	04:35:28 PM	Source water pressure low
84	07/09/04	04:35:28 PM	Source water pressure low
84	07/09/04	04:35:31 PM	Source water pressure low
84	07/09/04	04:35:39 PM	Source water pressure low
84	07/09/04	04:35:39 PM	Source water pressure low
84	07/09/04	04:35:41 PM	Source water pressure low
84	07/09/04	04:35:48 PM	Source water pressure low
84	07/09/04	04:35:48 PM	Source water pressure low
84	07/09/04	04:35:51 PM	Source water pressure low
84	07/09/04	04:35:58 PM	Source water pressure low
84	07/09/04	04:35:58 PM	Source water pressure low
84	07/09/04	04:36:01 PM	Source water pressure low
84	07/09/04	04:36:08 PM	Source water pressure low
84	07/09/04	04:36:08 PM	Source water pressure low
84	07/09/04	04:36:11 PM	Source water pressure low
84	07/09/04	04:36:18 PM	Source water pressure low
84	07/09/04	04:36:18 PM	Source water pressure low
84	07/09/04	04:36:22 PM	Source water pressure low
84	07/09/04	04:36:28 PM	Source water pressure low
84	07/09/04	04:36:28 PM	Source water pressure low
84	07/09/04	04:36:29 PM	Source water pressure low
84	07/09/04	04:36:30 PM	Source water pressure low
84	07/09/04	04:36:30 PM	Source water pressure low
84	07/09/04	04:36:33 PM	Source water pressure low
84	07/09/04	04:36:40 PM	Source water pressure low
84	07/09/04	04:36:40 PM	Source water pressure low
84	07/09/04	04:36:43 PM	Source water pressure low
84	07/09/04	04:36:50 PM	Source water pressure low
84	07/09/04	04:36:50 PM	Source water pressure low
84	07/09/04	04:36:53 PM	Source water pressure low
84	07/09/04	04:37:01 PM	Source water pressure low
84	07/09/04	04:37:01 PM	Source water pressure low
84	07/09/04	04:37:04 PM	Source water pressure low
84	07/09/04	04:37:11 PM	Source water pressure low

Figure 6.7 - 4 Alarm History Screen

Warnings Screen

This screen shows the current warning on the machine. Warnings highlight potential problems or reminders and do not result in the stopping of the machine or equipment on it.

The colour codes indicate the following:

- Light Green — Cleared Fault
- Light Yellow — Acknowledged Fault Present
- Light Red — New Warning

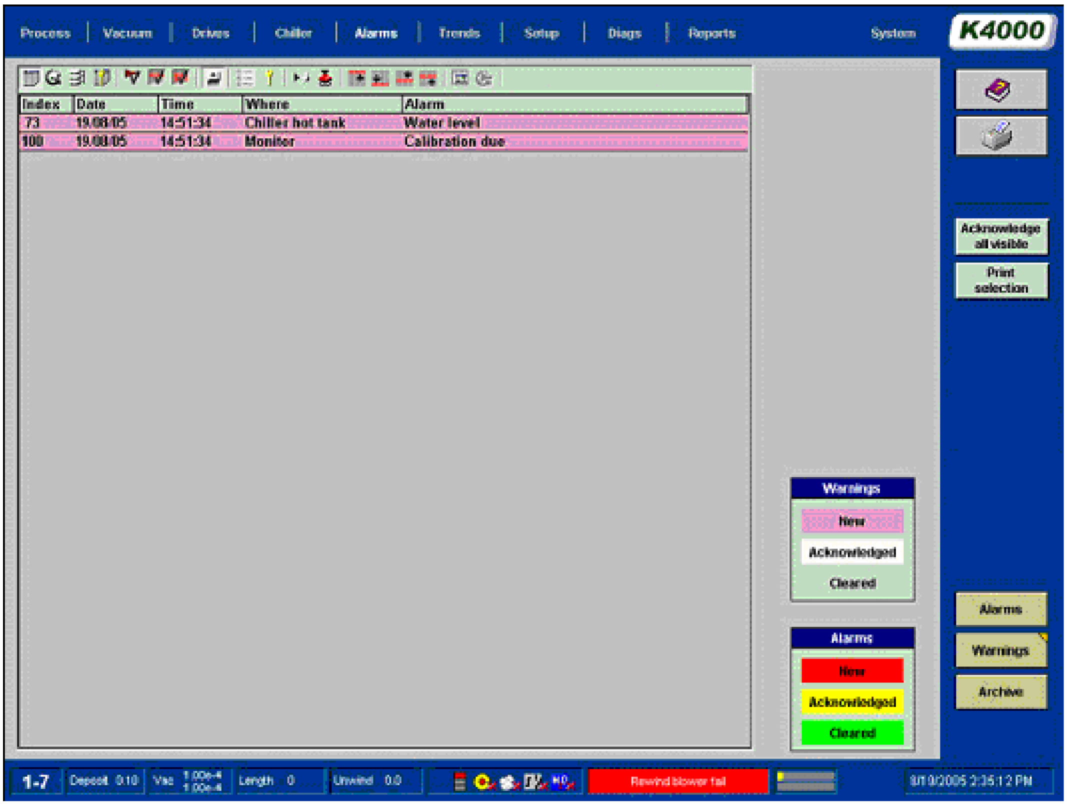


Figure 6.7 - 5 Warnings Screen

6.8 Deleting Files on the Computer (Supervisor Level)

If a material RECIPE file is no longer required, there is a facility within the software to allow this to occur. The DELETE FILE pop-up box is accessed from the SETUP screen by touching the FILE button

The supervisor has the ability to select an existing RECIPE file. Different product information can be stored in each of these RECIPES.

To select an existing recipe, select the RECIPE button and a drop down pop up screen will be shown.

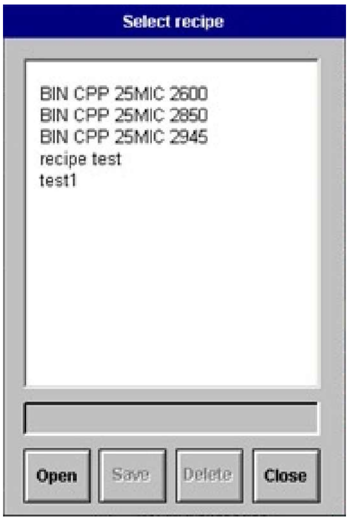


Figure 6.8 - 1 RECIPE Selections Pop Up Screen

Select a recipe from the list and then press the DELETE button to remove the file

Recipes can only be changed, deleted or created when the MMI system is at the Supervisor level.

6.9 Reporting Options/Data Logging

The computer can produce a number of reports of the machine in regard to the operation, alarm status and production cycle times.

Reports Function

The REPORT function screen allows the user to set up and generate a report on rolls of substrate previously coated and the running conditions of the machine. The screen is accessed from the REPORTS Tab at the top of the screen.

When opening the screen, the previous set up screen will be displayed.

Normally the screen below will be initially shown, this is the Select Screen.

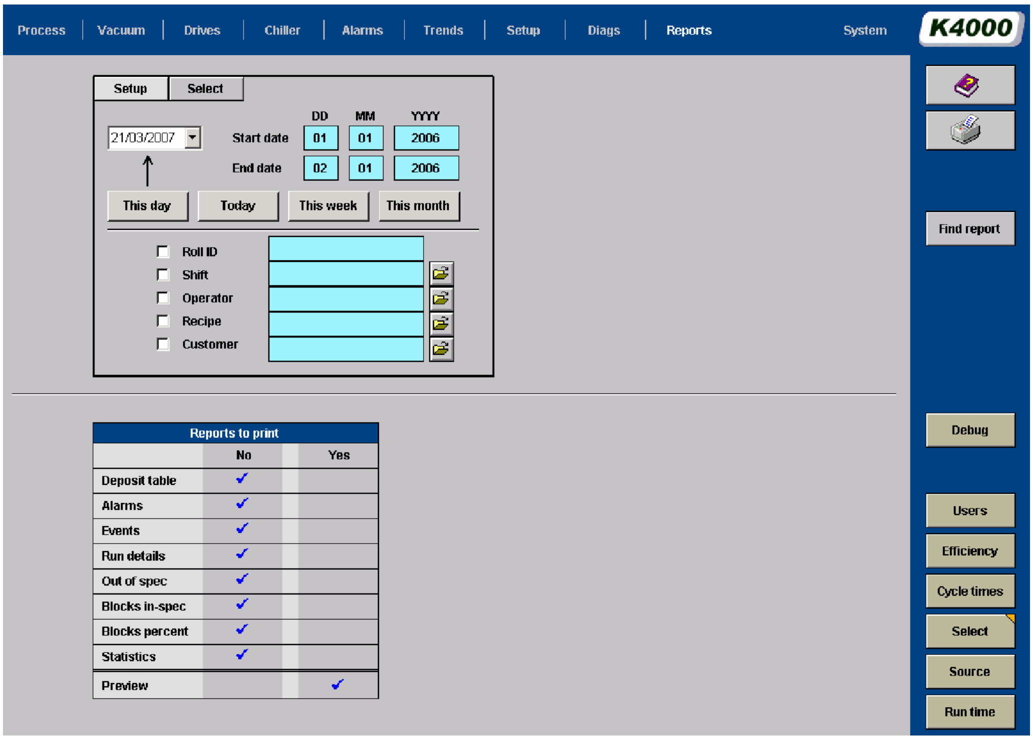
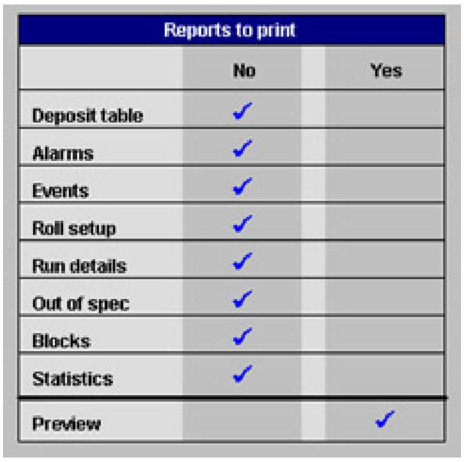


Figure 6.9 - 1 Reports Options Select Screen

This screen allows the user the choice of the following types of report on rolls that have been processed to be printed.



Reports to print		
	No	Yes
Deposit table	✓	
Alarms	✓	
Events	✓	
Roll setup	✓	
Run details	✓	
Out of spec	✓	
Blocks	✓	
Statistics	✓	
Preview		✓

Figure 6.9 - 2 Reports to Print Selection Screen

• Deposit Table	Deposition readings throughout the coating cycle.
• Alarms	The computer will list the alarm conditions that have occurred during the process cycle
• Events	The event reporting function is configured to write to a daily file and then save it for a preset time span. Event report is for the selected run only The event reporting function will report all aspects of the machine operation for the day in question. The user’s requirement should be typed in the relevant box.
• Roll Setup	The computer will print out the ROLL Setup as typed in. This prints the contents of the recipe file.
• Run Details	This report shows the cycle times and efficiency reports.
• Out of Specification	Whenever the deposition is outside the pre-set tolerance limits, the software will record the deposition and the position in the roll in length.
• Block	This report shows the deposition through the roll as numbers. 0 in specification, - light coating + Heavy coating.
• Statistical Report	The computer will produce a statistical report.

Table 6.9 - 1 Reports Table

Samples of Reports

Find below previews of the types of reports as described above

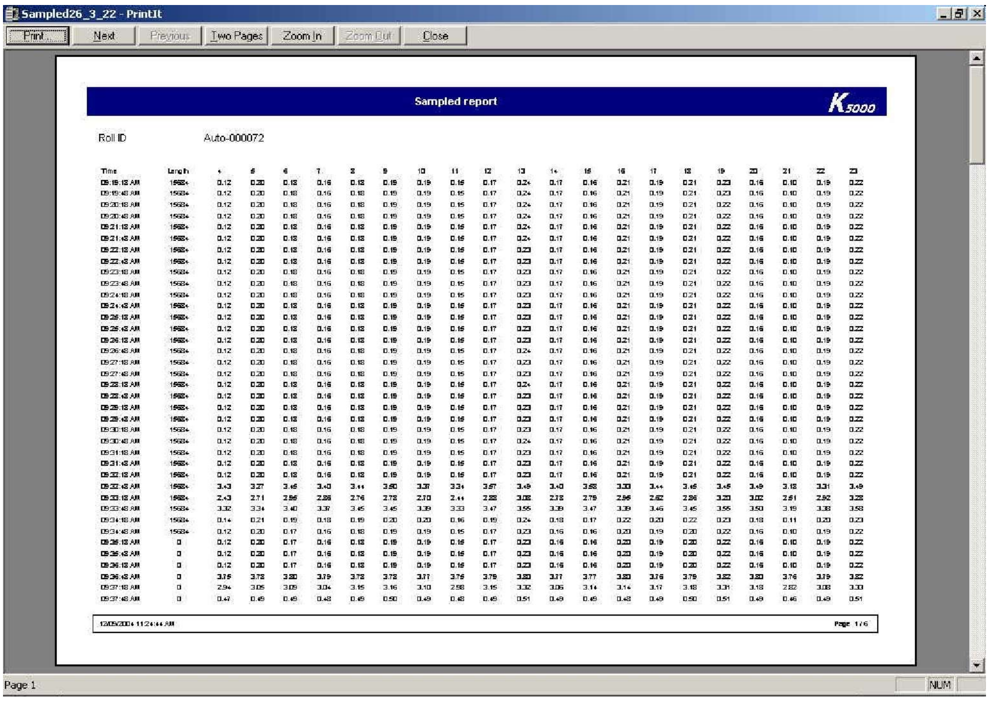


Figure 6.9 - 3 Deposit Report Preview

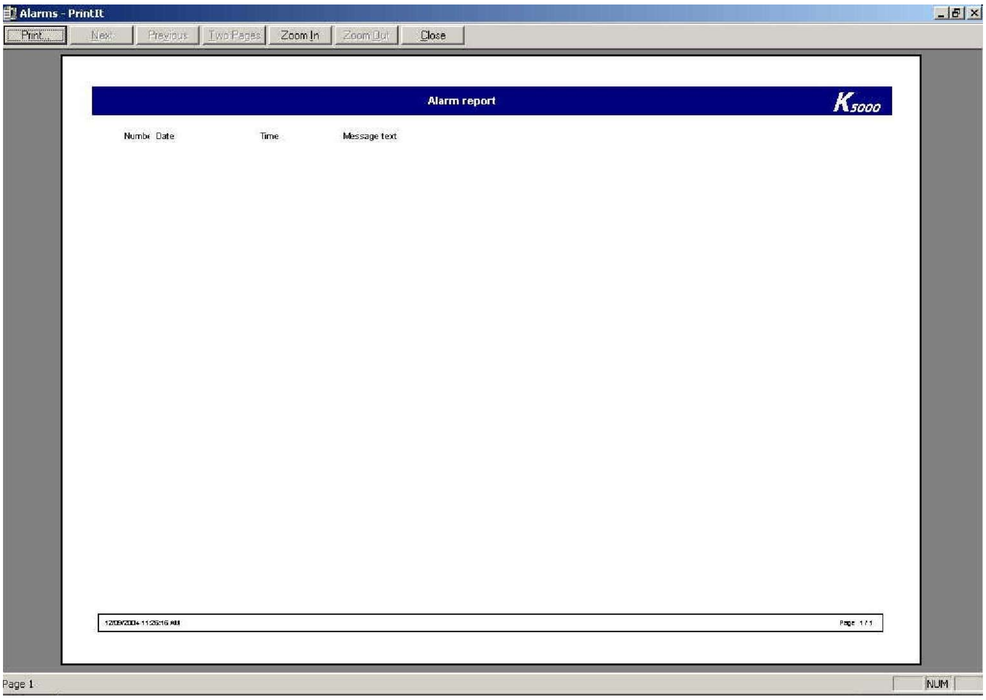


Figure 6.9 - 4 Alarms Report Preview

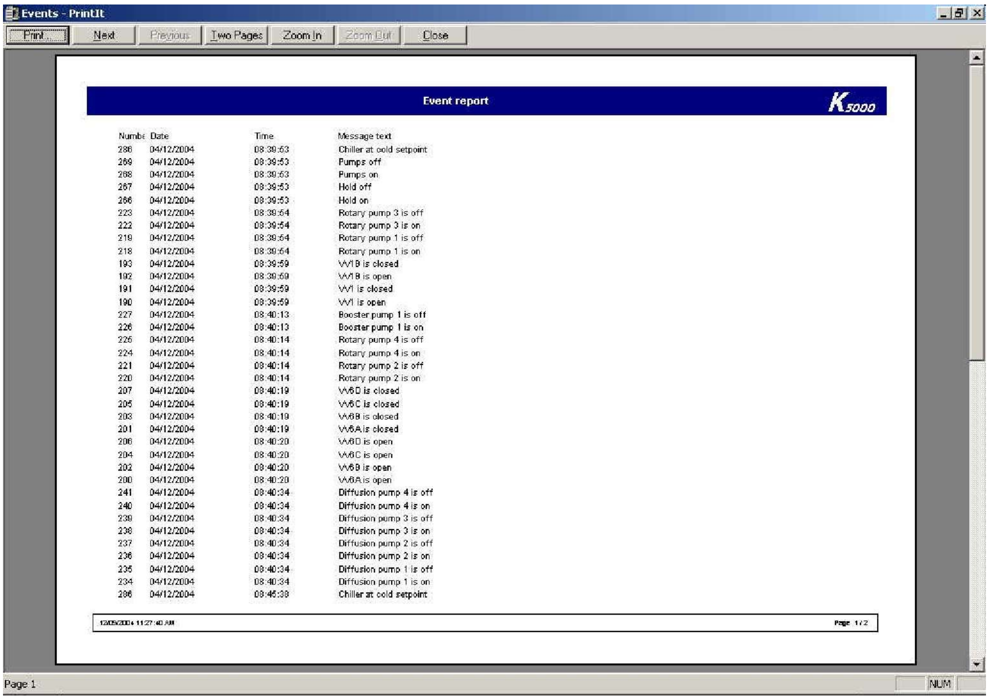


Figure 6.9 - 5 Events Report Preview

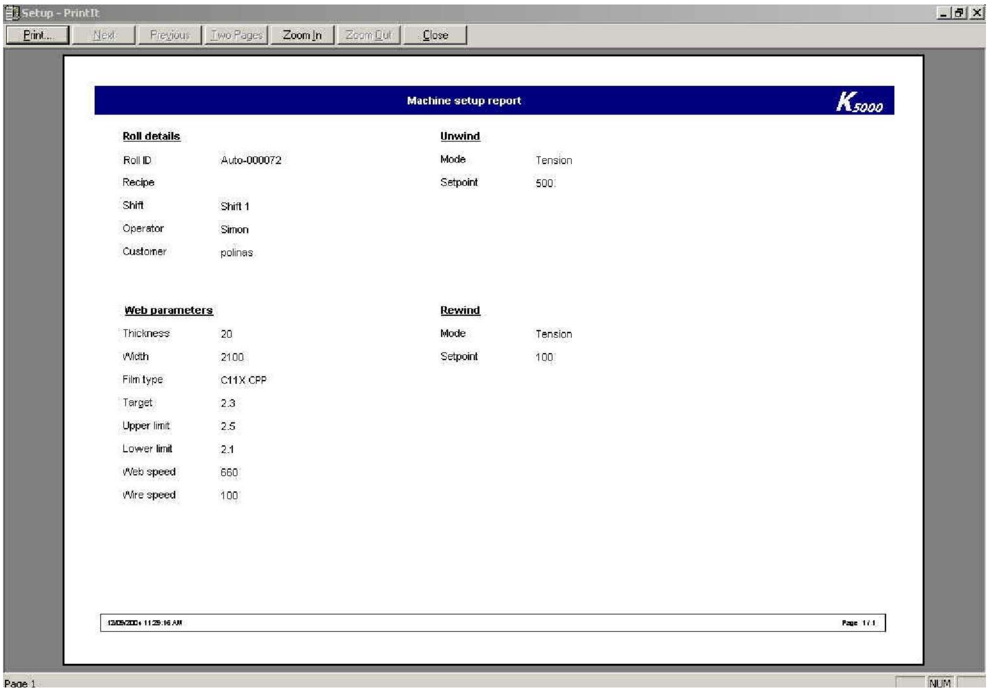


Figure 6.9 - 6 Roll Setup Report Preview

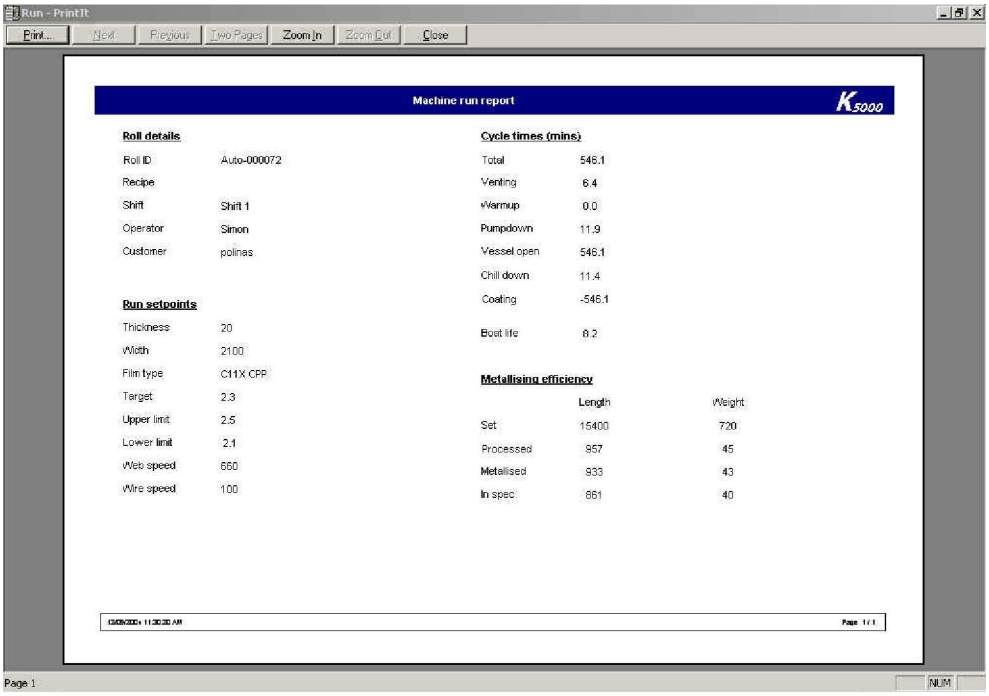


Figure 6.9 - 7 Run Details Report Preview

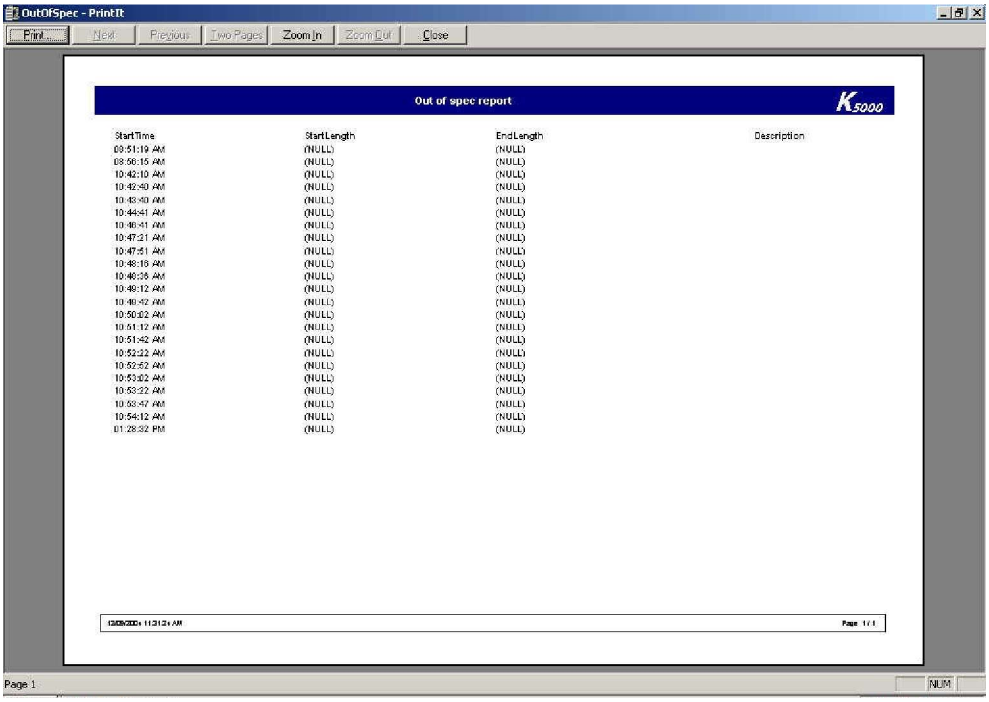


Figure 6.9 - 8 Out of Specification Report Preview

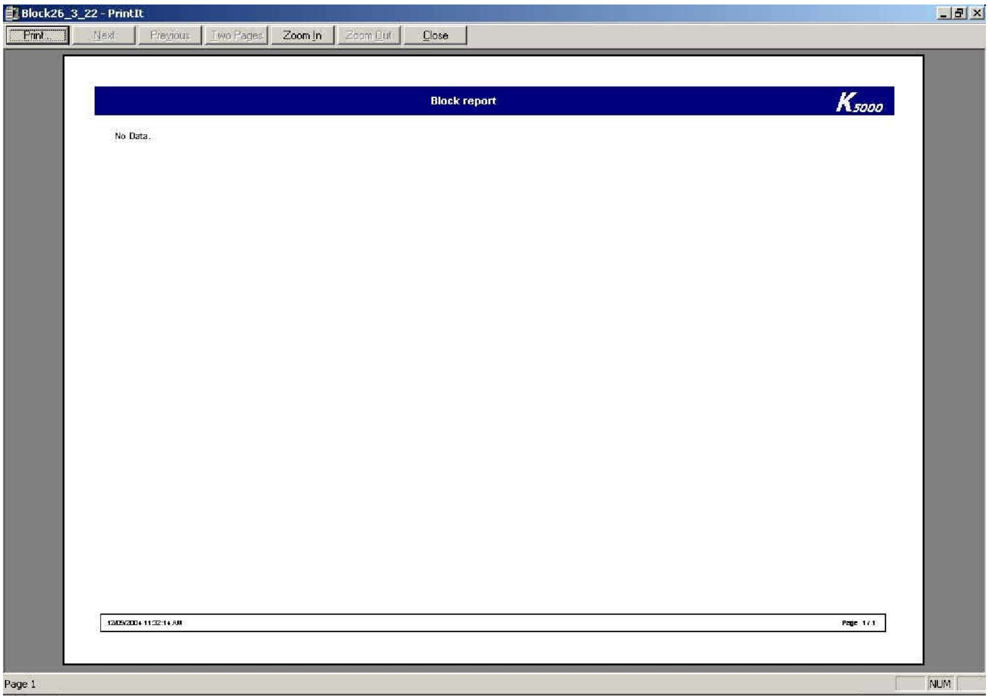


Figure 6.9 - 9 Block Report Preview

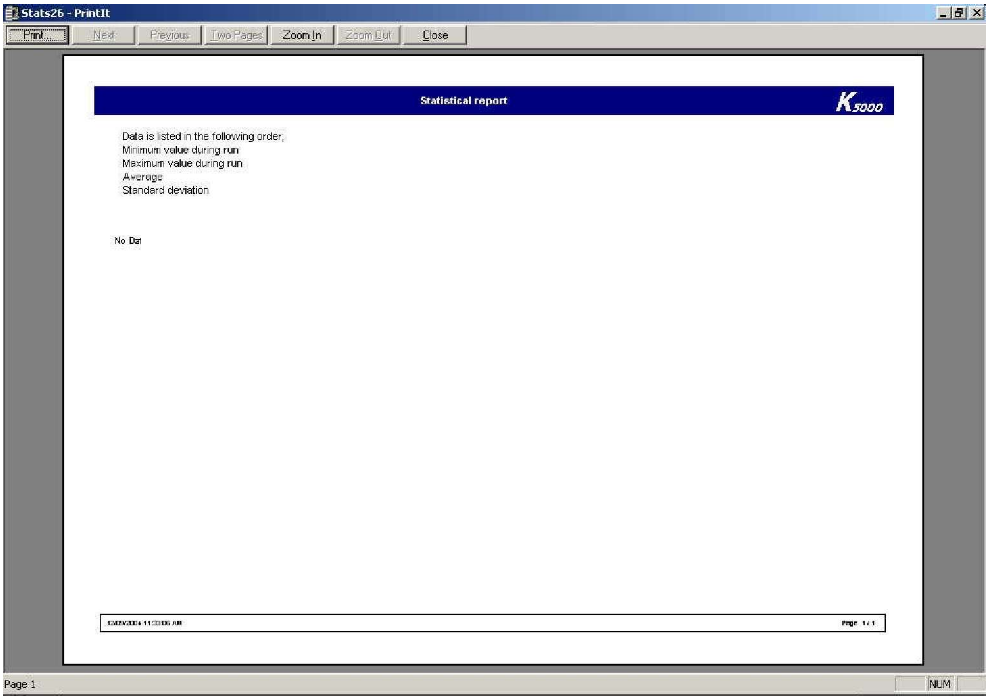


Figure 6.9 - 10 Statistical Report Preview

Viewing Archived Reports

There is an onboard archive function that allows the computer to store previous roll reports.

This cross reference of the main data log achieves containing different data. This archive is currently limited to six months, and then the data is deleted.

As the SETUP function is used and the user uses the LOAD function, they will be asked if they want to allow the computer to designate an automated reference roll number to the cycle or not.

These unique numbers can be used later to select reports.

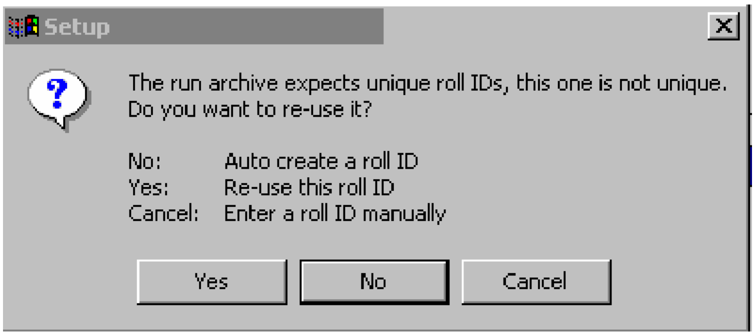
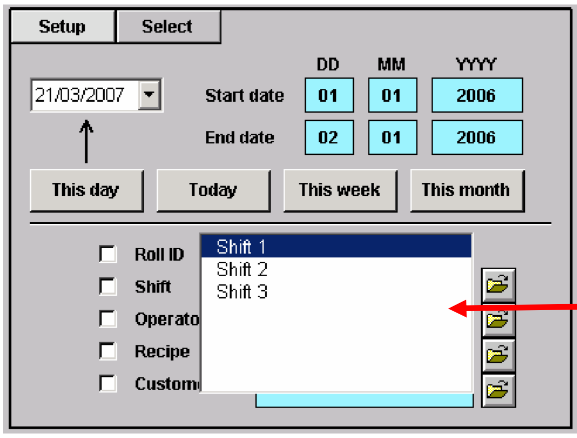


Figure 6.9 - 11 Setup Load Message



Select from data given in the setup screen, or type text into the field directly

Figure 6.9 - 12 Achieved Report Search parameters.

To find a previously achieved report, parameters can be set in the section of the screen shown above.

All reports are initially selected by the selected fields.

Fields:

Start Date	Allows the user to select a date from when the search should start.
End date	Allows the user to select a date to when the search should finish.
Roll ID	Allows the user to select a specific roll identification code
Shift	Allows the user to select a shift
Operator	Allows the user to select a selection of rolls based upon the operator login name.
Recipe	Allows the user to select a selection of rolls based upon the recipe selected.
Customer	Allows the user to select a selection of rolls based upon the customer name selected.

Table 6.9 - 2 Search Fields

Once the search fields have been selected, the SELECT button should be used.

The results from the search will be shown on the screen.

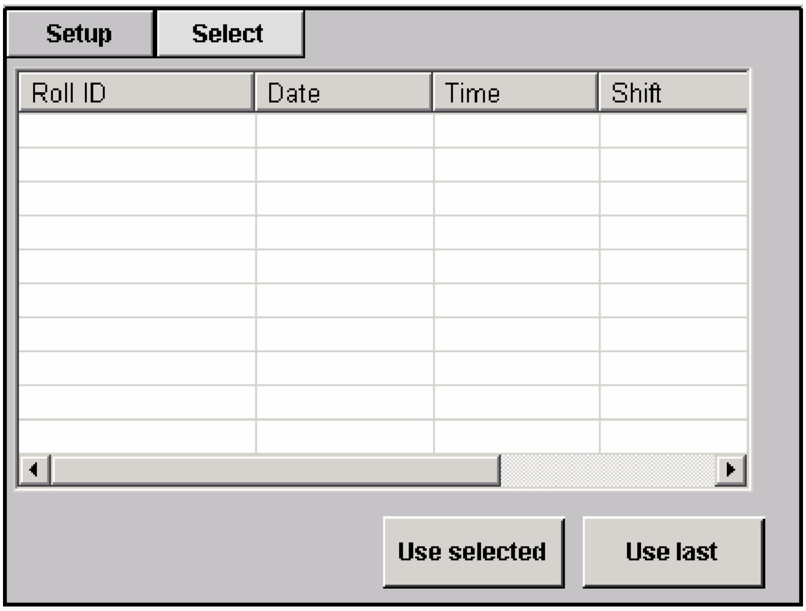


Figure 6.9 - 13 Report Search Results Table.

The details shown can be selected from the list content listing.

To select a file, it should be highlighted and then the USE SELECTION button be touched.

Alternatively, the last process run can be selected by touching the 'USE LAST' button.

Reports – Source

This report allows the user to monitor evaporator life cycle and also the aluminium wire consumption.

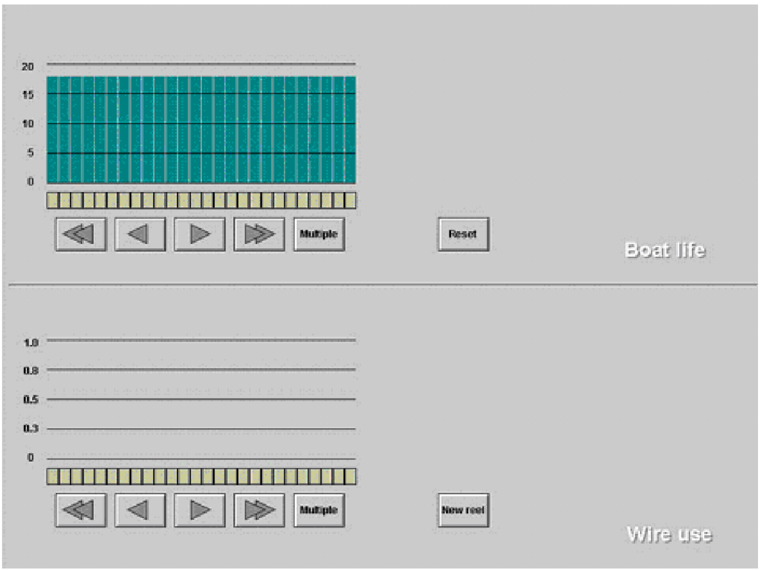


Figure 6.9 - 14 Source Report Screen

The numbers of individual channels can be selected using the buttons under the bar charts.

The reset button will reset one or more channels depending upon the selections made. The process of resetting channels is by manual operation only.

The amount of wire of a reel will depend upon the size of the reel installed.

Reports – Run Time

This screen allows the user to see how long major components have been operating on the machine.

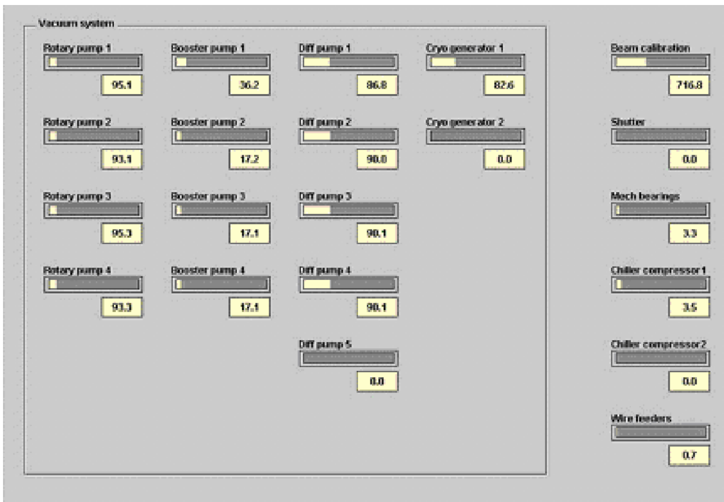


Figure 6.9 - 15 Run Time Report Screen

The bar chart will indicate as a percentage the amount of time ran against a predetermined maintenance period. These times are set in the maintenance section of the systems screens by GENERAL VACUUM EQUIPMENT.

Reports – Efficiency

This screen allows the user to quickly investigate how much of the substrate processed is actually within the preset specification.

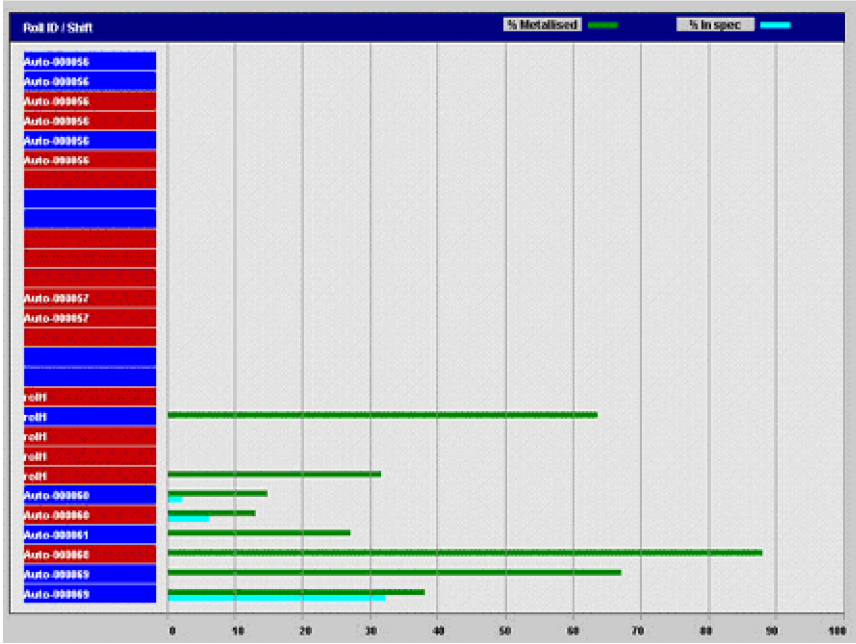


Figure 6.9 - 16 Machine Efficiency Report Screen

These reports are shown against rolls/cycles. This is usually the last 28 cycles with the last cycle at the bottom of the screen.

The left hand column is colour coded to the shifts team using the machine.

Reports – Cycle Times

This report allows the user to view the process cycles. The different periods in the overall production cycle are colour coded as per the guide at the top of the screen.

If the cycle time is long, the colour coded chart may be extended. This is indicated by the red X on the right-hand side.

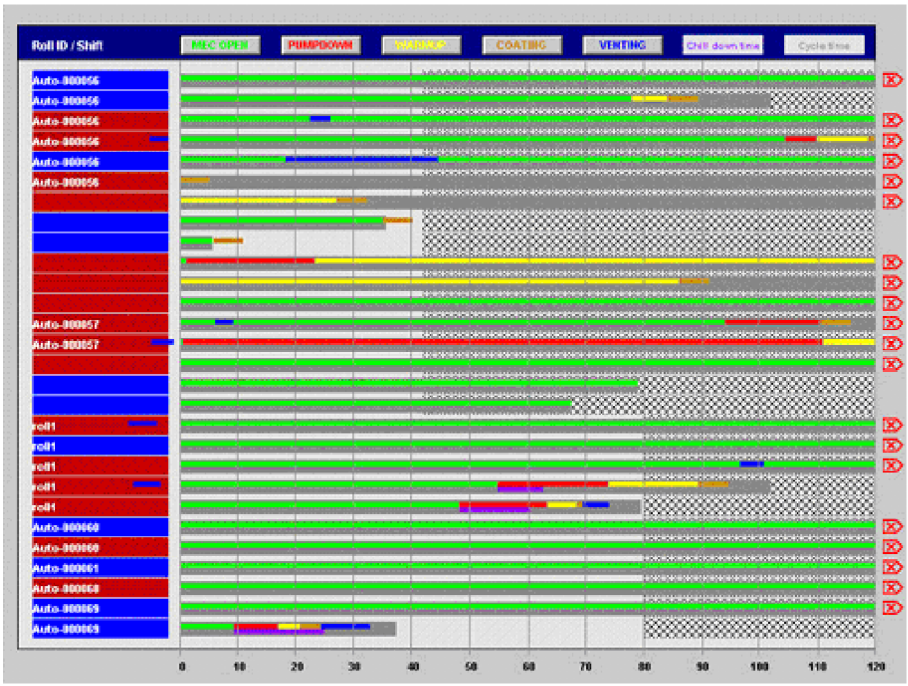


Figure 6.9 - 17 Machine Cycle Times Report Screen

Pressing the index buttons at the top of the screen will set the display order with the selected item being first.

The left hand column is colour coded to the shifts team using the machine.

Reports – Users

This report allows the person viewing the report to monitor who has logged into the machine, to operate it.

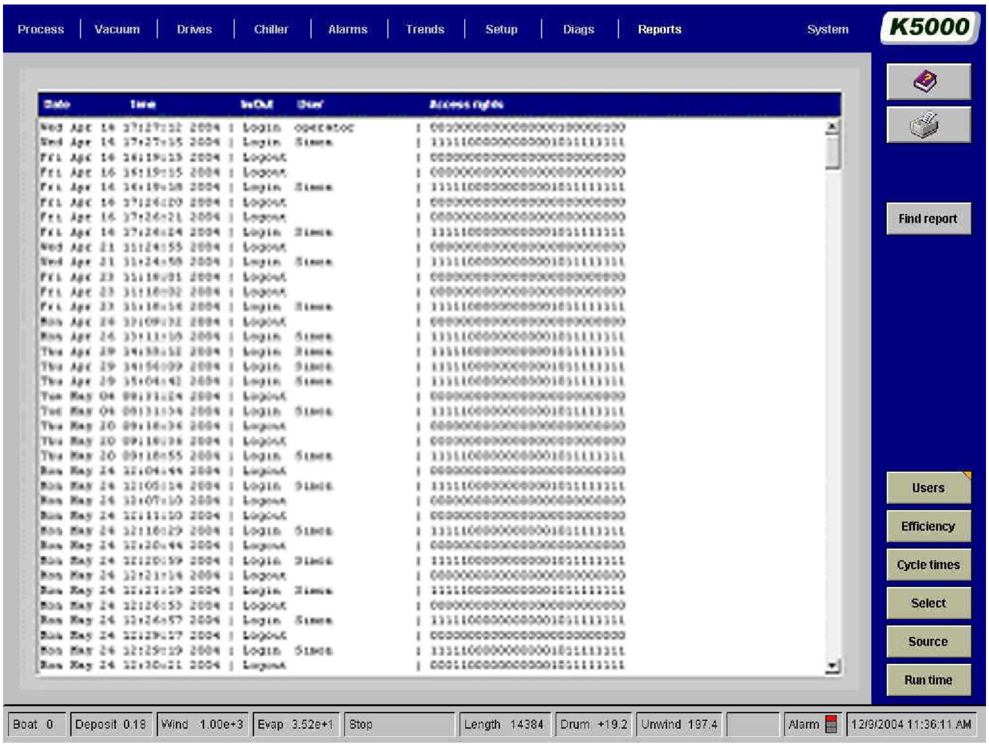


Figure 6.9 - 18 Users Report Screen

6.10 Inputting the Machine Operating Parameters

The machine relies upon the operator to enter certain data for it to perform its functions, as expected. The information is entered the SETUP screen. This screen is available at all times when the software is operating except when the evaporator power supply has been switched on.

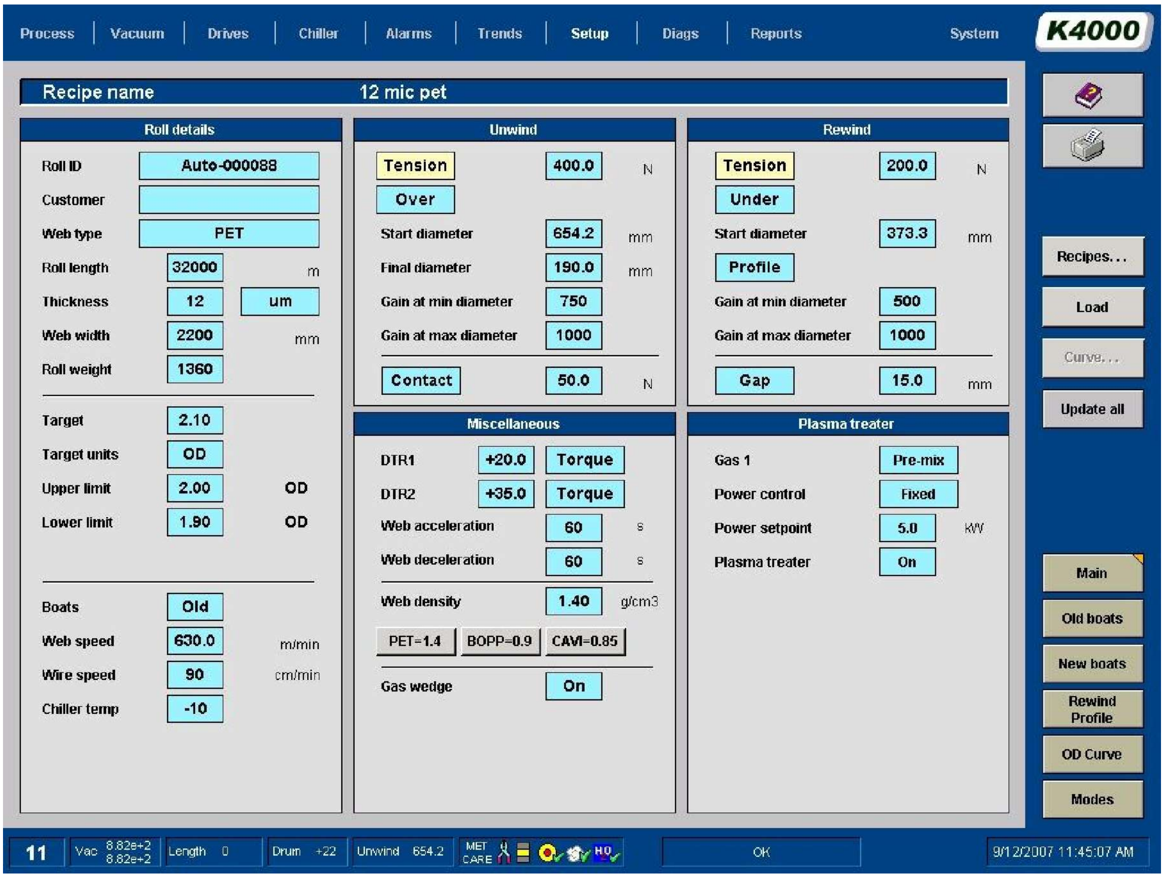
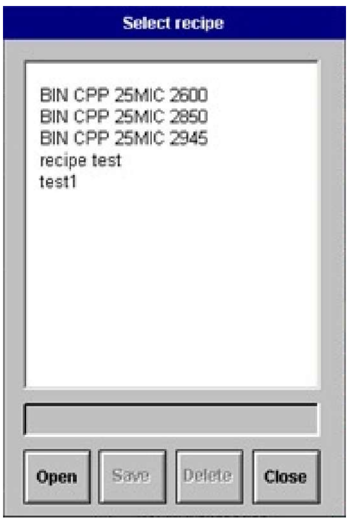


Figure 6.10 - 1 SETUP Screen

Recipes

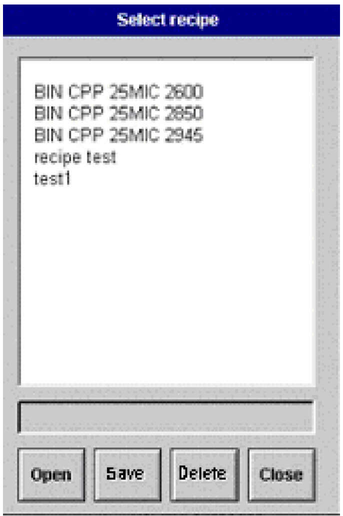


The operator has the ability to select an existing RECIPE to coat a certain material at predefined operating parameters. Different product information can be stored in each of these RECIPES.

To select an existing recipe, select the RECIPE button and a drop down pop up screen will be shown.

Figure 6.10 - 2 RECIPE Selections Pop Up Screen

The OPEN section of the screen should be touched. The computer will then display a list of previously entered recipes by name.



The user uses the raise or lower keys to move the highlighted cursor to the required RECIPE and then the ENTER key to select it.

Recipes can only be changed, deleted or created when the MMI system is at the Supervisor level.

Figure 6.10 - 3 RECIPE Screen – at Supervisor level

At this level the recipes can be saved using the SAVE button.

They can also be deleted using the DELETE button.

The SETUP screen allows the operator to set up various parameters including the following;

Parameter	Description
Roll ID	The roll identification reference can be used to find an archived report at a later date. As the recipe is loaded, the user will be questioned if to use a new reference or an automatically created code.
Customer	The customer name can be used to find an archived report at a later date.
Web Type	Information for reports
Roll Length	Information for reports
Thickness	Important that the drive system has this information to allow automatic diameter stopping.
Web Width	Important information for the machine to select the correct number of evaporators and aluminium wire feeders.
Roll Weight	Information for reports
Target	Aluminium deposition target
Target Units	Mode of measurement - Ohms or Optical density
Upper Limit	Declares the upper display limit on the deposition
Lower Limit	Declares the lower display limit on the deposition
Uncoated web transmission	Clear Tolerance (Used with optical monitor system)
Boats	Selects ‘Old’ or ‘New’ Evaporators
Web Speed	Sets required Line Speed
Wire Speed	Required aluminium wire feeder speed
Chiller Temp	Sets Chiller Unit Process Temperature
Unwind Tension	Sets Tension set point
Unwind Roll Direction	Sets roll winding direction
Unwind Start Diameter	Important information for the drive system
Unwind Final Diameter	Important information for the drive system
Unwind Gain at minimum diameter	Sets the drive response to changes in tension at the roll minimum diameter
Unwind Gain at maximum diameter	Sets the drive response to changes in tension at the roll maximum diameter
Unwind Layarm - mode of operation	Selects the mode of operation.
DTR1 Control Mode	Selects the mode of operation
DTR1 Set point	Set point value entered here
DTR2 Control Mode	Selects the mode of operation

Parameter	Description
DTR2 Set point	Set point value entered here
Web Acceleration	Set point value of time machine takes to accelerate to “full” speed
Web Deceleration	Set point value of time machine takes to decelerate to “full” speed
Web Density	Used by drive system. Select from default value buttons, or enter value found on material data sheet
Gas Wedge	Select ‘ON’ to allow usage
Rewind Spreader Position (Option)	Sets the position of the spreader roller.
Rewind Tension	Sets Tension set point
Rewind Roll Direction	Sets roll winding direction
Rewind Start Diameter	Important information for the drive system
Rewind Taper Profile Mode Selection	Important information for the drive system
Rewind Gain at minimum diameter	Sets the drive response to changes in tension at the roll minimum diameter
Rewind Gain at maximum diameter	Sets the drive response to changes in tension at the roll maximum diameter
Rewind Layarm - mode of operation	Selects the mode of operation.
Plasma – Gas 1	Select the type of gas being used.
Plasma – Power Control	Power control either fixed set-point or ramp with speed
Plasma – Power Set point	Power value set-point
Plasma – Plasma Treater	Select ‘ON’ to allow usage

Table 6.10 - 1 Set Up parameter Table

Operational Modes – Set Up

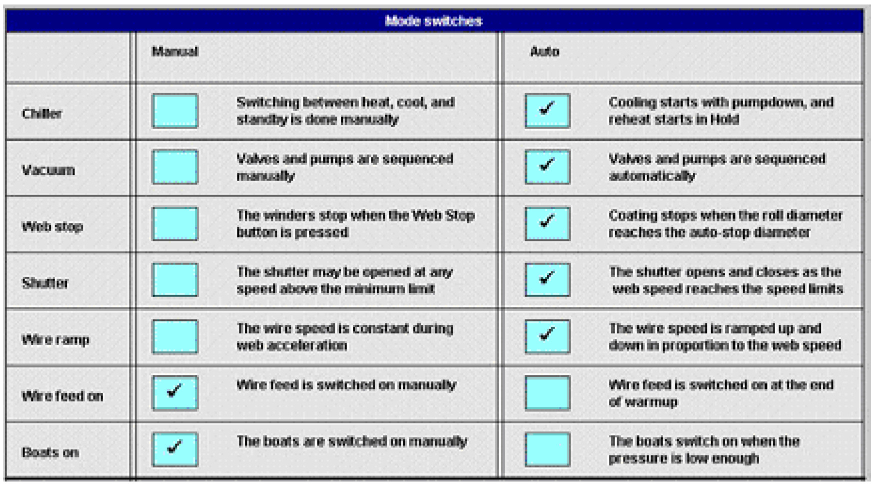


Figure 6.10 - 4 Operational Modes SETUP Screen

Chiller Operation	Select auto or manual control
Vacuum System	Select auto or manual control
Web Stop	Select auto slowdown feature
Shutter	Select auto or manual control
Wire Ramp	Select constant or varied speed
Wire Feed On	Selects auto wire feed on at the end of the warm up cycle
Boats On	Selects auto heat up of the evaporators when the pressure is low enough in the chamber.

Table 6.10 - 2 Set Up Mode Table

Other curves such as:

- Old Boats
- New Boats
- OD Curve

- Rewind Profile

Are activated from the setup screen. Descriptions of their functions and how to modify are in other relevant sections of this manual.

Loading the setup Parameters

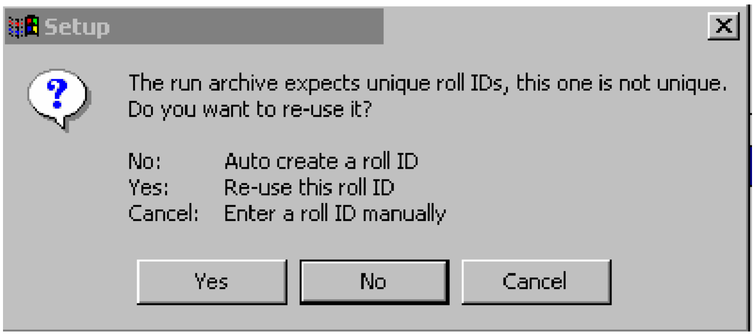


Figure 6.10 - 5 LOAD Pop Up Screen

When attempting to load the recipe, a pop up message screen will question the user as above. A selection has to be made.

6.11 Monitoring the Coating Deposition On The Substrate

There is a process requirement to monitor the amount of coating being deposited upon the substrate.

Low deposition rates will render the finished product unsuitable, if the coating deposit is not thick and uniform enough to meet the end users requirements...

Heavy deposition rates will allow the process to become inefficient.

The end user of the substrate will generally request a deposit measured in optical density or Ohms/ Square, between a minimum and maximum level.

Therefore the uniformity across the width of the substrate and down its length is important.

Depending upon the equipment that has been supplied with this machine, the deposition will be measured in either optical density using a light source/measuring head type of device or a non-contact resistivity monitoring system. The following section will describe the type installed on this machine.

6.12 Aluminium Deposition Non-Contact Monitor System

This monitor system uses sensors situated near to the film surface. These are located in a beam mounted in the winding mechanism. The sensors send signals to the computer to PROCESS screen. This displays the deposition coating value, numerically and on a bar chart.

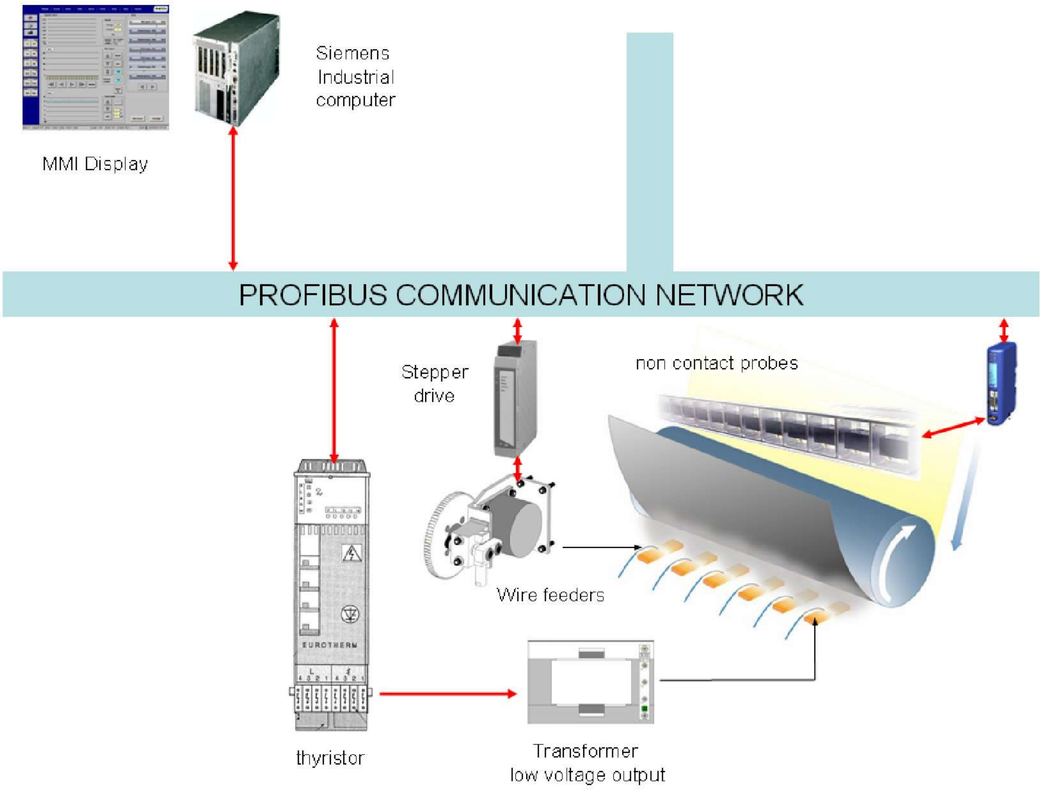


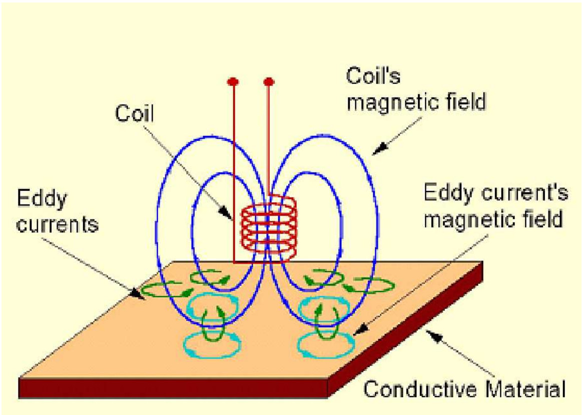
Figure 6.12 - 1 Non Contact Monitor System Schematic

There are a number of displays that can be seen on the PROCESS screen, including;

- Deposition levels in a bar chart format
- Wire feed rates in a bar chart format.
- Power levels supplied to the evaporators.

Probe Operation

In simple terms the computer sends signals from individual probes to the surface of the coated substrate. Sensor probes have coils within them, which a RF signal is passed through.



This RF signal drives all the probes at their resonant frequency to produce a maximum output signal.

This signal is transmitted to the buffer unit that then converts the signal and sends it on to the computer, where is displayed in an ohms reading. From these values it can determine the optical density of the aluminium on the substrate and then displays this value on the PROCESS screen.

Figure 6.12 - 2 Non Contact Probe Principals

When the substrate with aluminium is passed in front of the probes, it absorbs some of the radiated signal.

The probes are scanned every 0.5 seconds and the display is updated

Once the measured value, has been displayed the control signals to control the wire feeders will be altered to reduce the error between the desired and actual target deposition level by the computer.

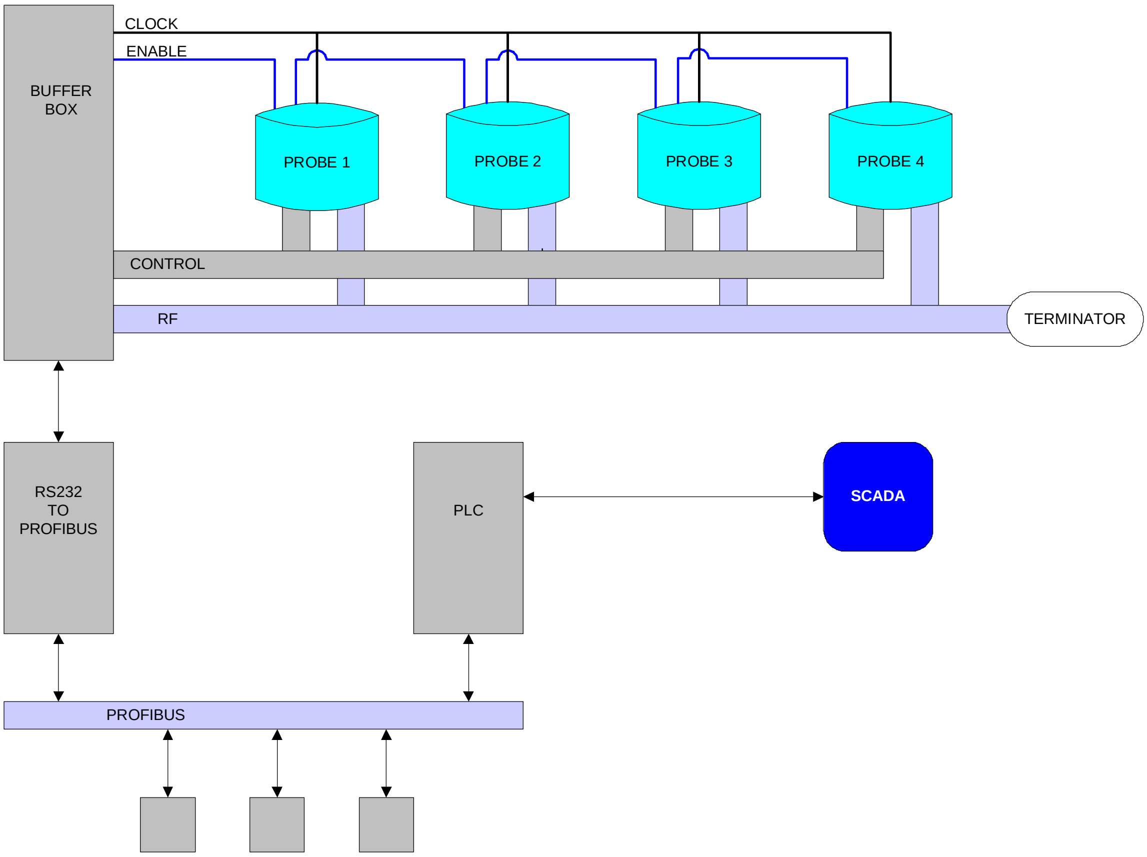


Figure 6.12 - 3 Non Contact Probe System Schematic Layout

6.13 Displaying the Deposition Thickness

The main PROCESS screen allows the user to monitor the deposition being coated on to the substrate.

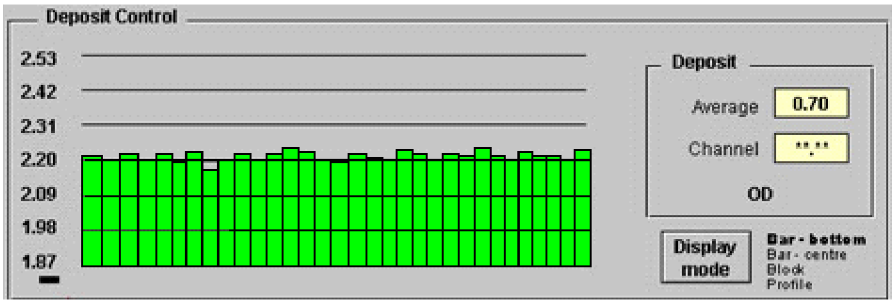


Figure 6.13 - 1 Deposition Section of the PROCESS Screen – OD Display Bar-Bottom

Display Modes:

The following modes of display are available:- Bar – Bottom (as above)

- Bar – Centre – This where the error bars extend from the centre.

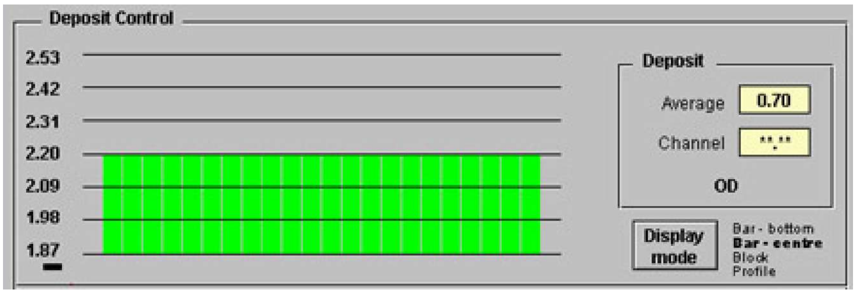


Figure 6.13 - 2 Deposition Section of the PROCESS Screen – OD Display Bar-Middle

- Block – This is a display showing the deposition as the roll of substrate progresses through the machine.

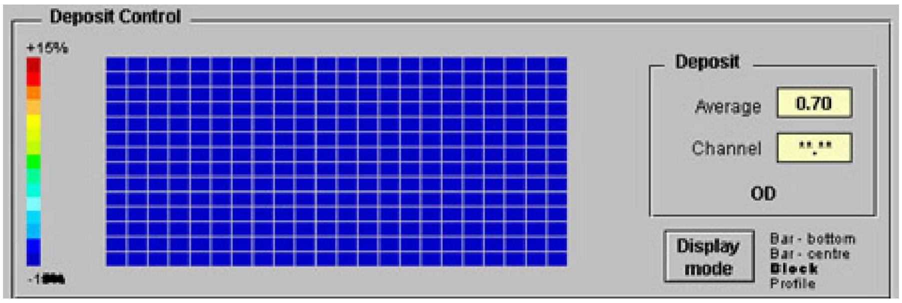


Figure 6.13 - 3 Deposition Section of the PROCESS Screen – OD Display Bar-Block

- Profile

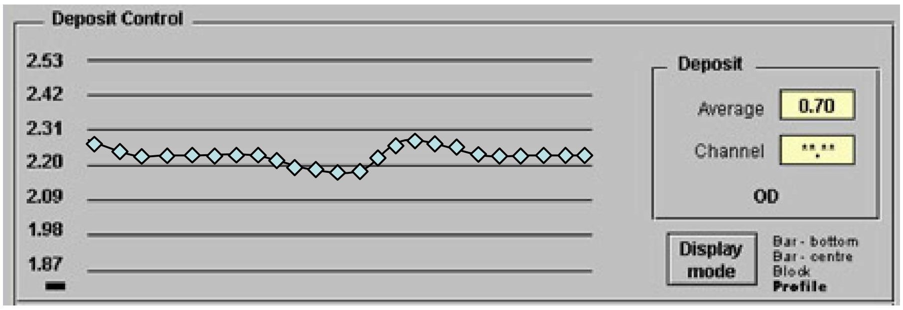


Figure 6.13 - 4 Deposition Section of the PROCESS Screen – OD Display Bar-Profile

These are selected by touching the display button.

The coating deposited on to the substrate can be displayed as a value of the following displays.

The choice of display is made in the SETUP screen.

Resistivity

The amount of coating measured by the monitoring system can be displayed as a resistance (Ohms/square) reading.

The display will show the bars at the top of the screen when there is no aluminium being coated onto the web. As the aluminium is started to be evaporated onto the web, the display will change and the bars will reduce in height.

If the coating is thick the bars will be at the bottom of the screen.

The target of the deposition previously requested is displayed in the left-hand side of the display.

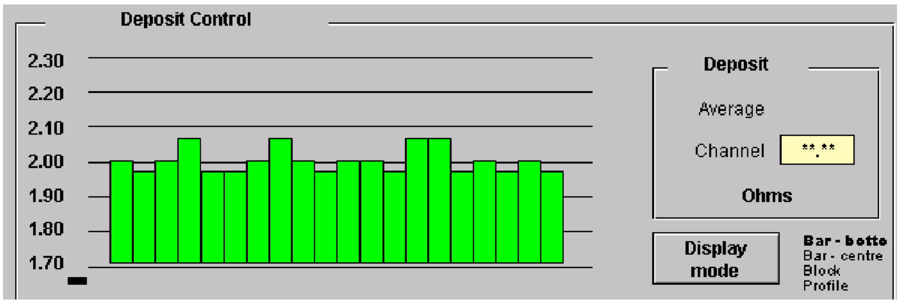


Figure 6.13 - 5 Deposition Section of the PROCESS Screen – Resistance Display

Optical Density Display

The system also has the ability to display the aluminium deposited as an optical density value.

This feature is selected in the SETUP screen.

The display will show the bars at the bottom of the screen when there is no aluminium being coated onto the web. As the coating is started to be evaporated onto the web, the display will change and the bars will increase in height.

If the coating is thick the bars will be at the top of the screen.

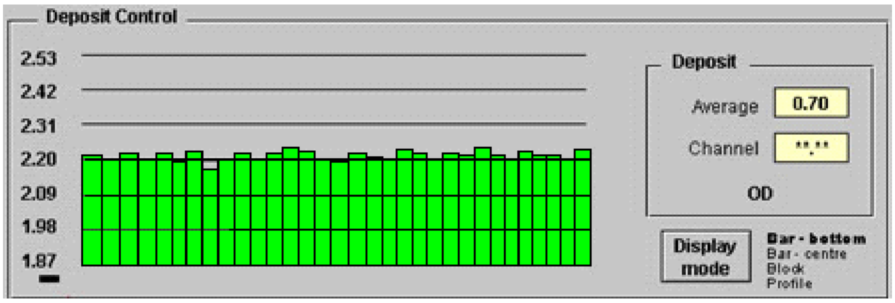


Figure 6.13 - 6 Deposition Section of the PROCESS Screen - Optical Density

Optical Density Curve

There is a conversion curve in the SETUP screen to allow for the conversion of the resistance readings to optical density display. This curve is recipe based.

This function is only available in the supervisor level.

The button should be touched and then the curve button.

Figure 6.13 - 7 OD V Resistivity Curve Selection Button

The optical density of individual substrates is different, for example paper in comparison with PET.

It has been also found that for the same aluminium thickness, as measured in resistance, the optical can be different between substrates.

Therefore individual optical density v resistance curves can be created for each RECIPE

General Vacuum Equipment supplies a base curve.

The curves must be compiled by the individual users based upon comparisons between direct resistance v optical density measurements taken off-line.

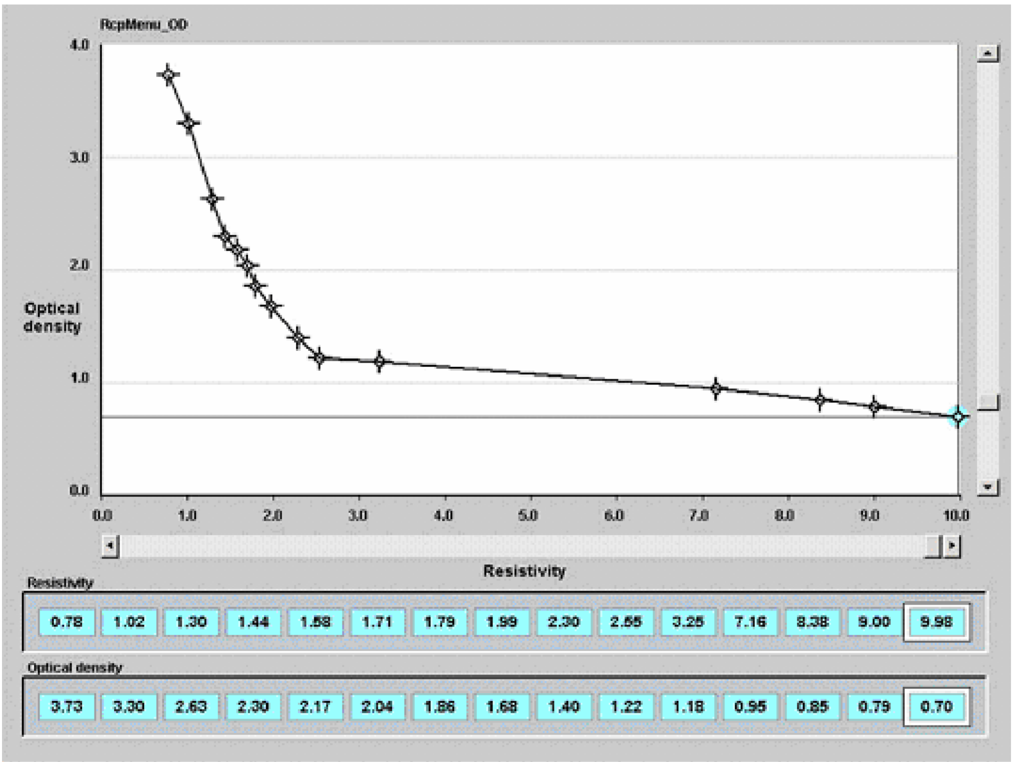


Figure 6.13 - 8 Optical Density V Resistance Set Up Screen

The curve is plotted by touching the screen on one of the points and by dragging it on the screen until the desired value is reached. The ohms and optical density values for each of the cross hairs are displayed at the bottom of the screen.

Alternatively, a keyboard can be used to key in the values.

Curve Editor

The curve editor can be used to simplify the building of the conversion curve.

The curve can be straightened between two points using the STRAIGHTEN button.

The curve can be copied using the COPY button, and then be pasted using the PASTE button.

To apply the curve to the recipe, use the APPLY button.

To close the curve editor, use the CLOSE button.

Figure 6.13 - 9 Curve Editor Pop Up Screen

6.14 Automatic Deposition Control

The machine computer has the following modes for controlling the evaporation of the aluminium deposition.

Stop Mode

In this mode there is no control signals sent to the wire feeder stepper motors and evaporator power supply thyristor units.

Therefore the evaporators are not heated up and no wire is fed to the evaporators.

This is the normal mode at start up of the machine.

Self Test

Touching the "Boats On" button on the PROCESS screen will energise the power supply to the evaporators, via the thyristors.

This action also triggers the computer to start the ramp up of the control signal to the thyristors.

During this period, the system will initially carry out a self-test of the hardware. This is indicated by the bar chart in the deposition section of the screen changing, as the testing is taking place.

After this is completed, the POWER bar chart will be displayed at the bottom of the screen indicating the pre-set evaporator power settings. A coloured bar chart will start to rise from the bottom indicating the power being supplied to the evaporators.

The individual power level settings are set by moving the relevant diamond shapes upwards.

These indicators are set between 0 and 100% of the output of the power source.

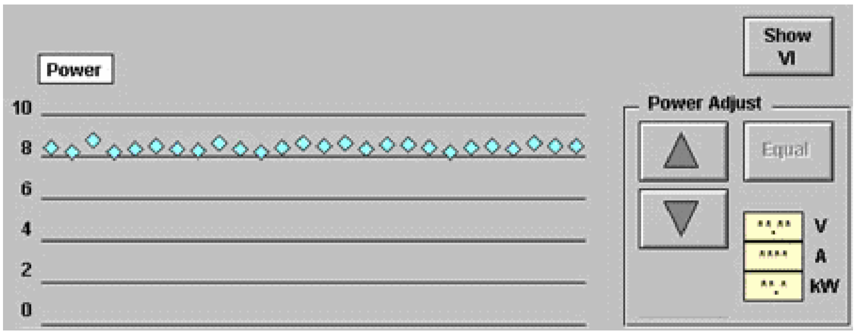


Figure 6.14 - 1 POWER Section of the PROCESS Screen

Warm-Up Mode (Standby)

This mode is normally used to apply a signal to the steppers to give a wire feed rate of 100 percent of the nominal base speed from each wire feeder.

For example, if the wire feed rate was set up for 105cm/min. (as displayed on PROCESS screen) and the output bars are at 100%, the actual wire feed rate would be 105 cm from each feeder.

The computer will place the wire feed outputs in this mode once the evaporators have reached their operating temperature. This occurs at the end of the evaporator ramp cycle.

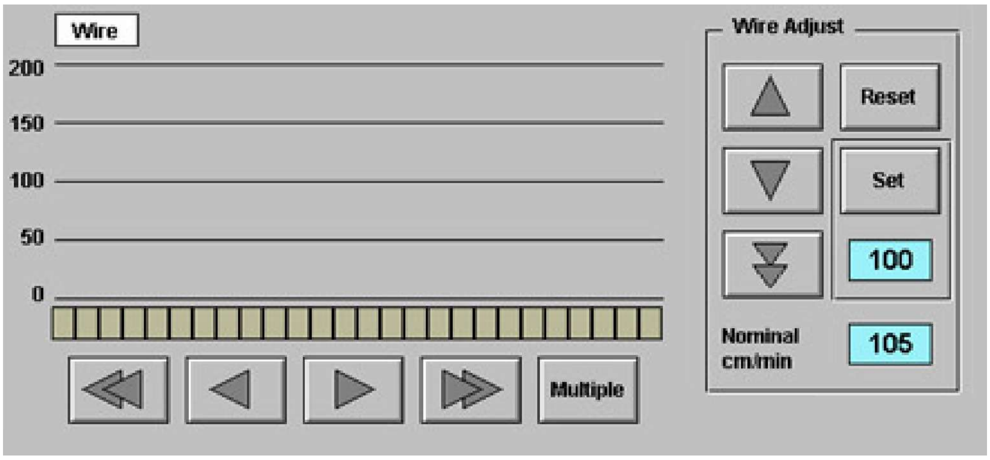


Figure 6.14 - 2 Wire Feed Output Section of PROCESS Screen.

The wire base speed is set up in the SETUP screen and can be modified.

All the wire speed feed set points can be adjusted at once by touching the SELECT ALL section of the PROCESS screen. The base wire feed can be adjusted by touching the raise or lower buttons by the side of the wire feed display.

Alternatively, if the base speed number is touched, a new set point can be entered using a keypad.

Optimum Set-point

The software has the ability to pre-set the output levels to give optimum coating.

This is based upon the output values at the end of the previous coating cycle. The warm up values will be set to the same output values. This only occurs if the AUTO mode was selected.

Resetting Wire Feed Rates to 100%

To reset the warm up values to a 100% output, a RESET button is provided. This can be used when new evaporators are to be supplied. This function can only be used when the controller is not in the AUTO mode

Note: The wire will not be fed onto the evaporators if the main evaporator power supply has not been supplied to the thyristors.

Using the Set Button

The SET button will set the selected channel output to the numeric value shown below the button

Manual Control Mode

If the user wants to alter the wire feed outputs while PROCESS manually, he has to use the cursor keys on the screen as described in section 3 of this manual. .

Automatic Control Mode

This mode of operation allows the computer to change automatically from standby mode (STAND-BY) to the control (AUTO) mode, once all of the monitored readings are within 30 % of the preset uniformity value required.

Normally, the user would select this mode once the actual line speed has matched the set speed.

Figure 6.14 - 3 MANUAL/AUTO Control Selection Button

The mode is selected by touching the CONTROL AUTO/MAN switch on the PROCESS screen.

Once the AUTO mode has been selected, the controller will adjust the wire feed rates via the steppers to give minimal error between the actual and required deposition target.

For any reason that the readings go outside the preset 50% band, the controller will switch back into the Warm up (Standby) mode.

The deposition may move outside the 50 % region for a number of reasons and reference should be made to relavent section of this manual for process related problem information.

Also at the end of coating period, the controller will remember and store the output values that were being sent to the wire feeders. This ensures better uniformity earlier into the coating run of the next roll of material.

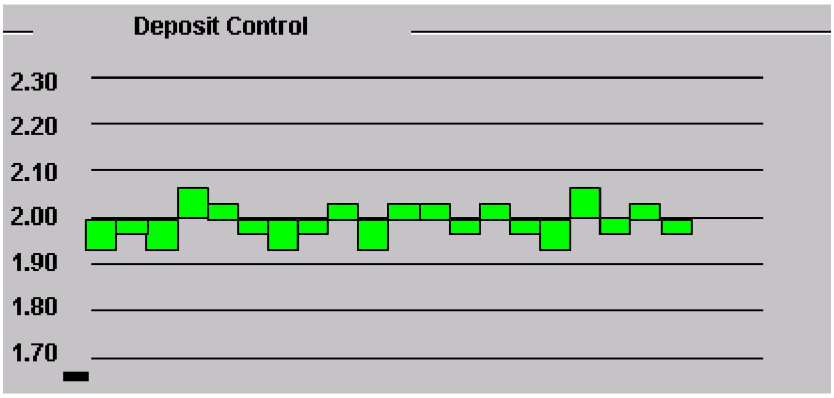


Figure 6.14 - 4 Deposition Control Display in AUTO Mode

The above screen indicates will show the probes in operation and the deposition rates measured on each of them. They will change colour, from red to green, when the actual measured rate is within the pre-set tolerance levels.

These tolerance limits are set in the SETUP screen.

The middle of the screen indicates zero error between the actual and set deposition levels.

Moving the cursor below each of the channels can monitor the individual deposition readings. The measurement is indicated in the box located in the below the channel selected.

The average deposit across the width of the substrate is shown on the right hand side of the screen.

Probe Calibration

The method of calibrating the deposition probes is described in section 14 of this manual.

6.15 Chamber Windows

Clear profiled polycarbonate material is provided on all viewing windows on the chamber to allow viewing. This material is a special optical grade and General Vacuum Equipment can supply replacements.

The source window has pieces of toughened glass placed in front of it in the chamber to protect them against aluminium coating.

	Warning !
	**Do not remove or modify the supplied viewing windows**

Refer to section 8 of this manual in regard to cleaning of the windows.

	Caution !
	**Do not use any solvents to clean the windows, as failure may occur**

6.16 Chiller System Operation

A self-contained chiller unit supplied to cool the medium that is fed to the process drum and rollers.

The chiller and auxiliary equipment are interlocked to carry out the following functions dependent upon the operating mode of the machine's vacuum system.

Chiller System Status	Machine Status
Automatic Start Up	With pumping group AUTO START function or “Pumps” function. The heating of the heat tank will occur.
Automatic Shutdown:	With pumping group shutdown function. The power to the heat tank and chiller system will be removed.
Heats the rollers:-	When pumping system is in either HOLD or VENT mode
Cooling: -	When pumping system is in the vacuum mode.

Table 6.16 - 1 Chiller Operational Status

The chiller is interlocked to the computer to indicate any fault condition that occurs.

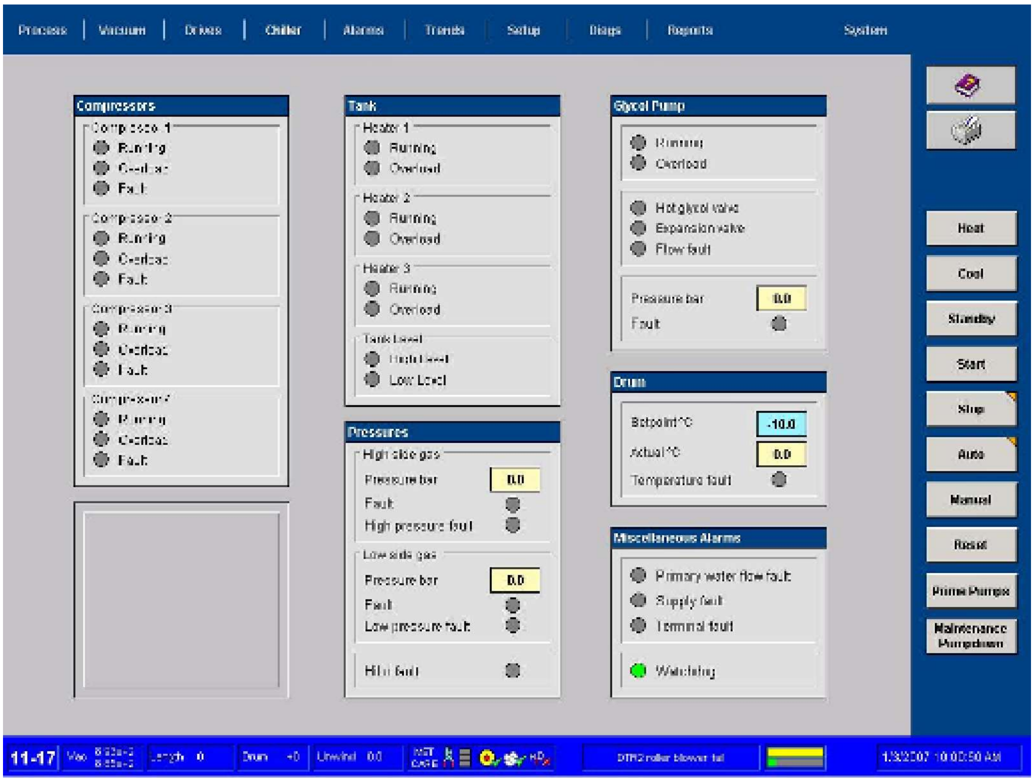


Figure 6.16 - 1 CHILLER Screen

The chiller process temperature is set from the computer either from the SETUP screen or from the CHILLER screen. Touching the set point will allow the user to change the set point within the operational range of the chiller.

Note: All chillers supplied by General Vacuum Equipment are designed to operate within a certain temperature range. Operating the unit outside the specified range will lead to cycling of the unit and a shortened life span of their components.

Leakages of the coolant mixture should be reported immediately to your maintenance department.

Operation

A simple explanation of the system operation follows.

Start Up.

On start up of the machine, the chiller unit will start.

The compressors are not energised at this time, however, the compressors external heaters are powered up

At the same time, the liquid in the hot tank will be heated up to the set point vale and will be kept at this temperature.

Do not operate the chiller with the lid removed off the top of the tank as liquid will be lost from it through evaporation.

Cool Mode

In the cool mode, the pump will run and the liquid will be circulated around the drum and cooled roller.

As the overall temperature gets lower, the cooling capacity will be reduced by switching off the individual compressors. This will be indicated by changes in the status of the compressors on the mimic.

Heat Mode

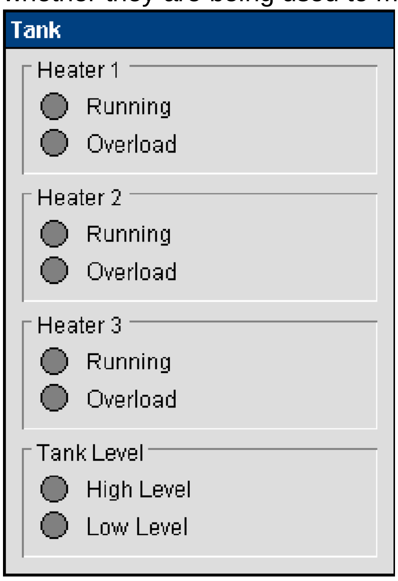
In the heat mode, the valve (SV1) will open and the liquid from the tank will be circulated around the drum and cooled roller by the pump. This will continue until the defrost set point of 25ºC is reached.

The pump will then switch off and valve will close..

Troubleshooting through the MMI Interface

The MMI for the chiller gives good diagnostic features, which indicates the chillers performance.

Each of the heaters are activated independently. The colour of the “Running” indicator shows whether they are being used to maintain the temperature in the tank.

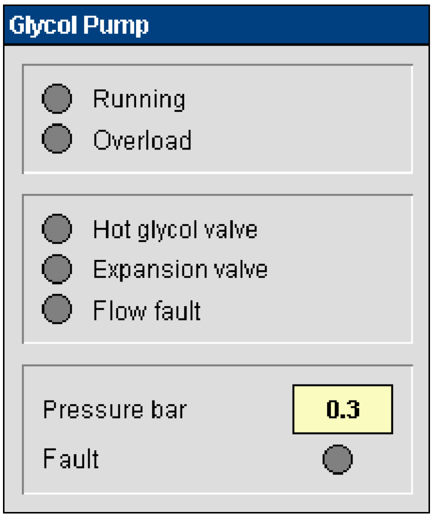


Faults:

- Overload — If the circuit breaker supplying the individual heater trips, this alarm will be indicated.
- Tank Level High — Indicates that the tank level is too high and needs reducing.
- Tank Level Low — Indicates that the tank level is too low and needs filling up.

Figure 6.16 - 2 Chiller – Water Tank Status

The colour of the "Running" indicator shows whether the pump is rotating.



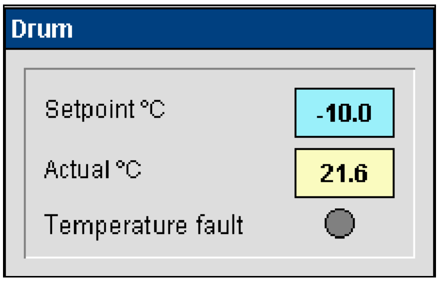
Hot Glycol Valve — Will indicate when the valve has opened allowing the hot glycol into the system

Expansion Valve — Will indicate when the expansion valve is opened during the cooling down of the system

Faults:

- Overload — If the circuit breaker supplying the pump has tripped, this alarm will be indicated.
- Pressure Fault — If there is either a under or over pressure situation monitored, this alarm will be highlighted. Refer to the vendor manual for further troubleshooting.
- Flow Fault — If there is a low flow situation monitored, this alarm will be highlighted. Refer to the vendor manual for troubleshooting.

Figure 6.16 - 3 Chiller – Glycol Pump Status



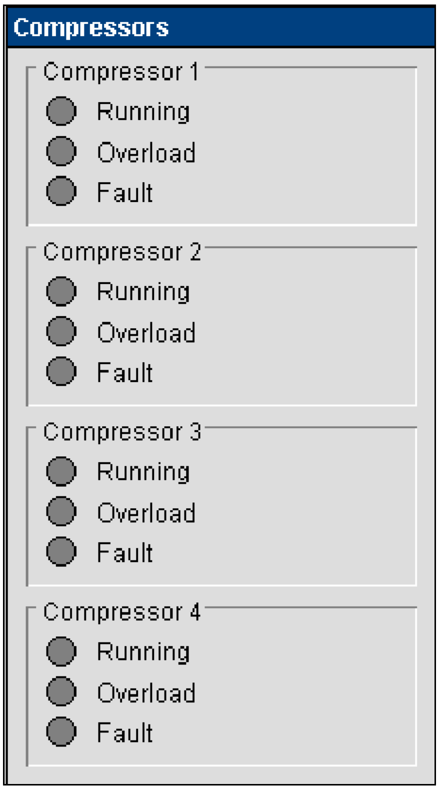
- Set-point Temp. — The blue block allows the user to input the required temperature setting.
- Actual Temperature — This is the temperature feedback as measured on the return to the chiller from the process rollers.

Faults:

- Temperature Fault — If the measured temperature exceeds the low (-30) or high (60) safety temperatures, this alarm will be shown. It can not be reset until the return temperature has returned to within normal levels.

Figure 6.16 - 4 Chiller – Drum Temperature Status

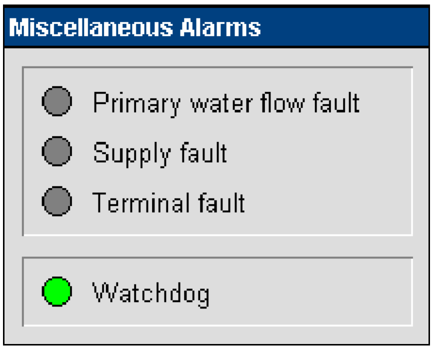
Each of the compressors are activated independently. The colour of the "Running" indicator shows whether they are being used.



Faults:

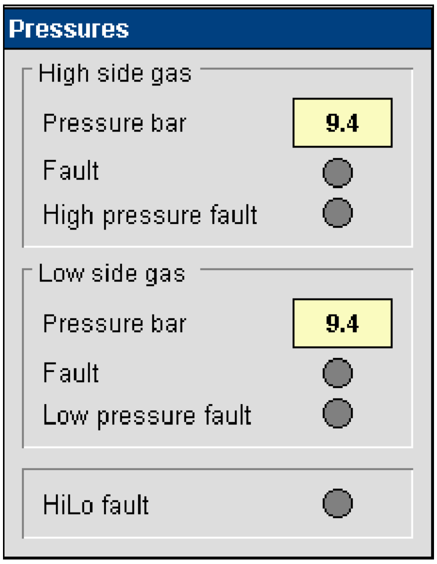
- Overload — If the circuit breaker supplying the individual compressor trips, this alarm will be indicated.
- Fault — If this indicator is illuminated, refer to the vendor manual to allow investigation. The first item to check is the oil level modulator which has its own condition status indicators.

Figure 6.16 - 5 Chiller – Compressors status



- Primary Water Flow — Lack of sufficient cooling tower water flow will highlight this alarm. Check the incoming water supply.
- Supply Fault — If the main power supply varies outside the operating limits of the preset monitor, this alarm will be indicated.
- Terminal Fault — Any critical alarm will also be indicated by this alarm.
- Watchdog — This indicator will flash a regular time span to indicate communications between the main machine PLC and the chiller PLC.

Figure 6.16 - 6 Chiller – Miscellaneous Alarms



High Side Gas

Pressure Value — This is an indication of the common discharge line pressure. This should normally be between 13 and 16 bar under normal running conditions.

- Fault — If there is a fault with the sensor this alarm will be highlighted.
- High Pressure Fault — If there is a high pressure fault, this alarm will be highlighted. Check the vendor manual for troubleshooting

Low Side Gas

- Pressure Value — This is an indication of the common suction line pressure. This should normally be between 1 and 3 bar under normal running conditions.
- Fault — If there is a fault with the sensor this alarm will be highlighted.
- Low Pressure Fault — If there is a high pressure fault, this alarm will be highlighted. Check the vendor manual for troubleshooting
- HiLo Fault — This alarm monitors the oil pressure within the chiller. If highlighted, refer to the vendor manual for troubleshooting.

Figure 6.16 - 7 Chiller – Pressure Monitoring

6.17 Coating Shield System

The coating shield is located on the winding cart can be lowered for cleaning and adjusting the width of the drum edge shields.

The shield is hinged from the bottom of the winding mechanism. A pneumatic actuator, located on the outside of the machine faceplate is used to raise and lower the shield via a vacuum rotary leadthrough.

Control push buttons are located on the front of the winding cart to allow this operation to take place when the winding cart is fully retracted.

Note: The shield cannot be lowered or raised if the E-stop circuit has been activated or the Traverse Lockout is in the OFF position.

Figure 6.17 - 1 Photograph Showing User Lowering The Coating Shield.

The following sequence should be followed to lower the shield;

1. With no weight being exerted downwards, pullout the mechanical support bolt, located at the rear side of the shield.
2. Press the lower push-button until the shield is fully lowered. It will stop automatically.
3. Turn the LOCKOUT isolator through 90 degrees and place the users padlock on it.

The shield is then safe to work upon.

	**Warning !**
	Do not become trapped between the coating shield and the machine at any time. Turn the isolator after lowering the shield and lock it out.

Figure 6.17 - 2 Photograph of Mechanical Retainer on Coating Shield

Figure 6.17 - 3 Photograph Showing User Placing A Lock On The Traverse Isolator

The following sequence should be used to raise the shield;

1. Remove the padlock off the LOCKOUT isolator and turn it to the on position
2. Ensure that the area is clear.
3. Press the raise push-button to raise the shield.
4. The mechanical retainer will automatically engage.

Figure 6.17 - 4 Photograph Showing Drum Shield Lowered.

6.18 Adjusting Coating Width Shields

The drum shield protection system is provided to allow adjustment for different substrate widths to be coated without coating the drum with the aluminium

The shielding is manufactured from water-cooled and non-cooled sections.

Some highlights of the system are;-

- The concave section that locates close to the drum is designed in this manner to stop coating of the drum. It is also concave to reflect any radiant heat away from the substrate.

- The aluminium edge shield section has an extended plate that is a non-cooled. The design allows for slight adjustments of the width, without having to replace any of the non cooled sections of the assembly

The non-cooled sections are located away from the evaporators that are heated up.

The machines are specified for maximum and minimum widths. Sections of shielding are provided to allow covering of the drum for any substrate width between these minimum and maximum widths.

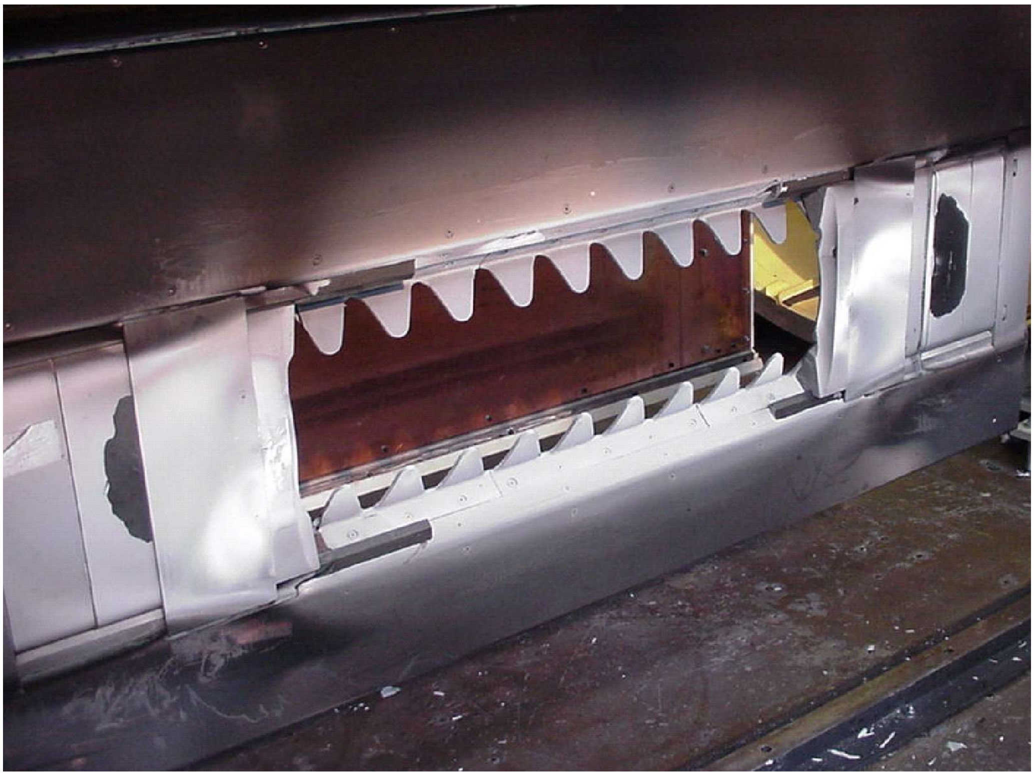


Figure 6.18 - 1 Photograph of Water Cooled Frame in its Lowered Position

Cleaning the Shield

Before attempting to change the shields, the shield requires to be cleaned.

Figure 6.18 - 2 Photograph of operator cleaning the cooled shields with a scraper

Figure 6.18 - 3 Photograph of operator cleaning the cooled frame with a scraper

Attempting to move the shield without cleaning it first will cause problems with items binding.

Adjusting Drum Coating Width Shields

The following instructions must be used when changing the shields;

1. Lower the shield assembly as described previously.
2. Lockout the movement isolator.
3. Clean the shield
4. Slide the cooled sections (aluminium) and the non-cooled (steel) to the centre of the shield for removal. The steel flexible hoses have been manufactured to allow removal of the shields without having to disconnect the water.

Warning !

DO NOT BECOME TRAPPED BETWEEN THE COATING SHIELD AND THE MACHINE AT ANY TIME. TURN THE ISOLATOR AFTER LOWERING THE SHIELD AND LOCK IT OUT.

Figure 6.18 - 4 Photograph of operator removing the water-cooled section

Figure 6.18 - 5 Photograph of operator removing the non-cooled section

5. Insert the non-cooled sections required to protect the drum from stray evaporation.

Figure 6.18 - 6 Photograph of Operator Inserting Non Cooled Sections

Figure 6.18 - 7 Photograph of Operator Replacing the Cooled Section

Move the cooled section, by-hand to the position where it will cover the edge of the film

Figure 6.18 - 8 Photograph of Operator Replacing the Cooled Section

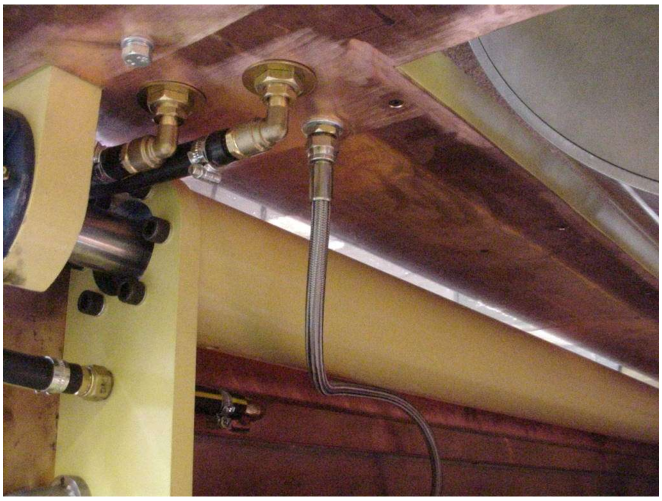


Figure 6.18 - 9 Photograph Showing That Coating Shield Flexible Hoses are Clear

6. Check that the hoses are clear before raising the shield, and check that they are not trapped
7. Raise the assembly using the push-button.
8. With the substrate in the machine (cantered). Mark the inner edges of the substrate with a "marker pen" and inch the drive system. This will indicate the amount of substrate that will be covered by the shields and therefore uncoated.

Figure 6.18 - 10 Photograph Showing Operator Marking Substrate To Check If Edge Shields Are In The Correct Location

9. If the clear edge is excessive or too small, adjust the shield slightly moving it, by hand. The clear edge should be normally 6mm.

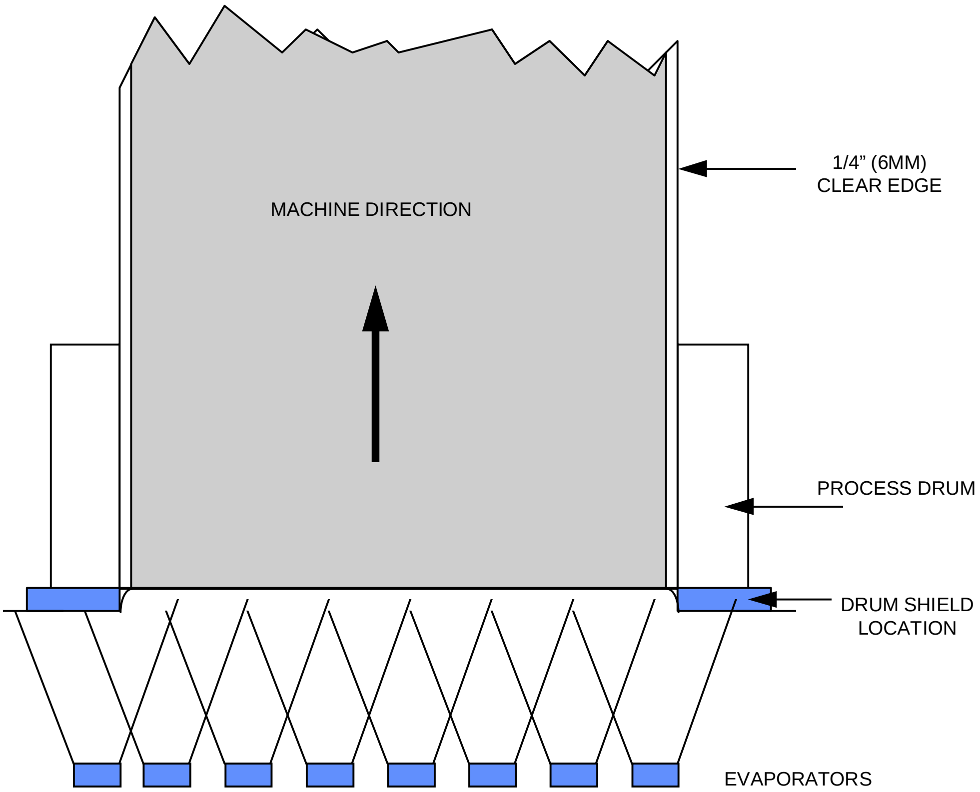


Figure 6.18 - 11 Drawing Showing Position of Drum Side Shields In Relationship To Substrate Edges

	**Warning !**
	DO NOT BECOME TRAPPED BETWEEN THE COATING SHIELD AND THE MACHINE AT ANY TIME. TURN THE TRAVERSE ISOLATOR AFTER LOWERING THE SHIELD AND LOCK IT OUT.

6.19 Adjustment/Removal of Coating Shield Process Sine Waves

To reduce heavy deposition areas on the substrate at high evaporation rates, a sine wave shield has been designed to be attached to the coating shield. This is split into a number of smaller sections for ease of handling.

Dependant upon the deposition rate (wire feed), the sine waves selected will vary. As a guide, with wire feed rates over 90 cm/min (2.0mm OD); Sine waves should be fitted to both sides of the opening.

At wire feed rates between 60-90 cm/min, a single set should be fitted on the lower side (Coating shield in is lowered position)

At any wire feed rate less than 60 cm/min, the sine wave shield can be removed totally.

Where the sine wave shield has been removed, copper blanks have been supplied to replace them and protect the mating surfaces.

	Caution !
	**Always use either a sine or a copper blank plate to stop potential damage to the mating surfaces.**

Hexagonal head screws have been provided to ensure that they can be cleaned of aluminium and removed.

Figure 6.19 - 1 Photograph of Operator Removing Holding Bolts from Sine Wave Shield.

On the rear of the shield, at the mating surface, a piece of 0.5mm thick graphite tape is used to ensure good heat transfer from the copper shield to the cooled zoning shield. It is essential that this material is used at all times.

Caution !

Do not operate the machine without a graphite layer between the mating surfaces or without all the required bolts being installed and tightened

Figure 6.19 - 2 Photograph of the operator removing the shield

Figure 6.19 - 3 Photograph of the operator showing the graphite on the rear of the shield.

Solar Film Applications

There maybe times, such as when light optical density coatings are required for solar film, etc.

At these times, the sine wave maybe removed and a straight edge cover can be installed.

6.20 Internal Lighting

The internal lighting within the chamber is switched on at all times when the power is supplied to the machine.

If a bulb requires changing, isolate the main machine power isolator before carrying out the operation.

	**DANGER** **Beware Live Electrical Circuits, Isolate before working in this area**

There is a function to switch off the background chamber lighting. This is achieved by touching the FRONT LIGHT button on the process screen.

This can only be activated when the vacuum level is below 1E-3 mbar or at atmosphere.

6.21 E-Stop Circuit

The emergency E-STOP circuit is provided to de-energise all the machine rotating equipment that is supplied by electrical power, when the E-STOP button on the operator panel is pressed.

Figure 6.21 - 1 Photograph of Operator Interface Panel

The following equipment will be isolated.

- Vacuum Pumps
- Chiller
- Plasma Treater (Option)
- Drive System
- Wire Feeders
- Evaporation Power supply

Resetting after a E-STOP

There is a pushbutton on the operator panel which requires pressing to reset the E-stop circuit and to enable the electrical circuits.

	Warning !
	**Ensure that the machine is in a safe condition before carrying out this procedure**

It will flash when it requires to be activated.

Resetting on Machine Start Up

The RESET push button operation is the same on machine start up

6.22 Guard-Stop Circuit

The Guard -STOP circuit is provided to de-energise the following equipment:-

- Cart Traverse Movement
- Coating Shield Movement
- Valve VV7 Movement
- Plasma Treater Power
- Shutter Movement
- Viewing Flap Movement
- Winding System Movement

It is triggered by rotating the traverse movement isolator to its OFF position

Resetting

There is a pushbutton on the operator panel which requires pressing to reset the Guard-stop circuit and to enable the electrical circuits.

	Warning !
	**Ensure that the machine is in a safe condition before carrying out this procedure**

It will flash when it requires to be activated.

Figure 6.22 - 1Photograph of Operator Interface Panel

Resetting on Machine Start Up

The RESET push button operation is the same on machine start up

6.23 Auto- Slowdown Function

The automatic stopping of the drive system at the end of the process cycle is classified as an “auto slowdown”. The machine will sense when the machine is approaching the ‘FINAL DIAMETER’ entered by the operator in the ROLL SET UP screen.

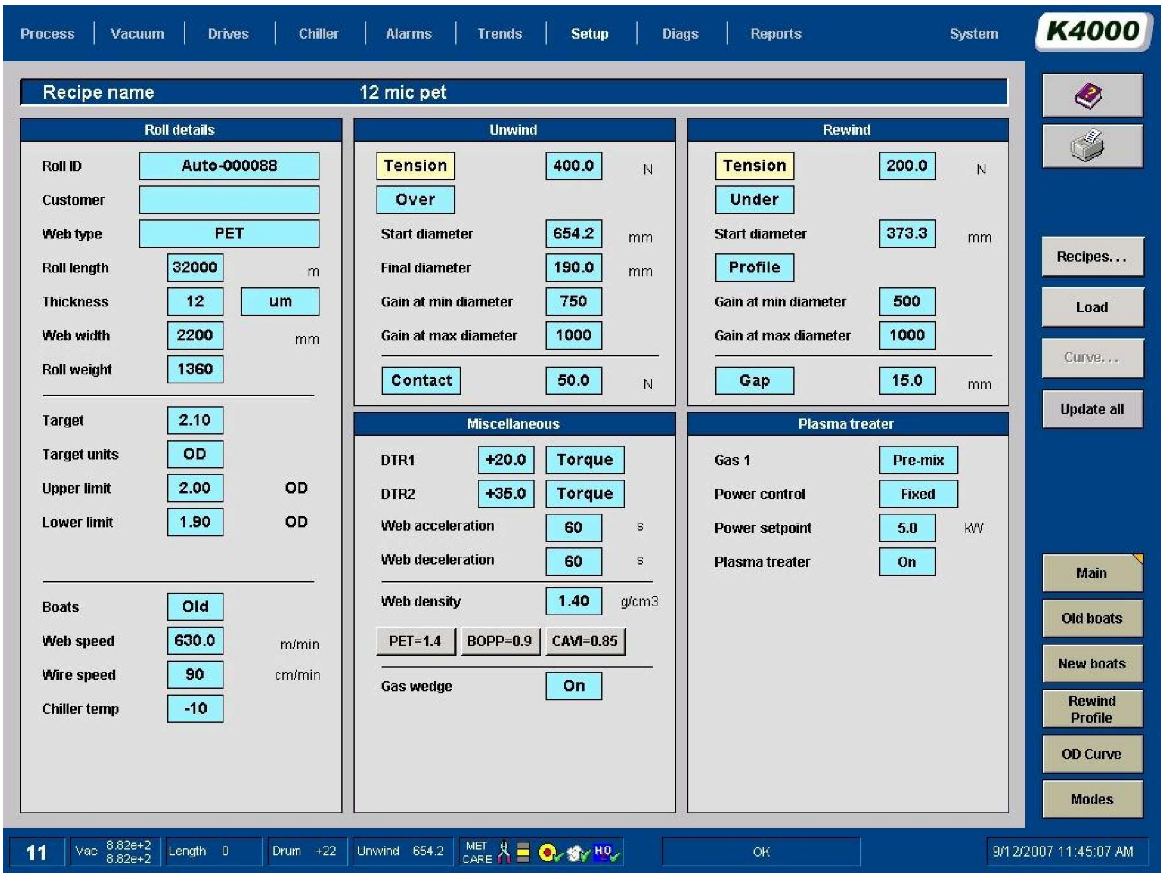


Figure 6.23 - 1 SETUP Screen

At this point, the drive system will start to slow down and the shutter will close. The wire to the evaporators will be turned off and after a further 30 seconds the power to the evaporators will be removed.

Further to this the chiller will switch into a reheat mode and the machine will automatically vent approximately 4 minutes later.